

# Validation of converted wrist model

The converted MJC model has been tested under the three categories below and with the accuracy of:

## - Step 1: XML Conversion Validation

Check multi-body forward kinematics (using endpoints), approximation of custom/coupling joints & conditional/moving path points

*Mean error:nan cm; std: nan cm*

## - Step 2: Muscle Kinematics Validation

Check muscle moment arms as indication how muscle wrap over joints

*Mean error:0.7382 cm; std: 0.5528 cm*

## - Step 3: Muscle Kinetic Validation

Check muscle force-length relationship as indication of how similar of them in generating forces

*Mean error:0.0926 Fmax; std: 0.0664 Fmax*

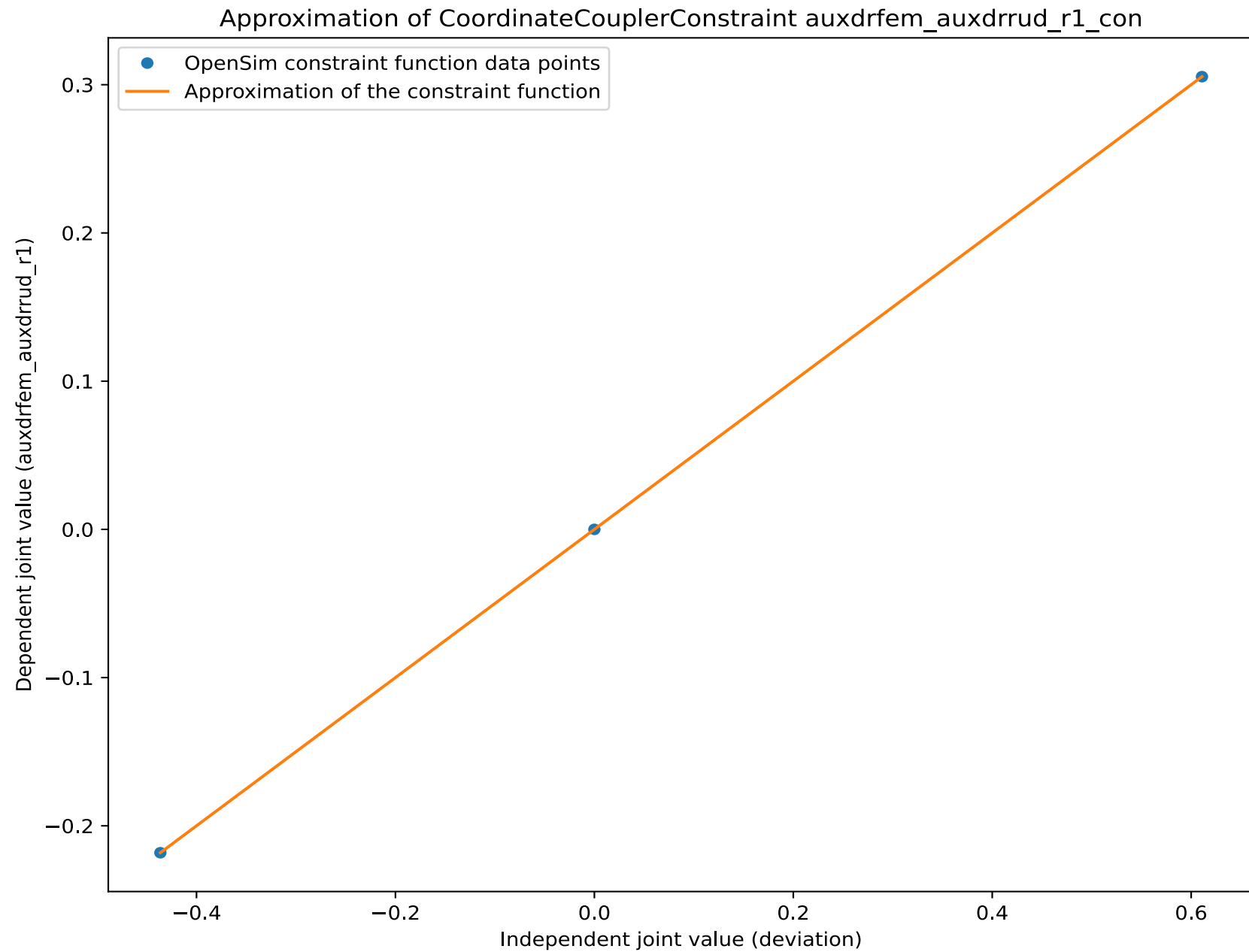
# Step 1: xml Conversion Validation

Randomly pose the model with 10 configurations within the joint limits. In each posture, the endpoints(markers) global locations of Osim and Mjc models are extracted and compared. Box plot of their mean-std errors are plotted together. Individual endpoint differences of these 10 postures are also plotted in the VLT folder, but not included inside this report.

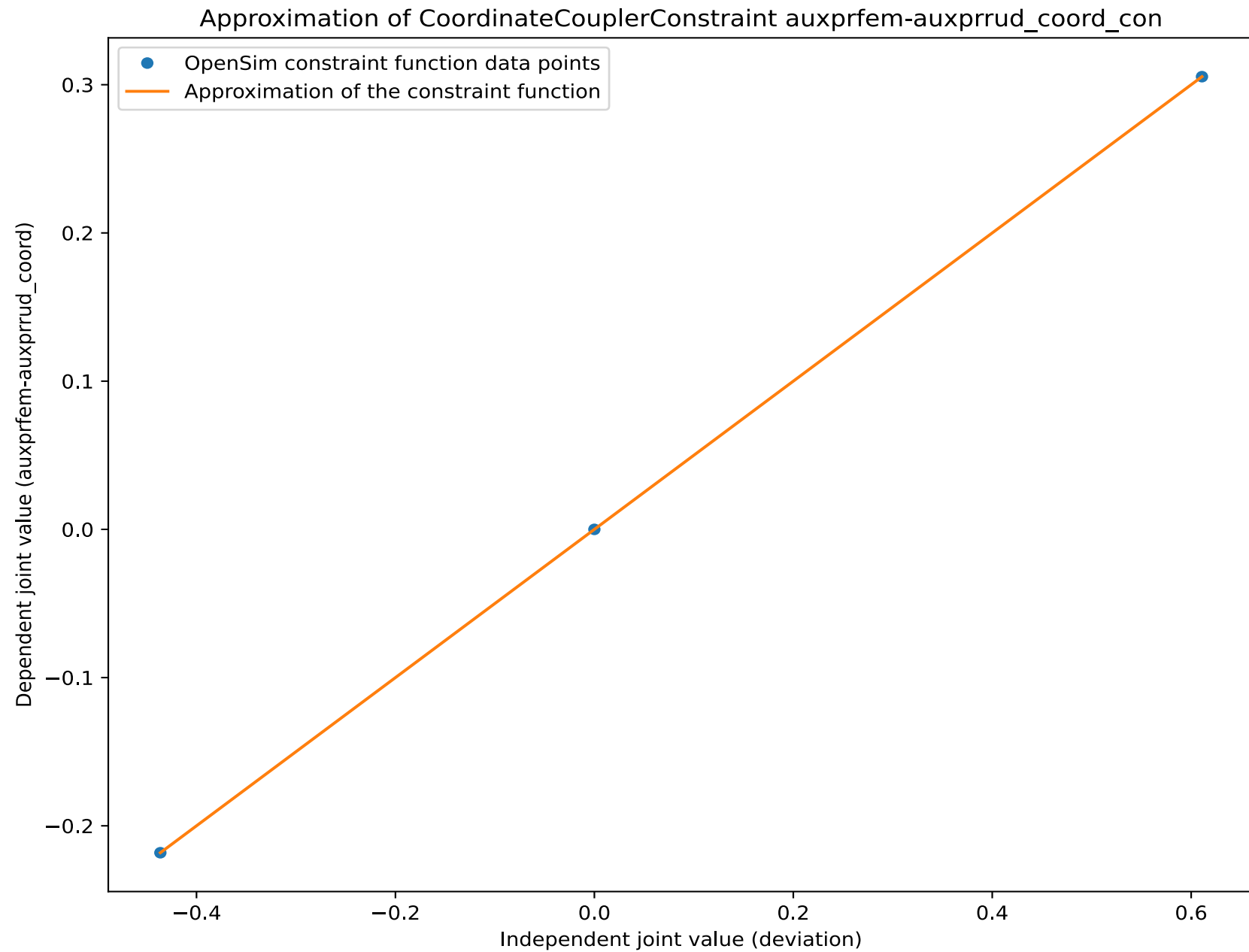
Besides the endpoint check, the approximation of customer joints, coupling joints, conditional/moving path points are plotted and attached. In these plots, blue dots/lines represent their setup in the OpenSim model. Yellow dots/lines represent the approximations in the MuJoCo model.

**NO End Points found in the model.**

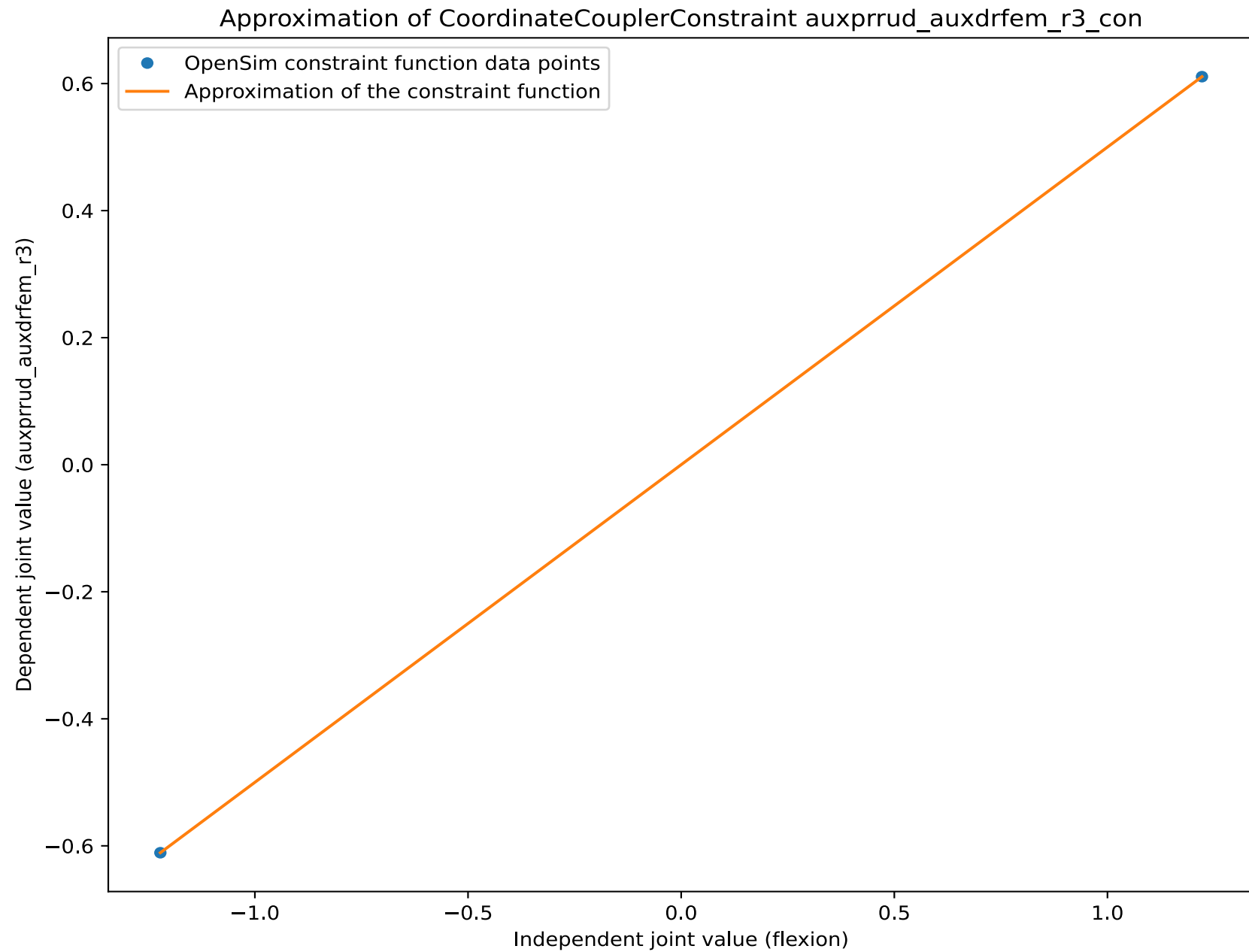
# Approximation of coupling joints



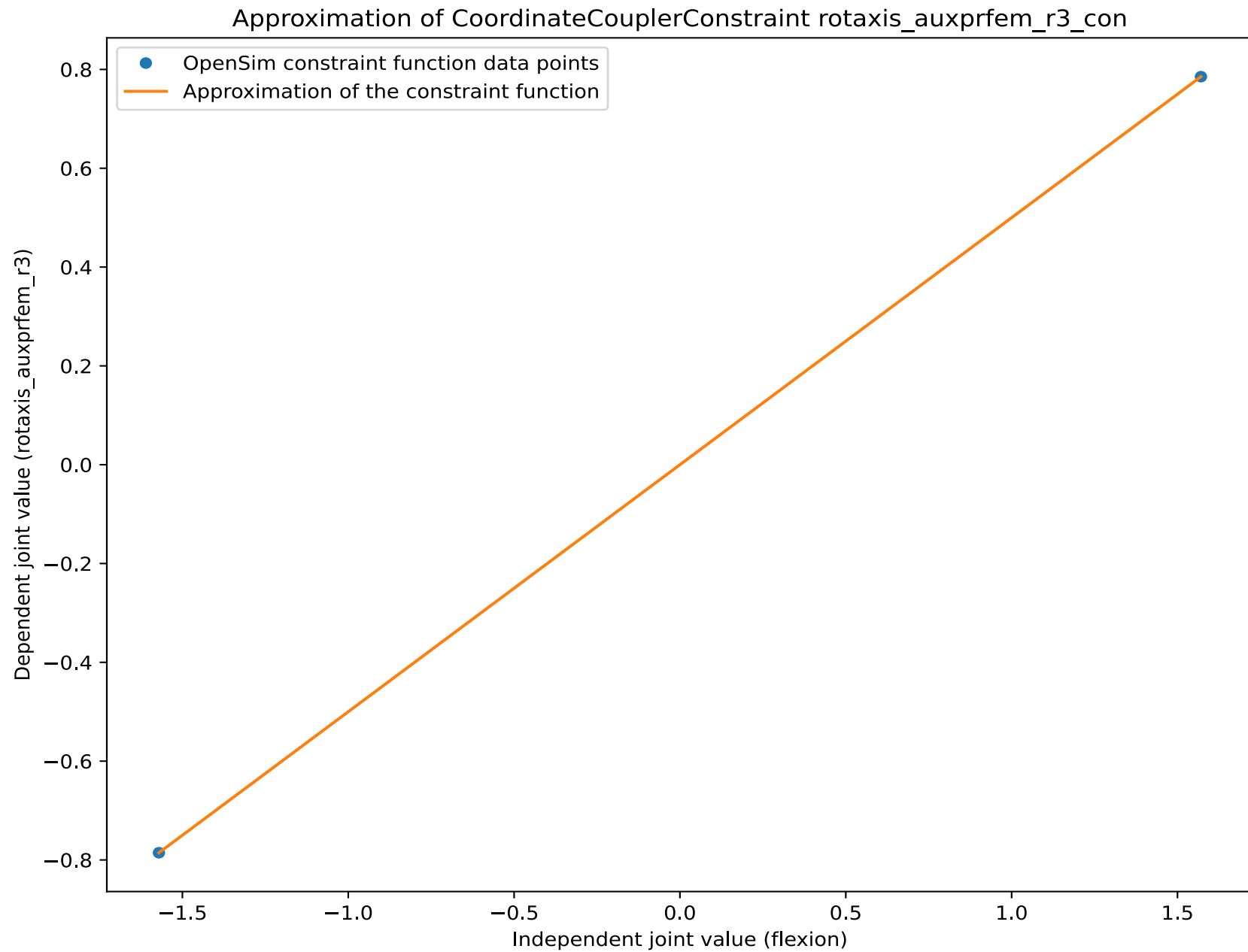
# Approximation of coupling joints



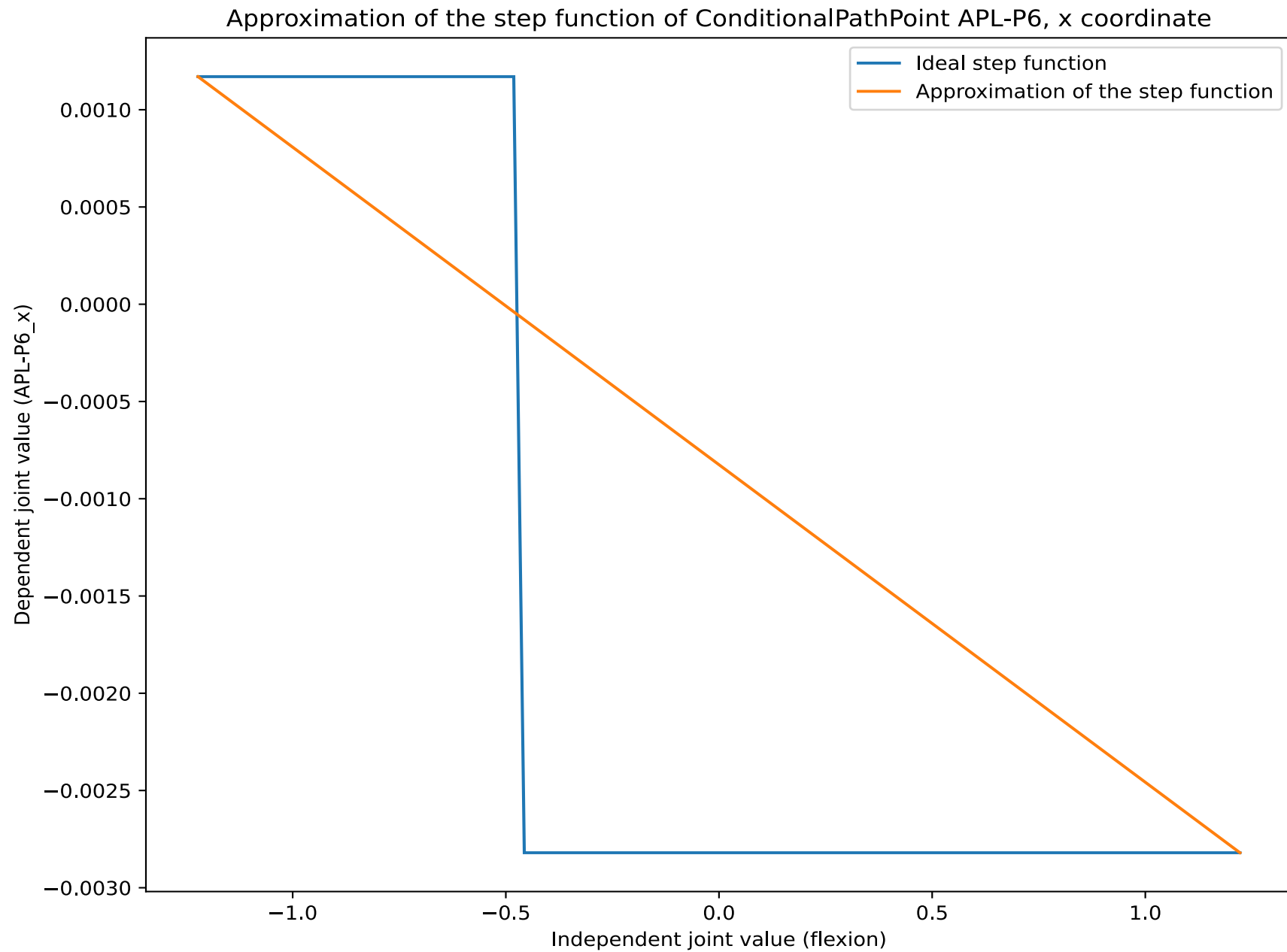
# Approximation of coupling joints



# Approximation of coupling joints

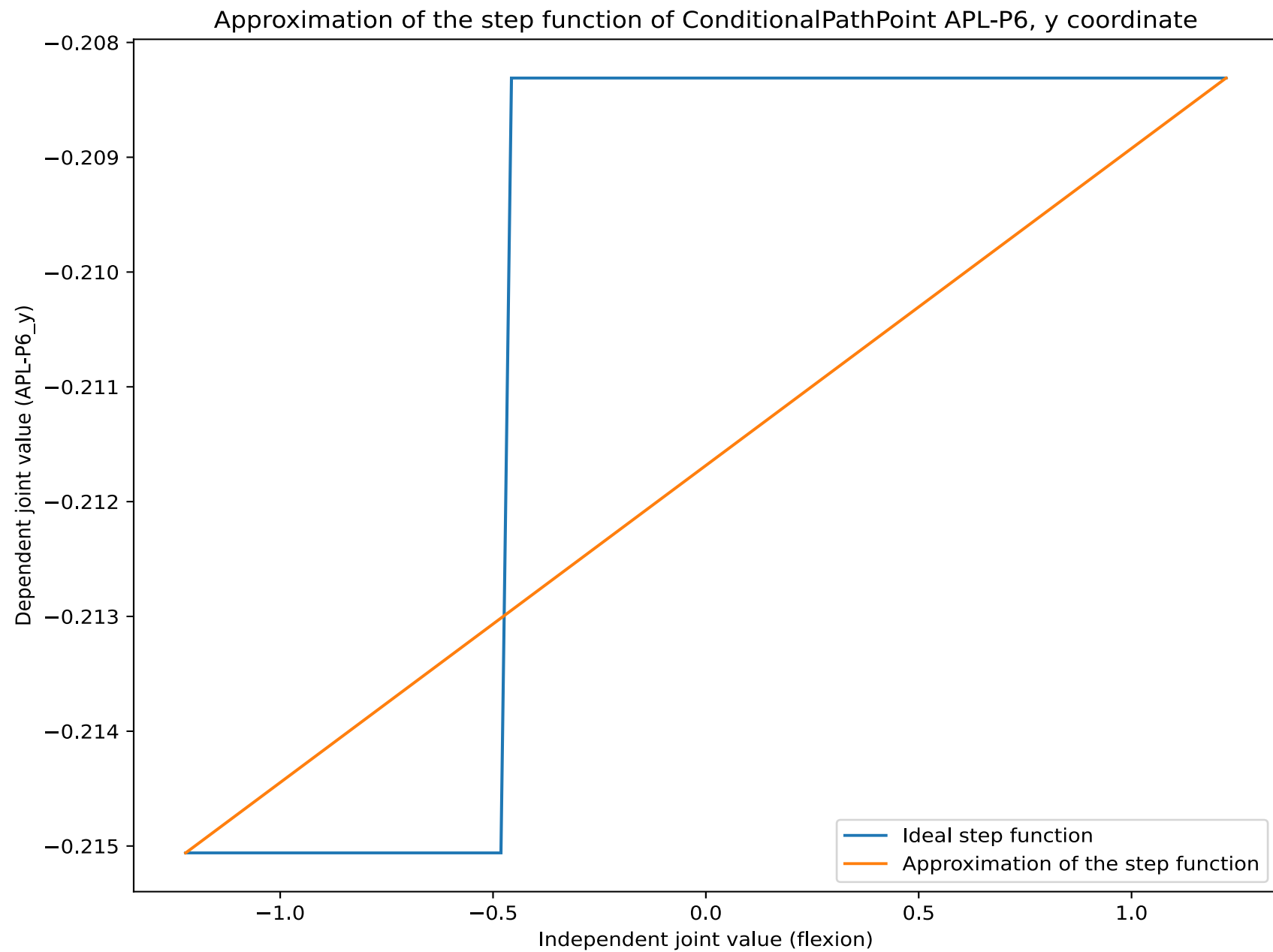


# Approximation of conditional path points

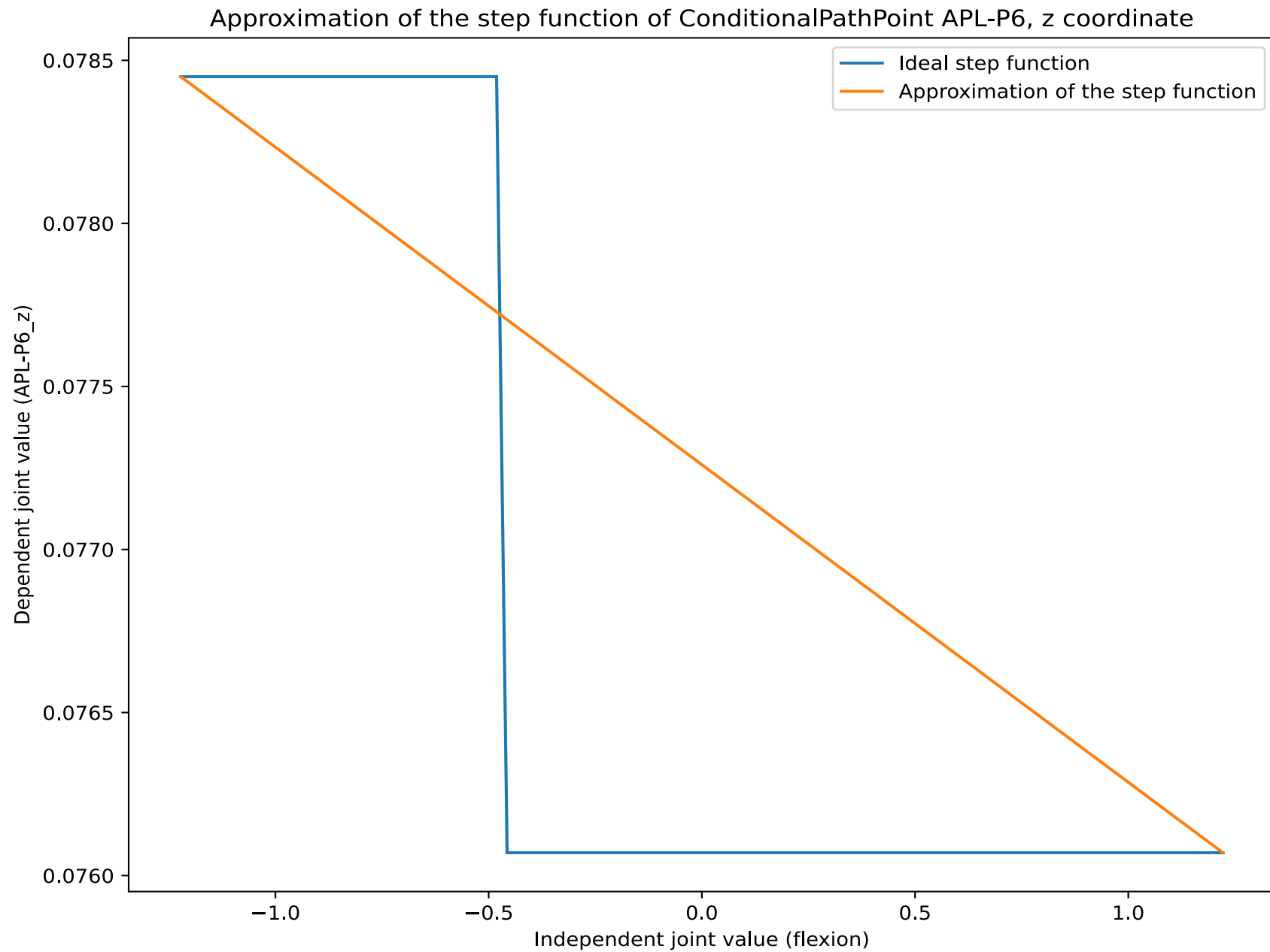




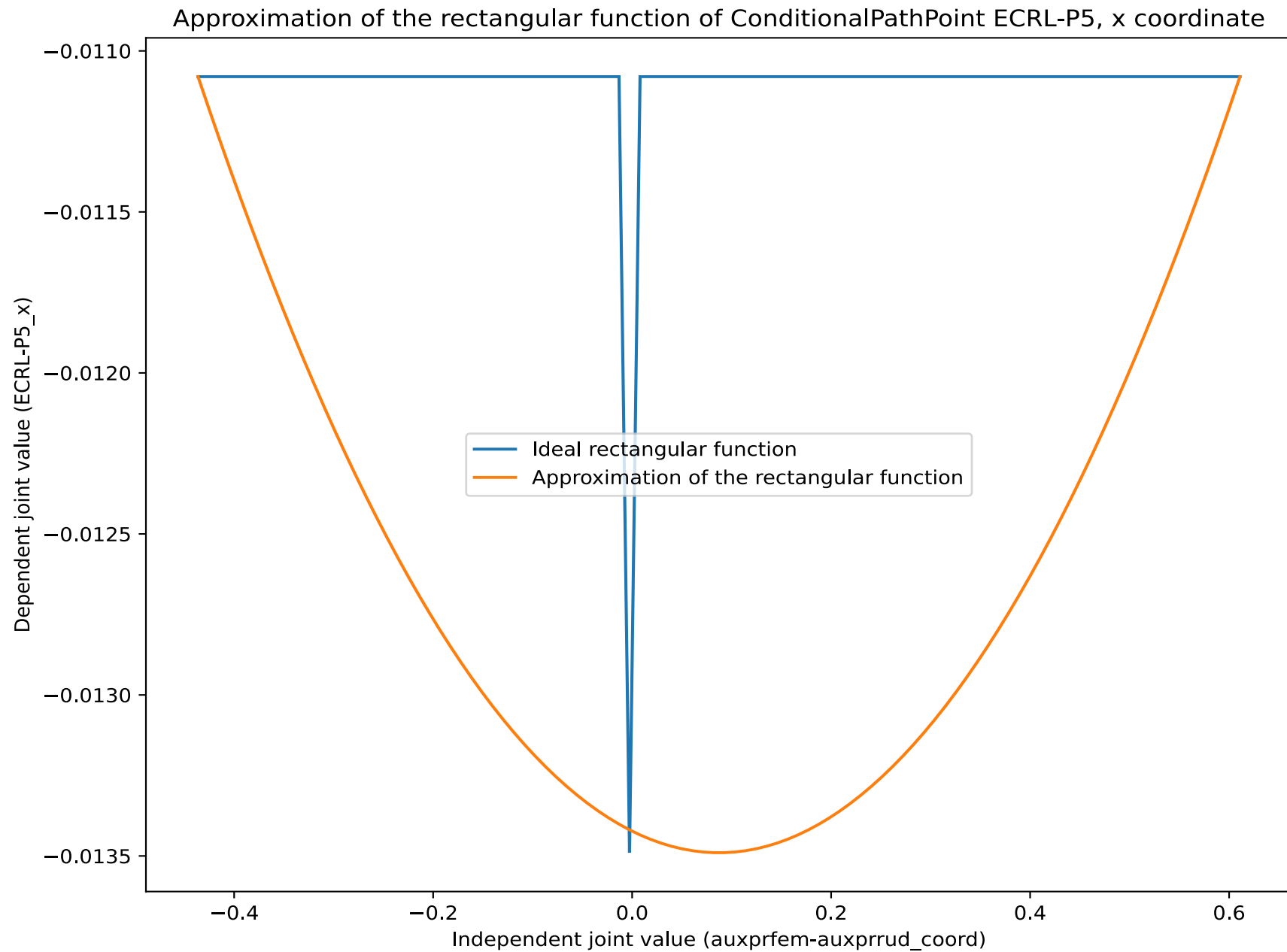
# Approximation of conditional path points



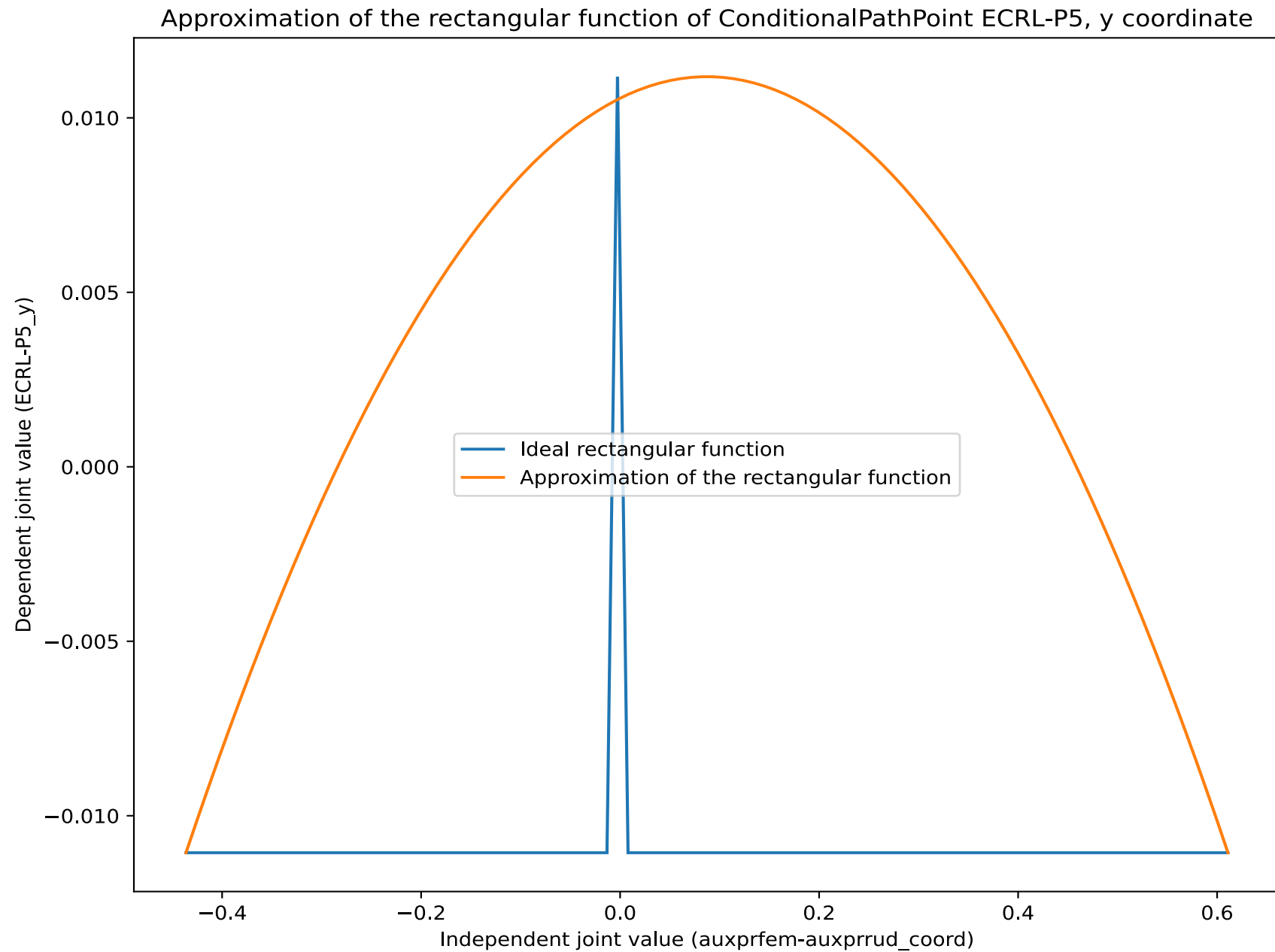
# Approximation of conditional path points



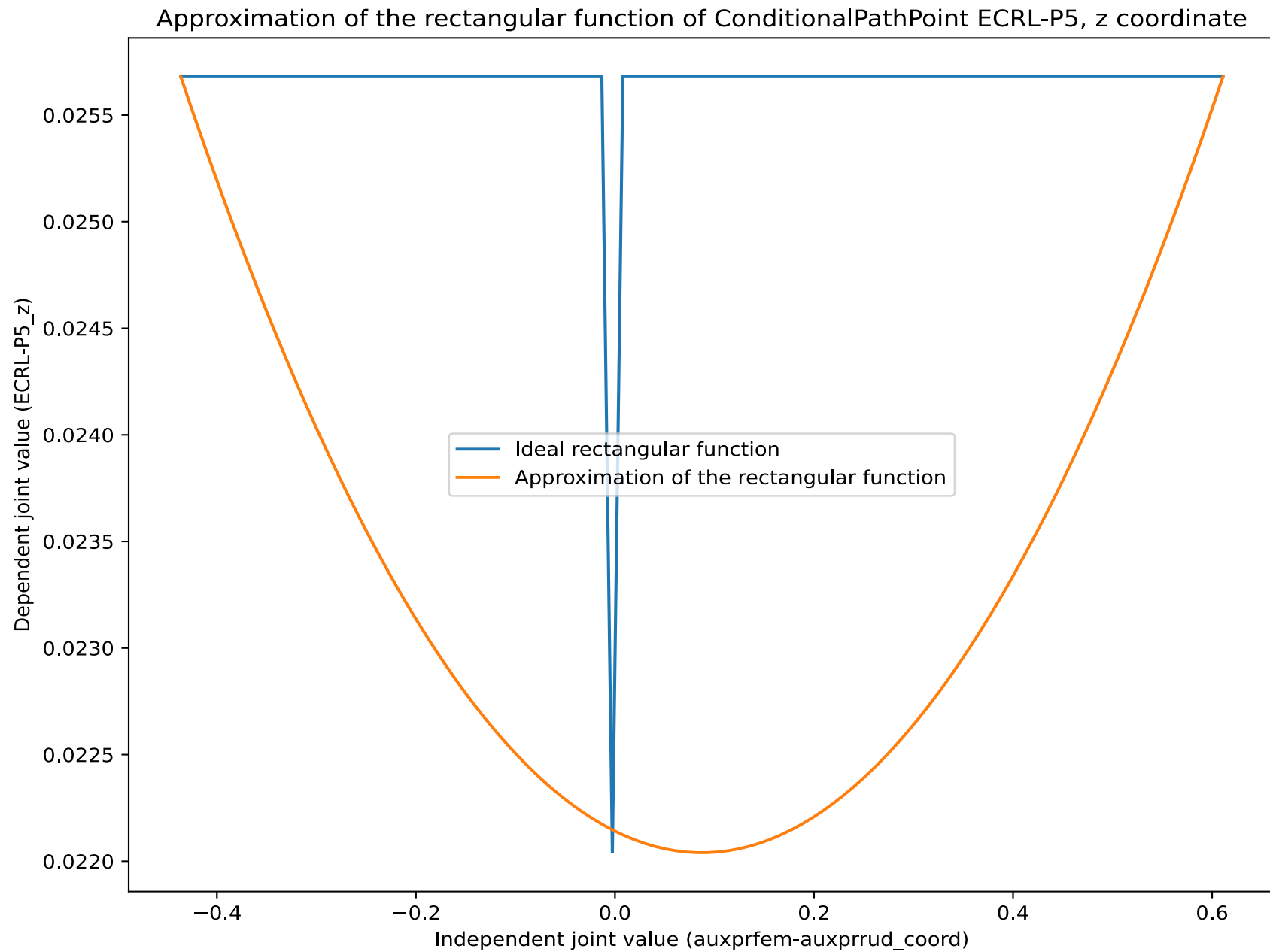
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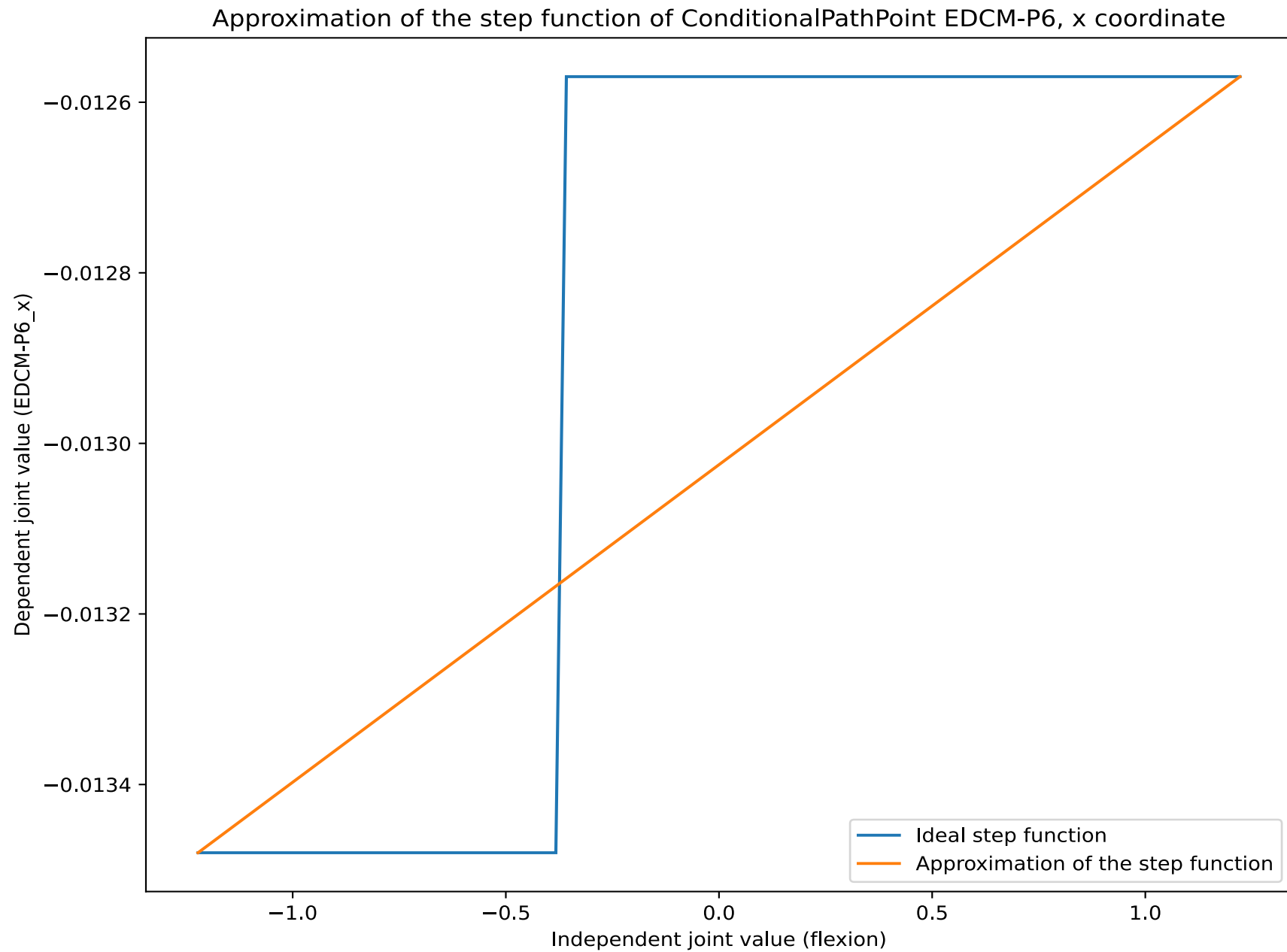
# Approximation of conditional path points



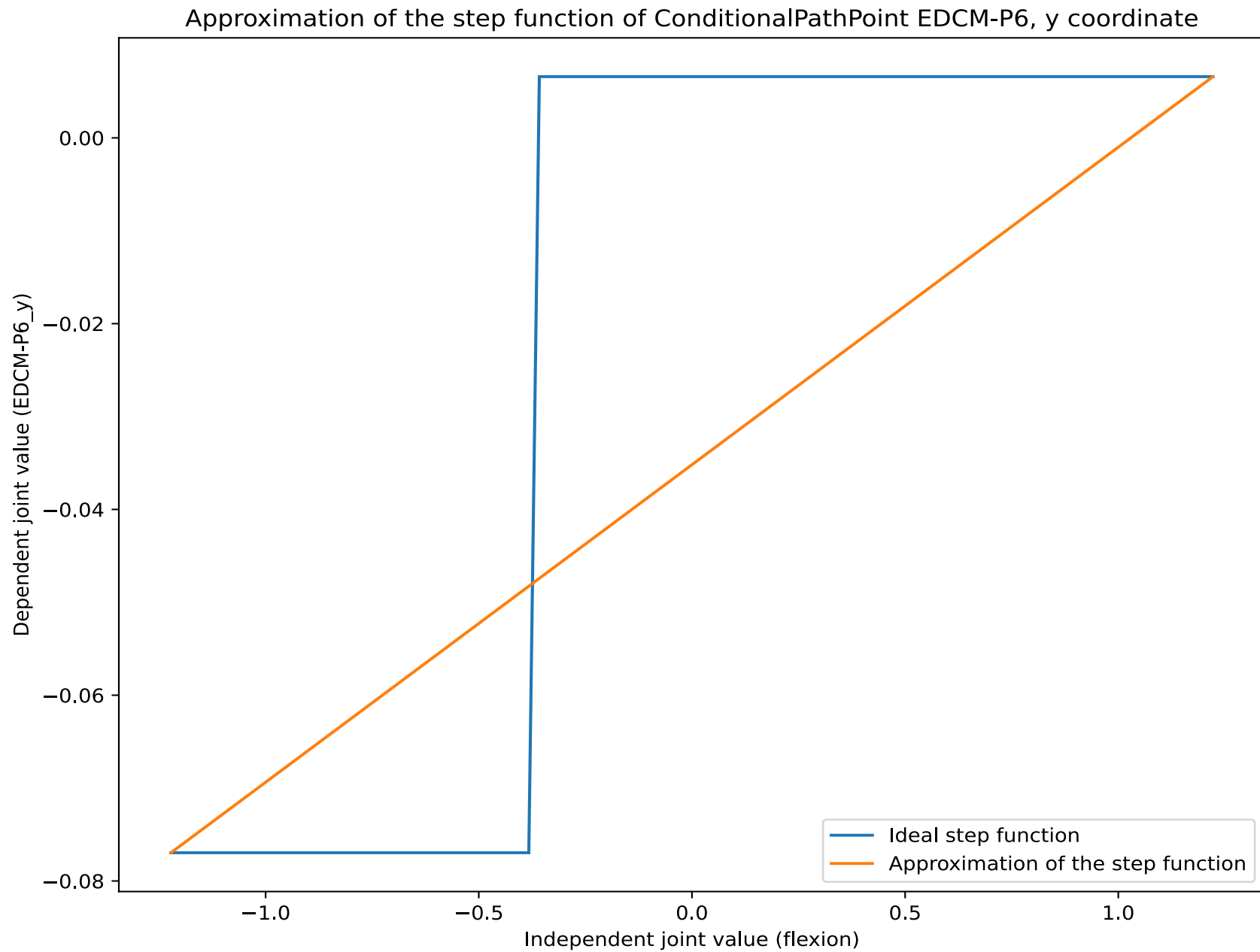
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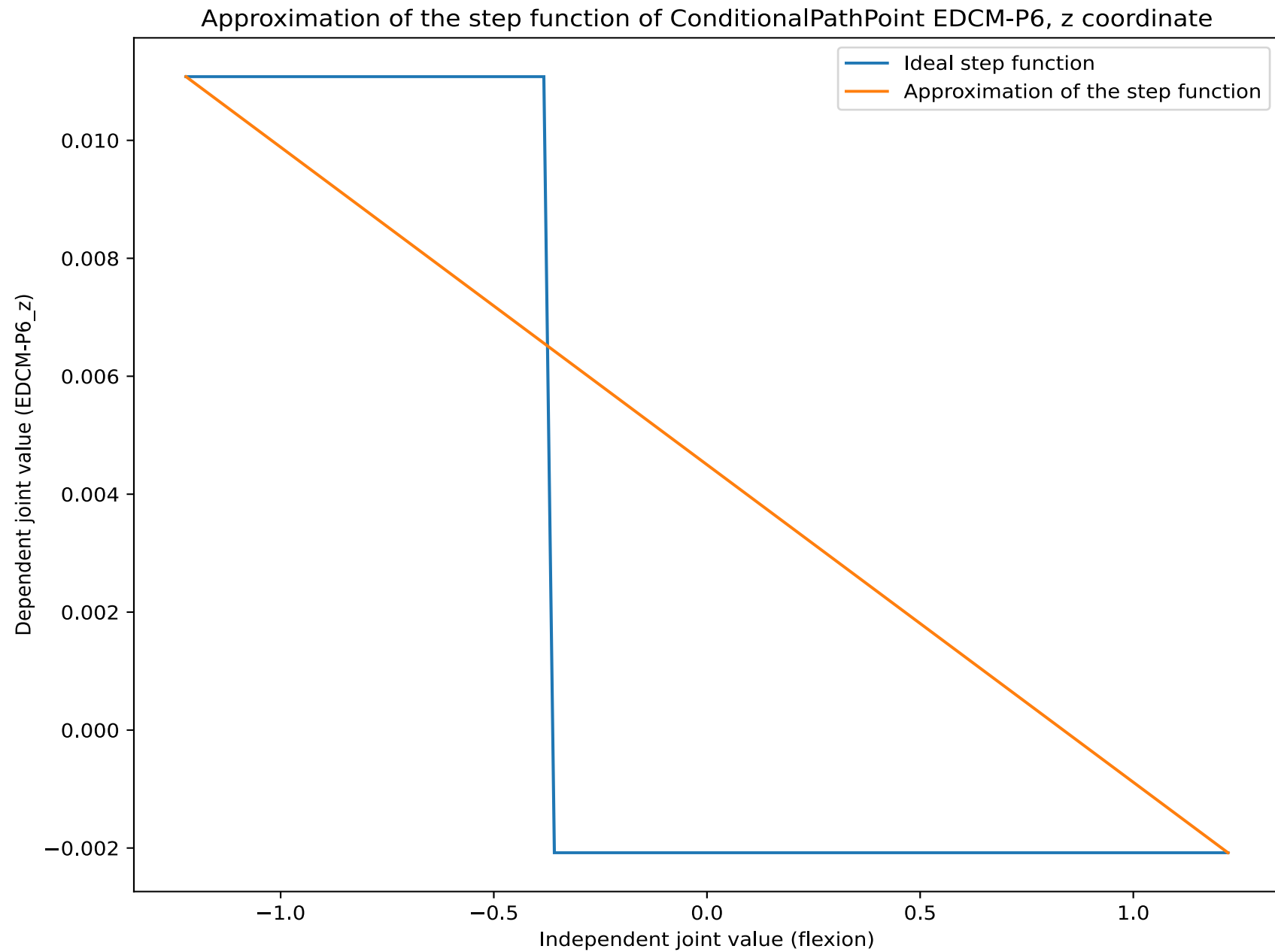
# Approximation of conditional path points



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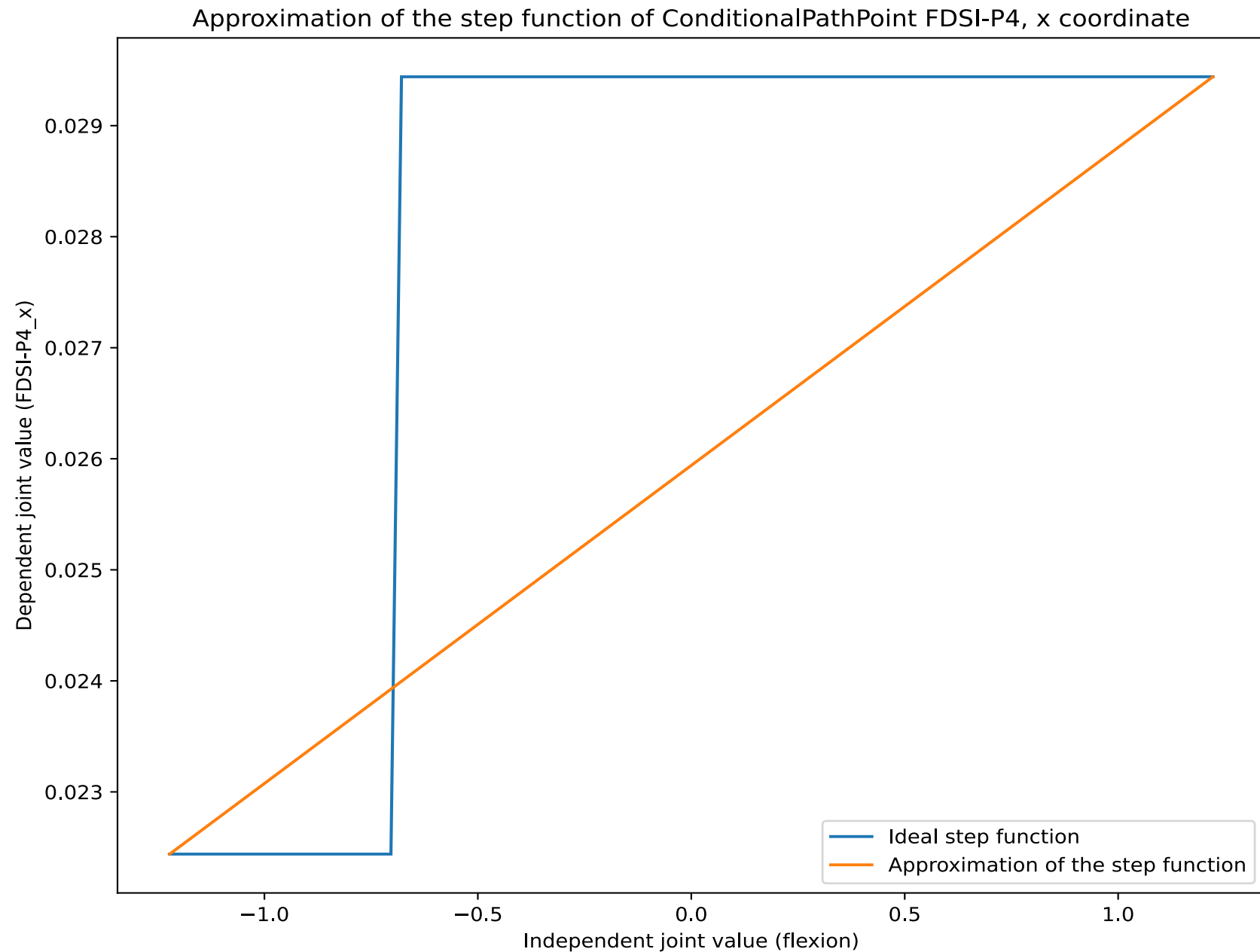


# Approximation of conditional path points

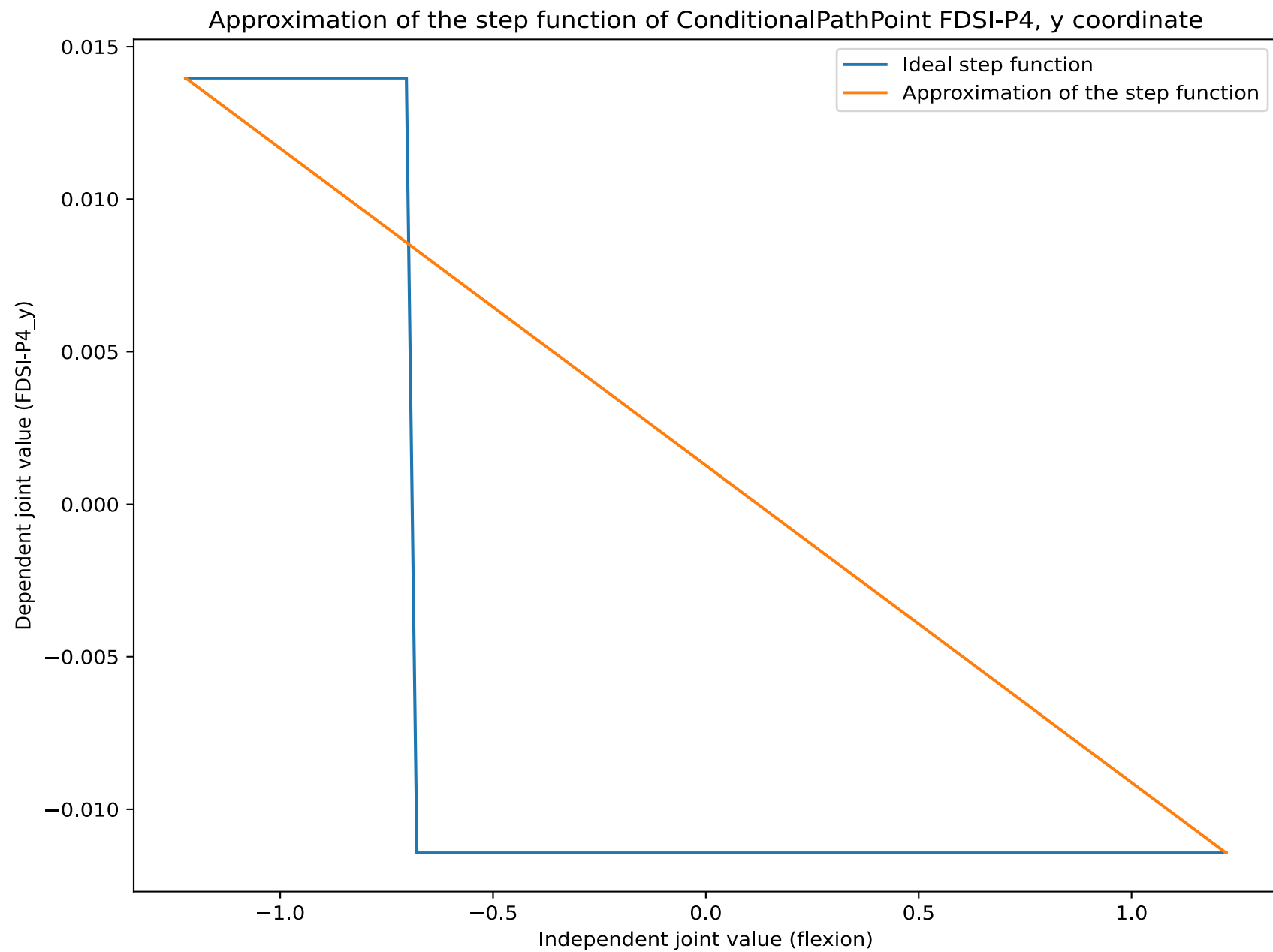




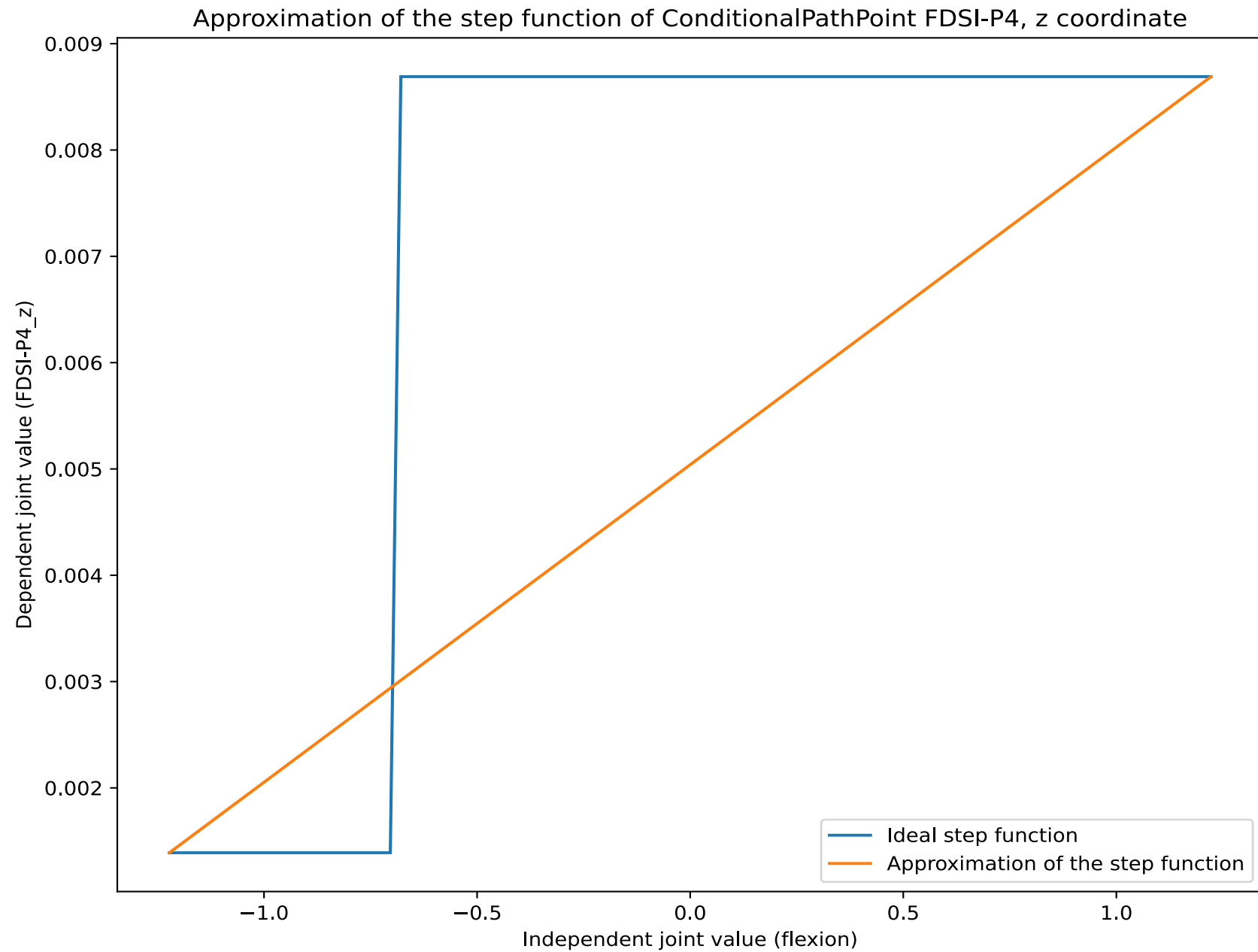
# Approximation of conditional path points



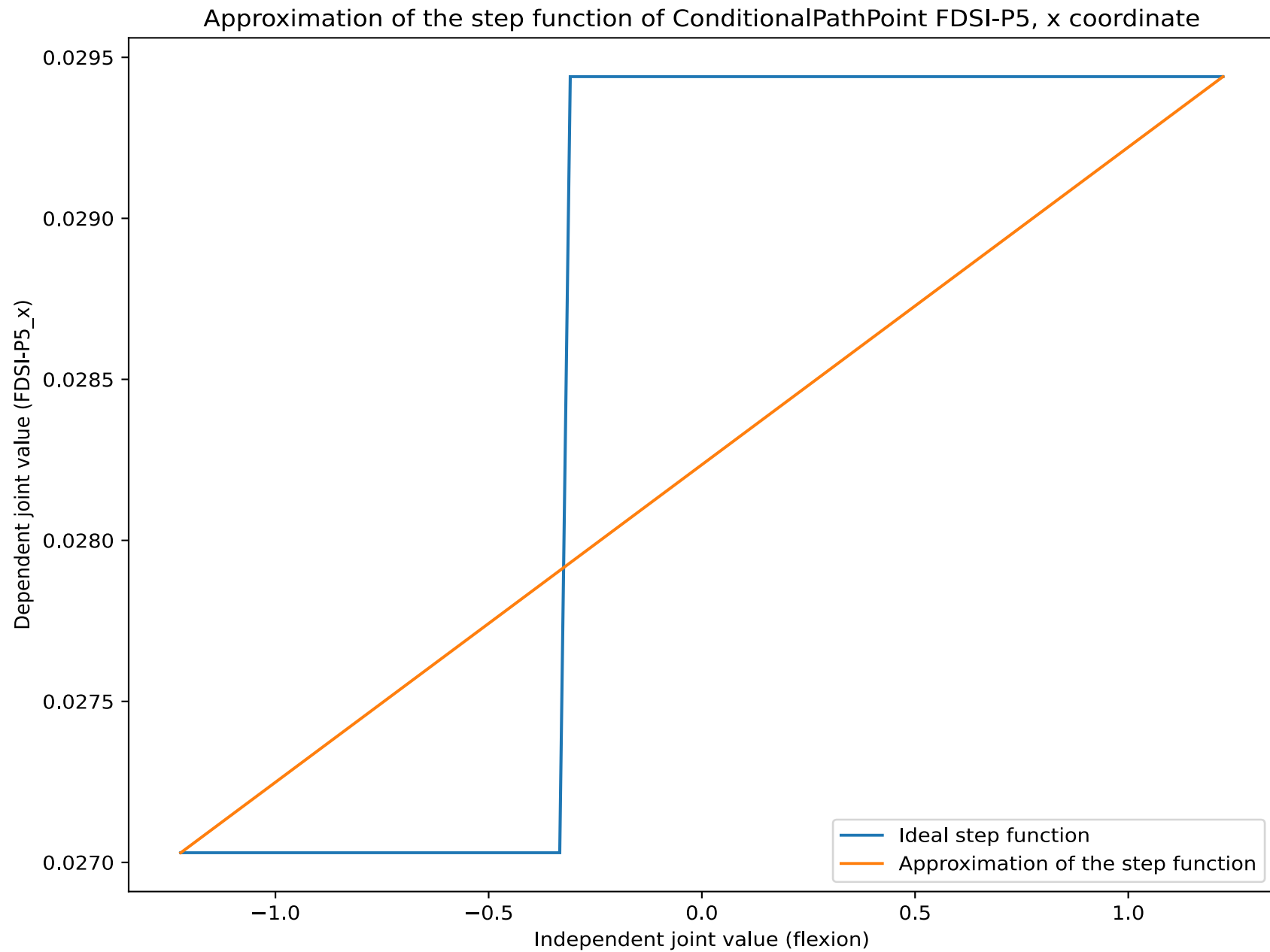
# Approximation of conditional path points



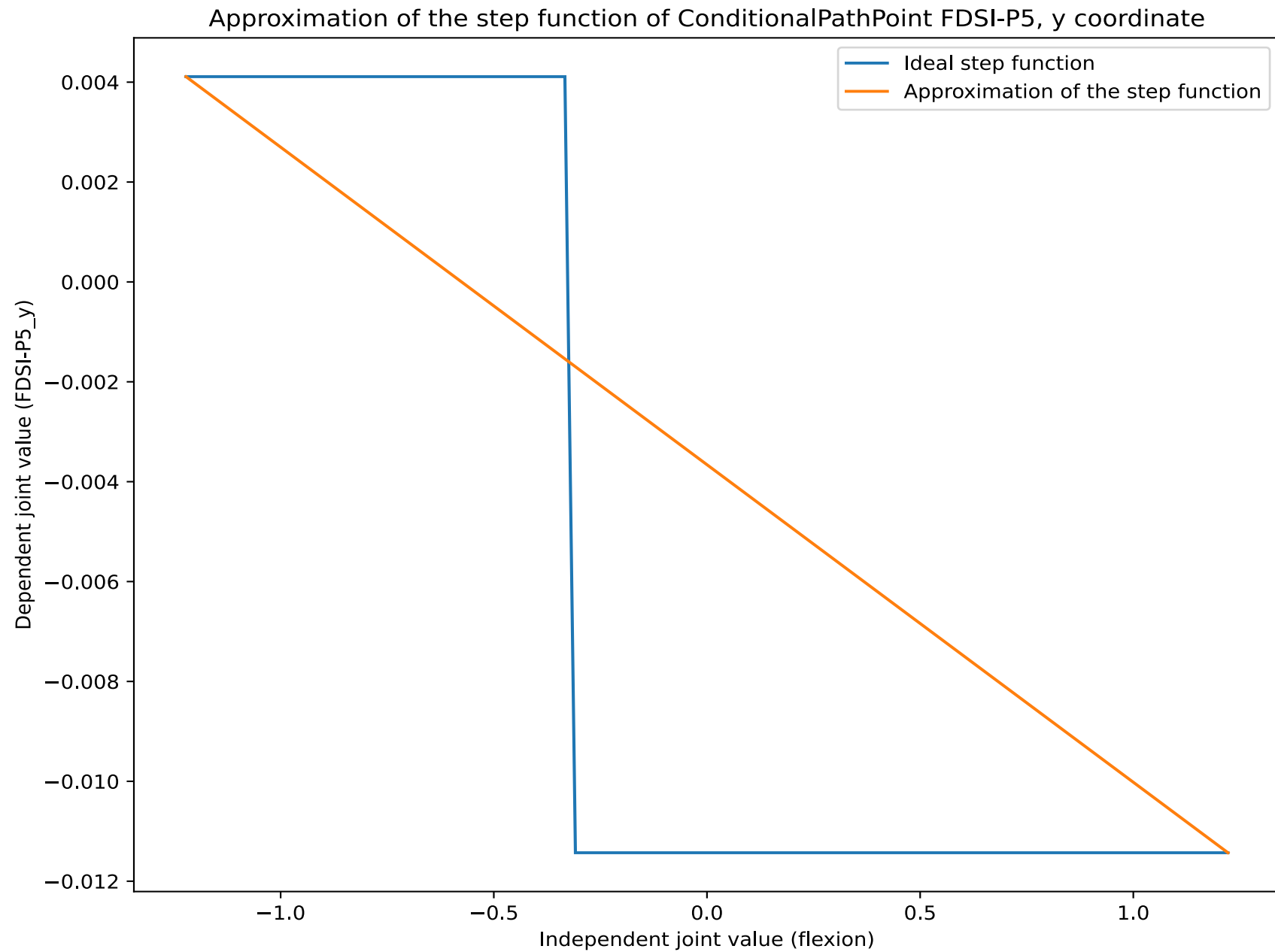
# Approximation of conditional path points



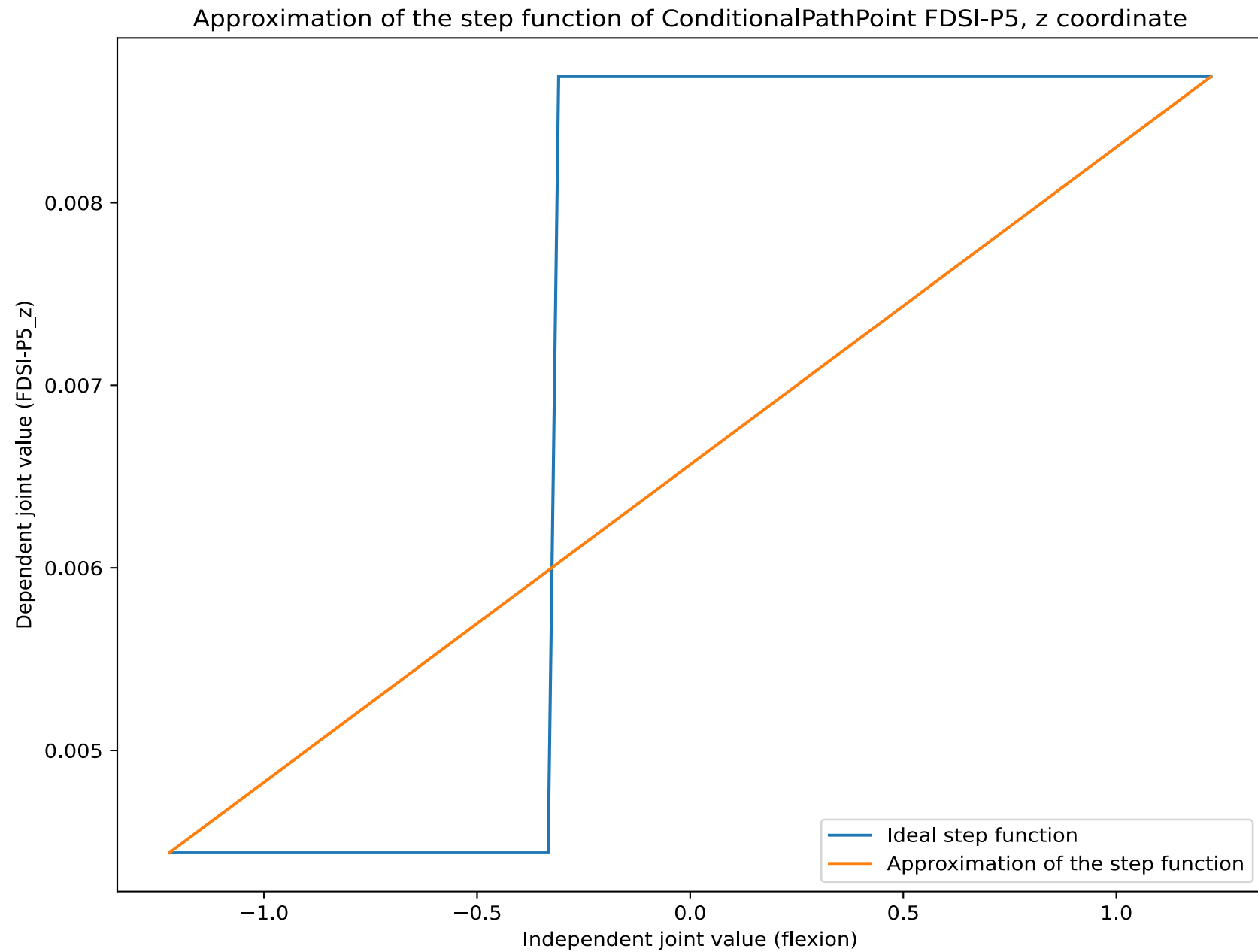
# Approximation of conditional path points



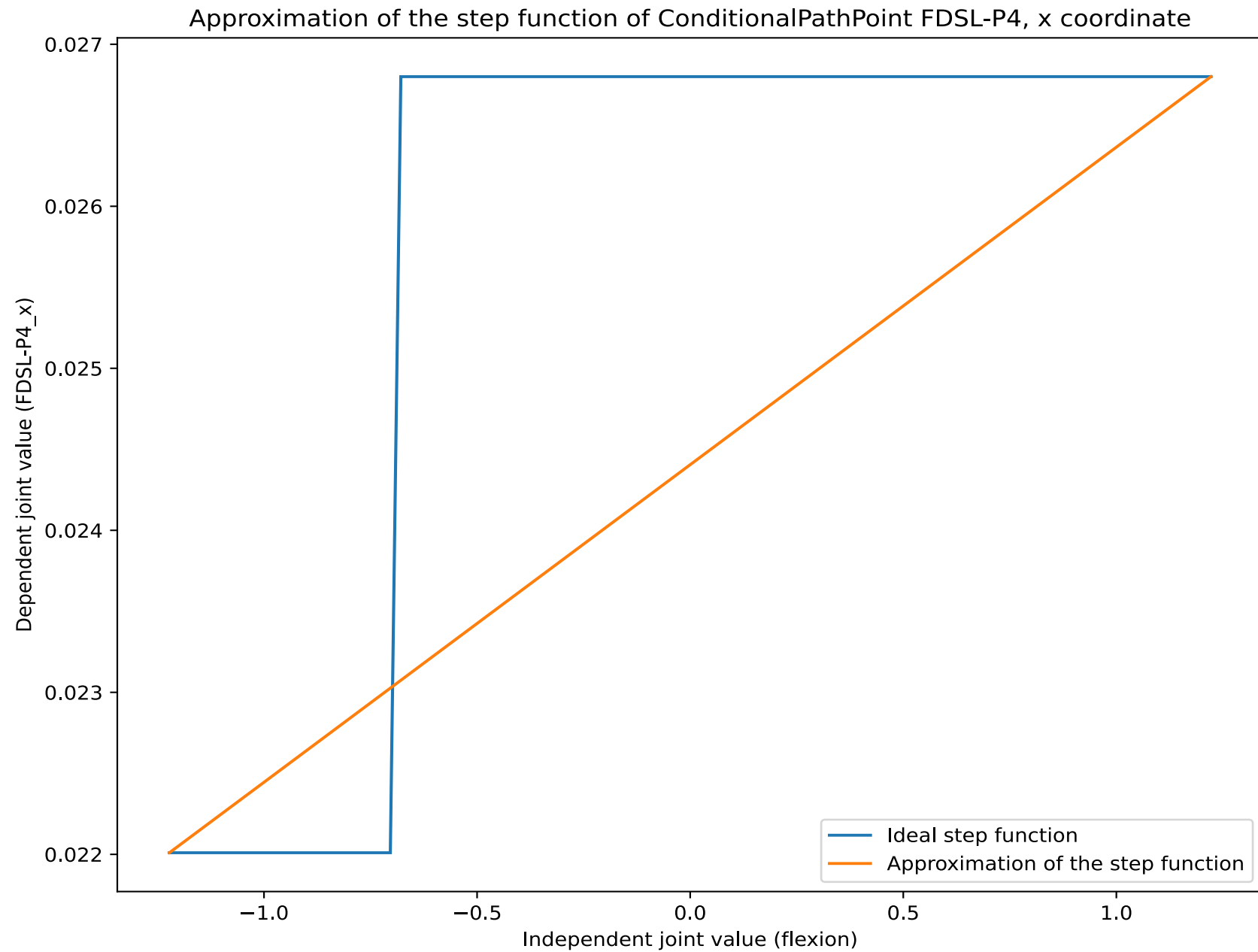
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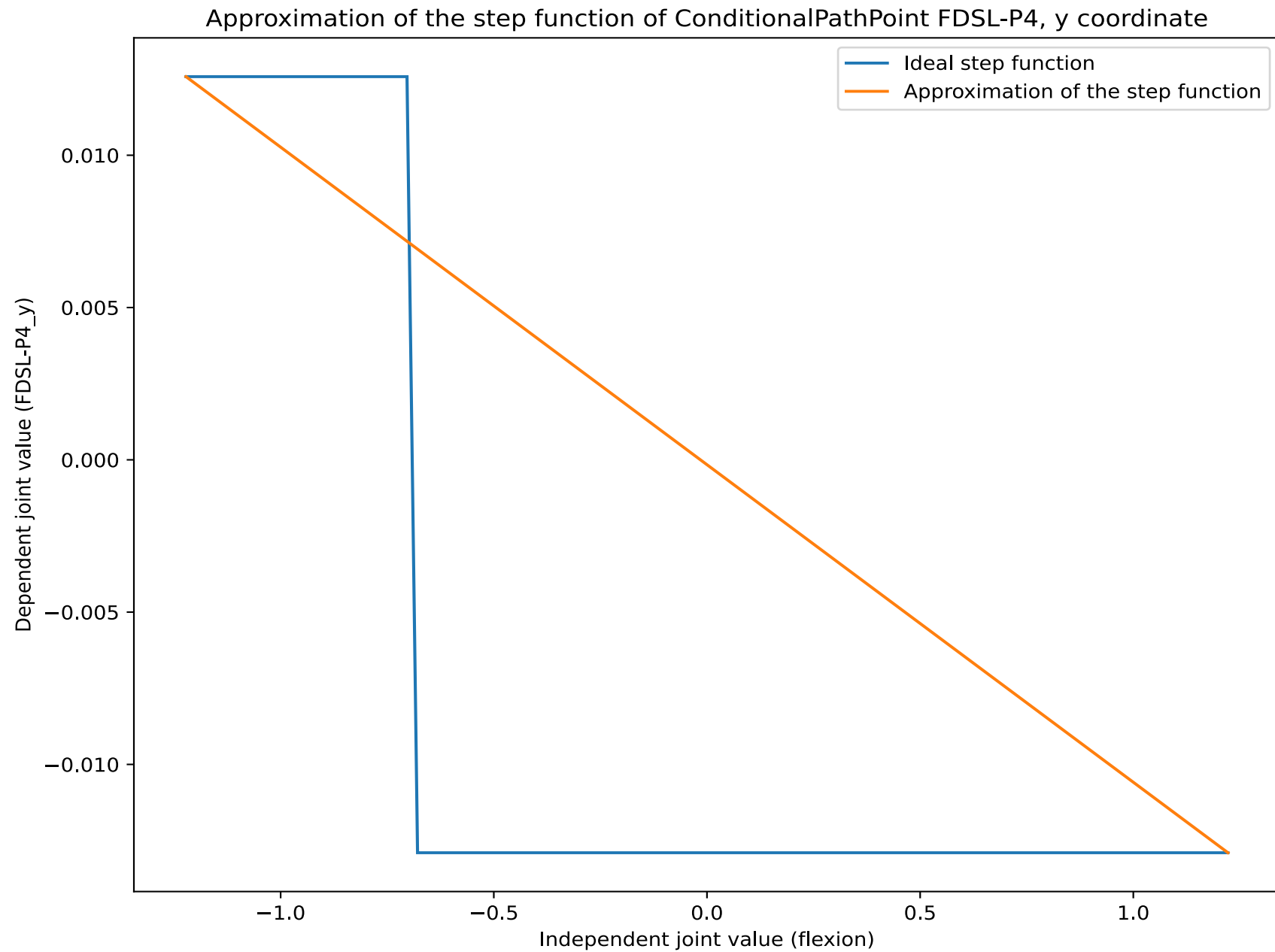
# Approximation of conditional path points



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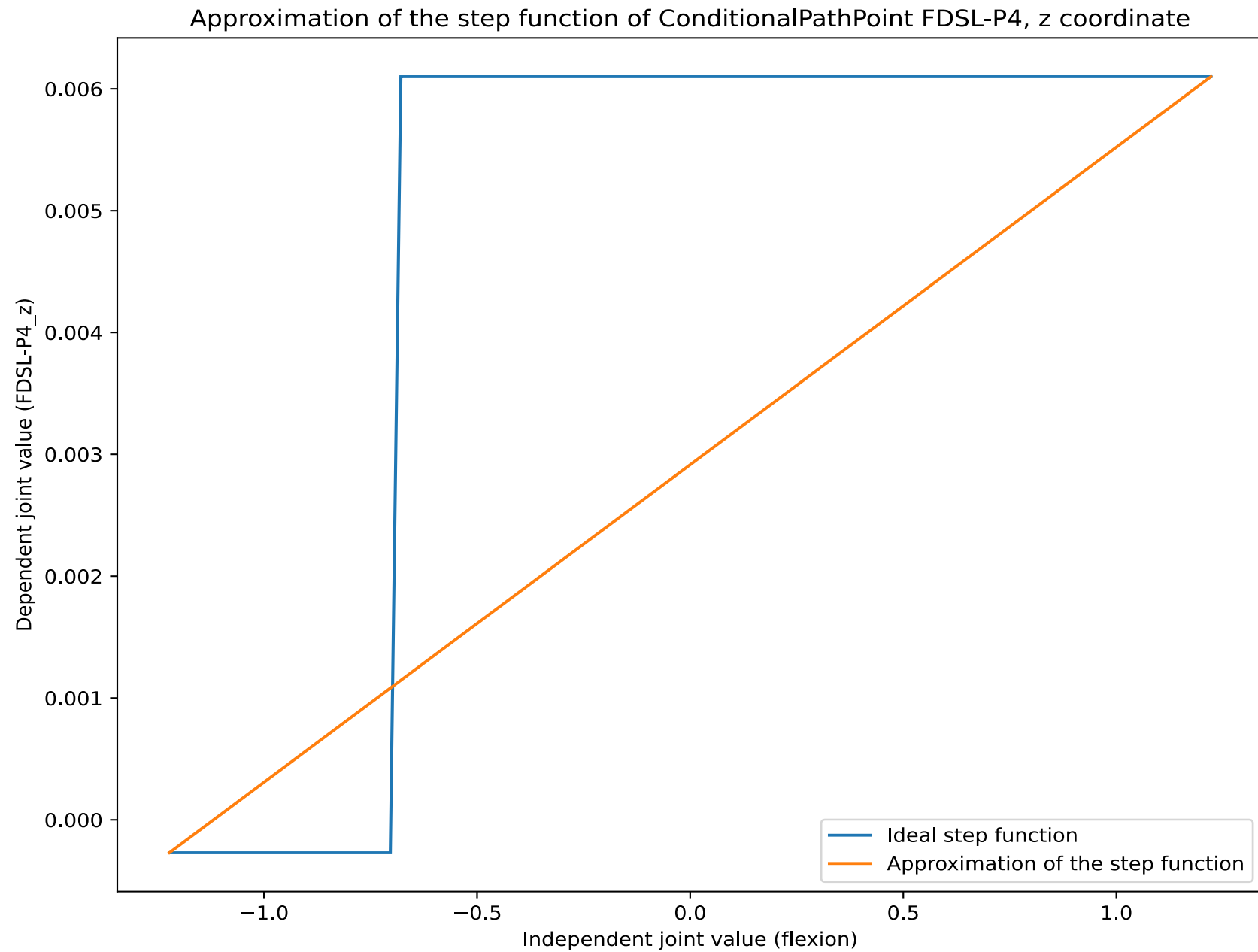


# Approximation of conditional path points

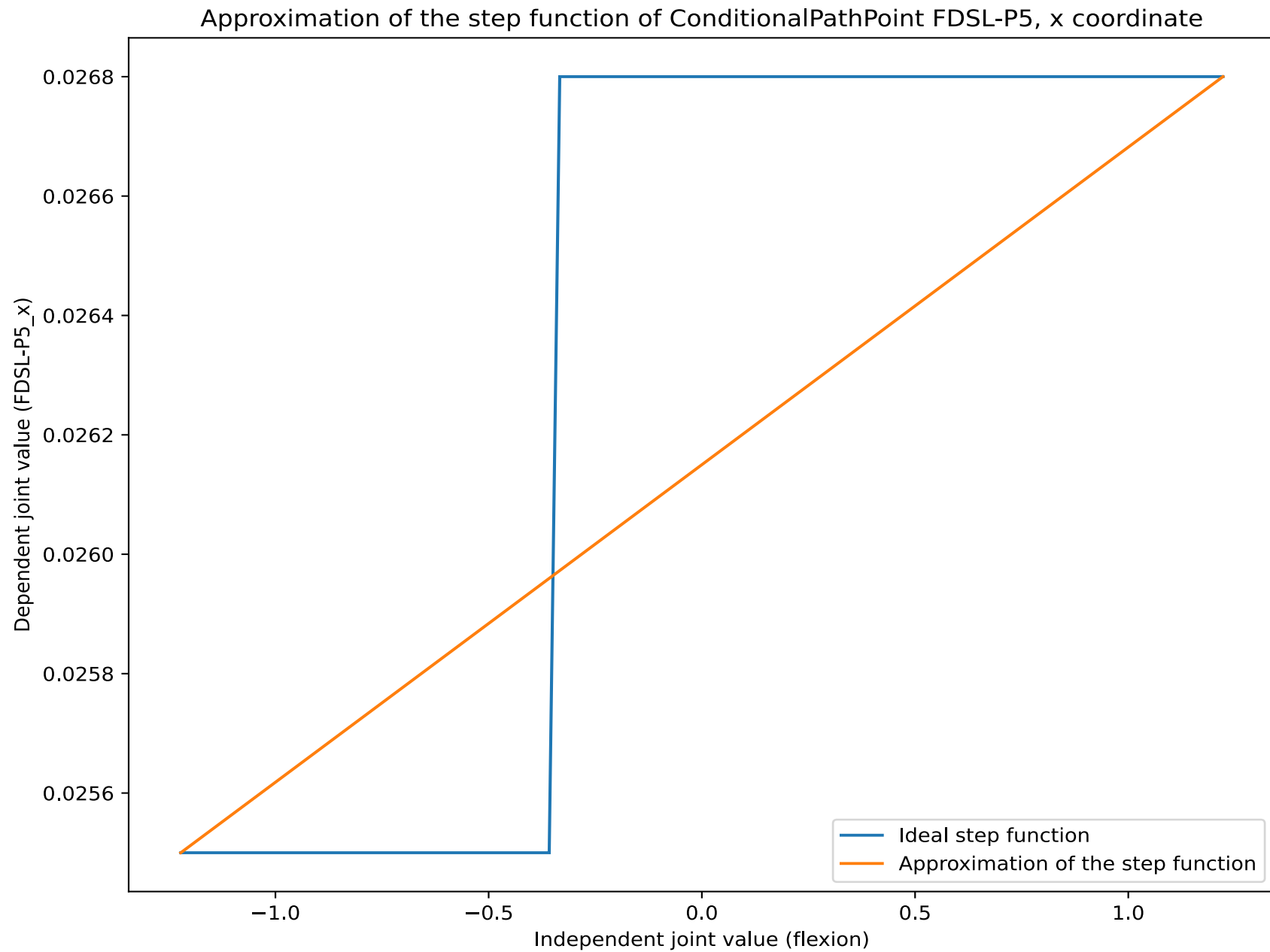




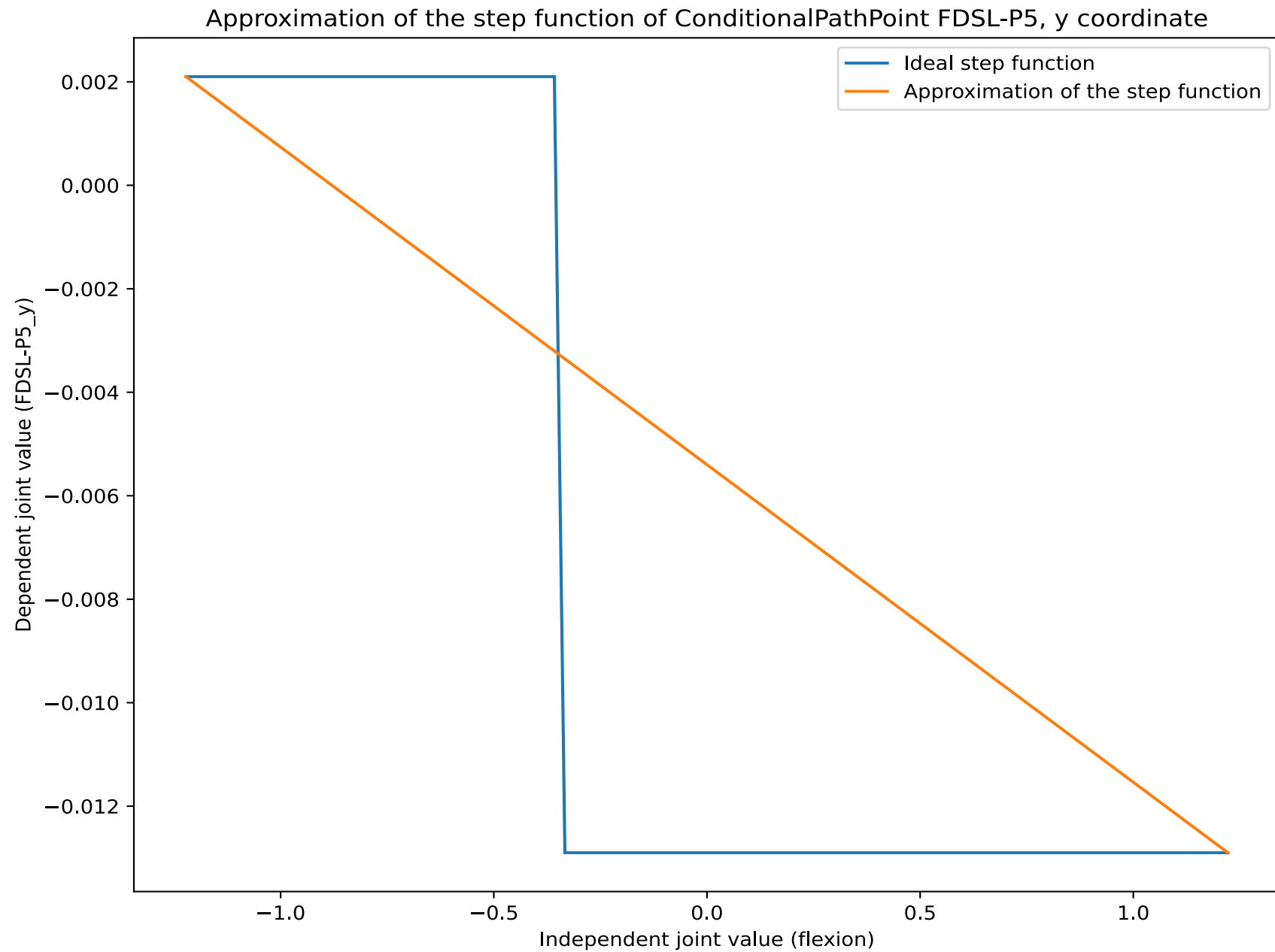
# Approximation of conditional path points



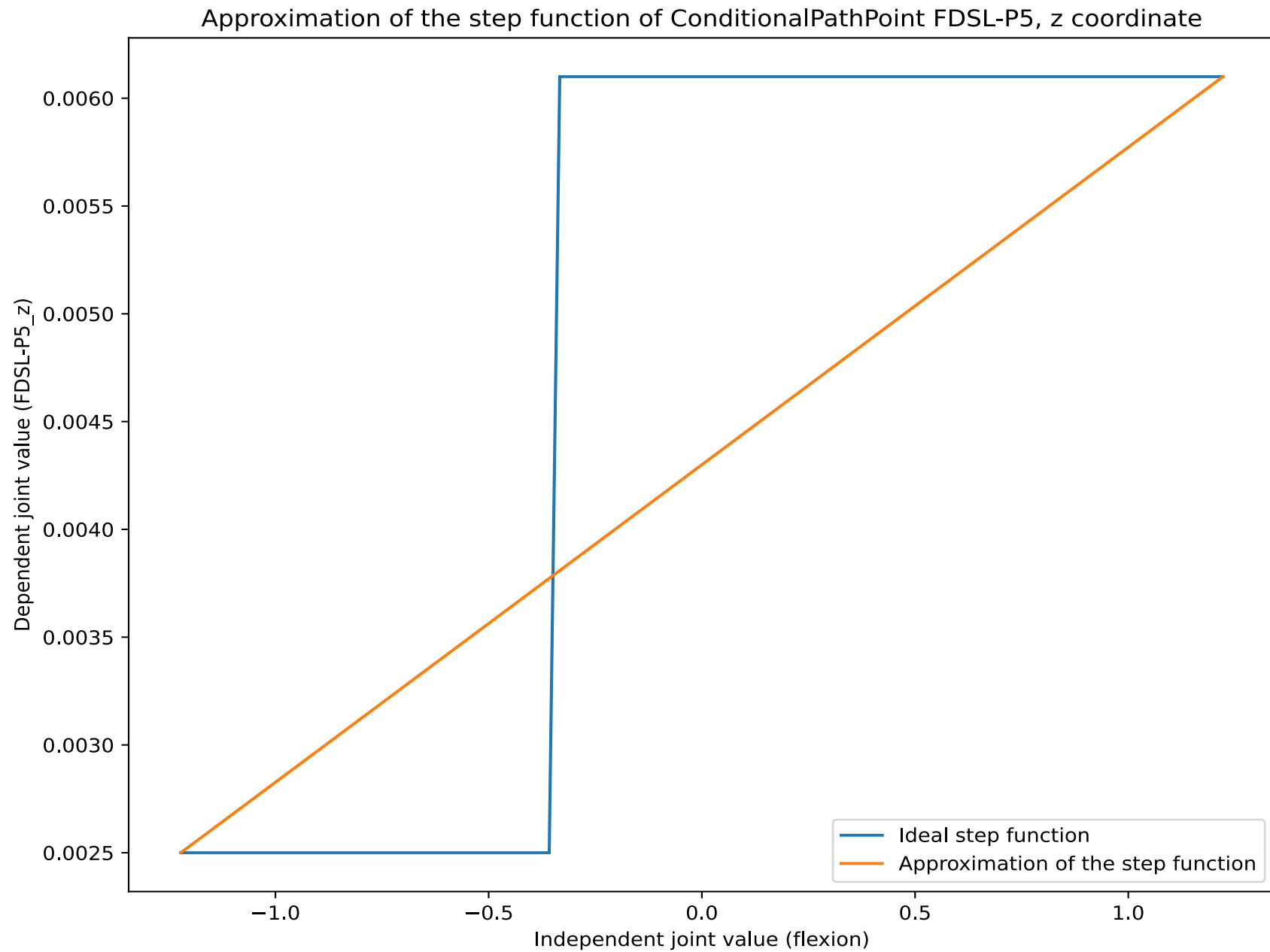
# Approximation of conditional path points



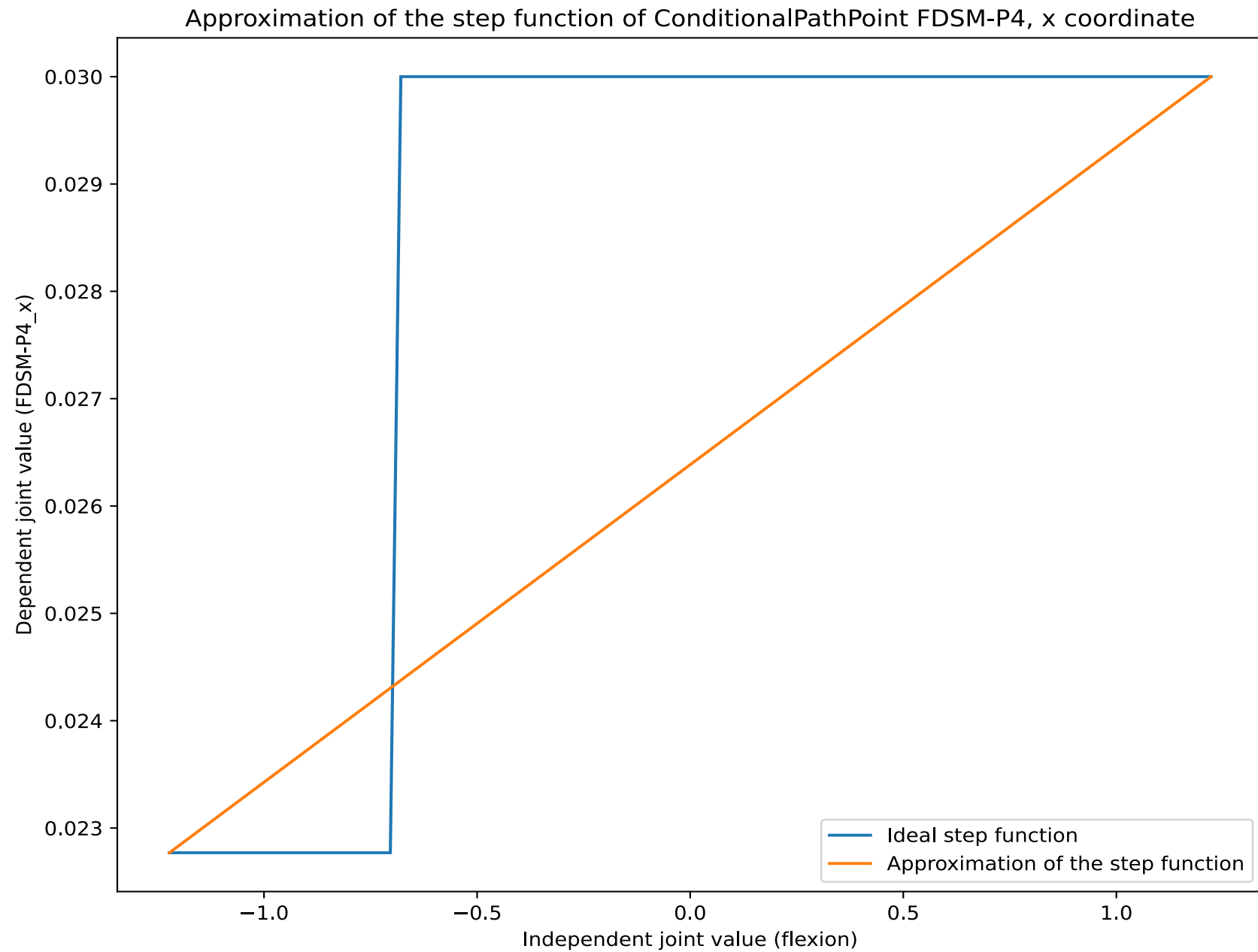
# Approximation of conditional path points



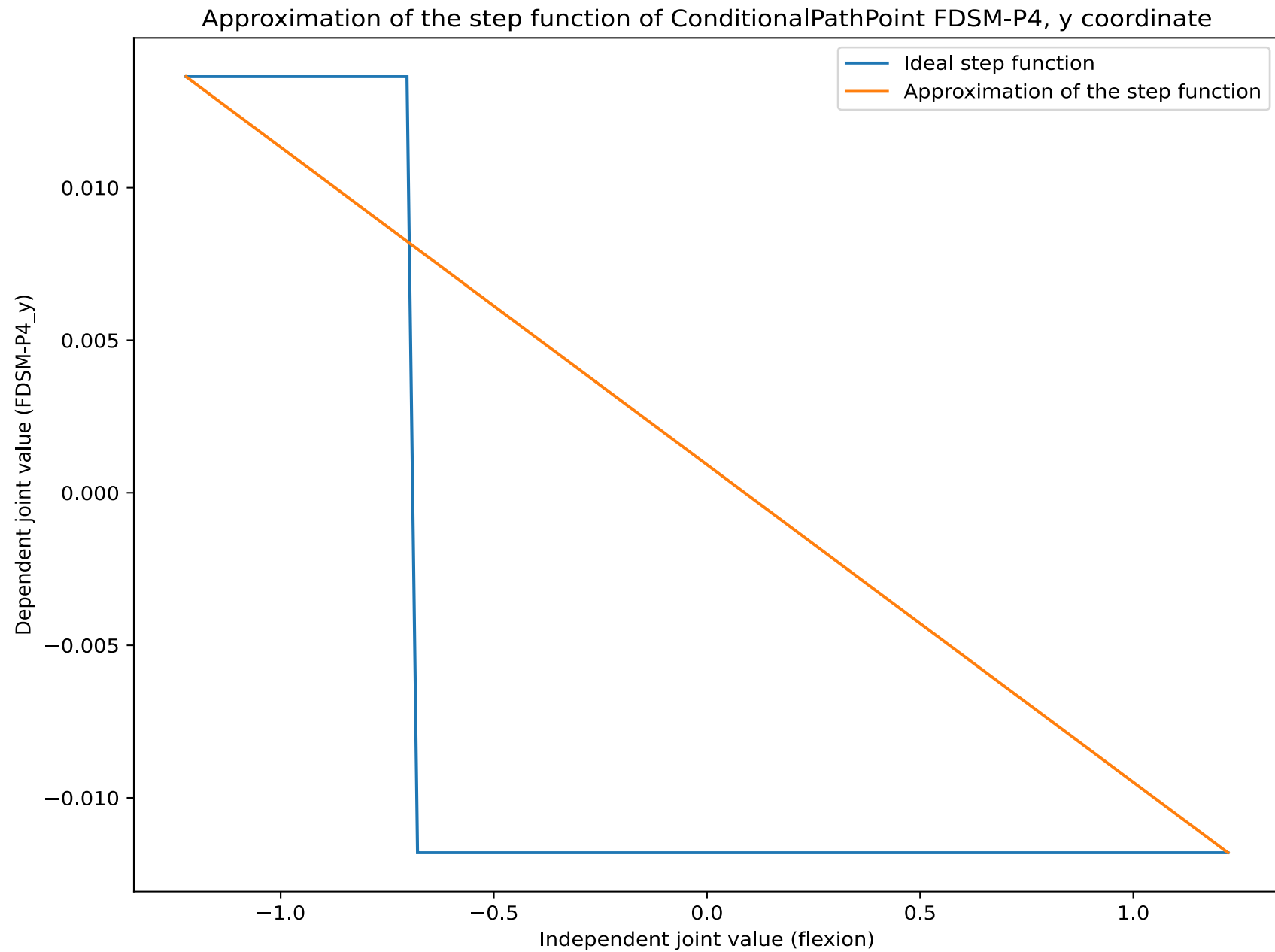
# Approximation of conditional path points



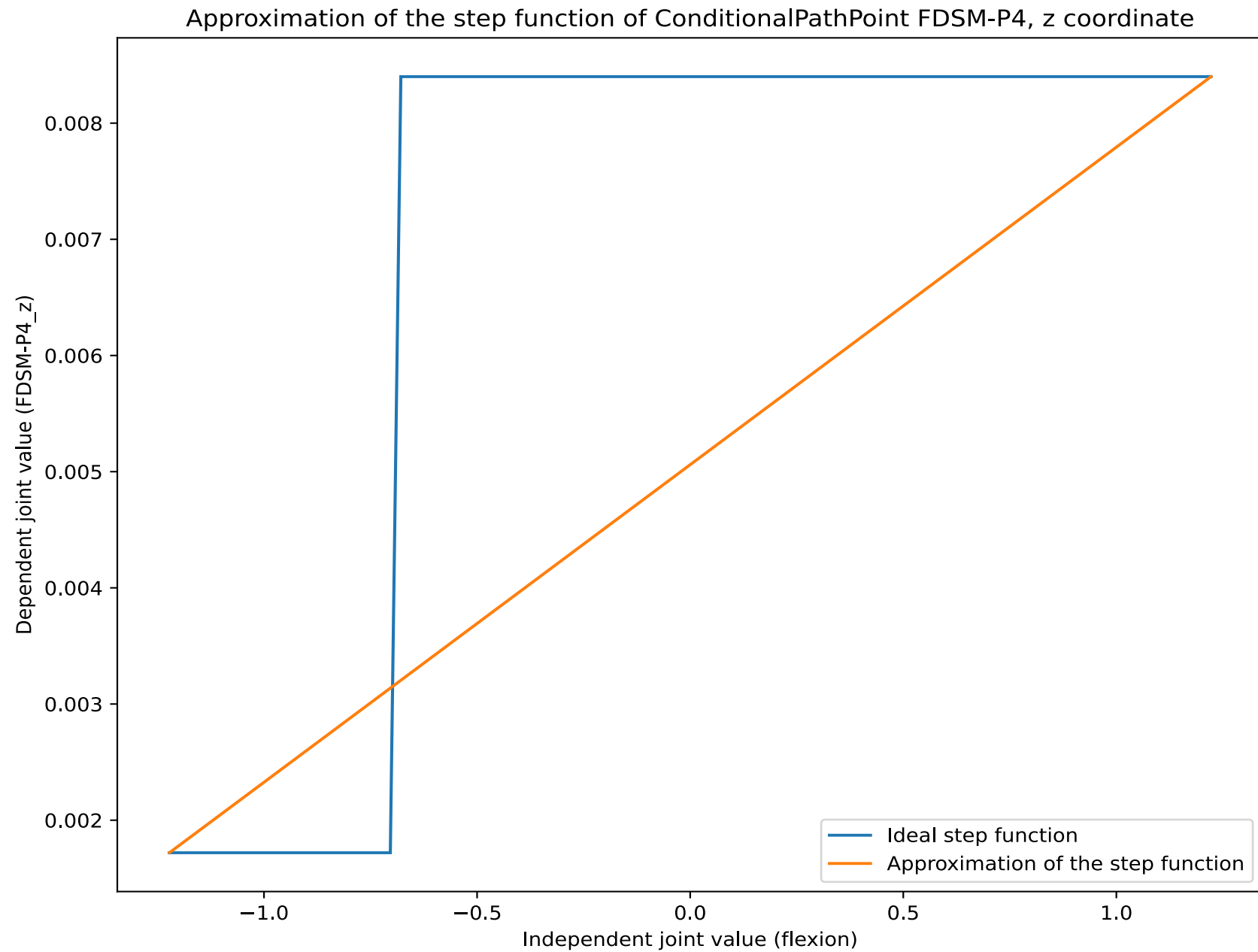
# Approximation of conditional path points



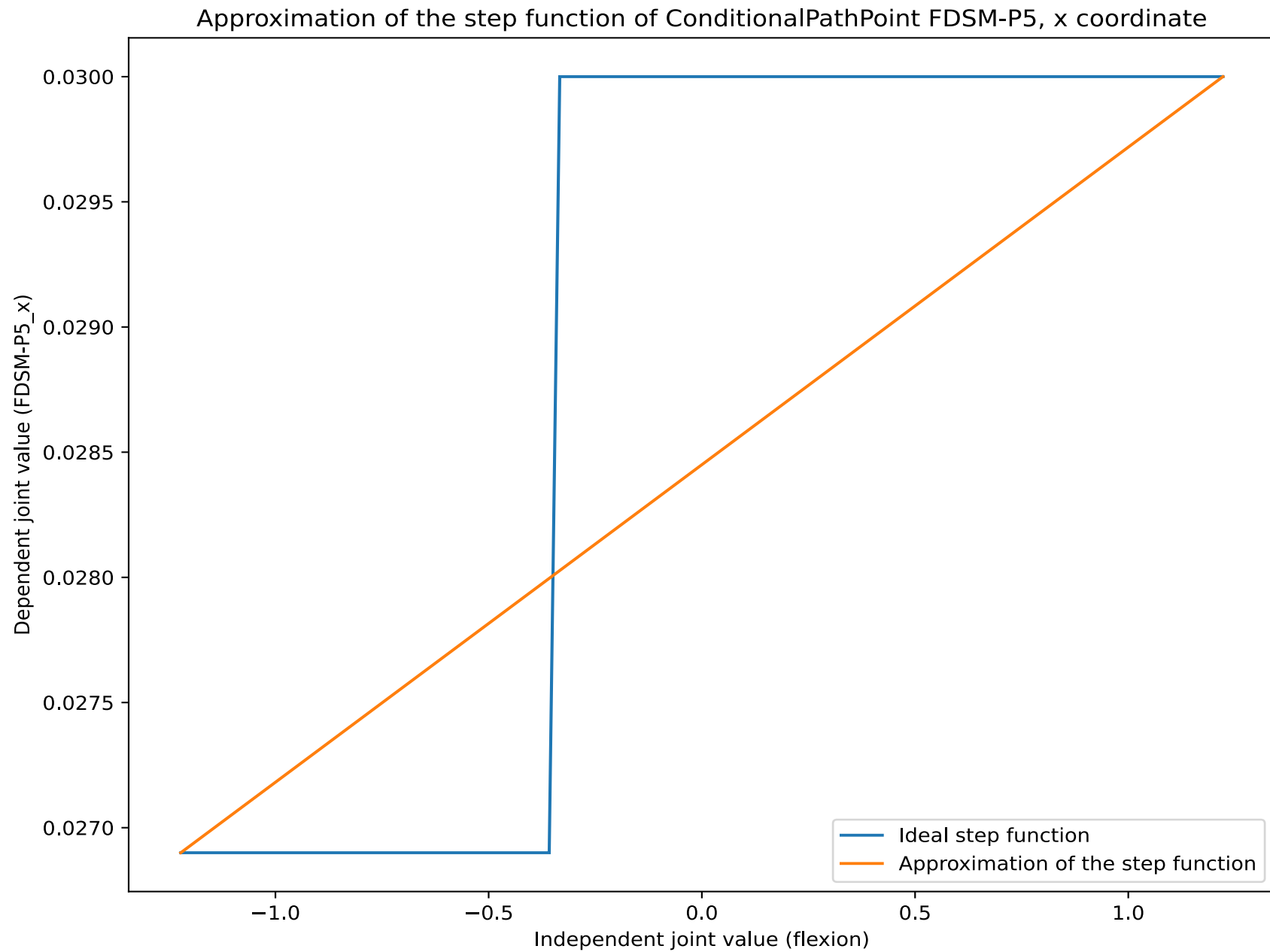
# Approximation of conditional path points



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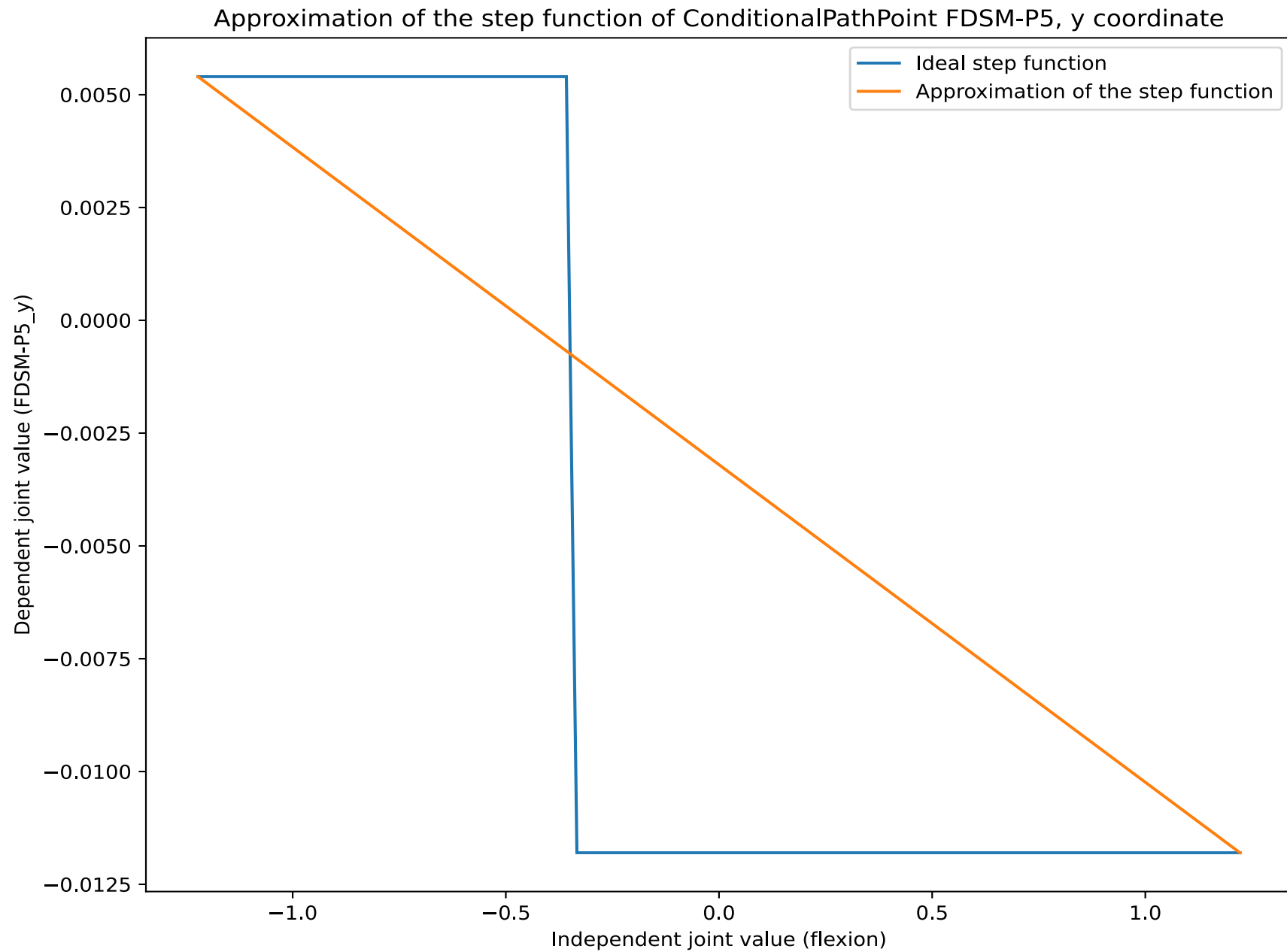


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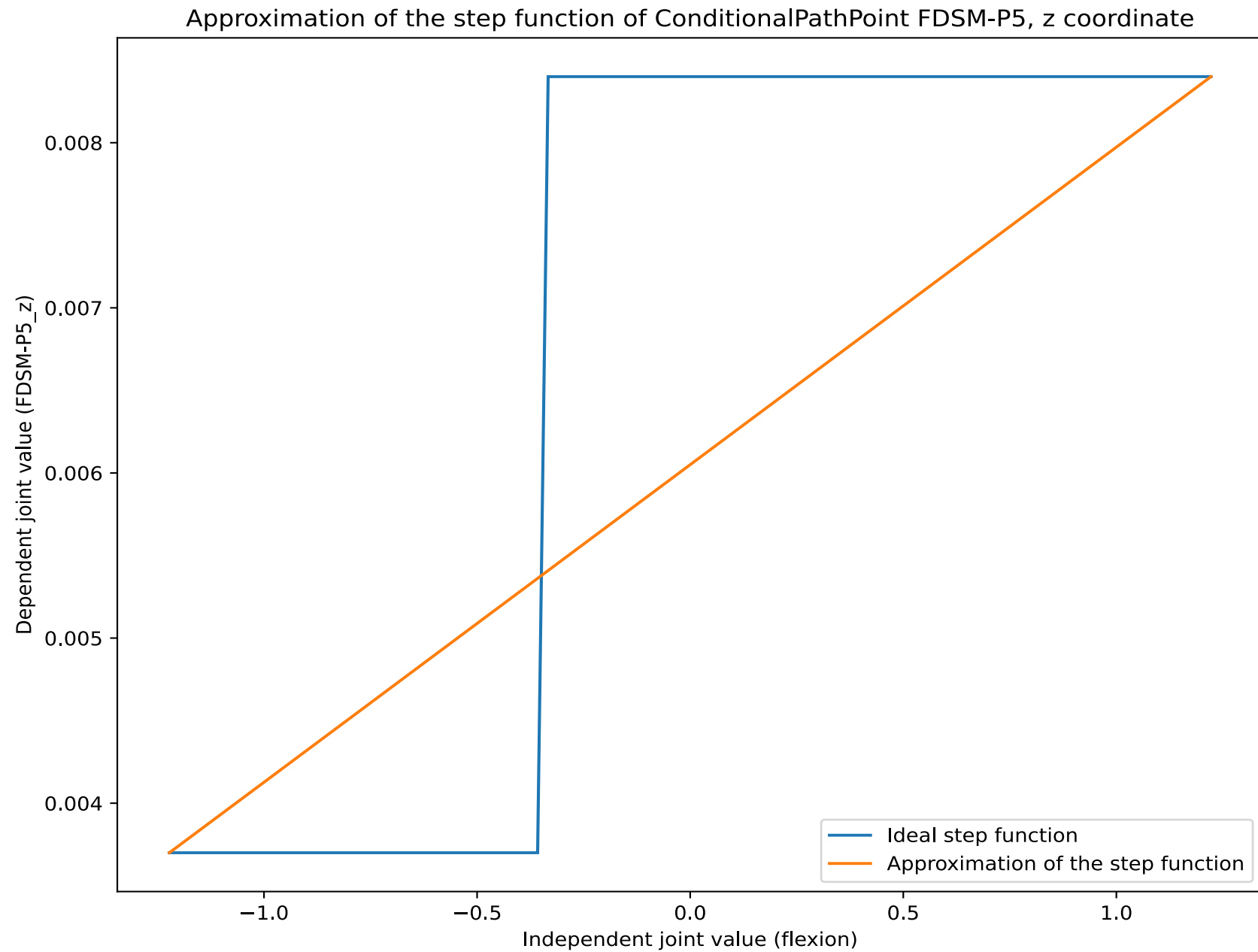




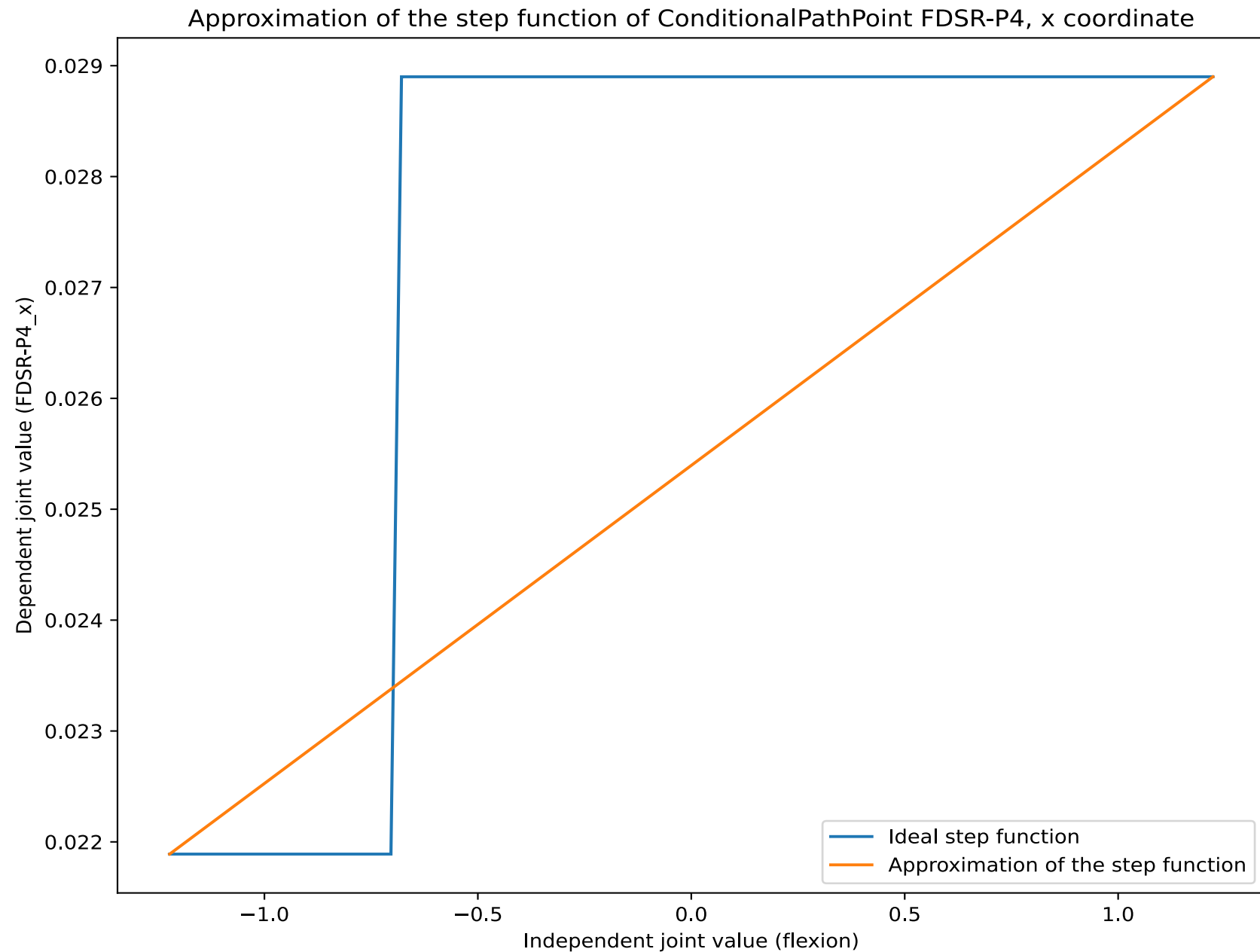
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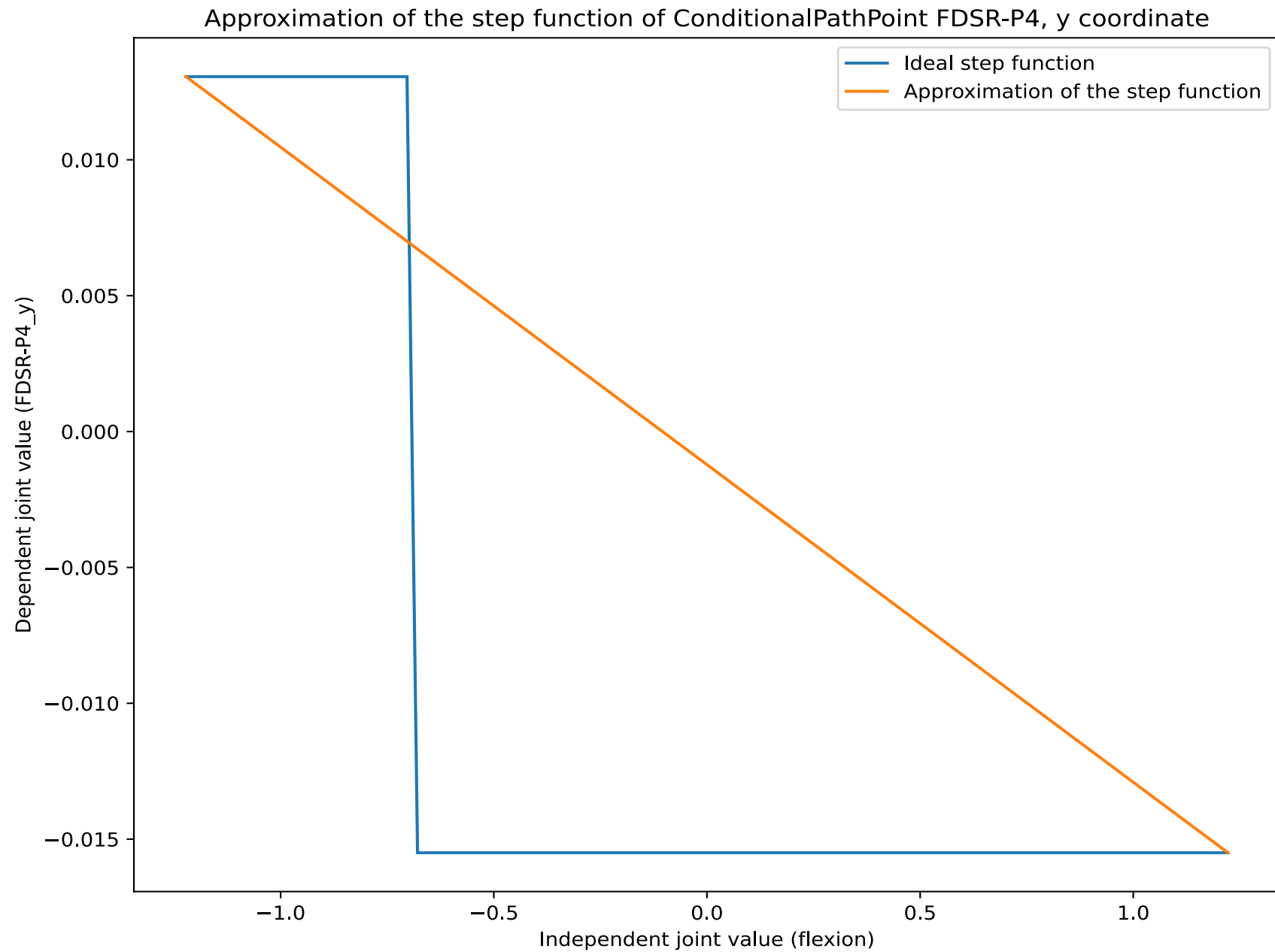
# Approximation of conditional path points



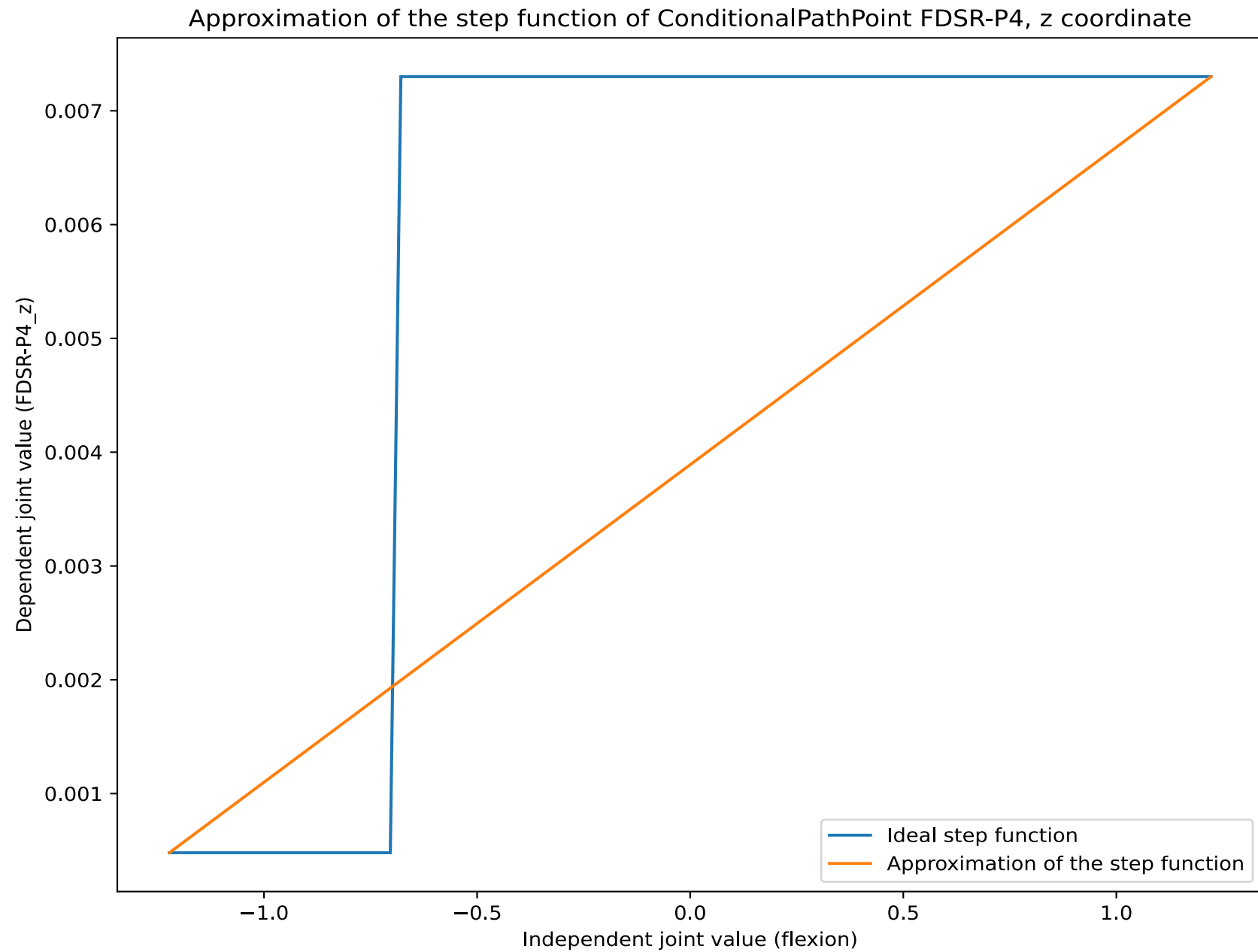
# Approximation of conditional path points



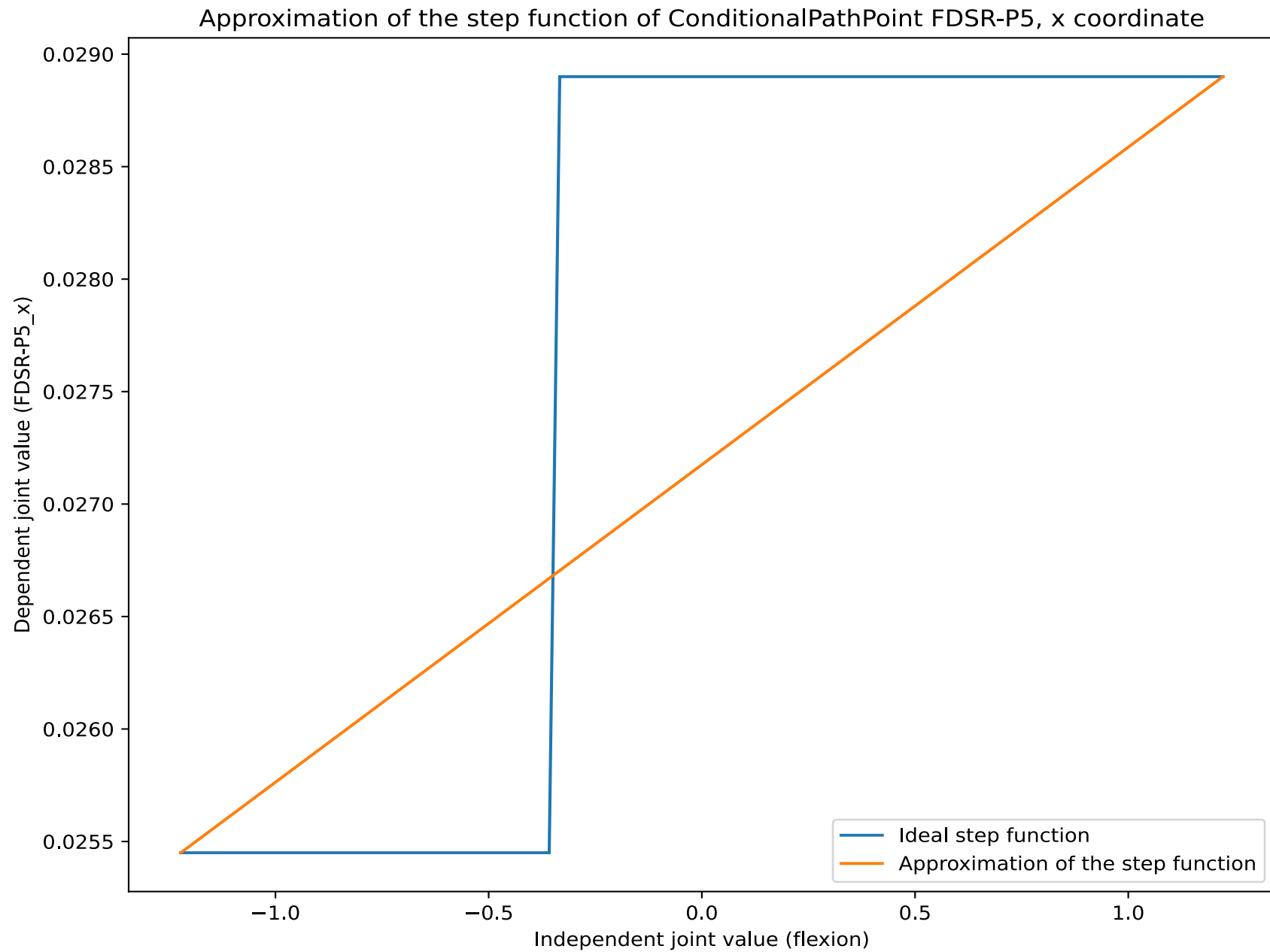
# Approximation of conditional path points



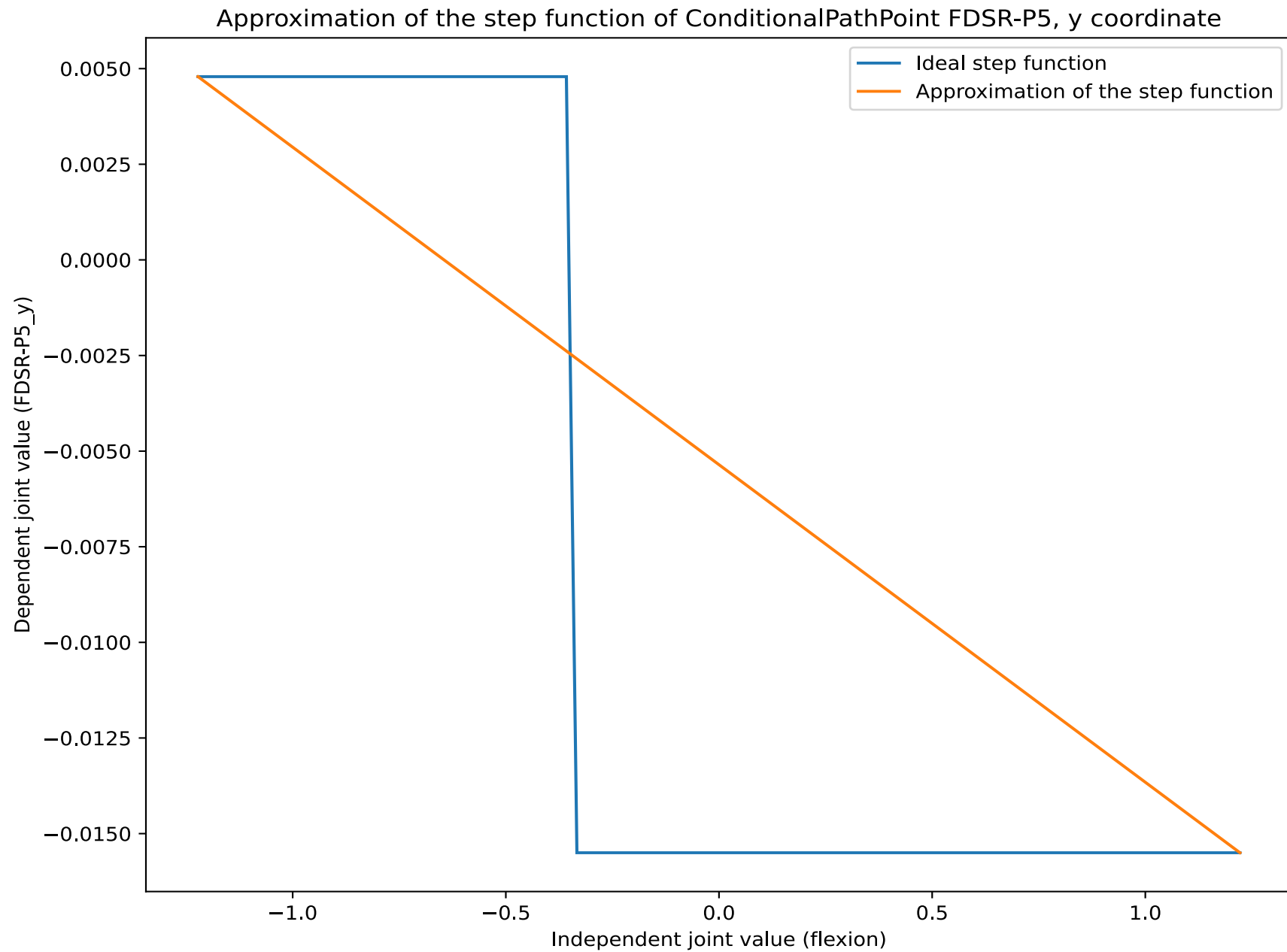
# Approximation of conditional path points



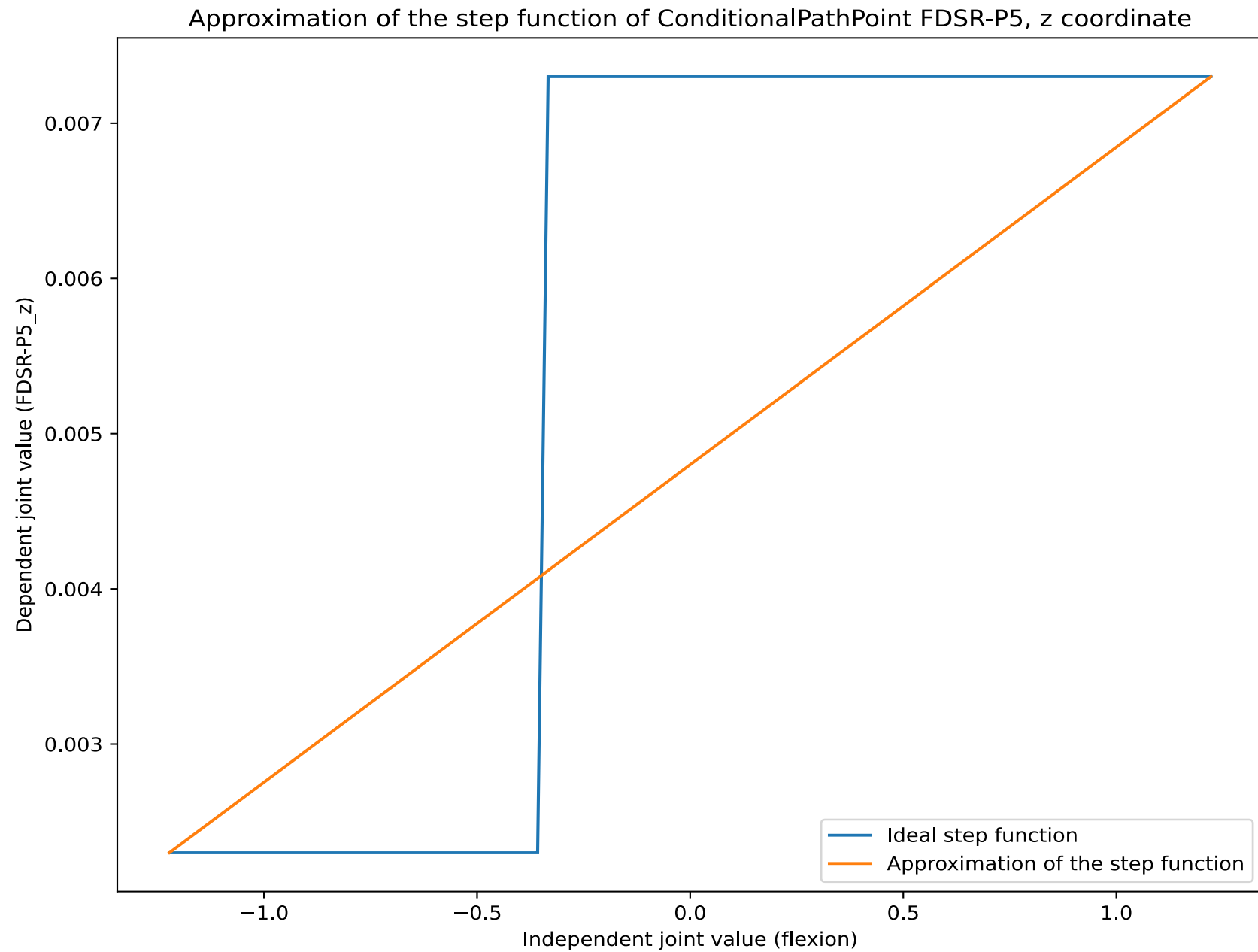
# Approximation of conditional path points



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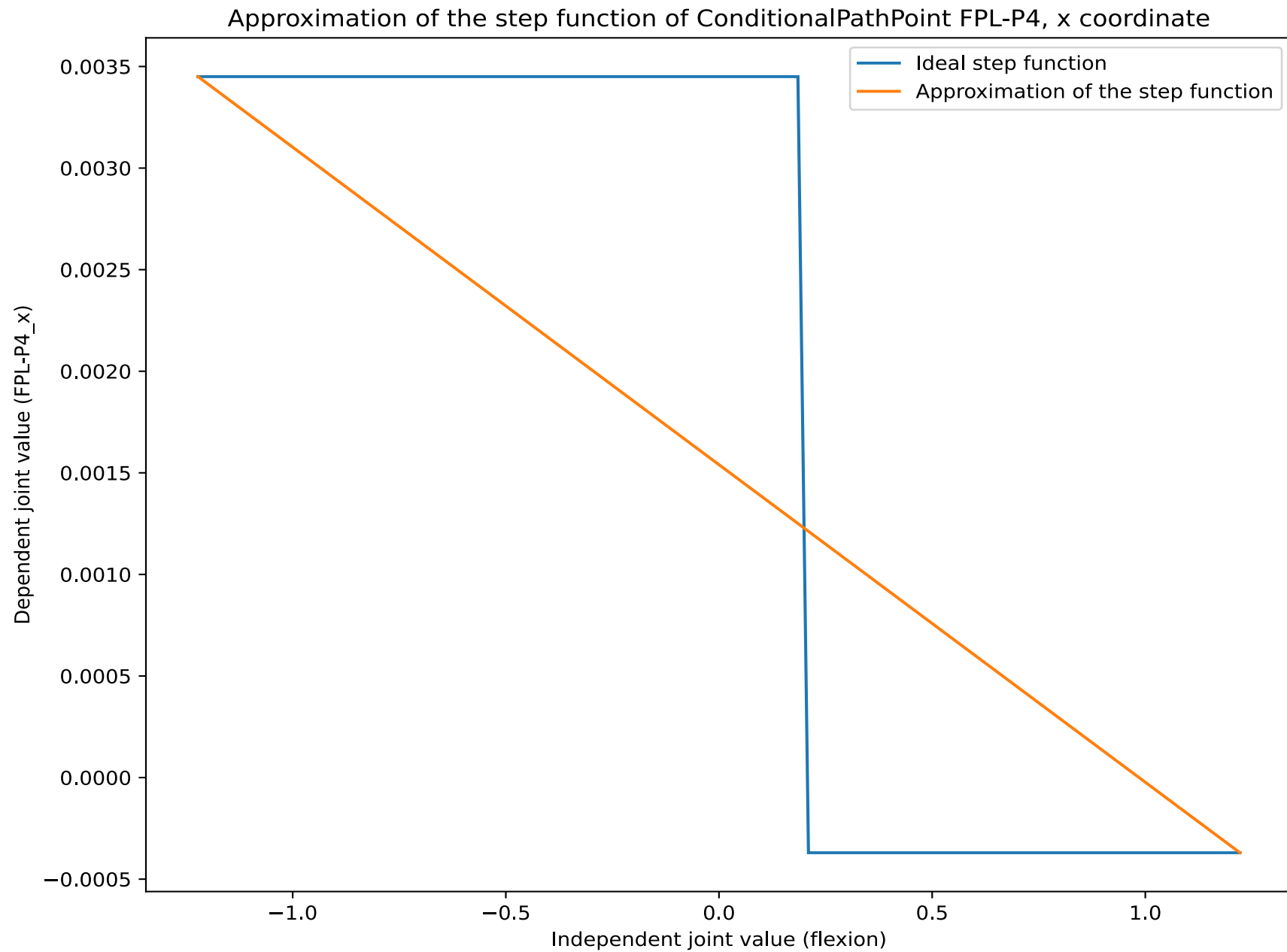


# Approximation of conditional path points

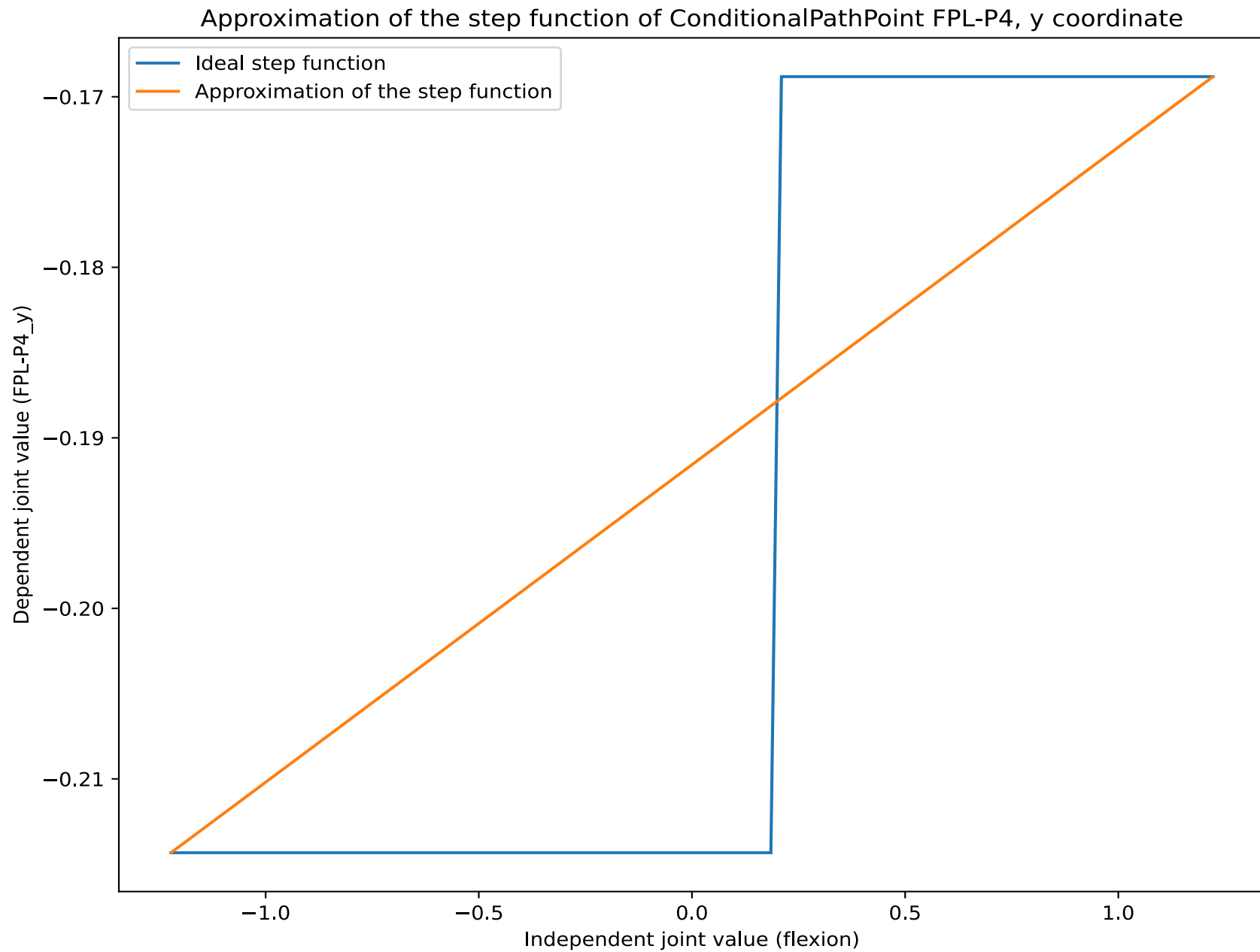




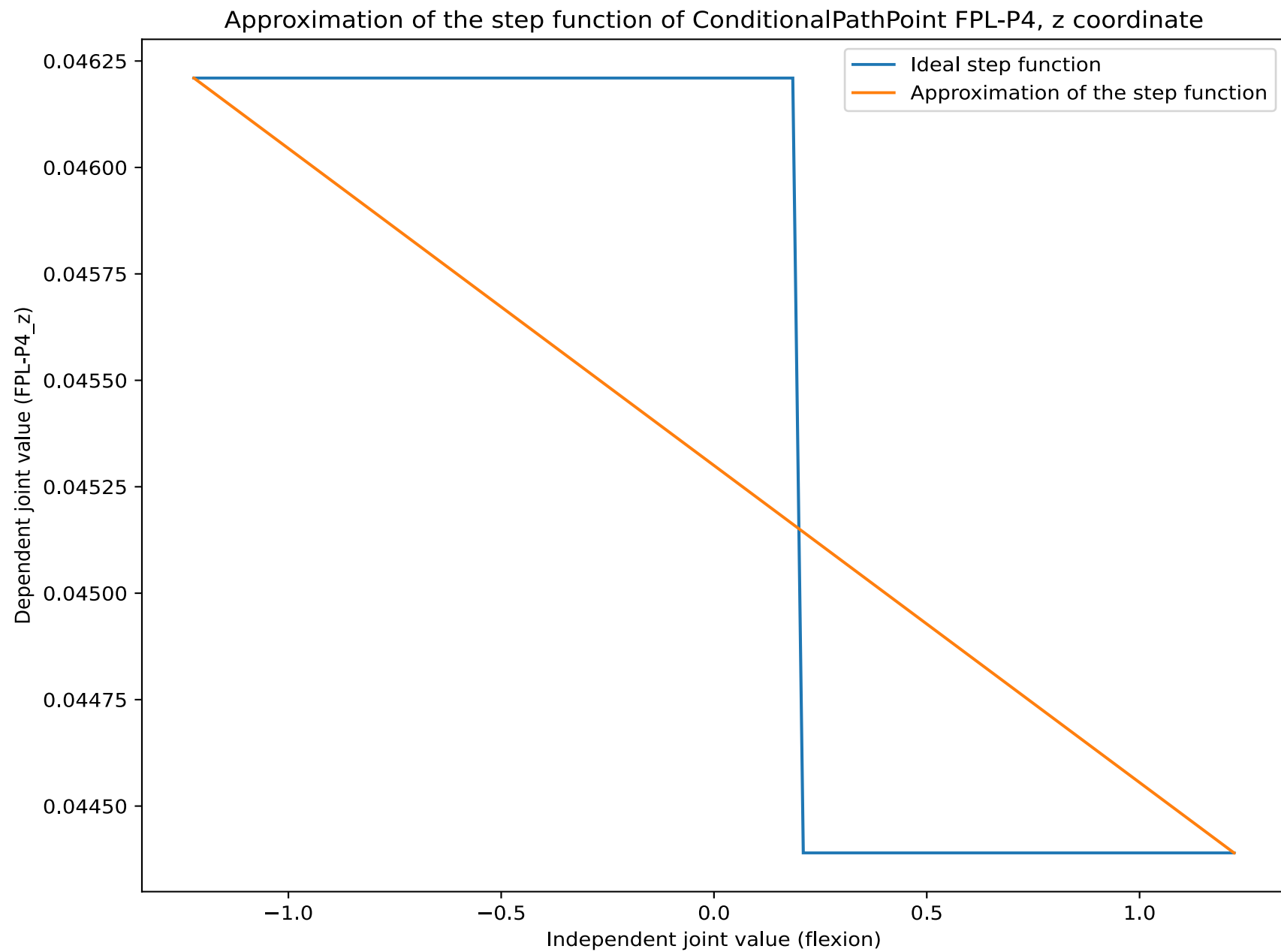
# Approximation of conditional path points



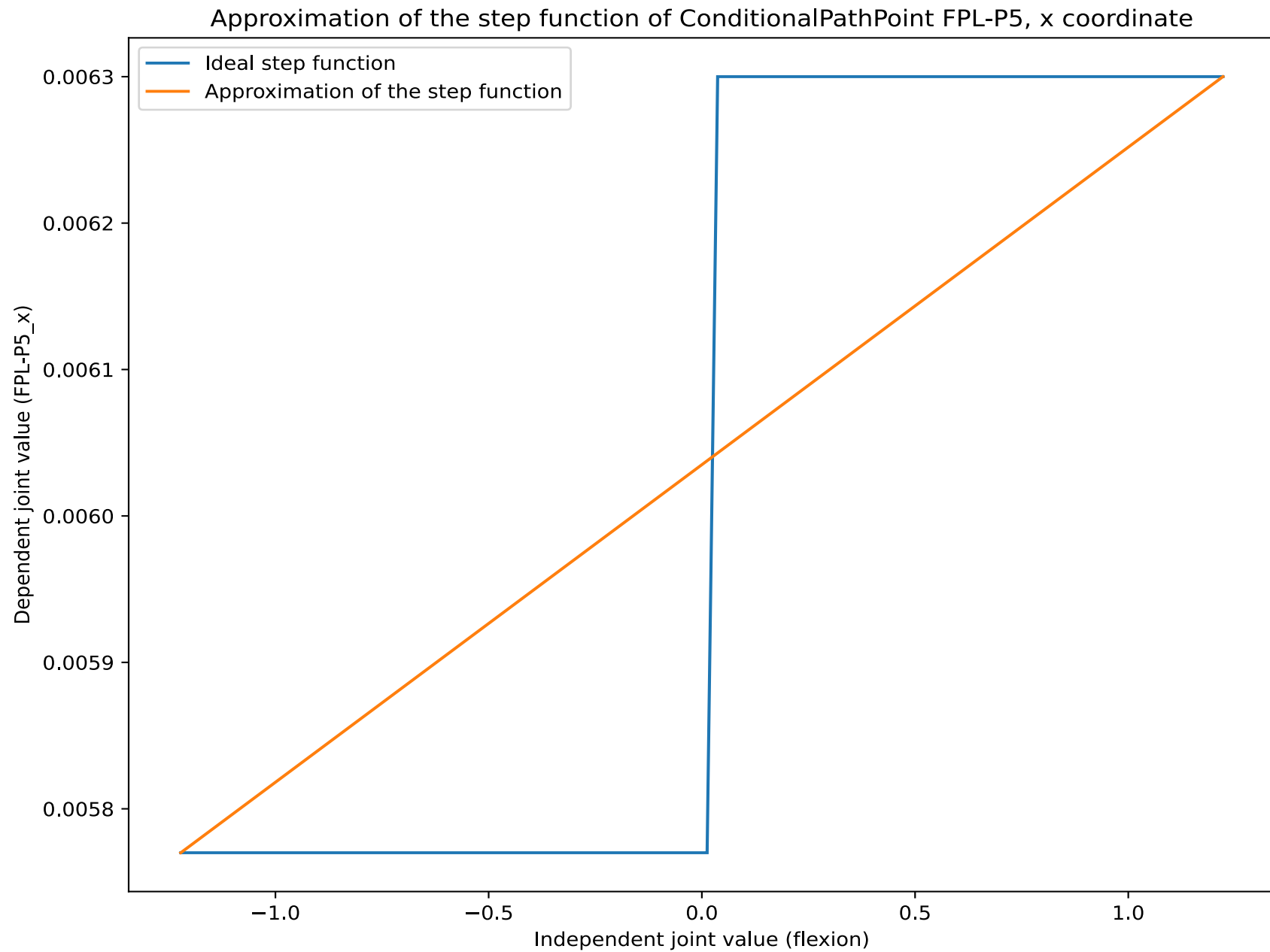
# Approximation of conditional path points



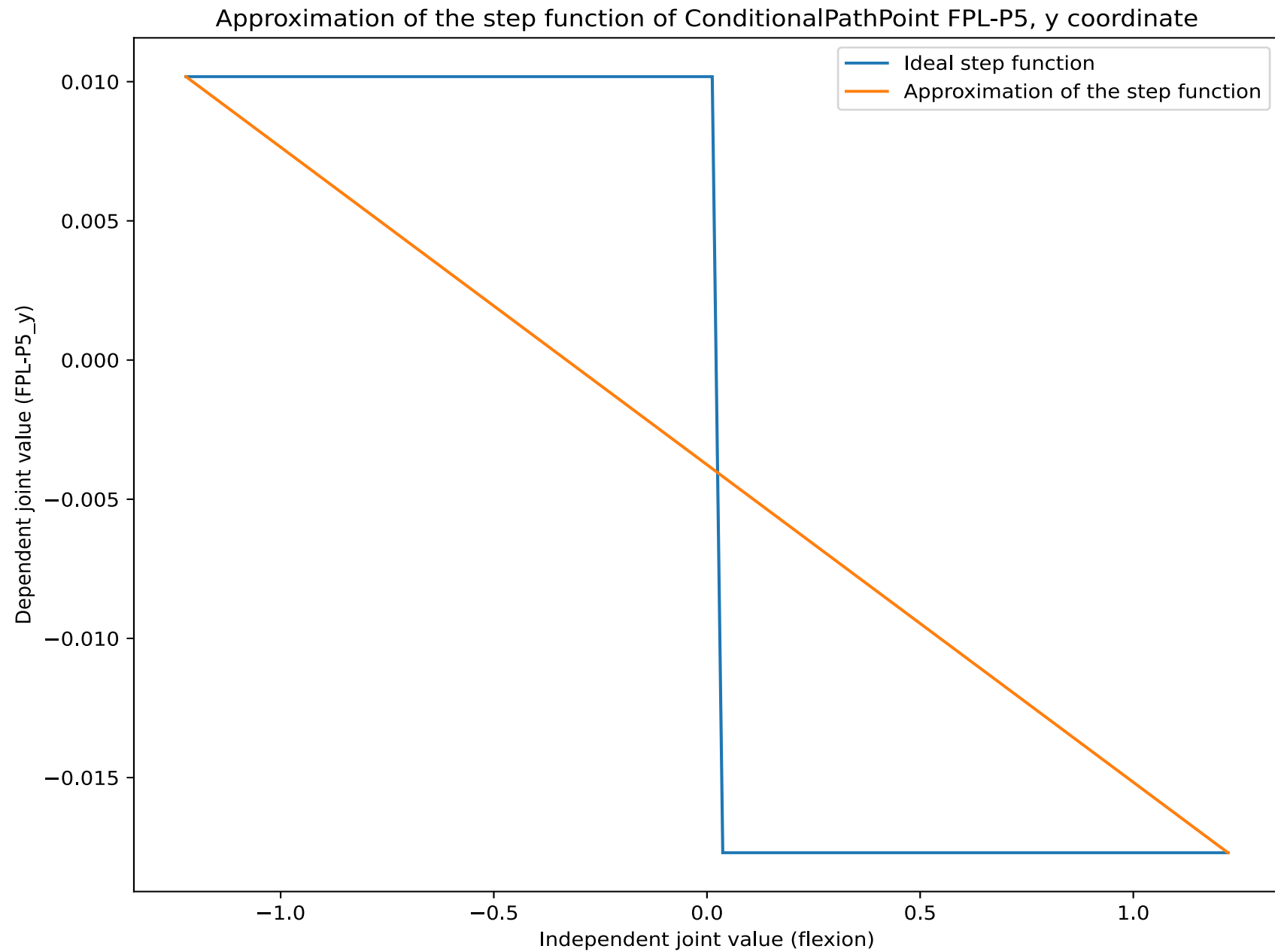
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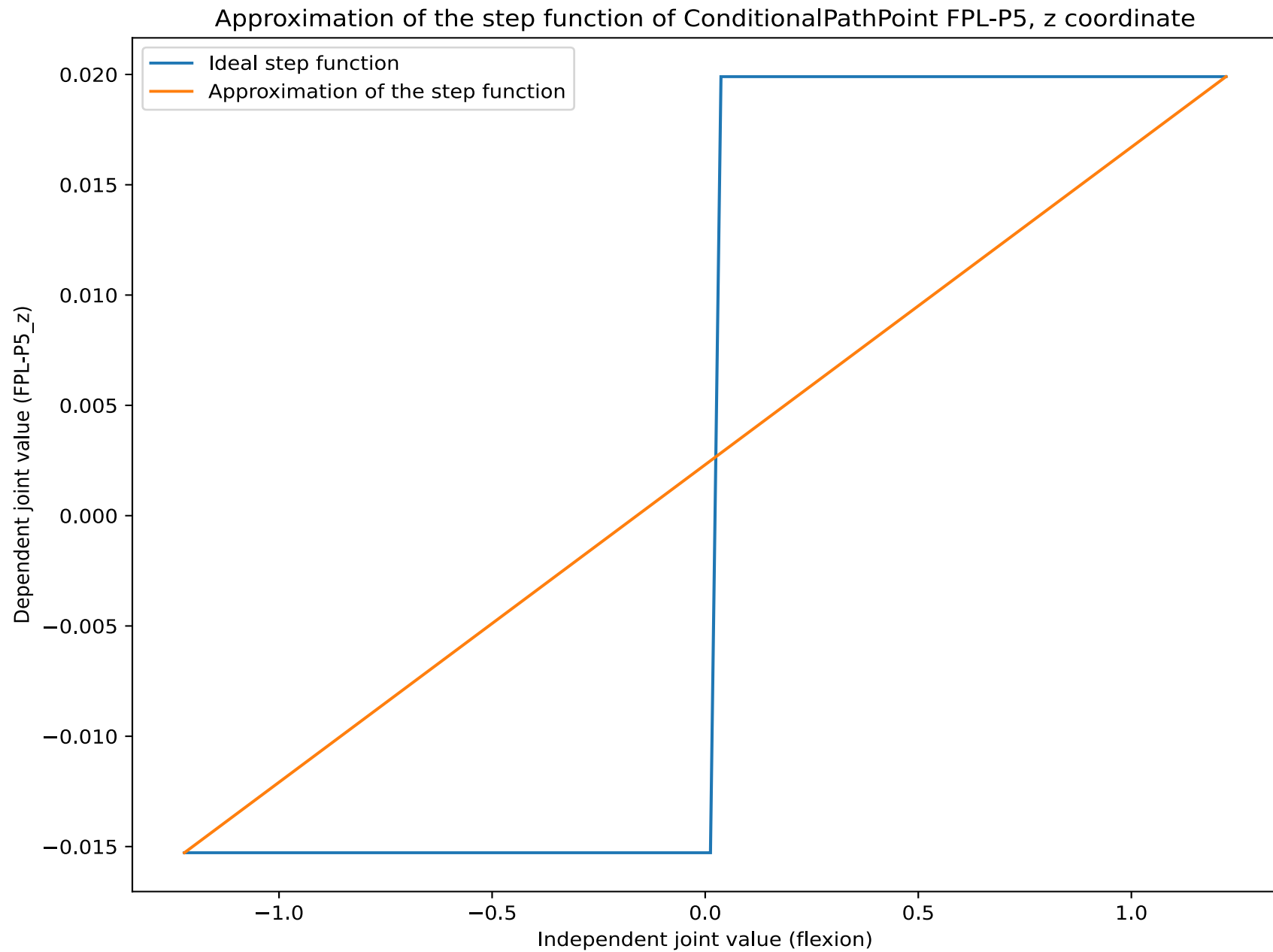
# Approximation of conditional path points



# Approximation of conditional path points



# Approximation of conditional path points



## Step 2: Muscle Kinematics Validation

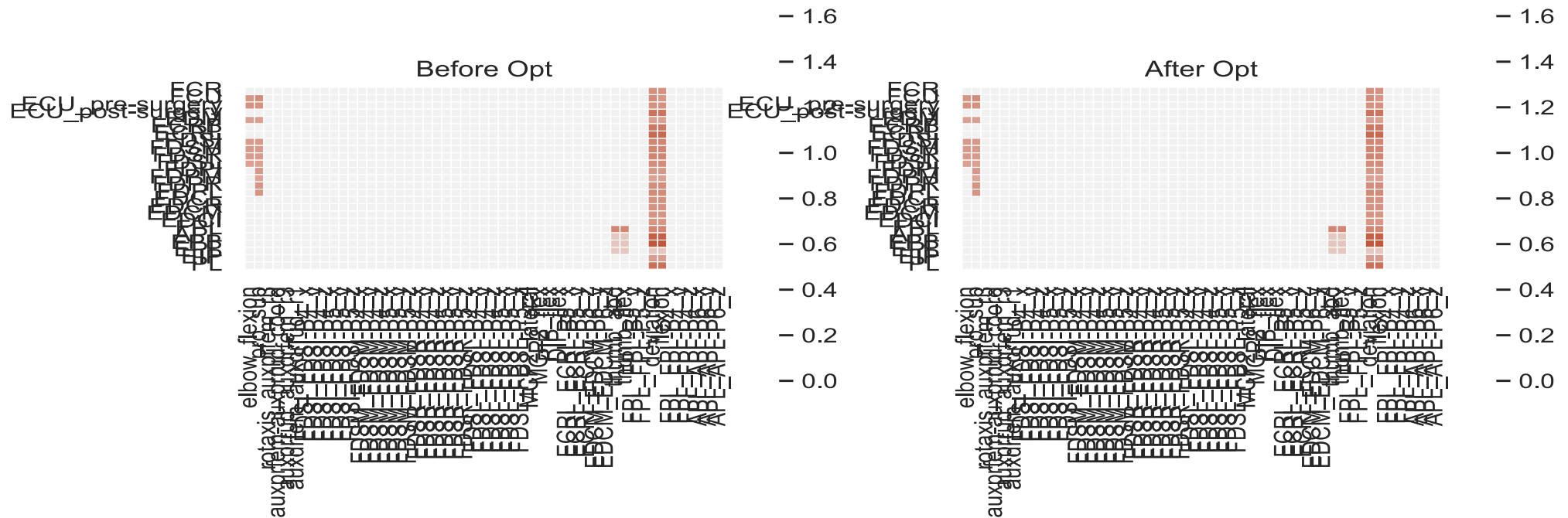
Moment arm of each muscle at each joint are compared between Osim and converted MJC model. A overall heatmap is included to indicate the overall moment arm errors before and after optimization. Then detail moment arm curves are plotted for comparison. For the muscles that wrap over multiple joints, moment arms with respect to one joint maybe affected by several other joints. In this case, several mesh points were check of these affecting joints, when plotting moment arms at one joint. This is why there are multiple lines (with different grey levels) plotted for one muscle on one joint.

How to interpret the plot:

Global title indicate the muscle and joints that affecting the moment arms in the plots. X axis indicate the joint that moment arms were extracted. Grey level of the lines indicate the mesh postures of other relevant joints (in the global tile, but not the x axis)

## Overall comparison of muscle moment arms before/after optimization

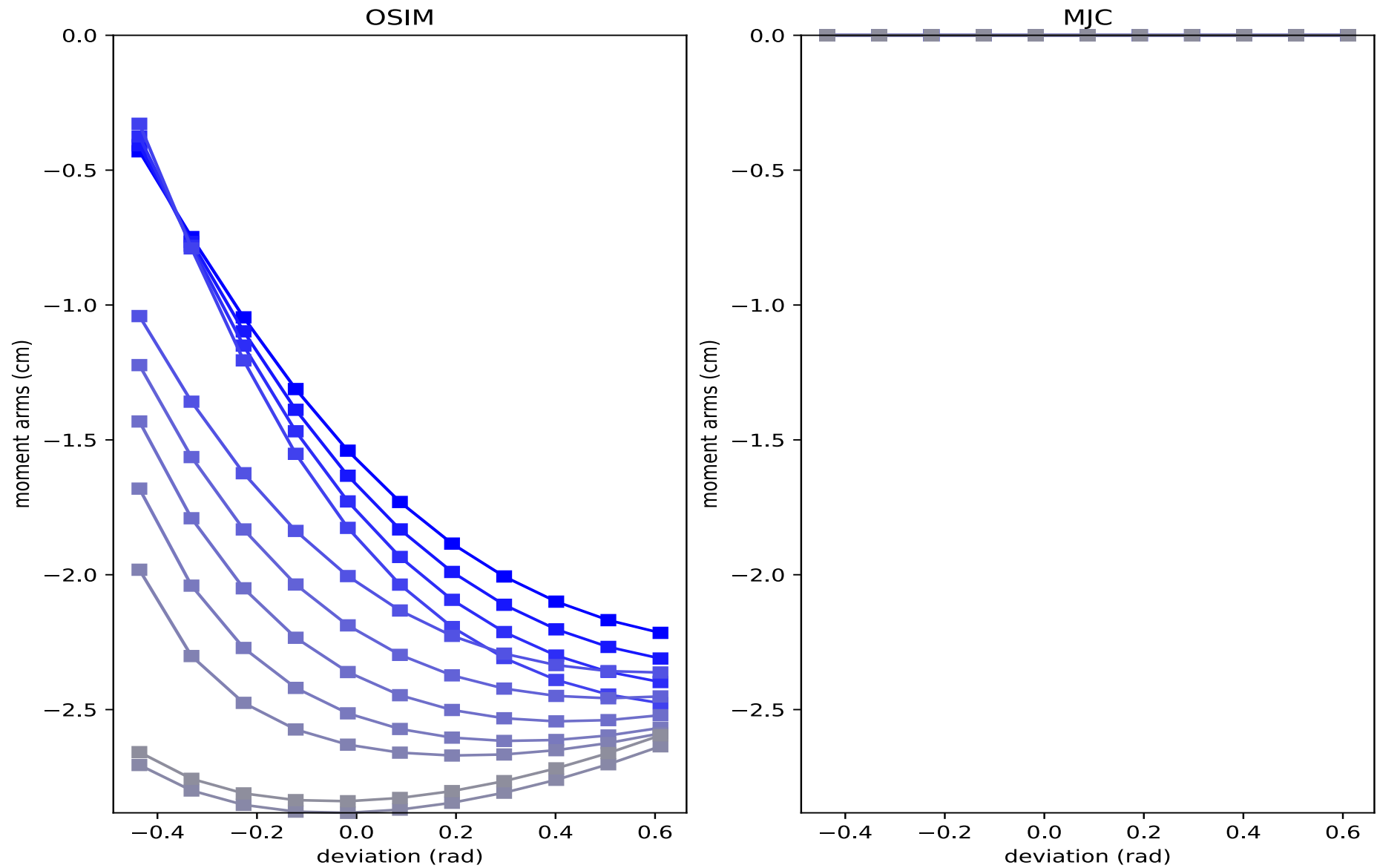
### Moment arm comparison of all muscles (cm)





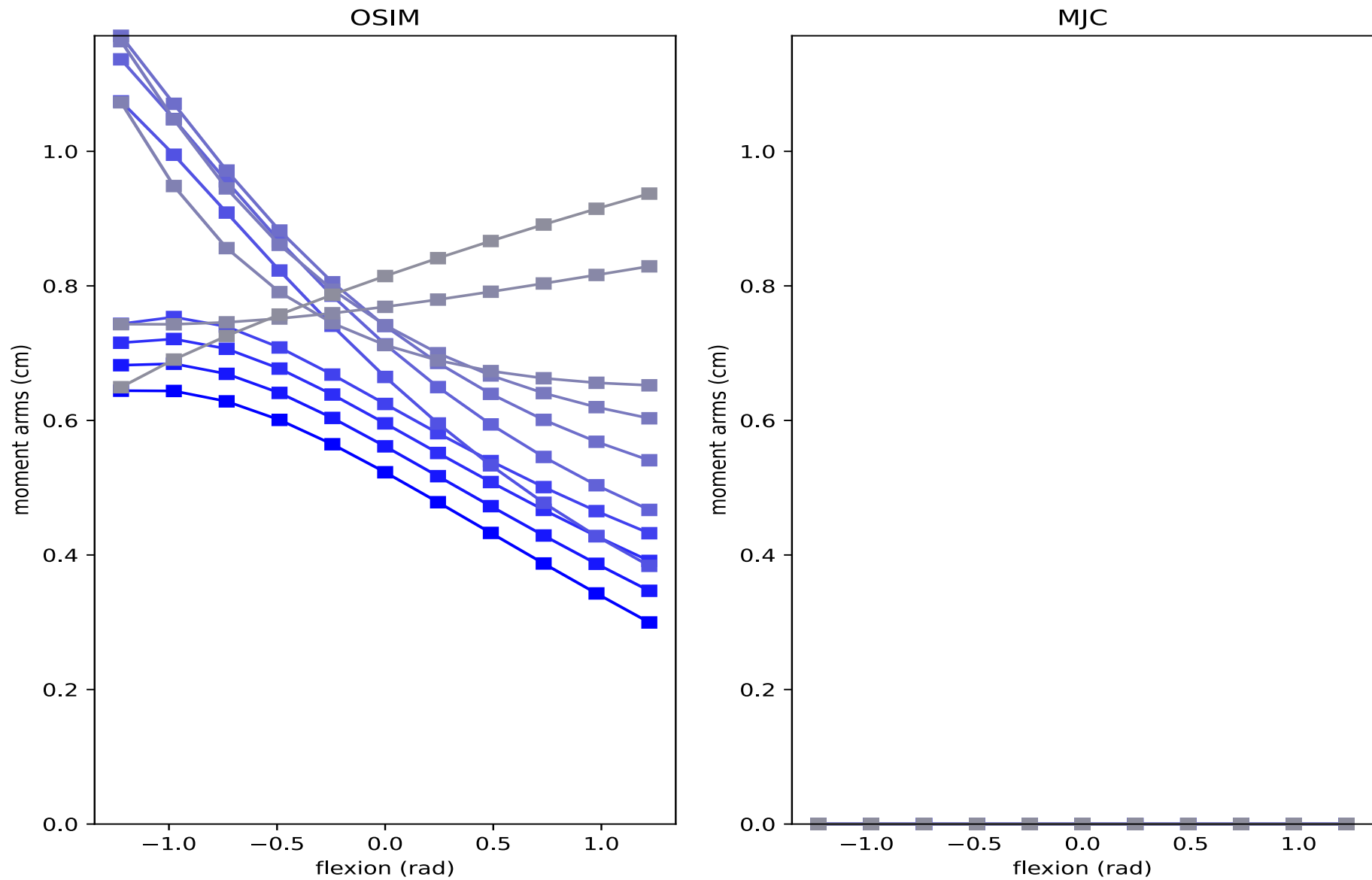
# Muscle specific moment arm comparison with Osim model after optimization

APL - deviation - flexion



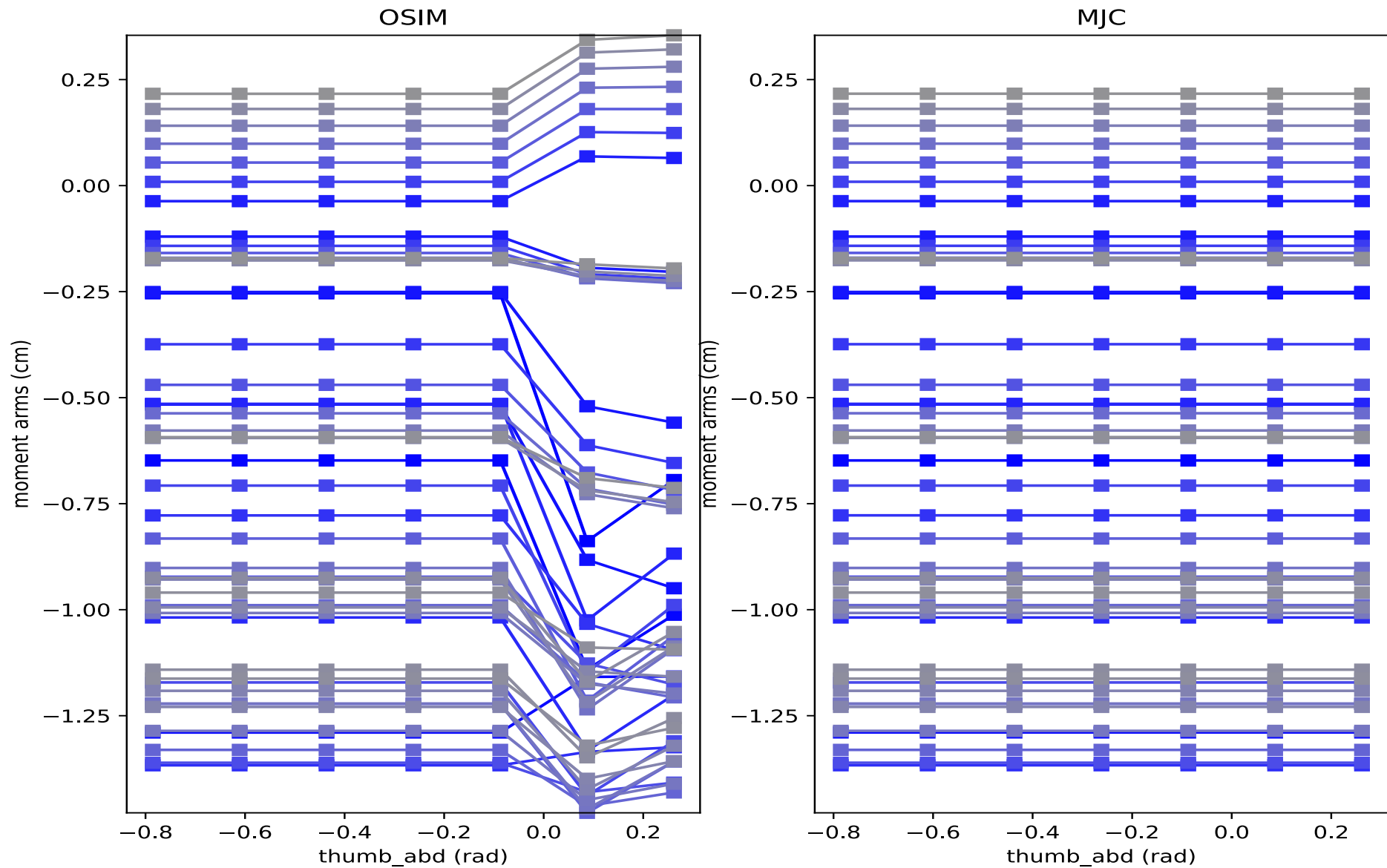
# Muscle specific moment arm comparison with Osim model after optimization

APL - deviation - flexion



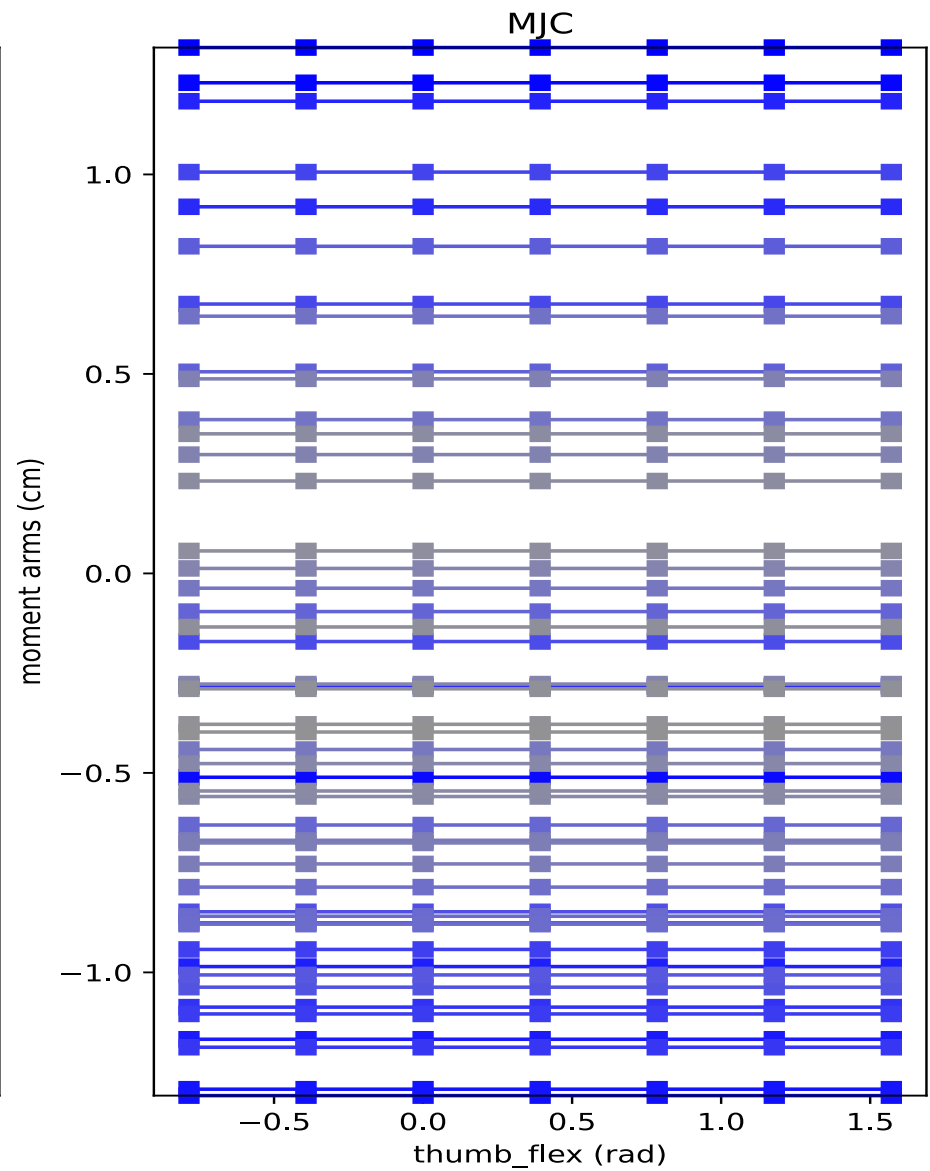
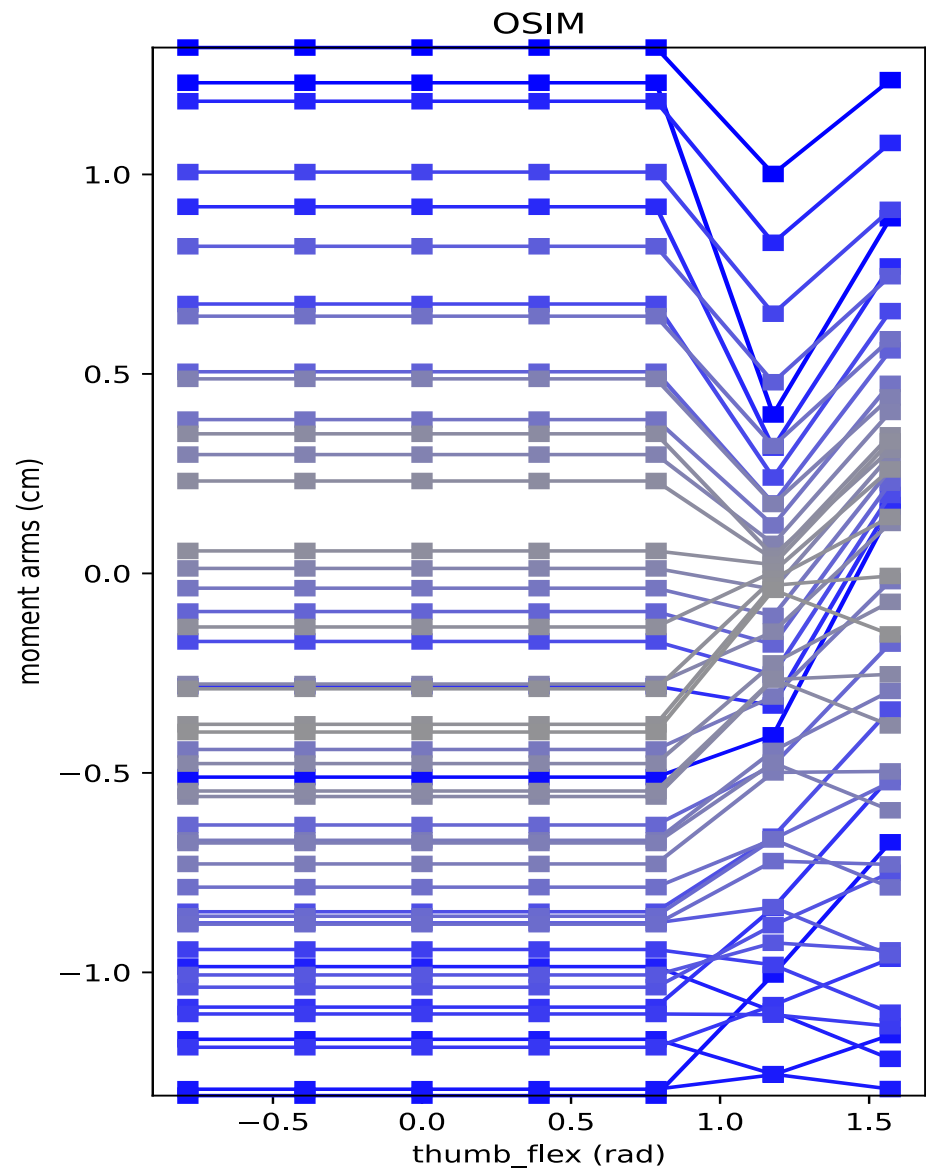
# Muscle specific moment arm comparison with Osim model after optimization

APL - flexion - thumb\_abd - thumb\_flex



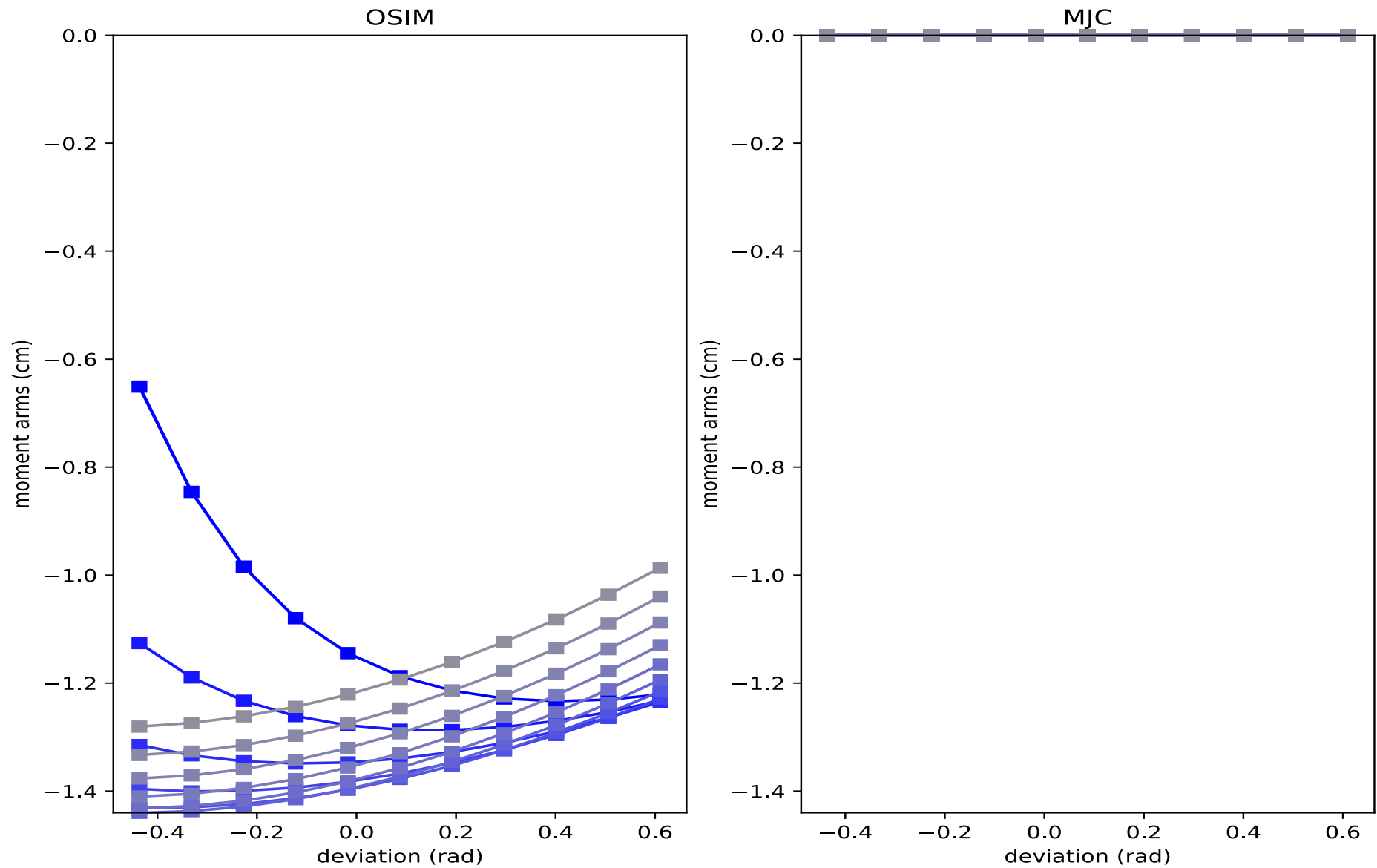
# Muscle specific moment arm comparison with Osim model after optimization

APL - flexion - thumb\_abd - thumb\_flex



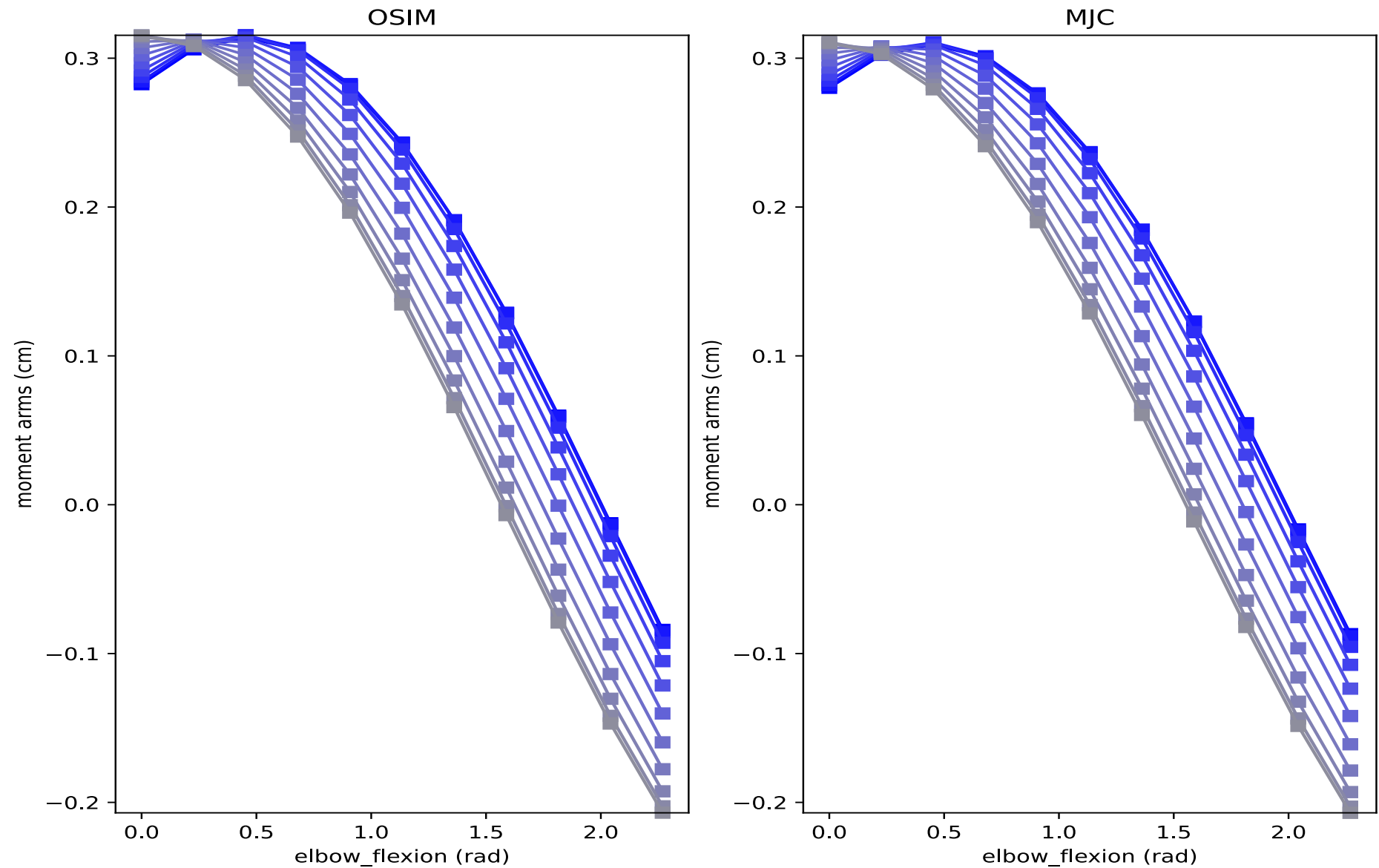
# Muscle specific moment arm comparison with Osim model after optimization

ECRB - deviation - flexion



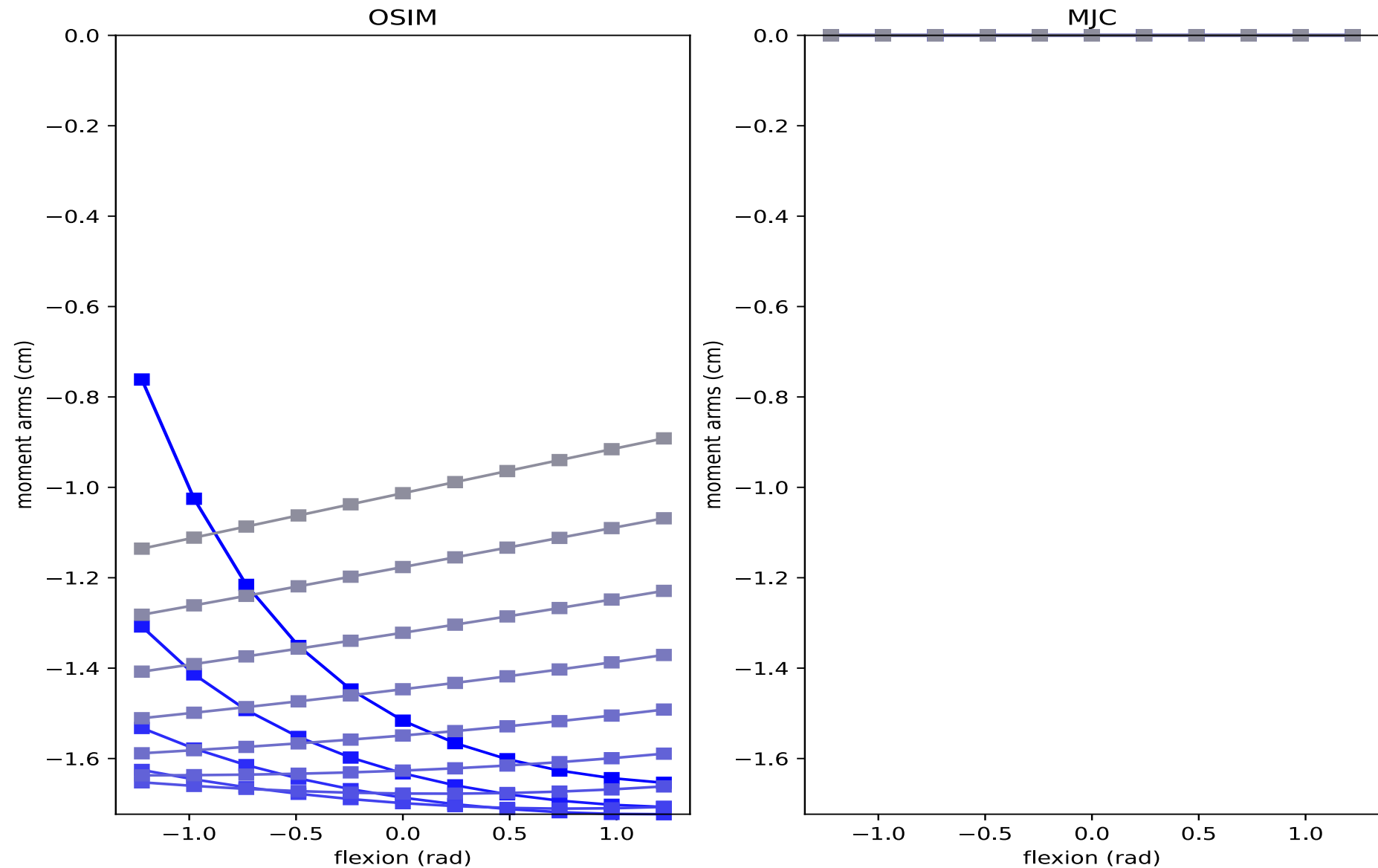
# Muscle specific moment arm comparison with Osim model after optimization

ECRB - elbow\_flexion - pro\_sup



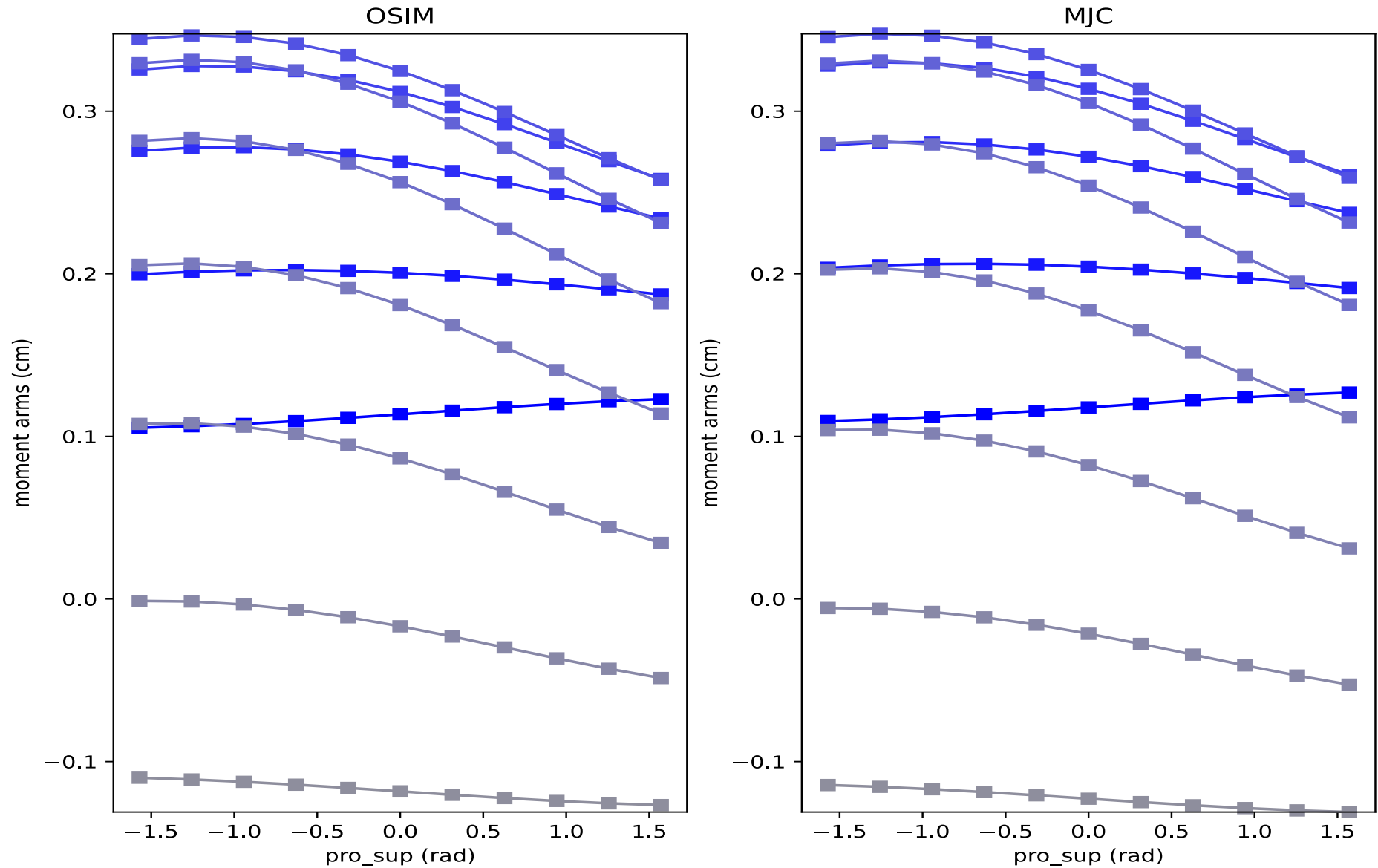
# Muscle specific moment arm comparison with Osim model after optimization

ECRB - deviation - flexion



# Muscle specific moment arm comparison with Osim model after optimization

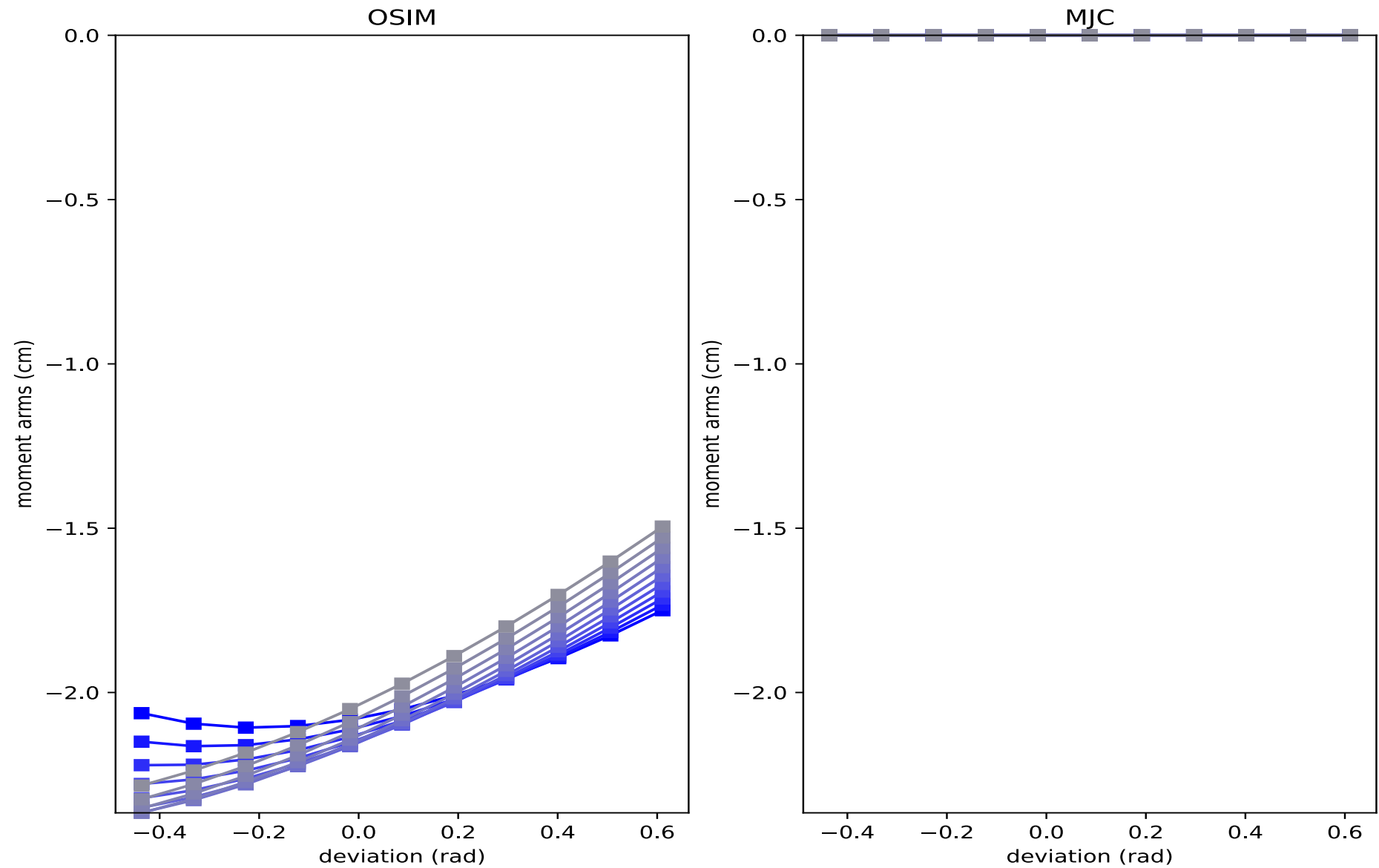
ECRB - elbow\_flexion - pro\_sup





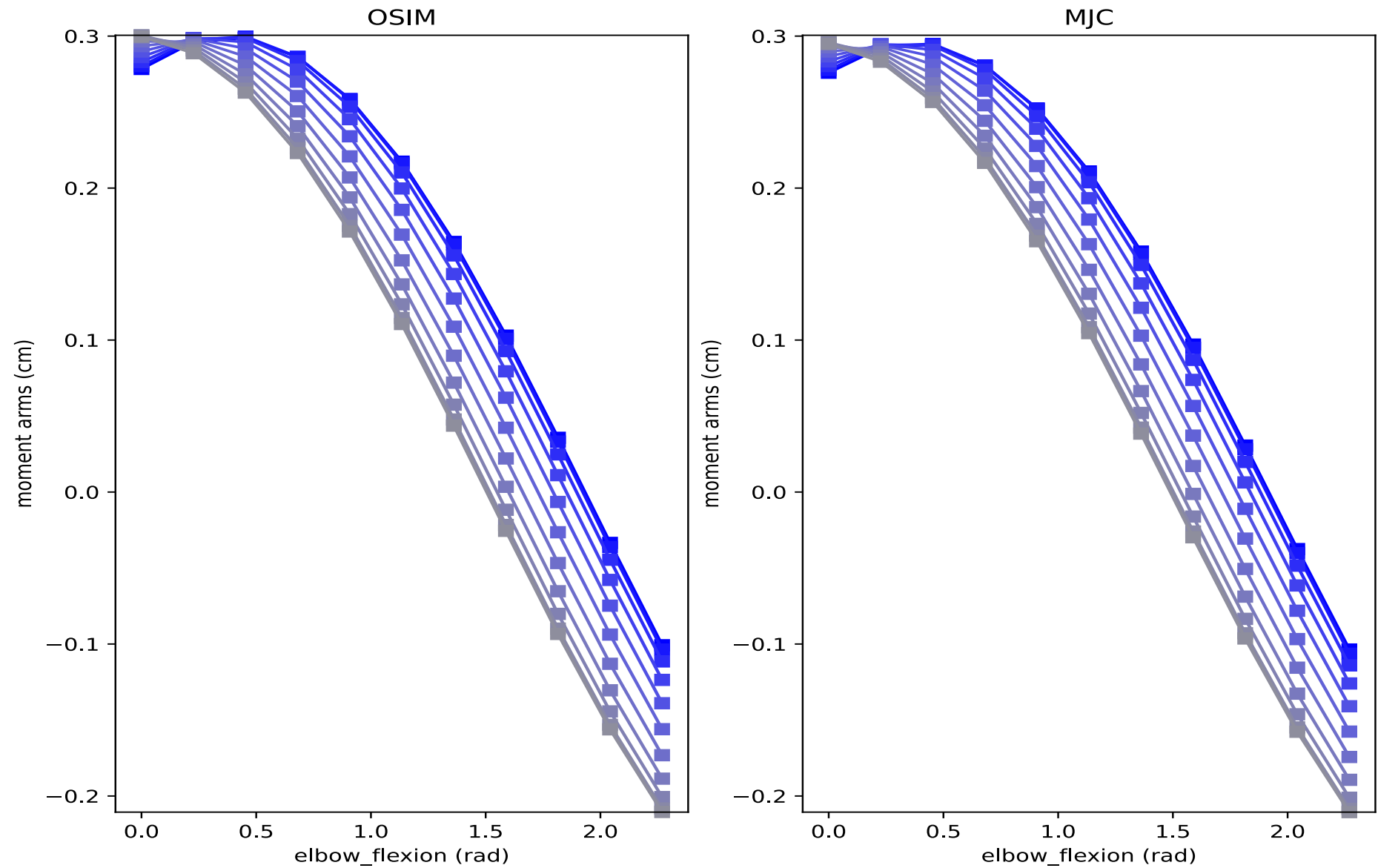
# Muscle specific moment arm comparison with Osim model after optimization

ECRL - deviation - flexion



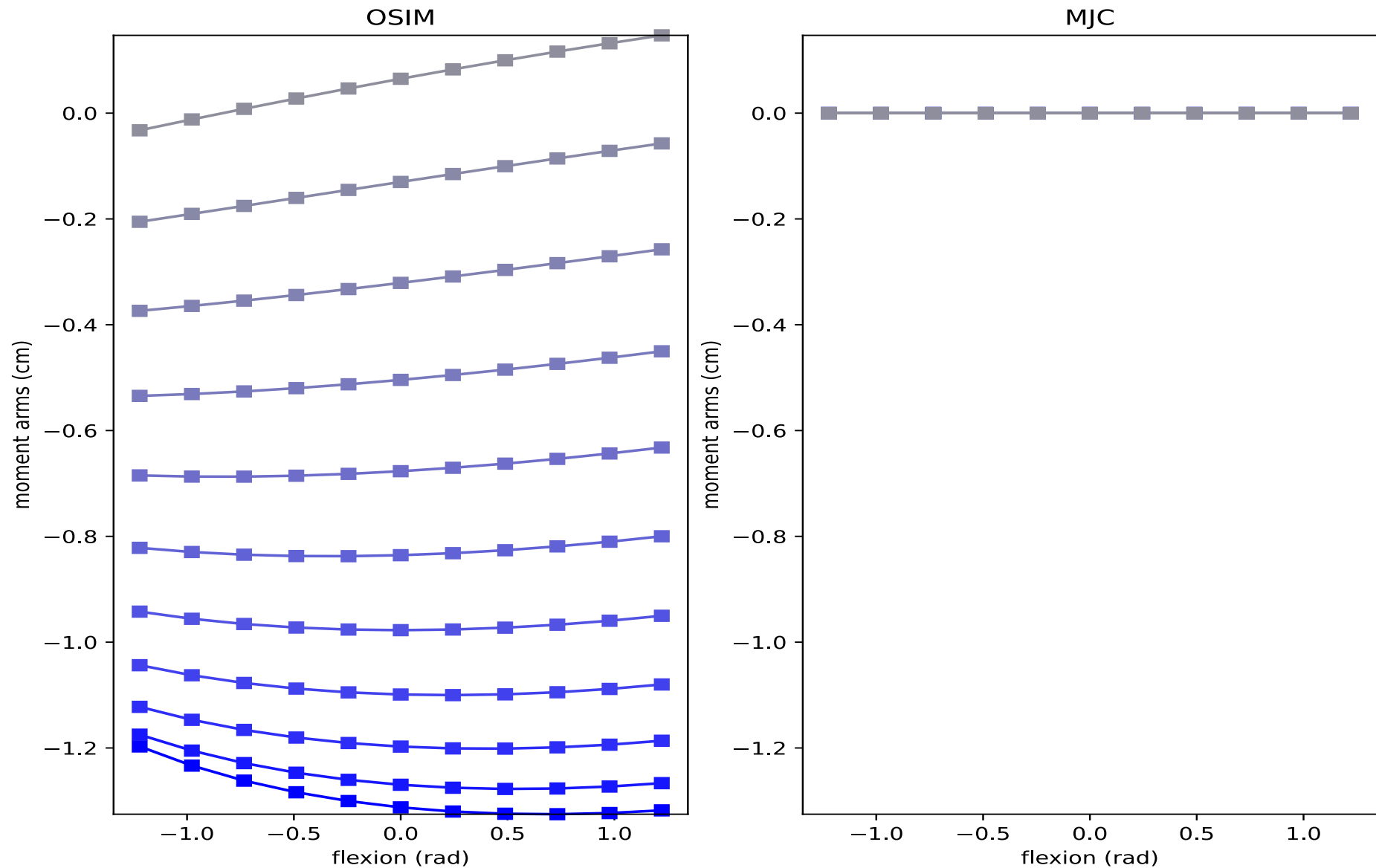
# Muscle specific moment arm comparison with Osim model after optimization

ECRL - elbow\_flexion - pro\_sup



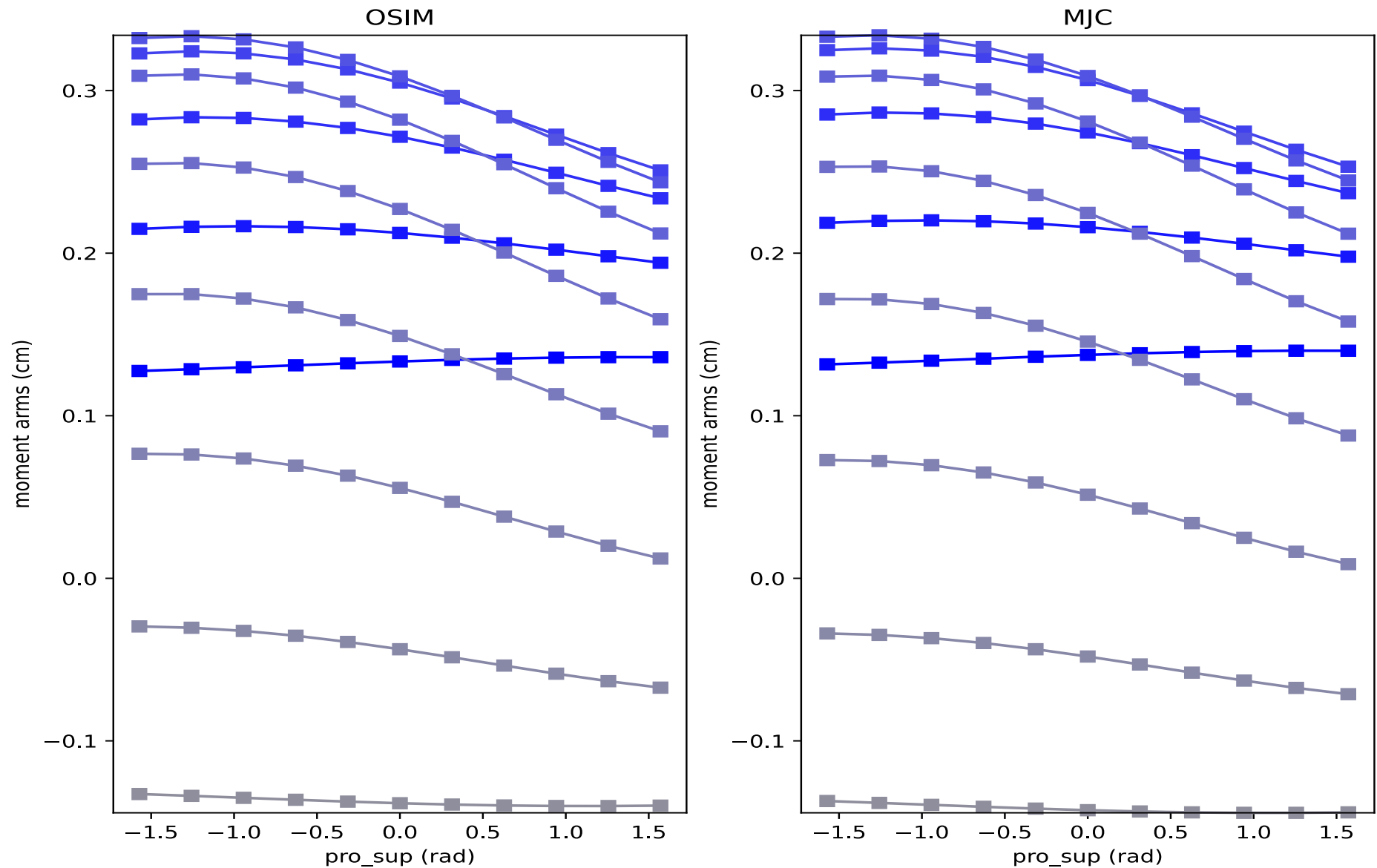
# Muscle specific moment arm comparison with Osim model after optimization

ECRL - deviation - flexion



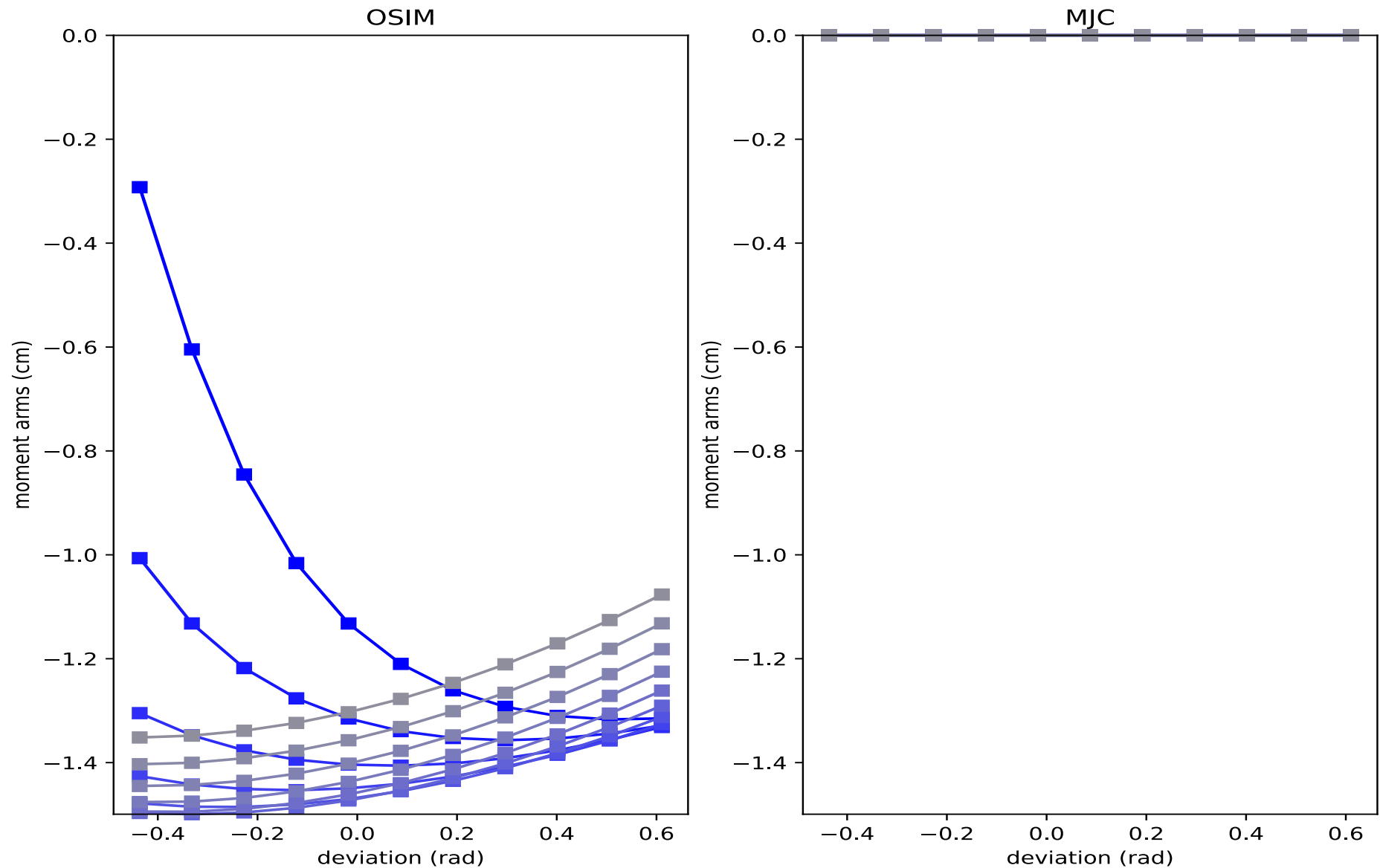
# Muscle specific moment arm comparison with Osim model after optimization

ECRL - elbow\_flexion - pro\_sup



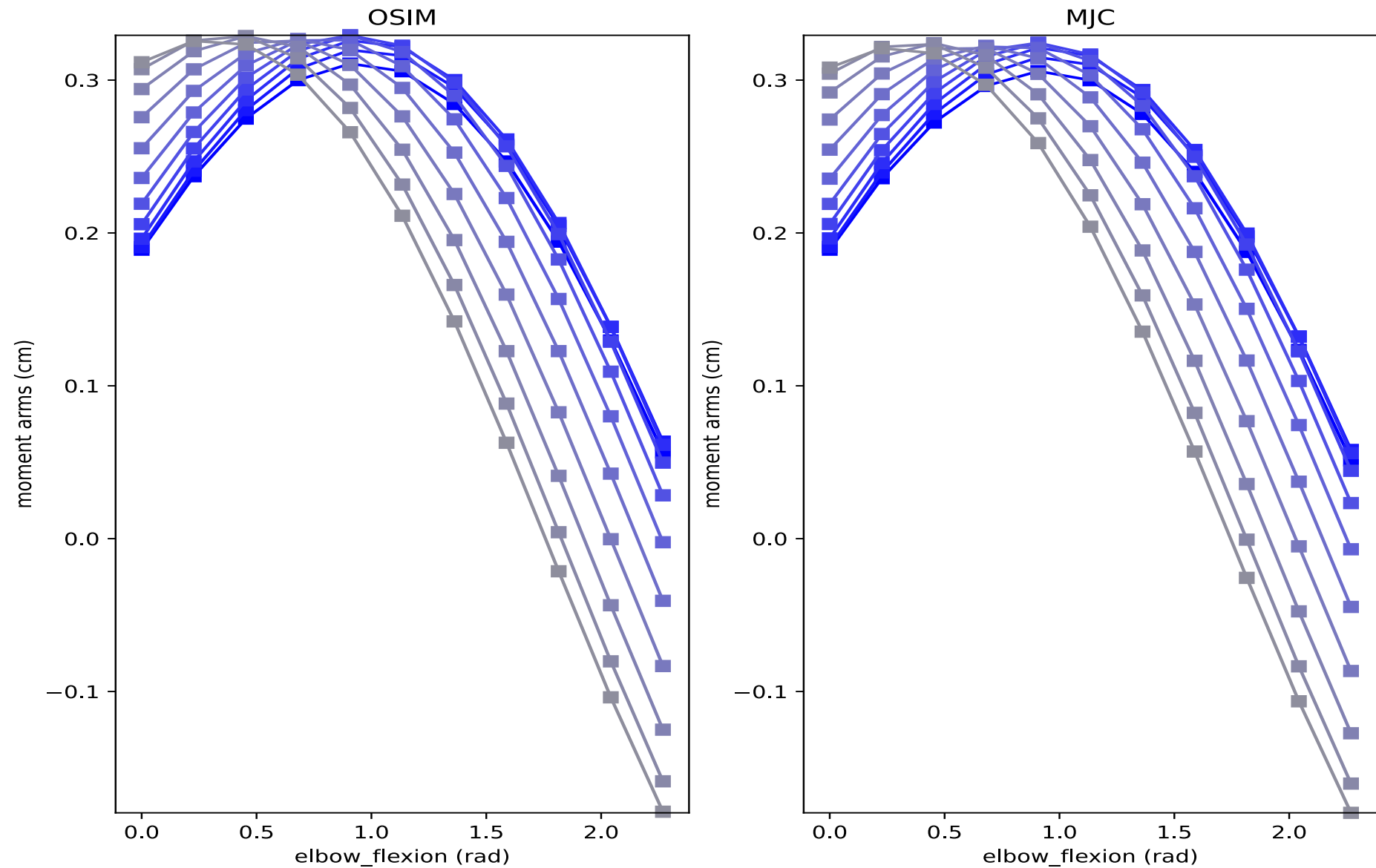
# Muscle specific moment arm comparison with Osim model after optimization

ECU\_post-surgery - deviation - flexion



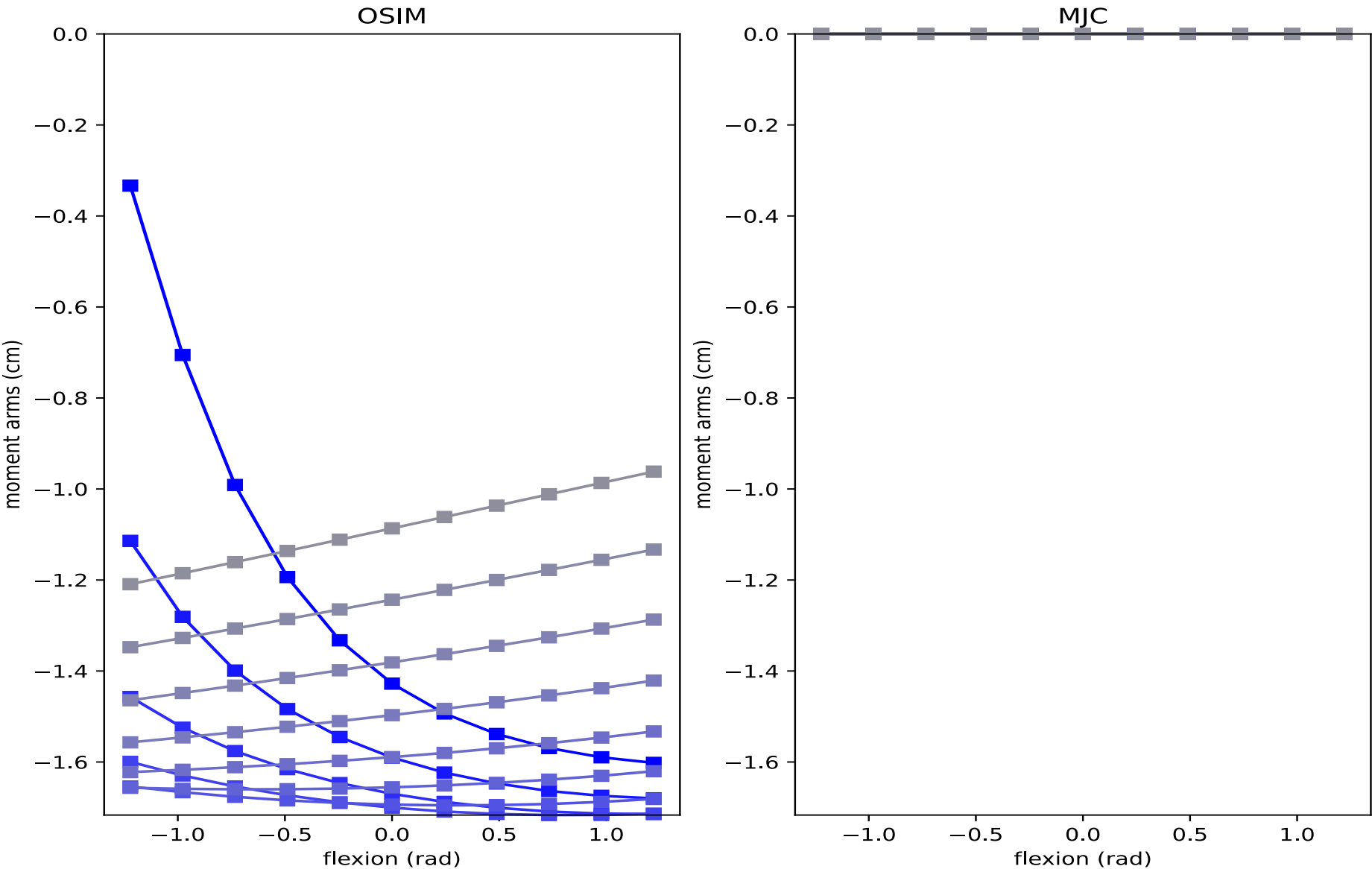
# Muscle specific moment arm comparison with Osim model after optimization

ECU\_post-surgery - elbow\_flexion - pro\_sup



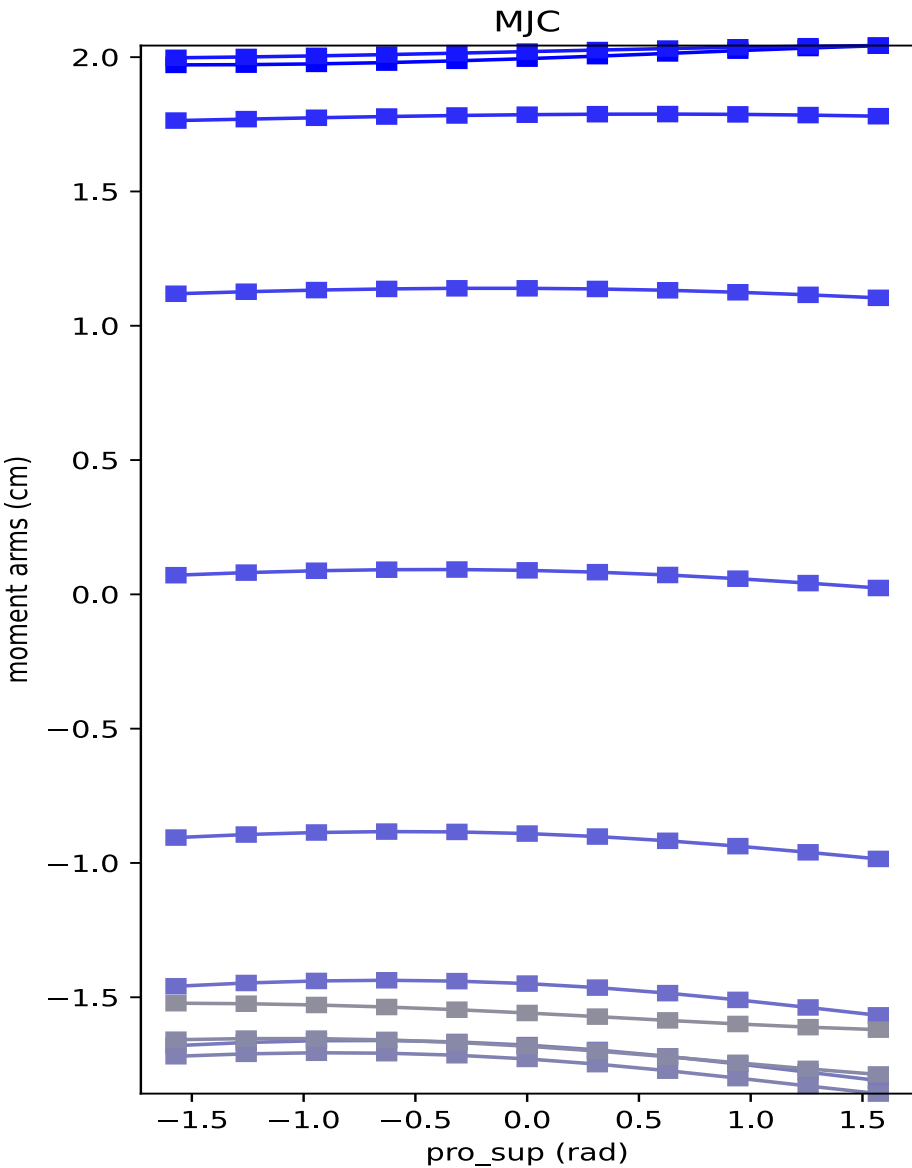
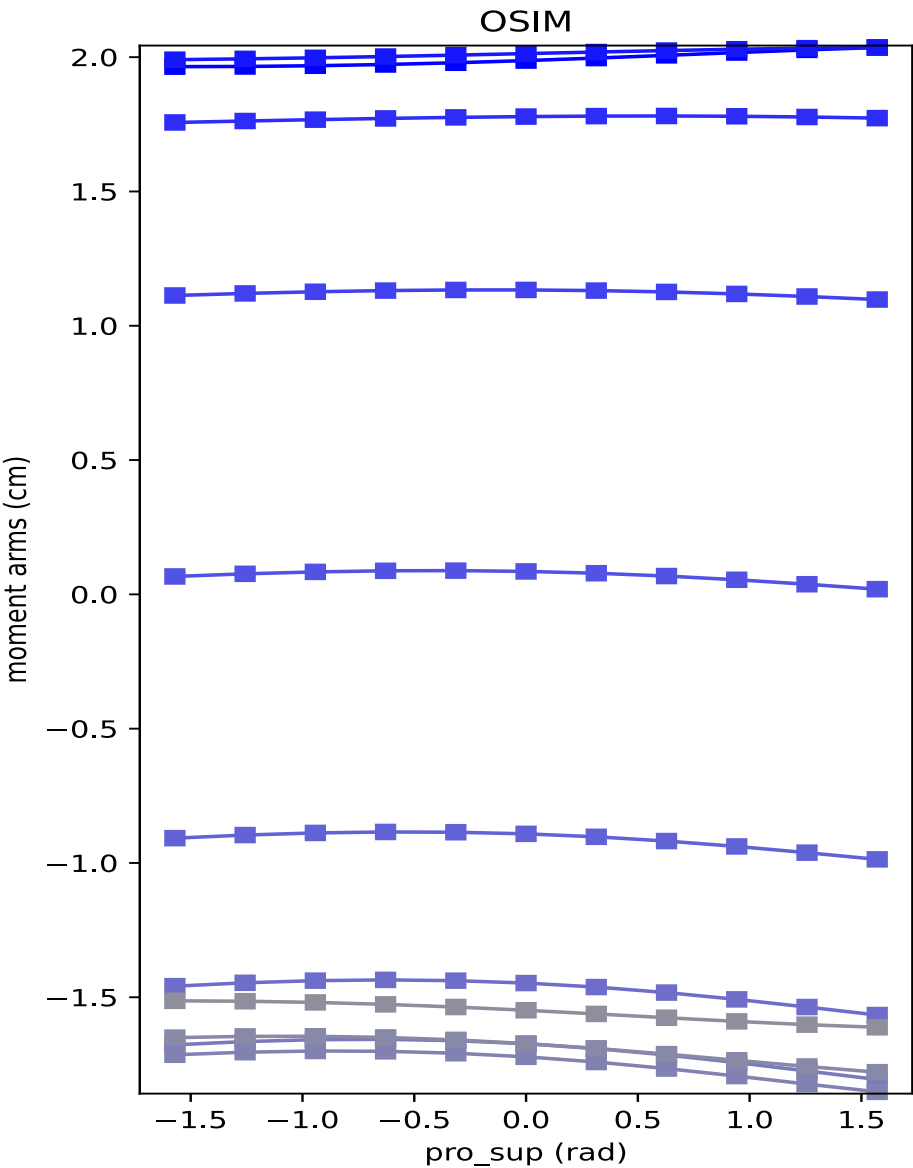
# Muscle specific moment arm comparison with Osim model after optimization

ECU\_post-surgery - deviation - flexion



# Muscle specific moment arm comparison with Osim model after optimization

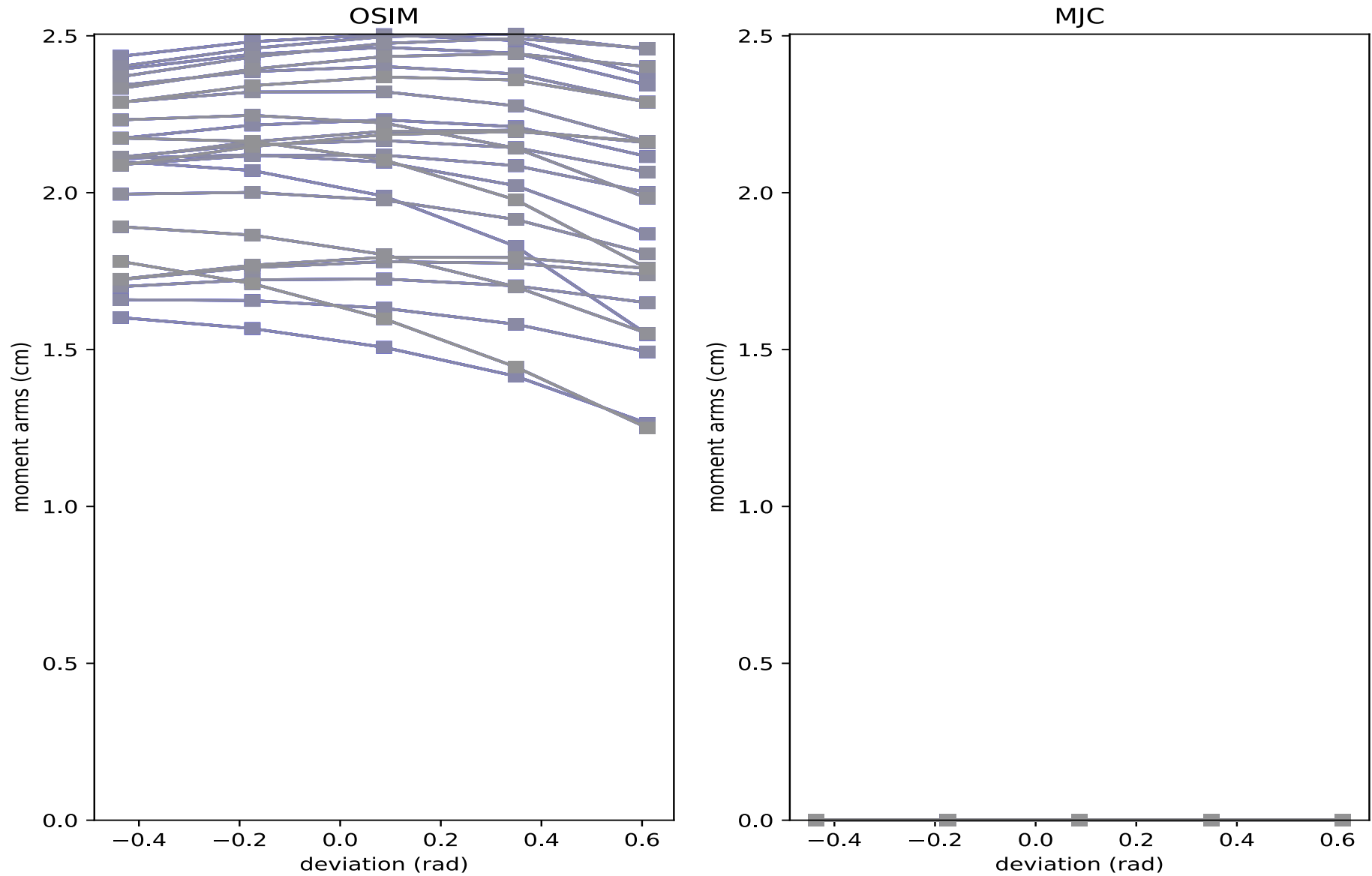
ECU\_post-surgery - elbow\_flexion - pro\_sup





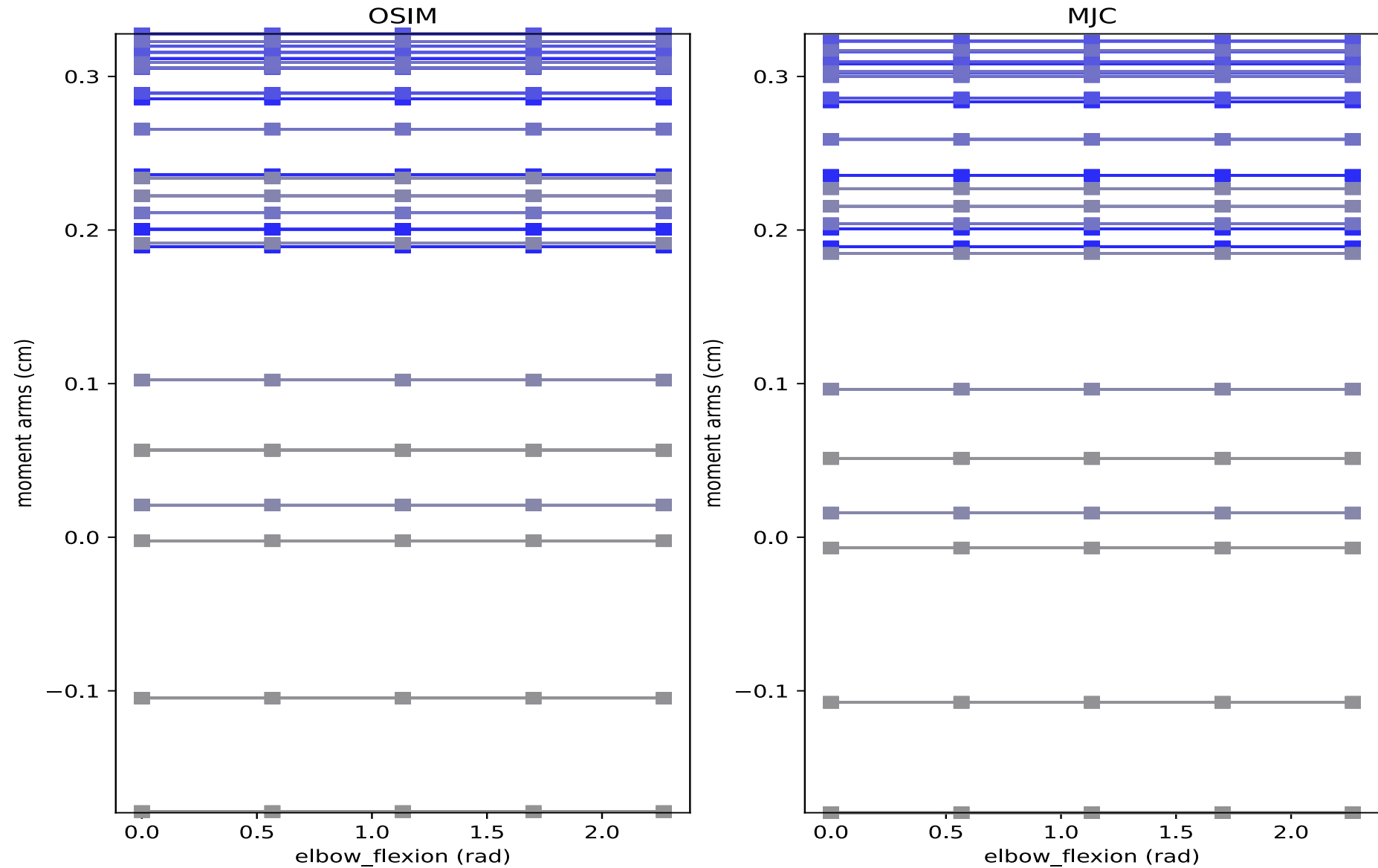
# Muscle specific moment arm comparison with Osim model after optimization

ECU\_pre-surgery - deviation - elbow\_flexion - flexion - pro\_sup



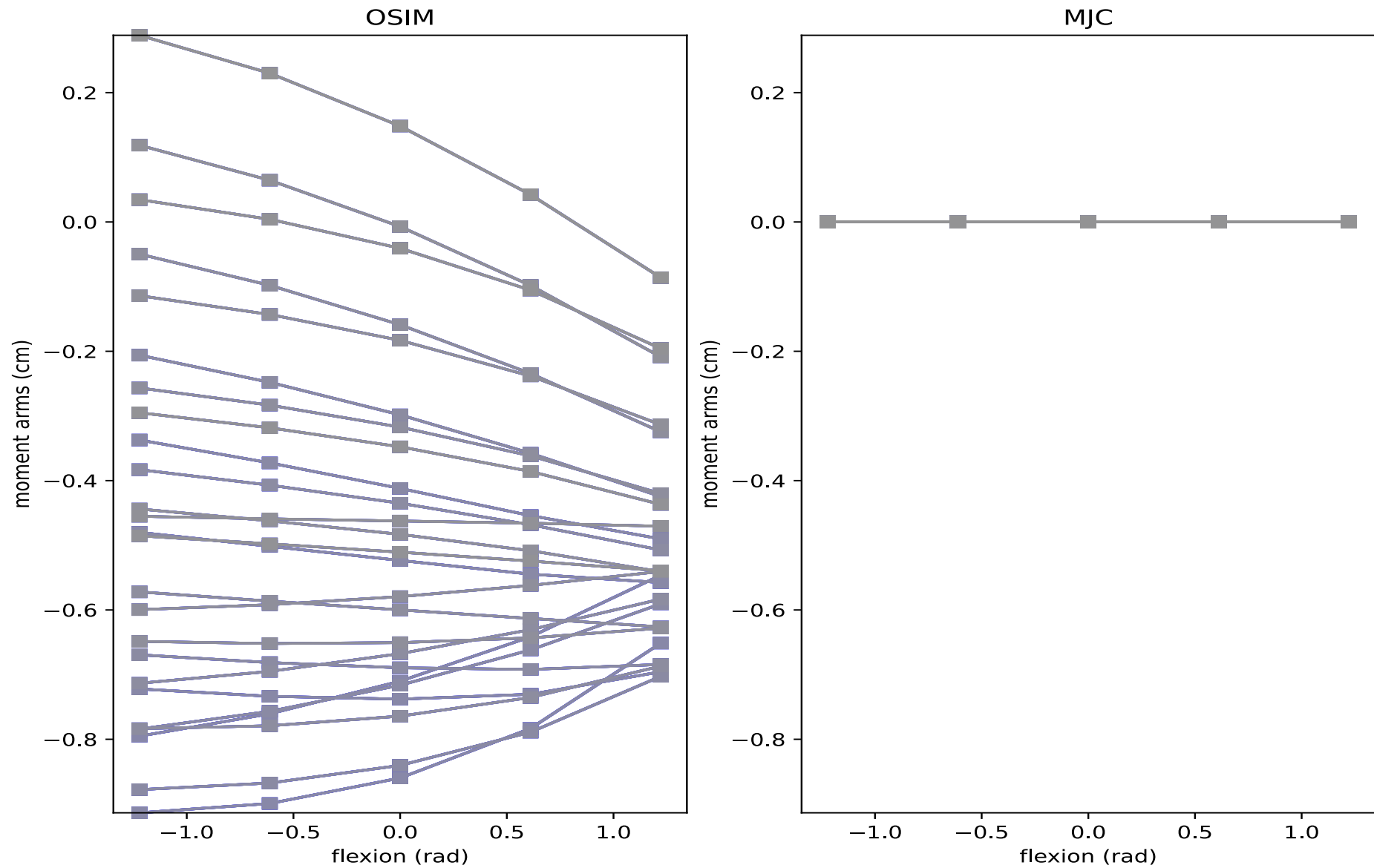
# Muscle specific moment arm comparison with Osim model after optimization

ECU\_pre-surgery - deviation - elbow\_flexion - flexion - pro\_sup



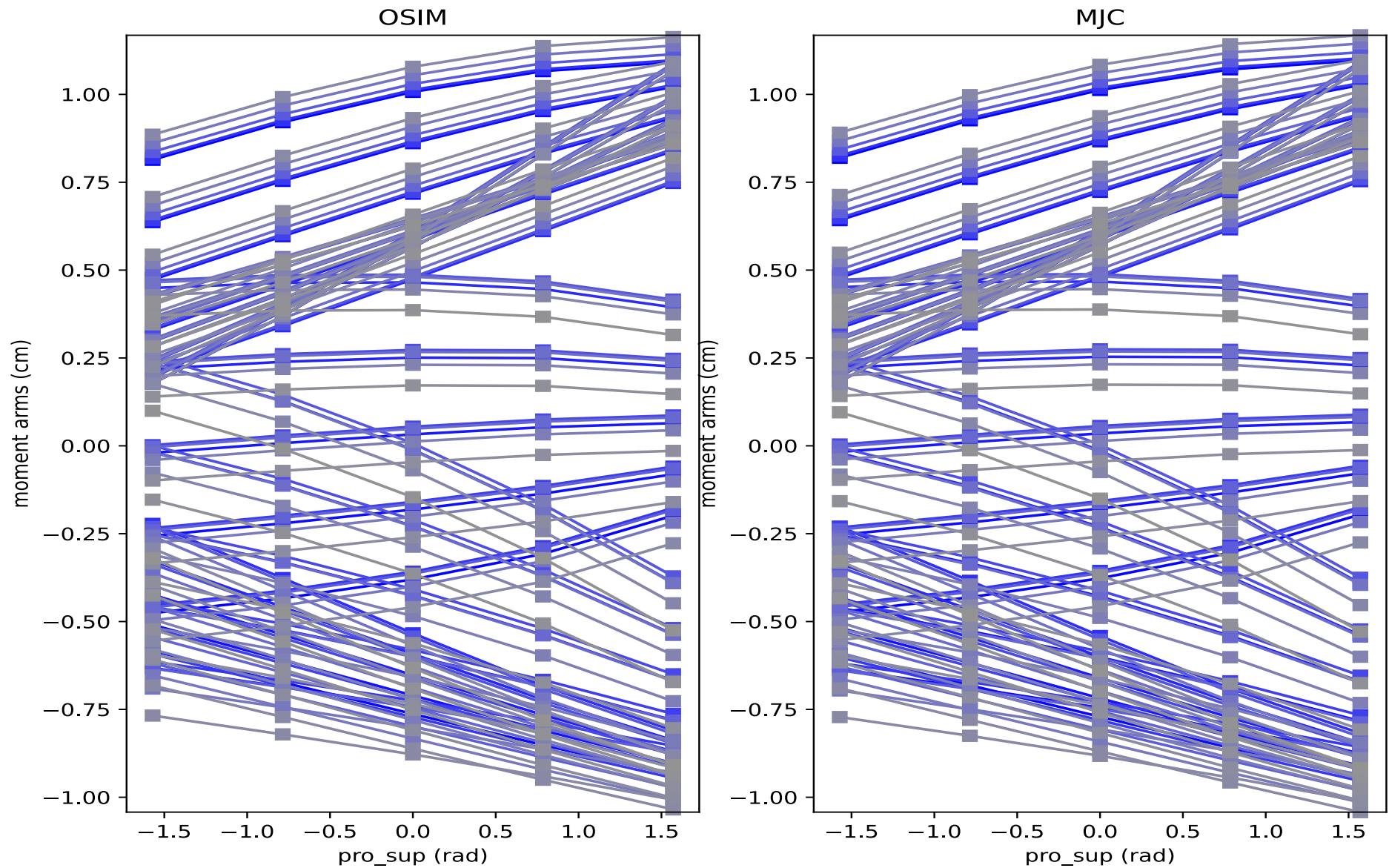
# Muscle specific moment arm comparison with Osim model after optimization

ECU\_pre-surgery - deviation - elbow\_flexion - flexion - pro\_sup



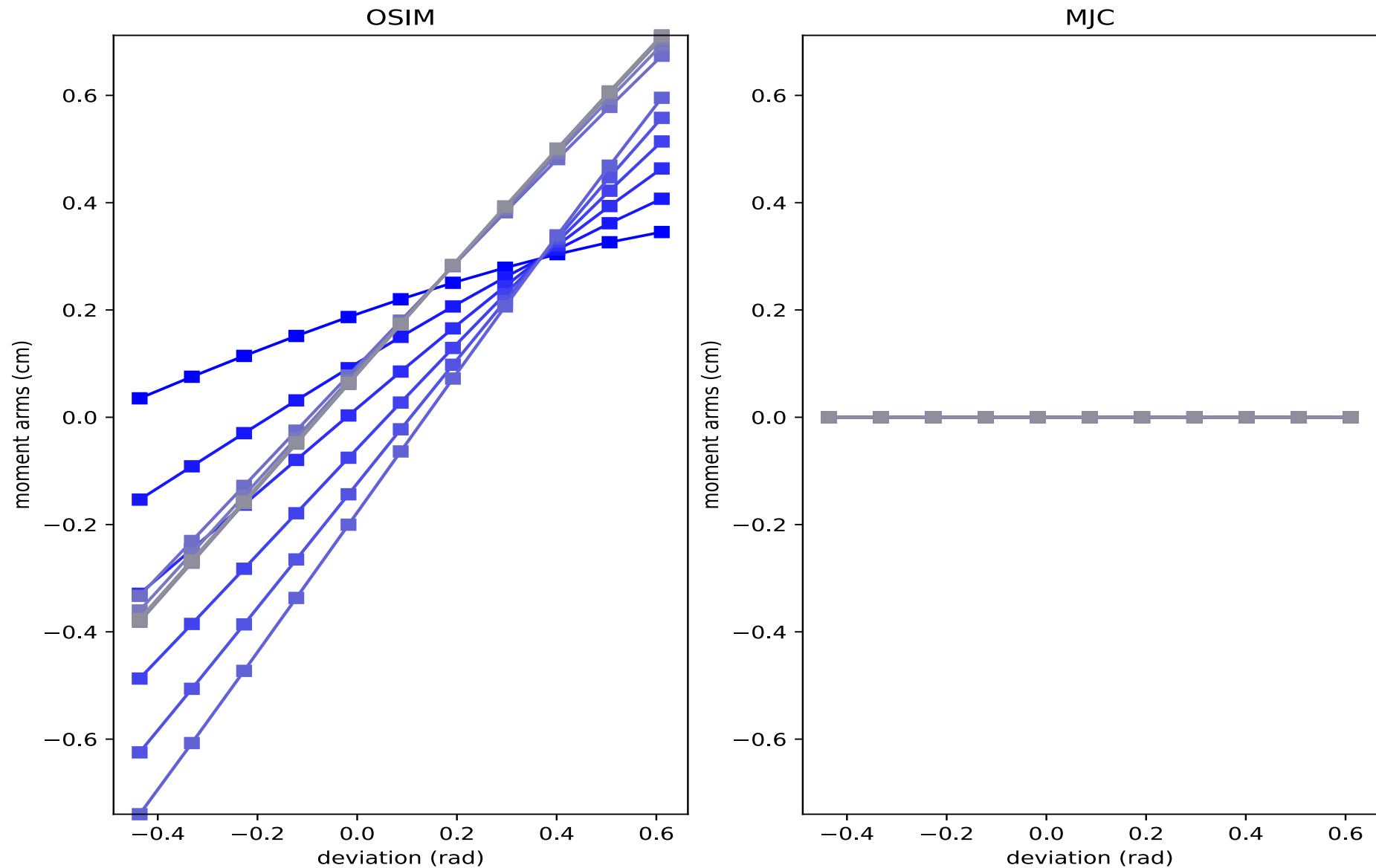
# Muscle specific moment arm comparison with Osim model after optimization

ECU\_pre-surgery - deviation - elbow\_flexion - flexion - pro\_sup



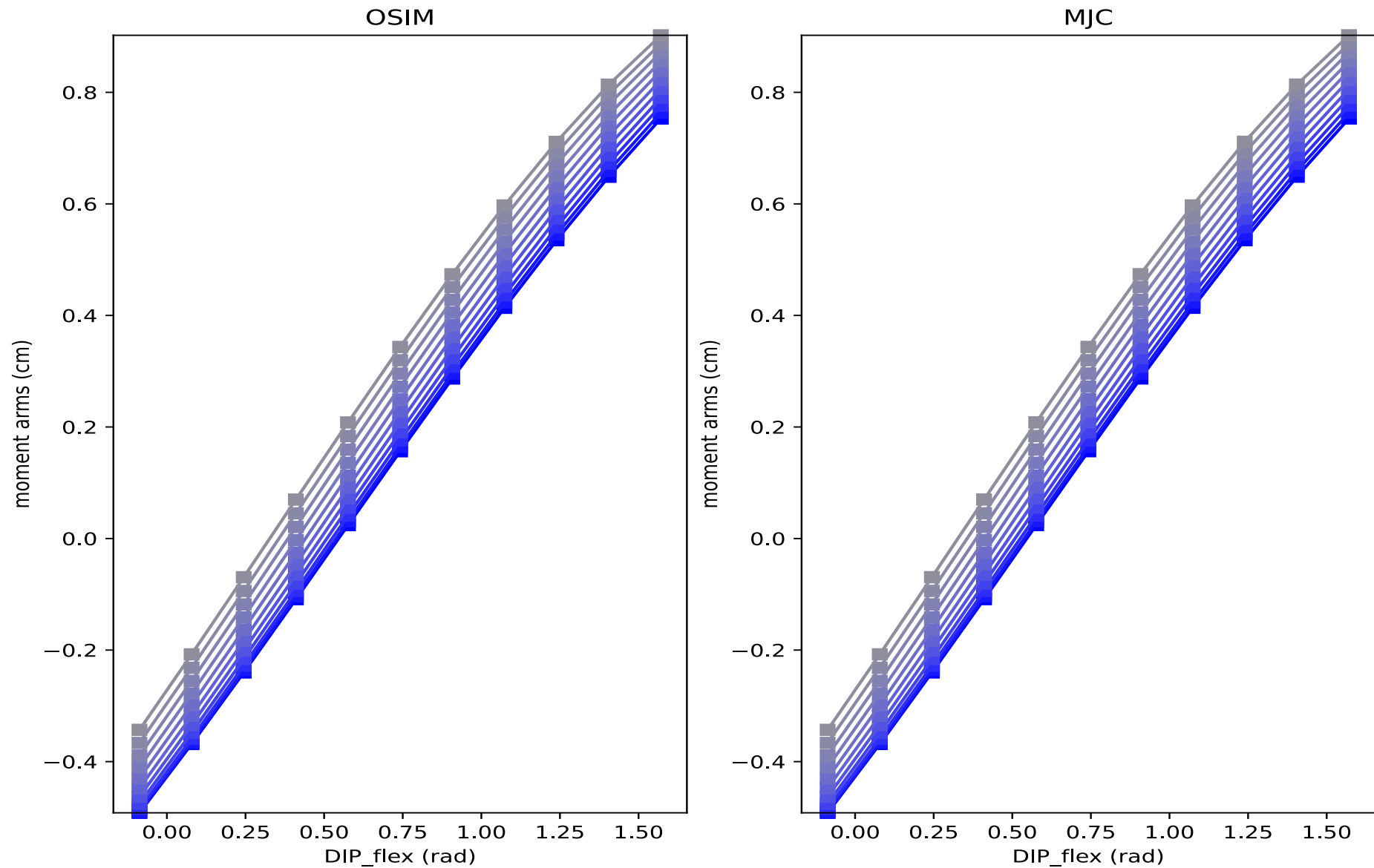
# Muscle specific moment arm comparison with Osim model after optimization

EDCI - deviation - flexion



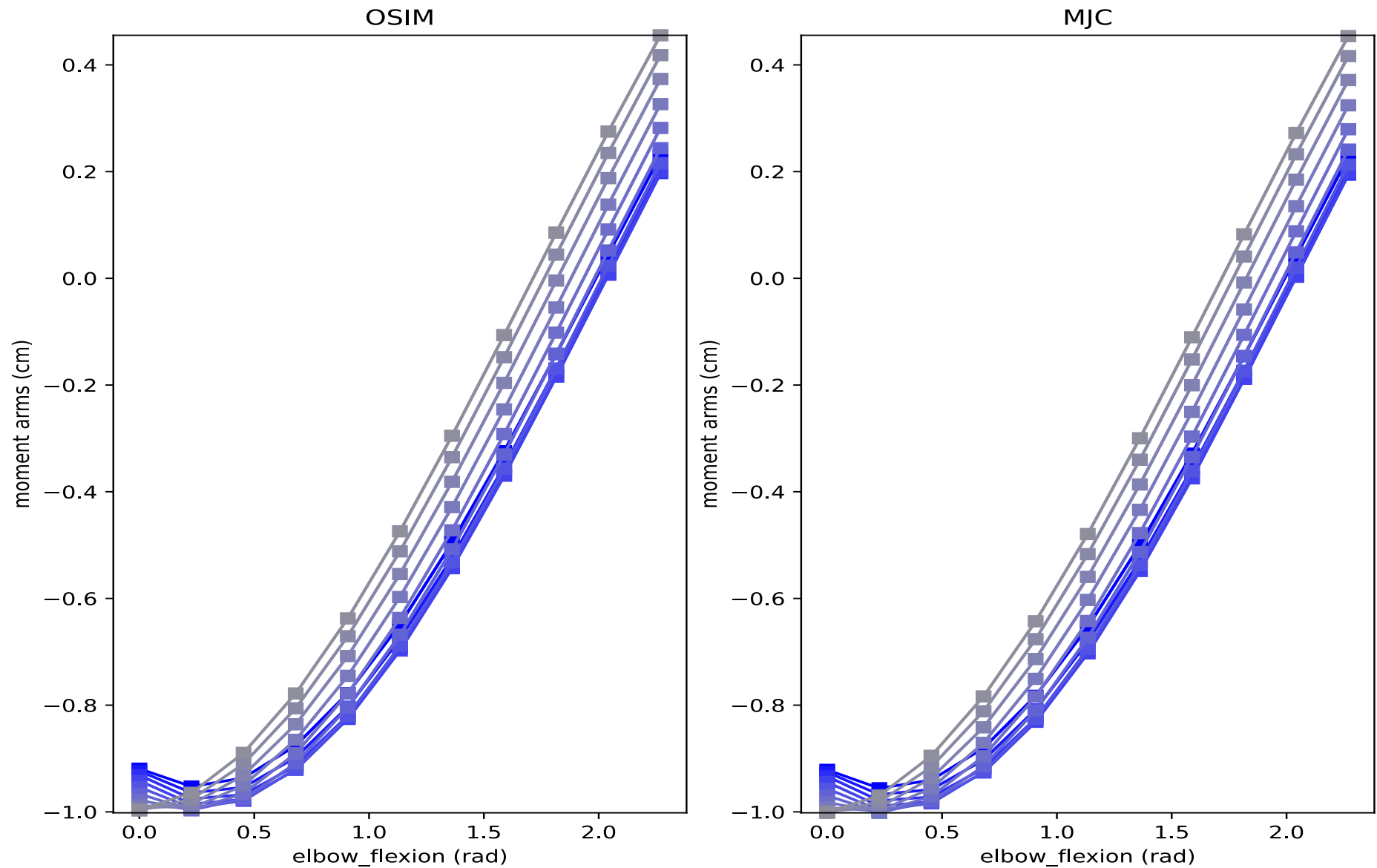
# Muscle specific moment arm comparison with Osim model after optimization

EDCI - DIP\_flex - PIP\_flex



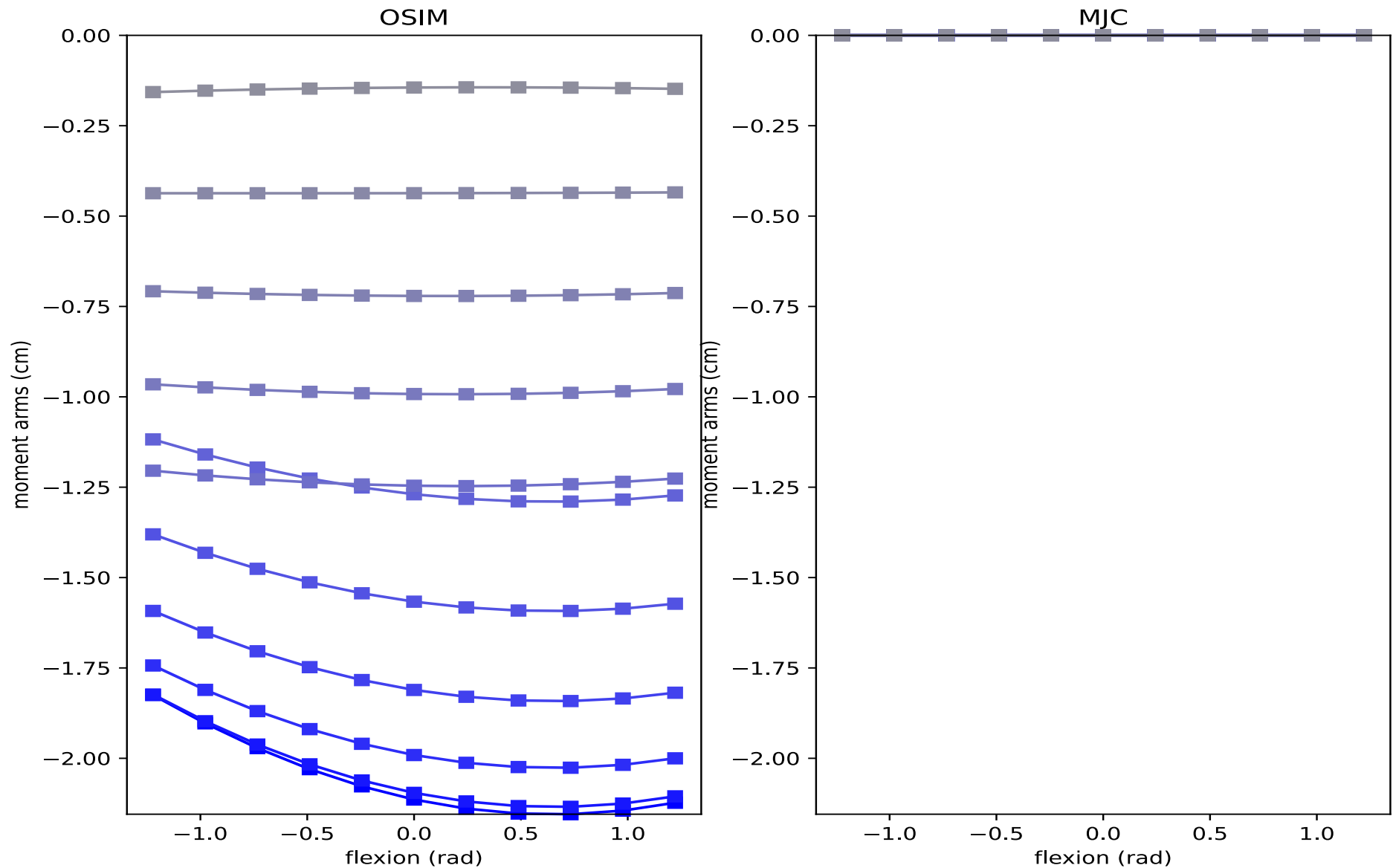
# Muscle specific moment arm comparison with Osim model after optimization

EDCI - elbow\_flexion - pro\_sup



# Muscle specific moment arm comparison with Osim model after optimization

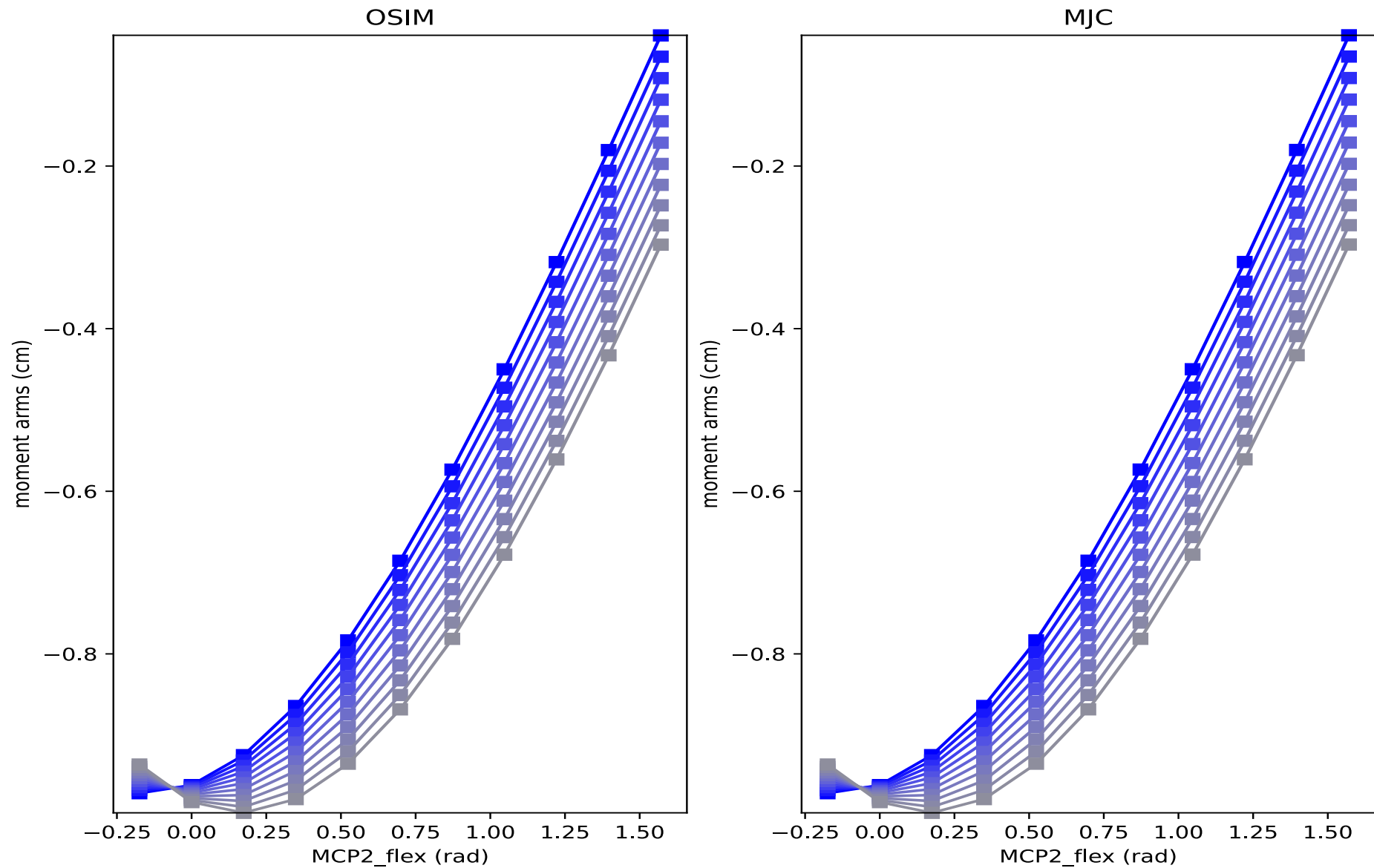
EDCI - deviation - flexion





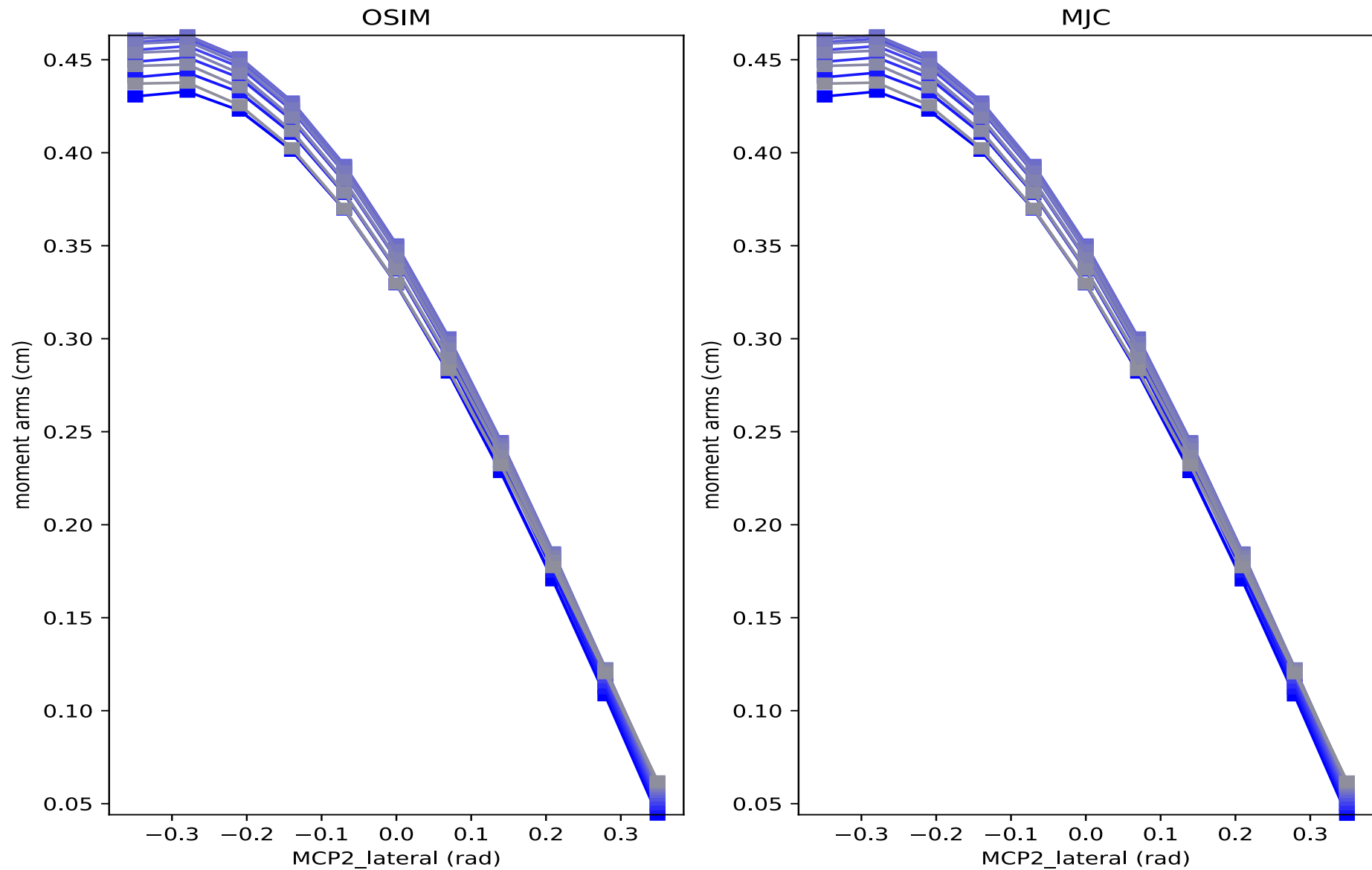
# Muscle specific moment arm comparison with Osim model after optimization

EDCI - MCP2\_flex - MCP2\_lateral



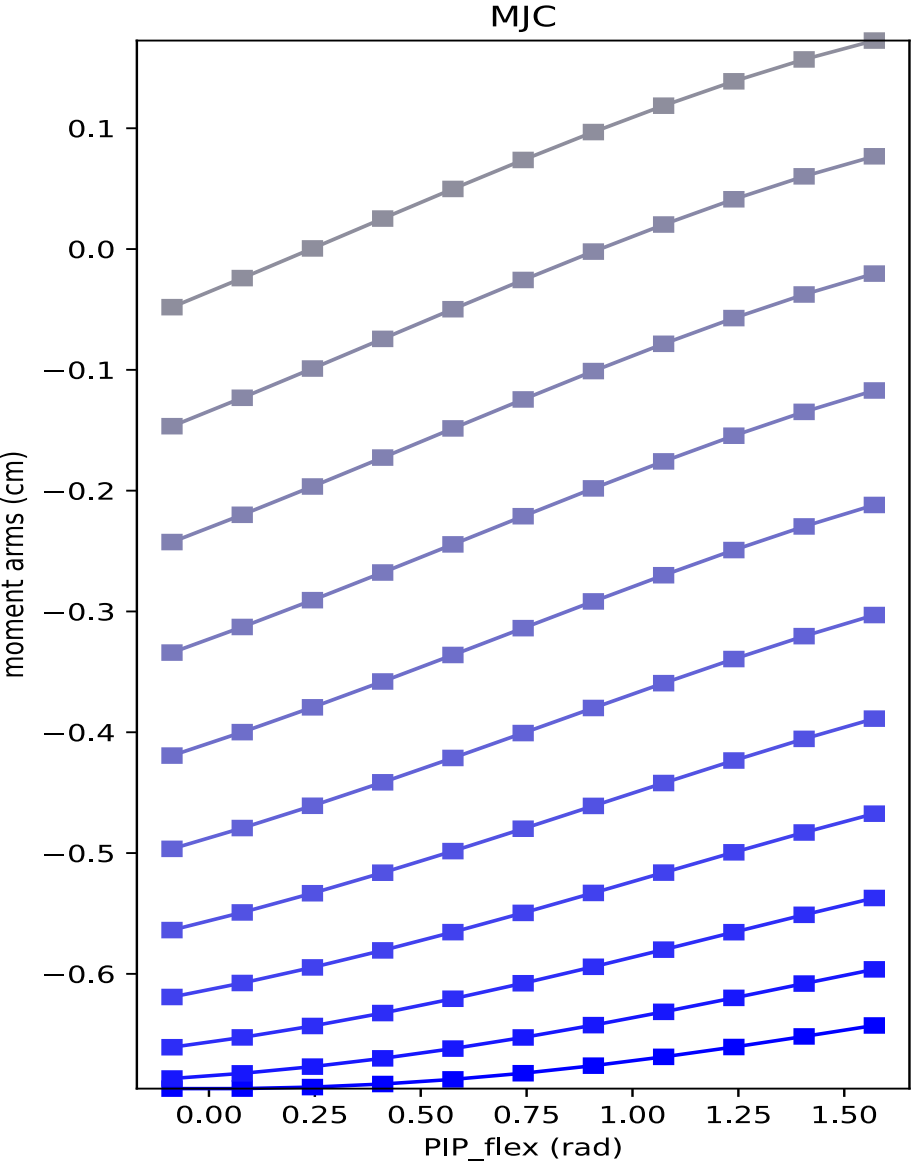
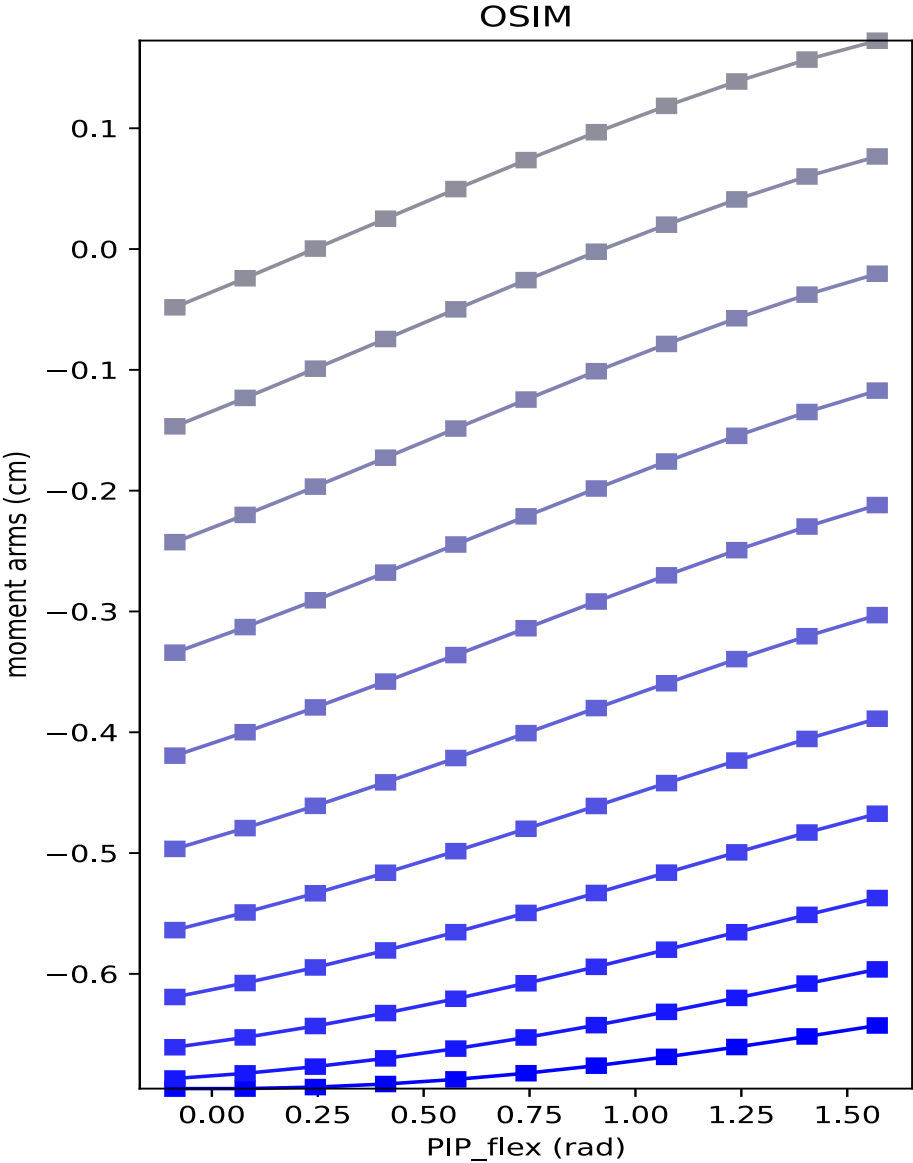
# Muscle specific moment arm comparison with Osim model after optimization

EDCI - MCP2\_flex - MCP2\_lateral



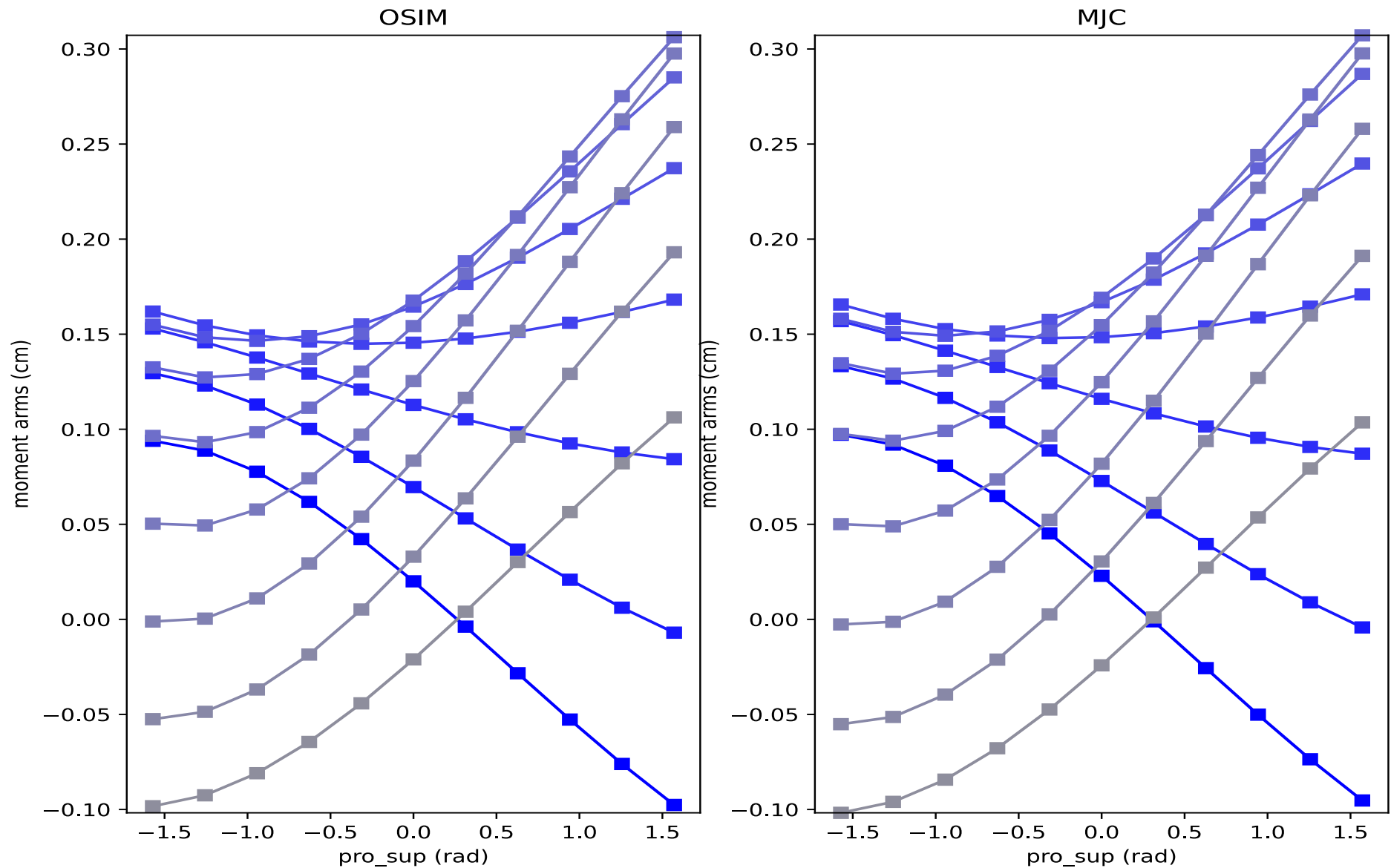
# Muscle specific moment arm comparison with Osim model after optimization

EDCI - DIP\_flex - PIP\_flex



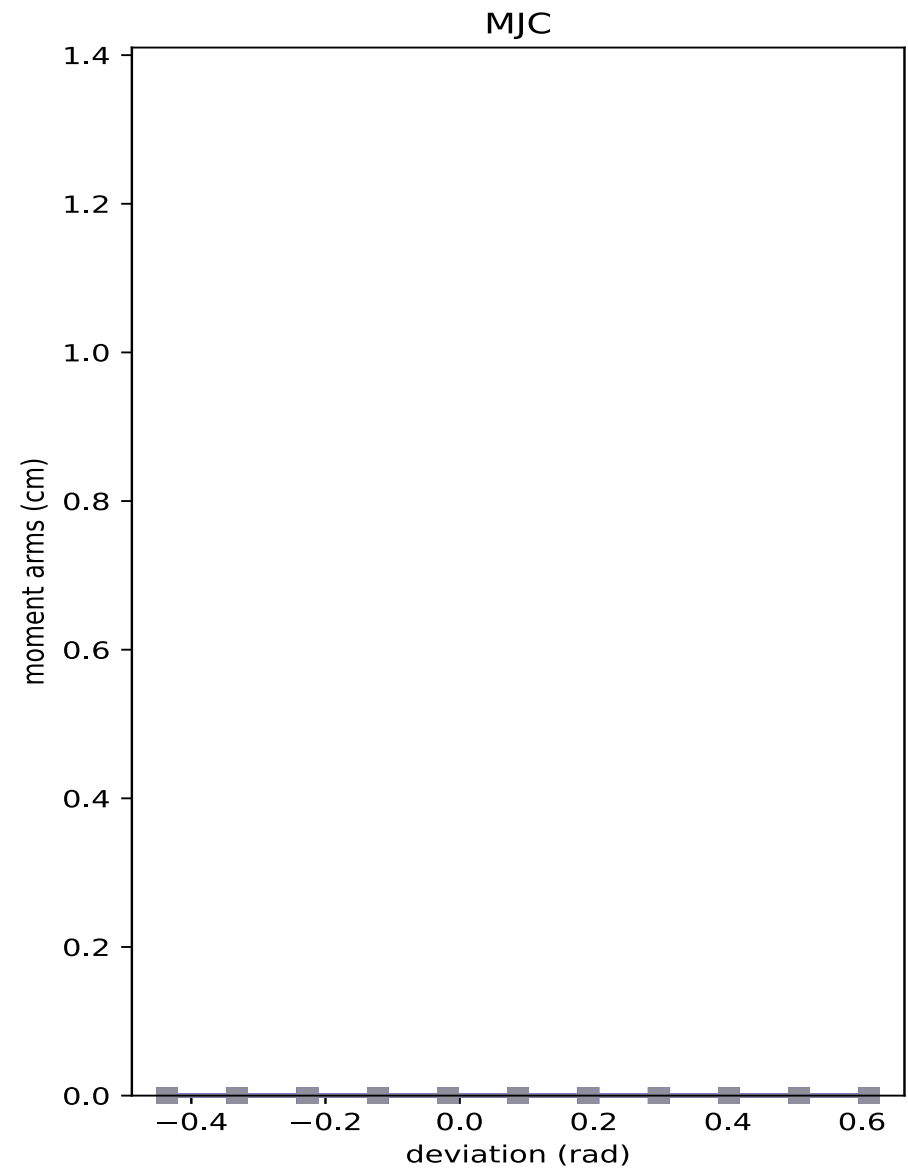
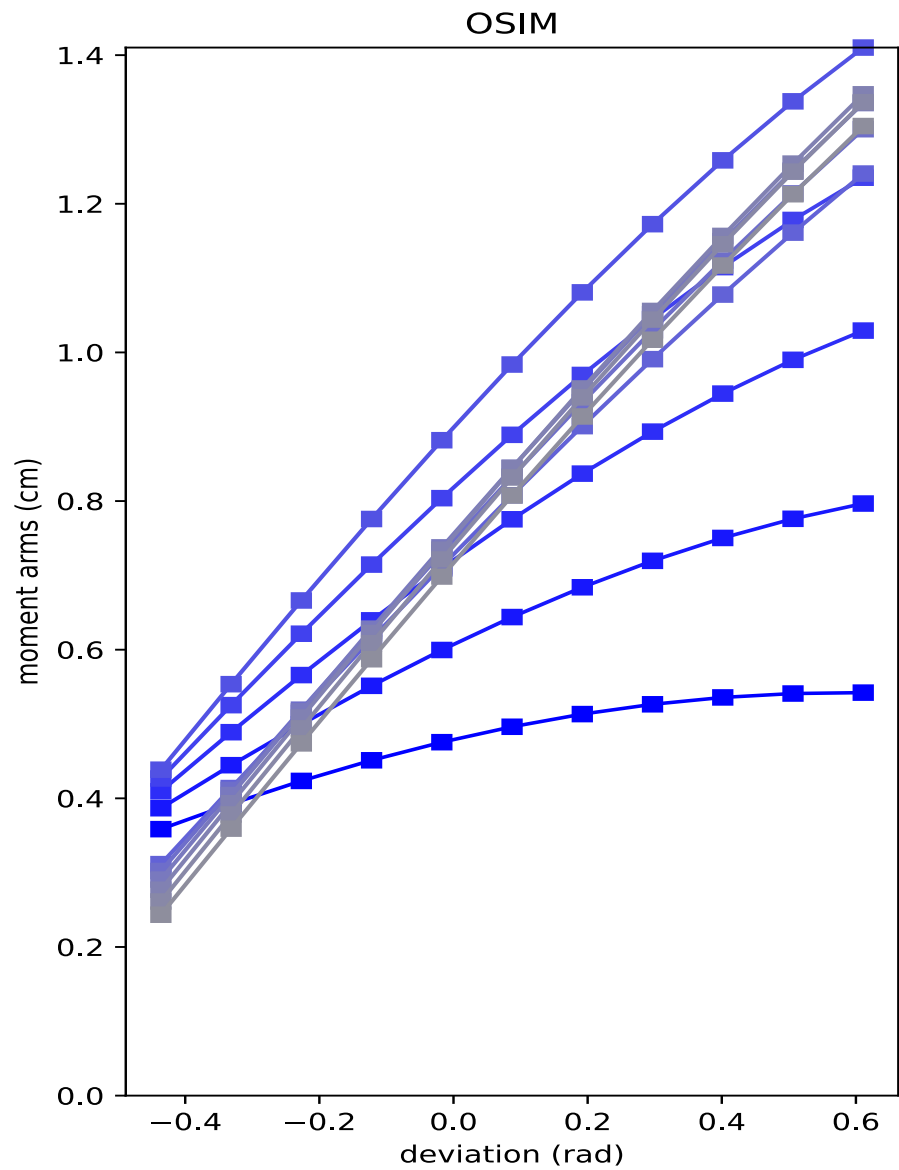
# Muscle specific moment arm comparison with Osim model after optimization

EDCI - elbow\_flexion - pro\_sup



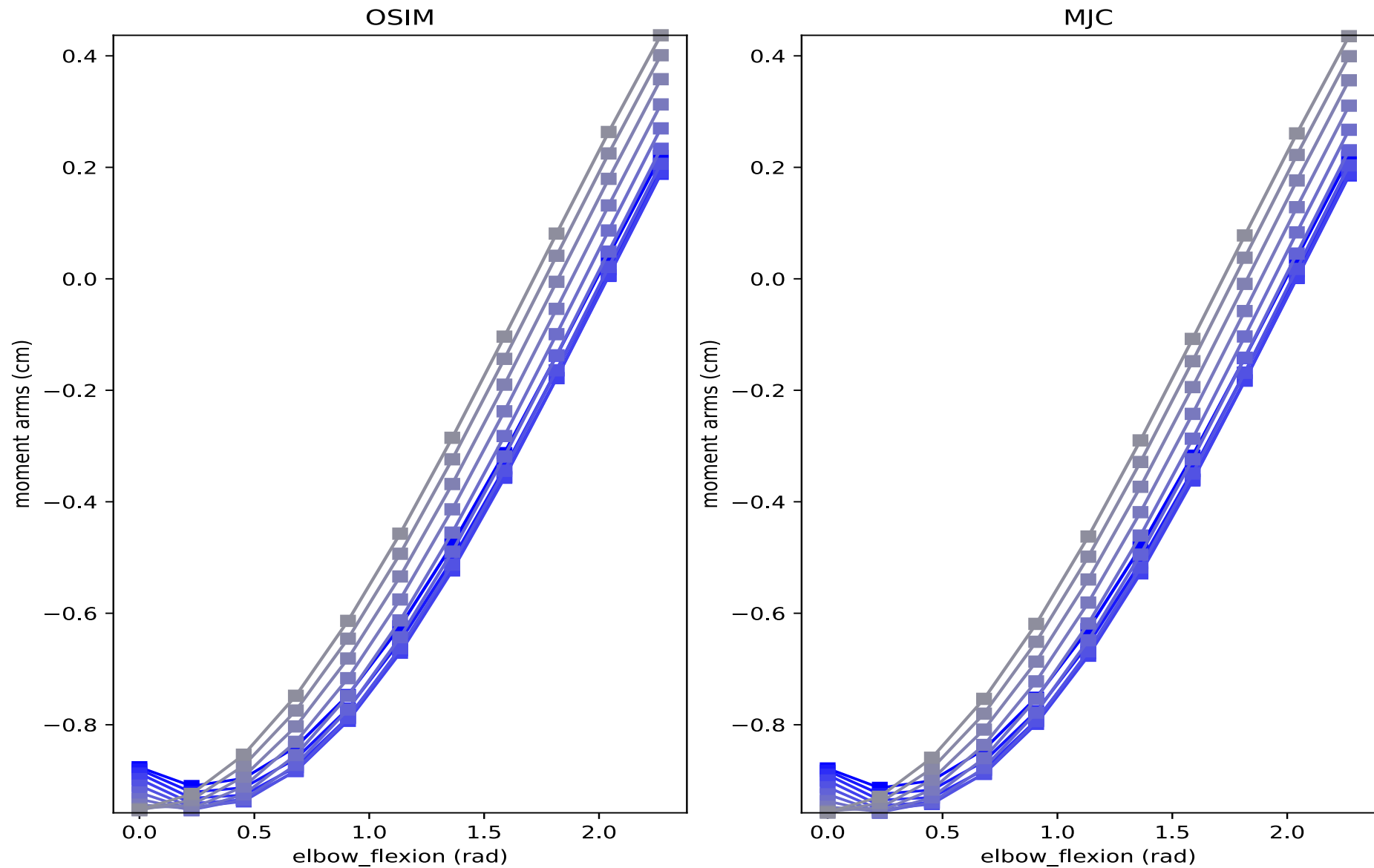
# Muscle specific moment arm comparison with Osim model after optimization

EDCL - deviation - flexion



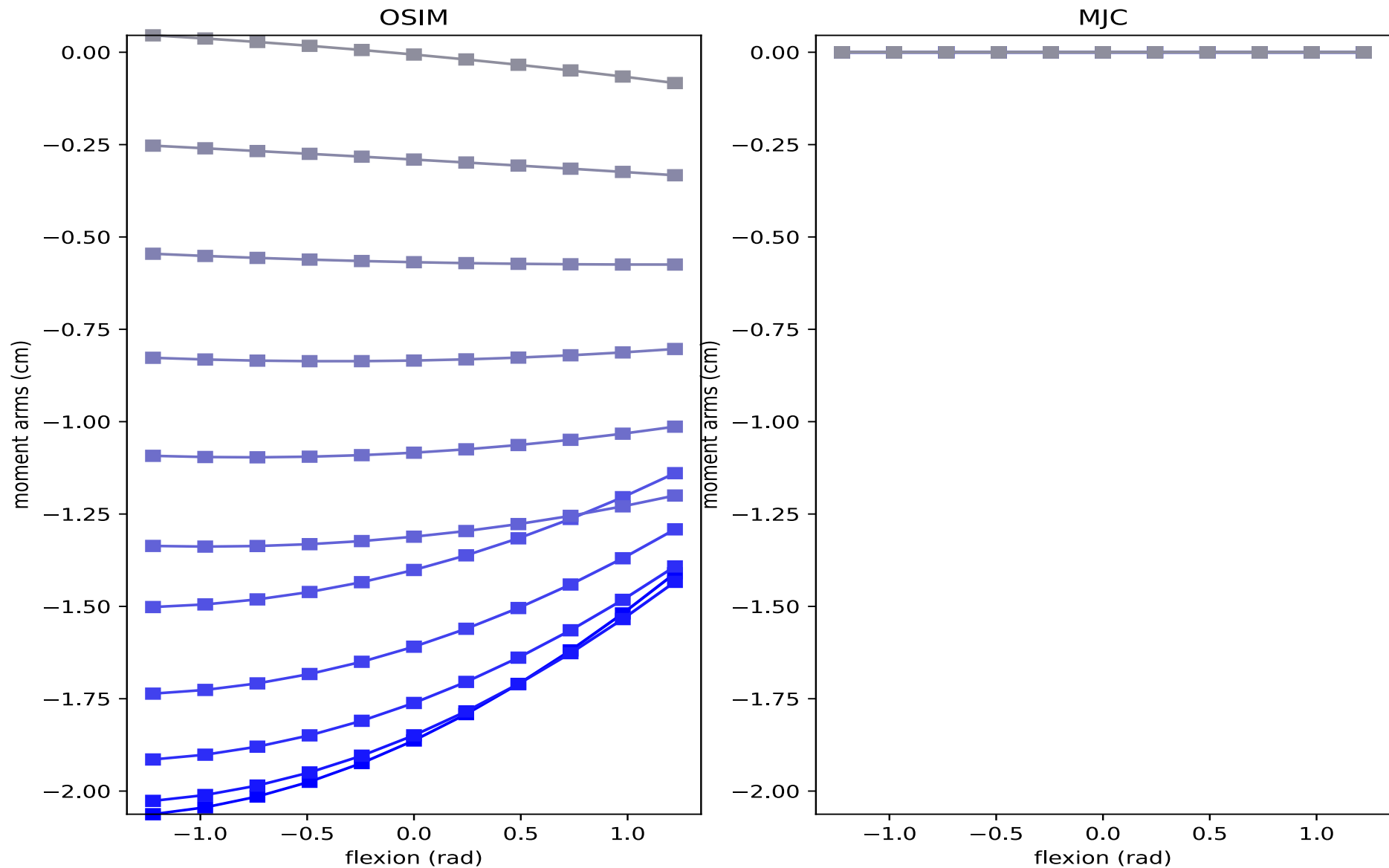
# Muscle specific moment arm comparison with Osim model after optimization

EDCL - elbow\_flexion - pro\_sup



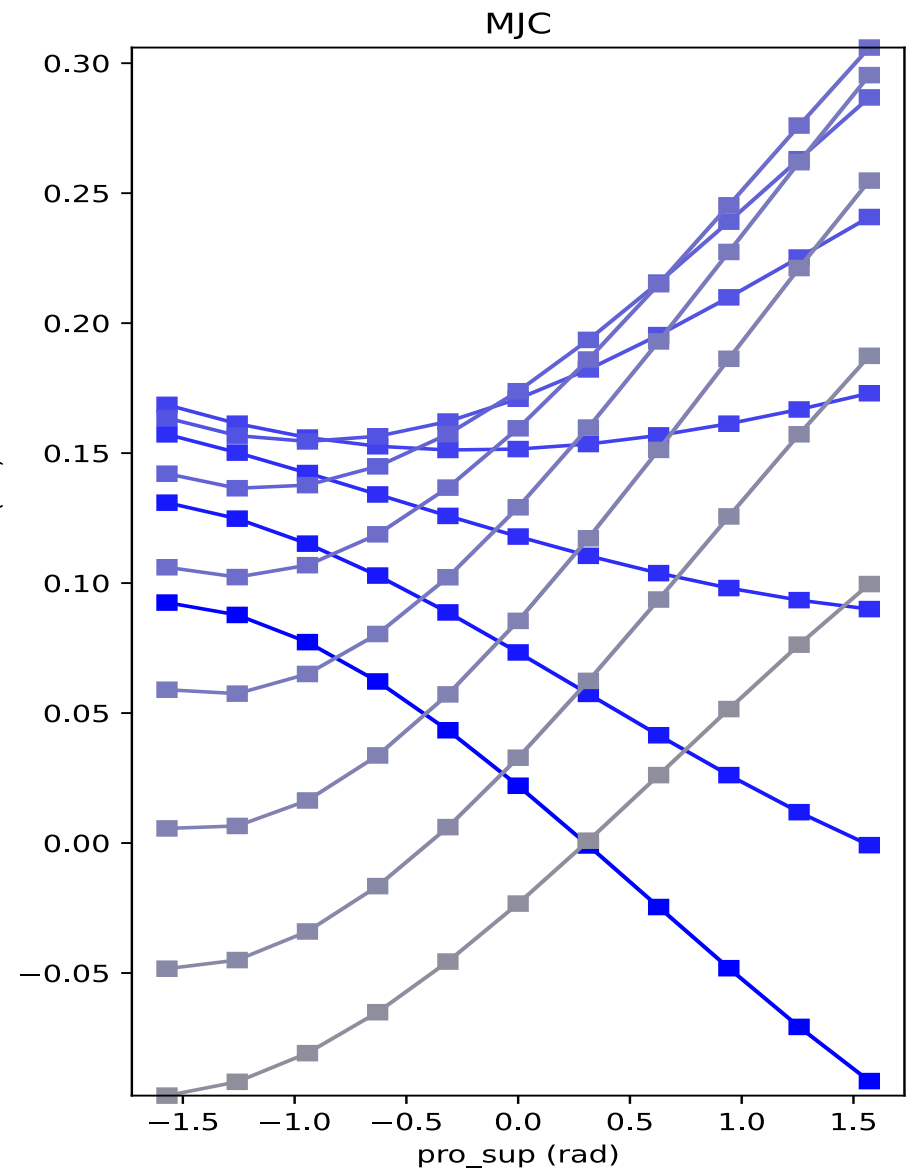
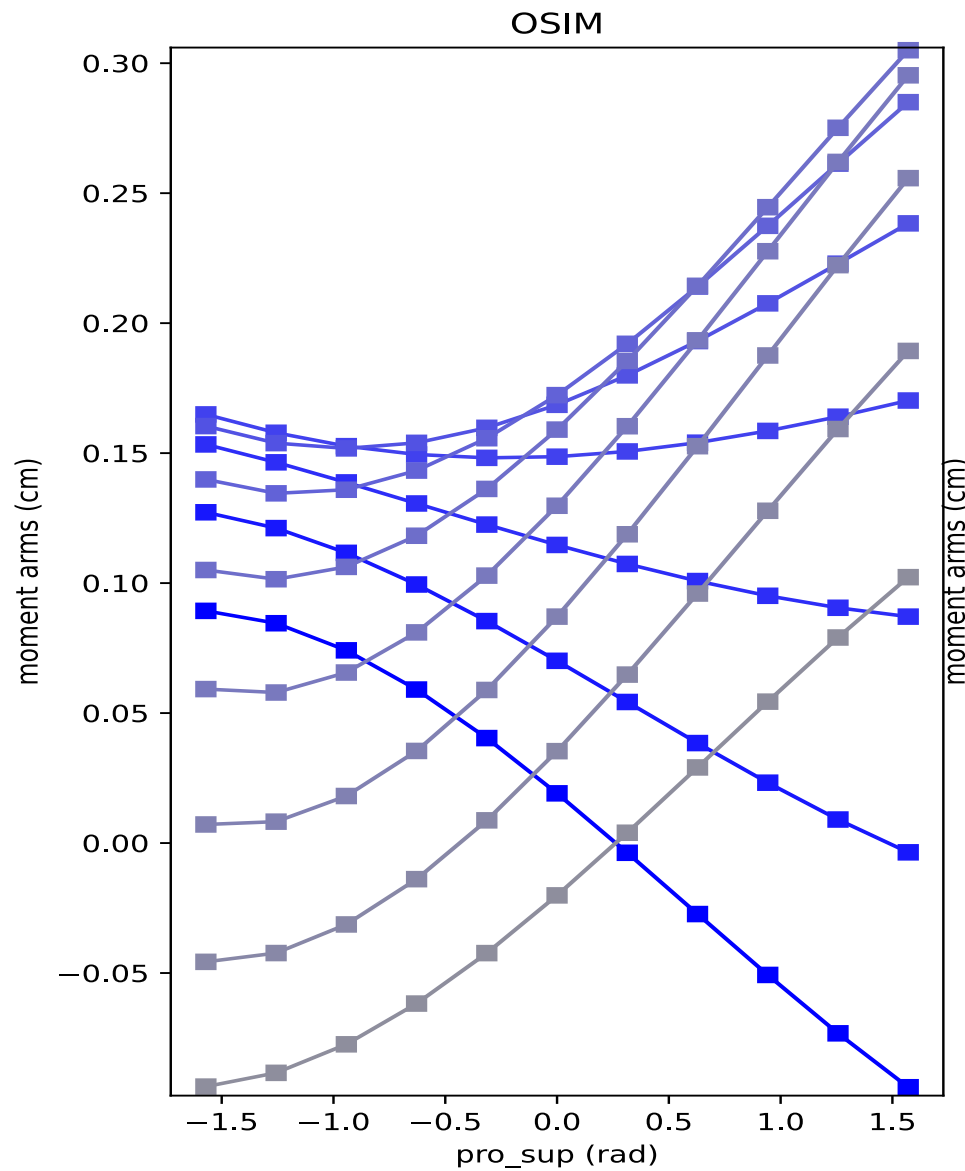
# Muscle specific moment arm comparison with Osim model after optimization

EDCL - deviation - flexion



# Muscle specific moment arm comparison with Osim model after optimization

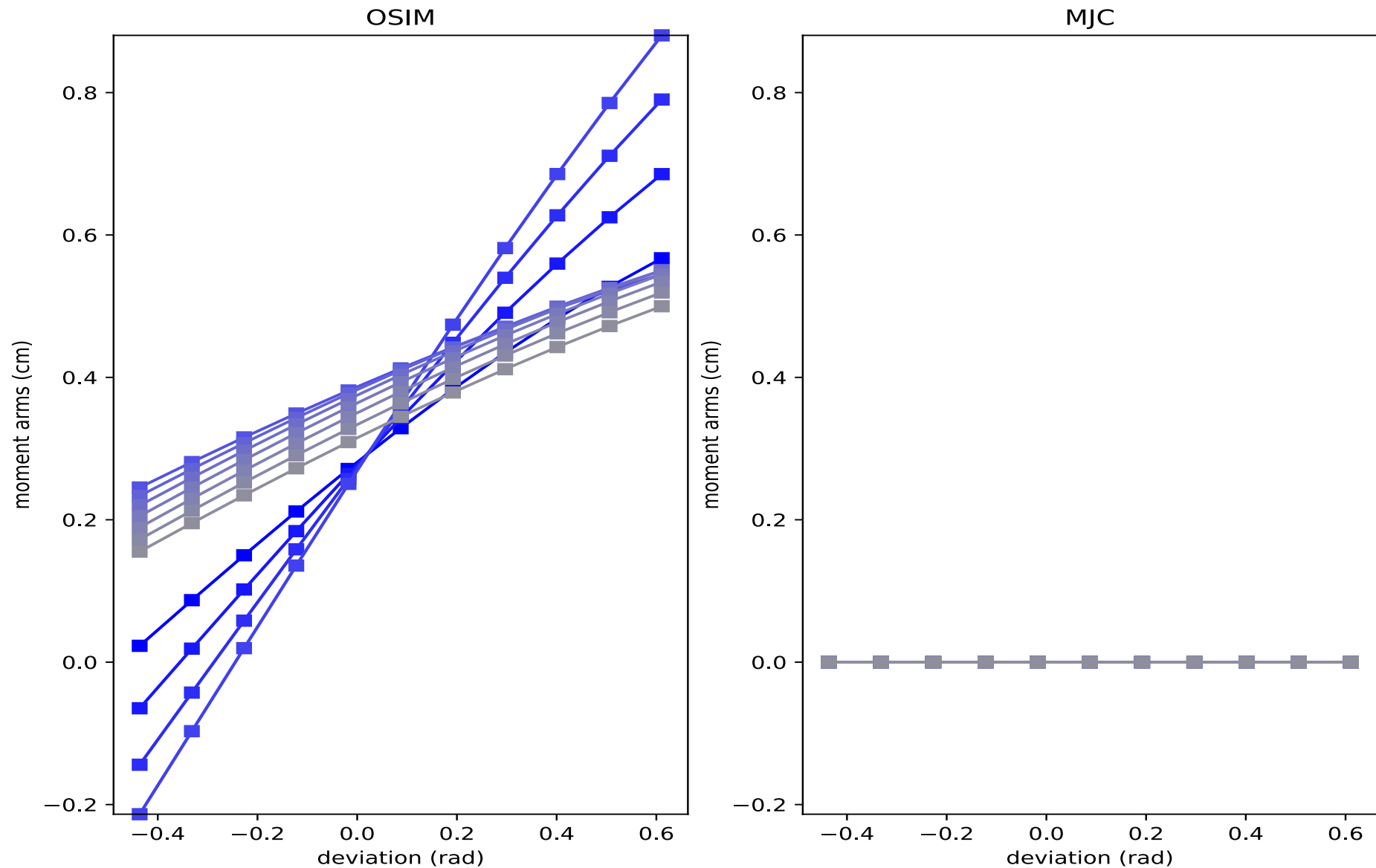
EDCL - elbow\_flexion - pro\_sup





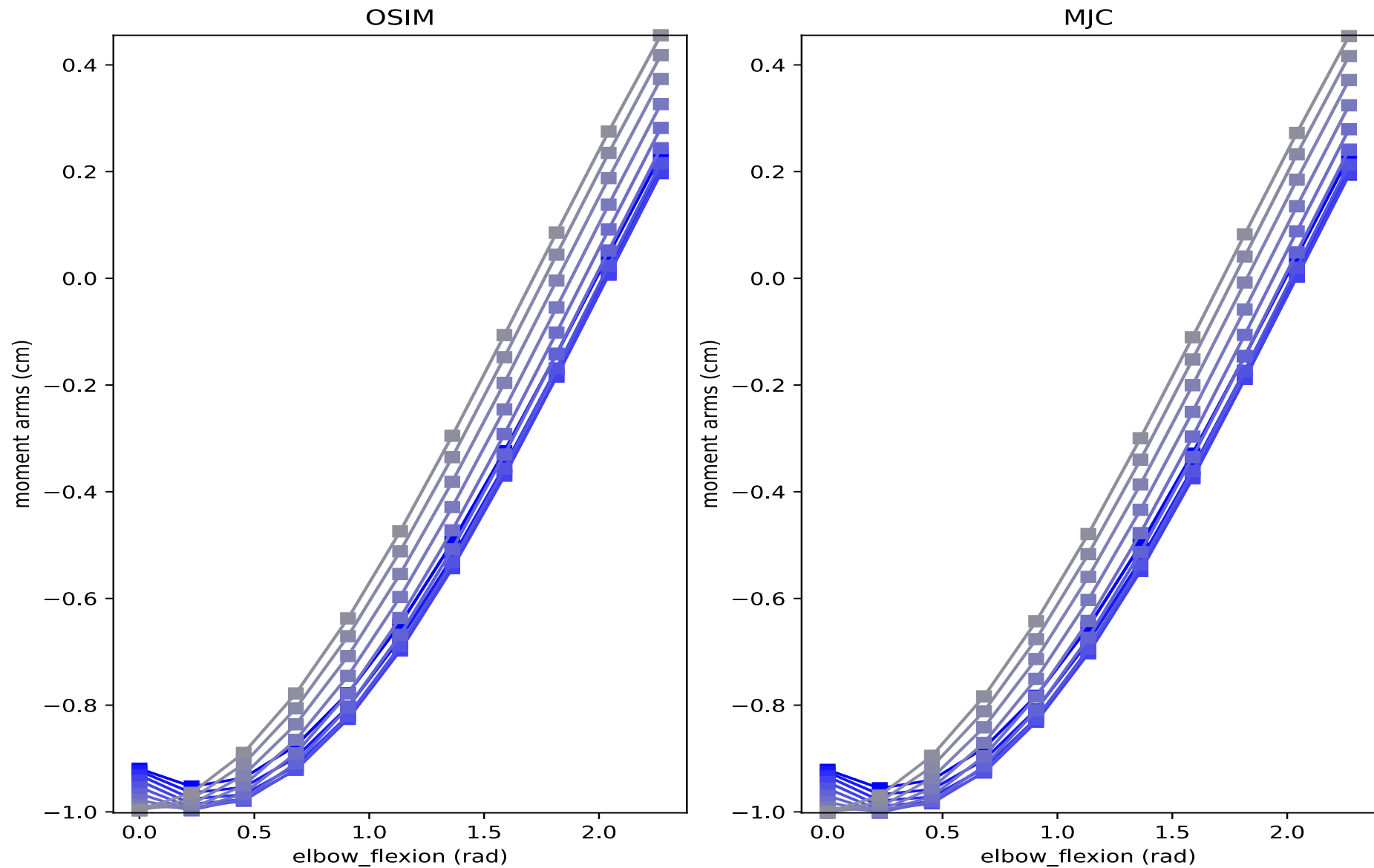
# Muscle specific moment arm comparison with Osim model after optimization

EDCM - deviation - flexion



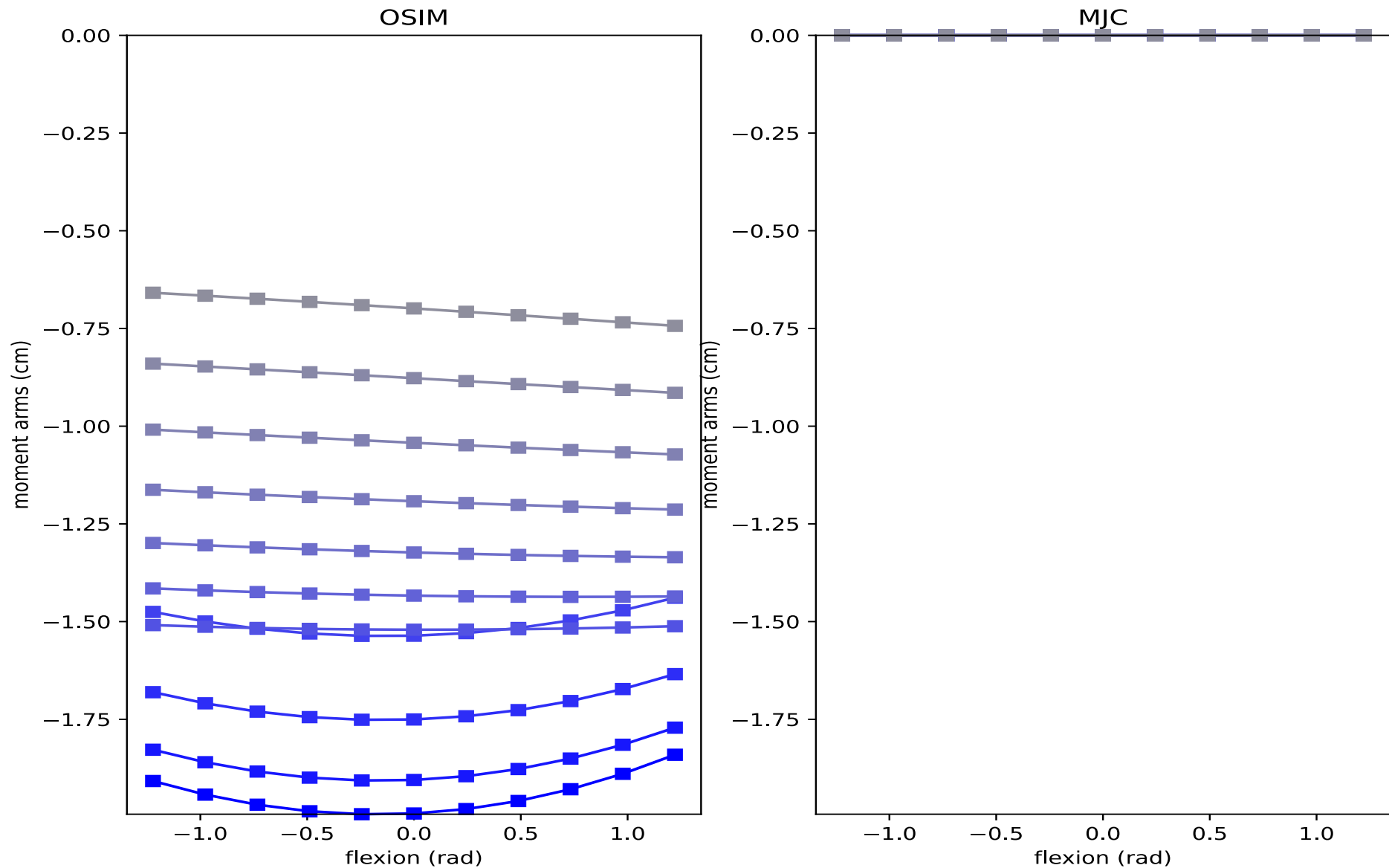
# Muscle specific moment arm comparison with Osim model after optimization

EDCM - elbow\_flexion - pro\_sup



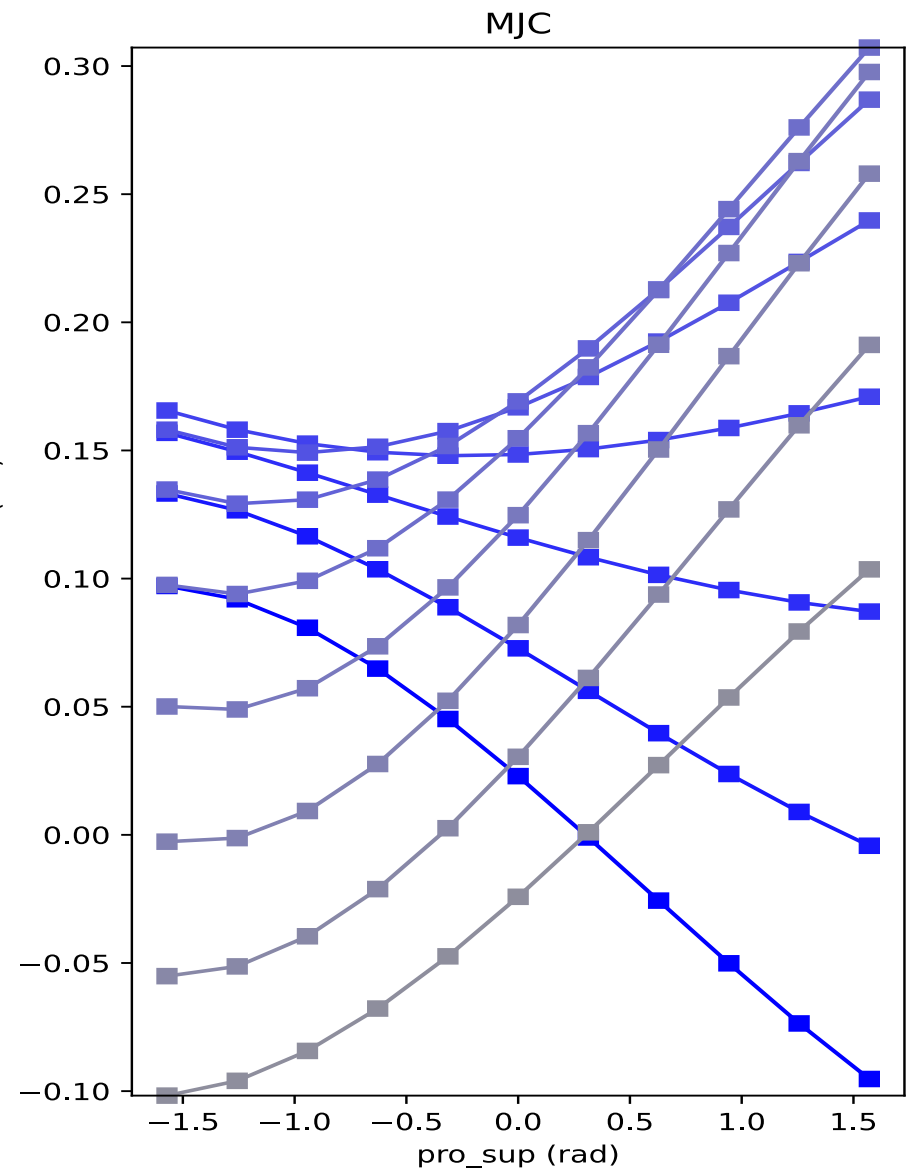
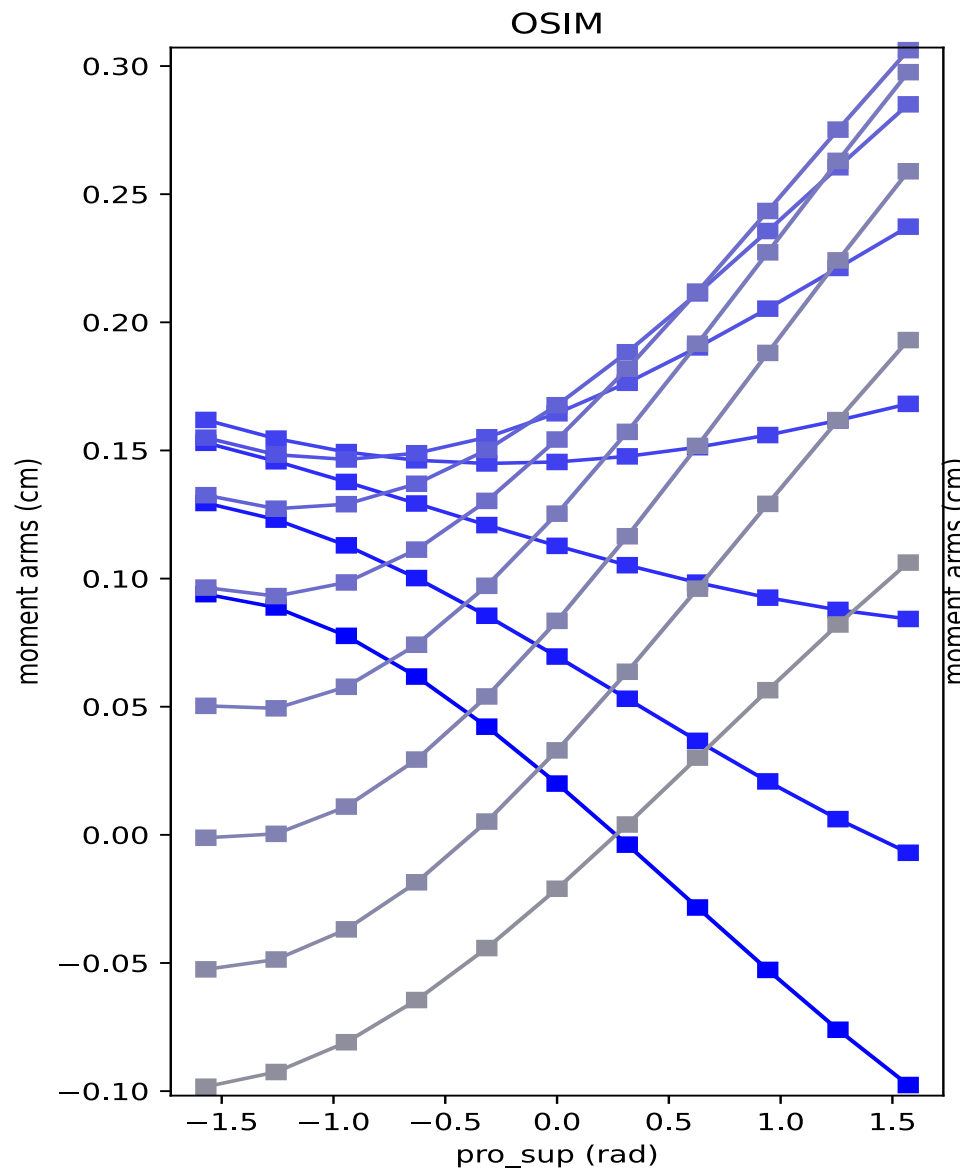
# Muscle specific moment arm comparison with Osim model after optimization

EDCM - deviation - flexion



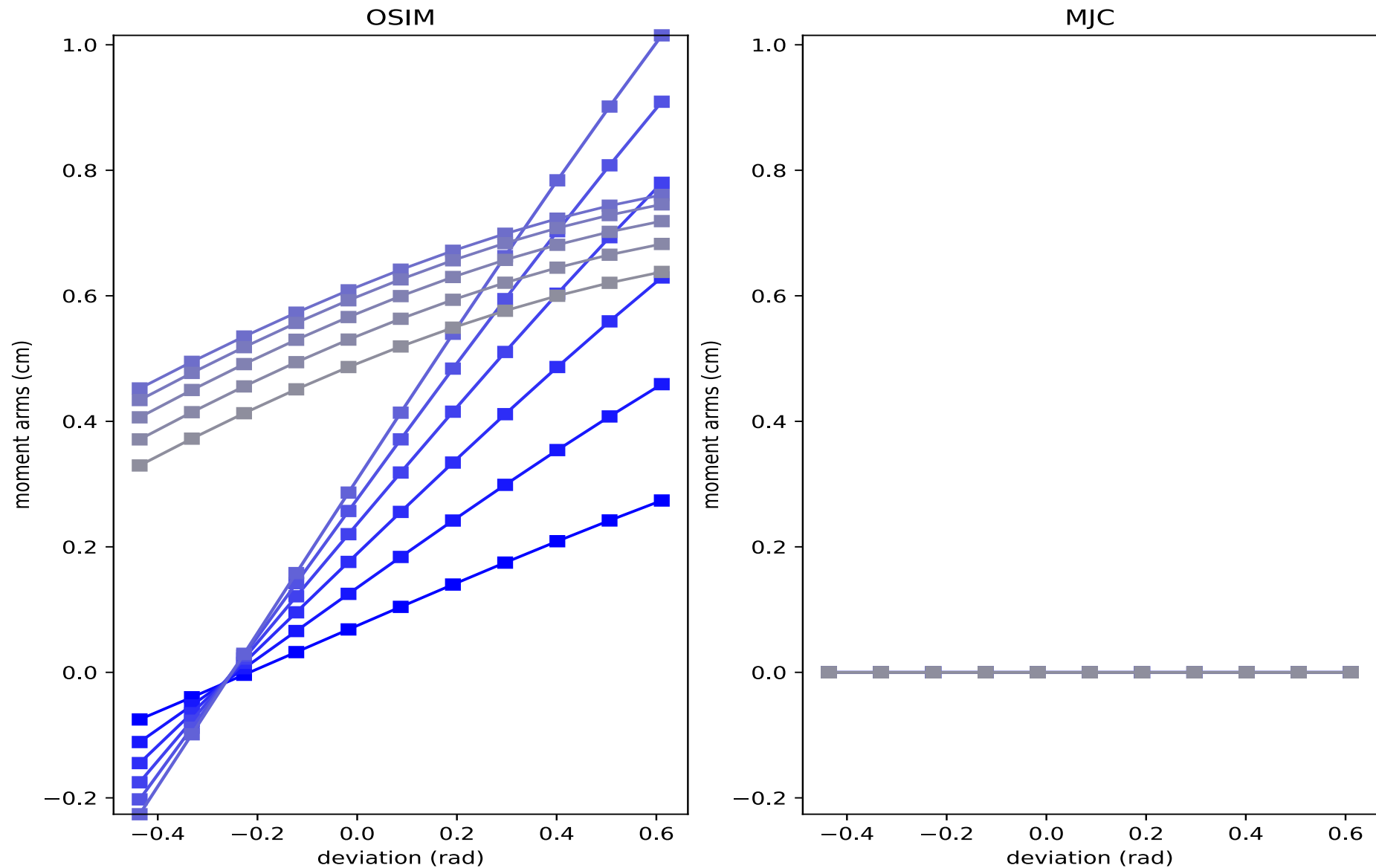
# Muscle specific moment arm comparison with Osim model after optimization

EDCM - elbow\_flexion - pro\_sup



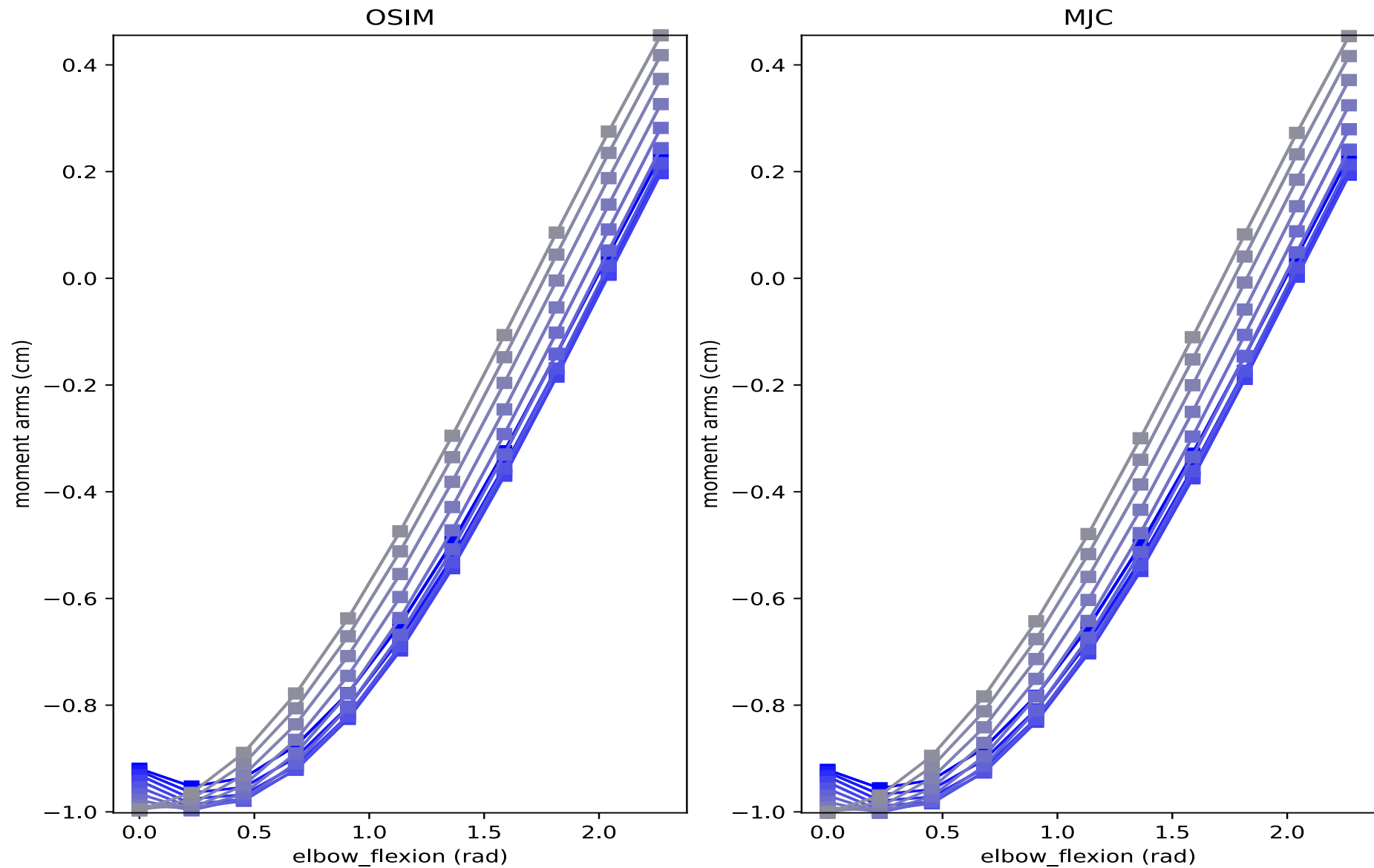
# Muscle specific moment arm comparison with Osim model after optimization

EDCR - deviation - flexion



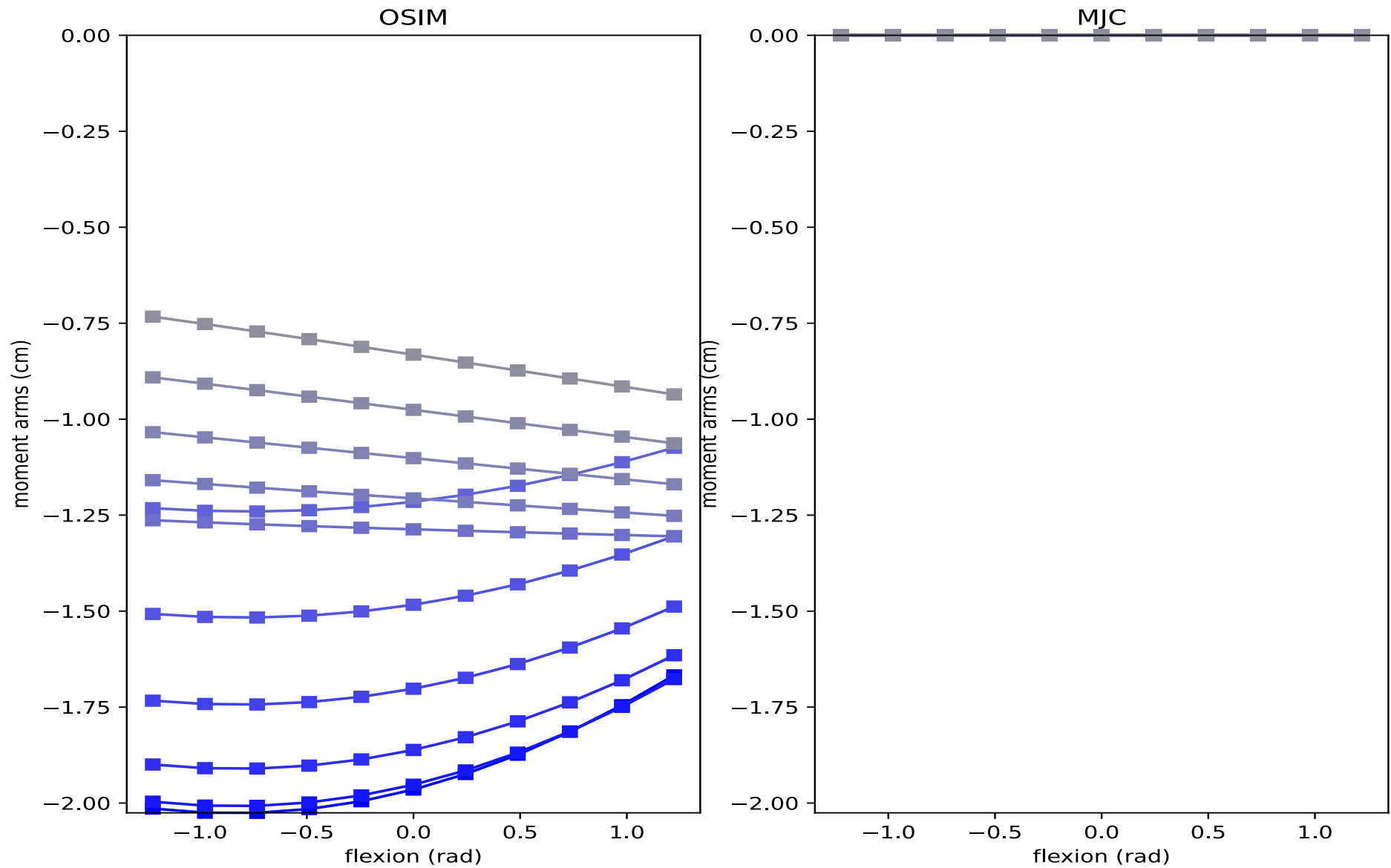
# Muscle specific moment arm comparison with Osim model after optimization

EDCR - elbow\_flexion - pro\_sup



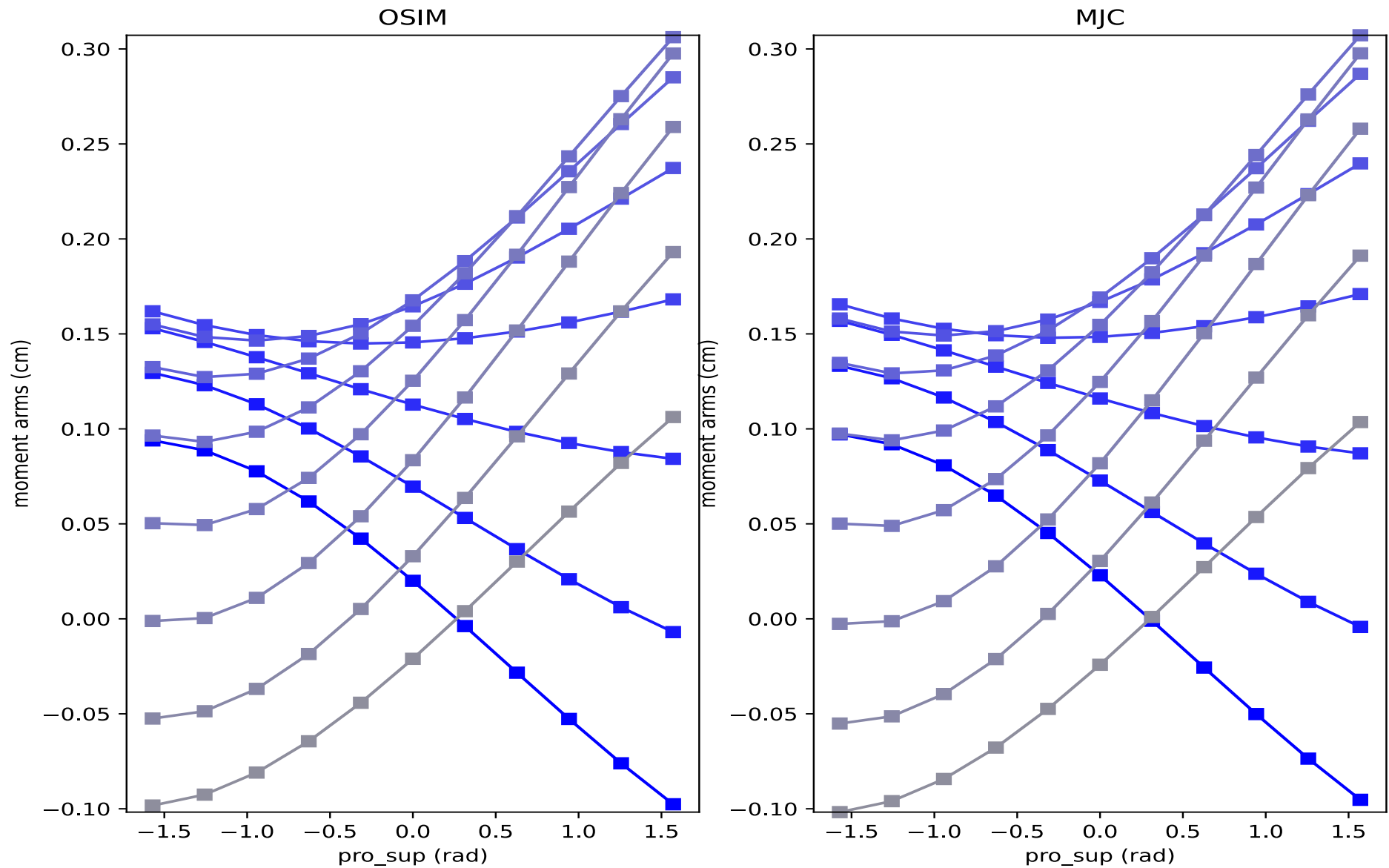
## Muscle specific moment arm comparison with Osim model after optimization

EDCR - deviation - flexion



# Muscle specific moment arm comparison with Osim model after optimization

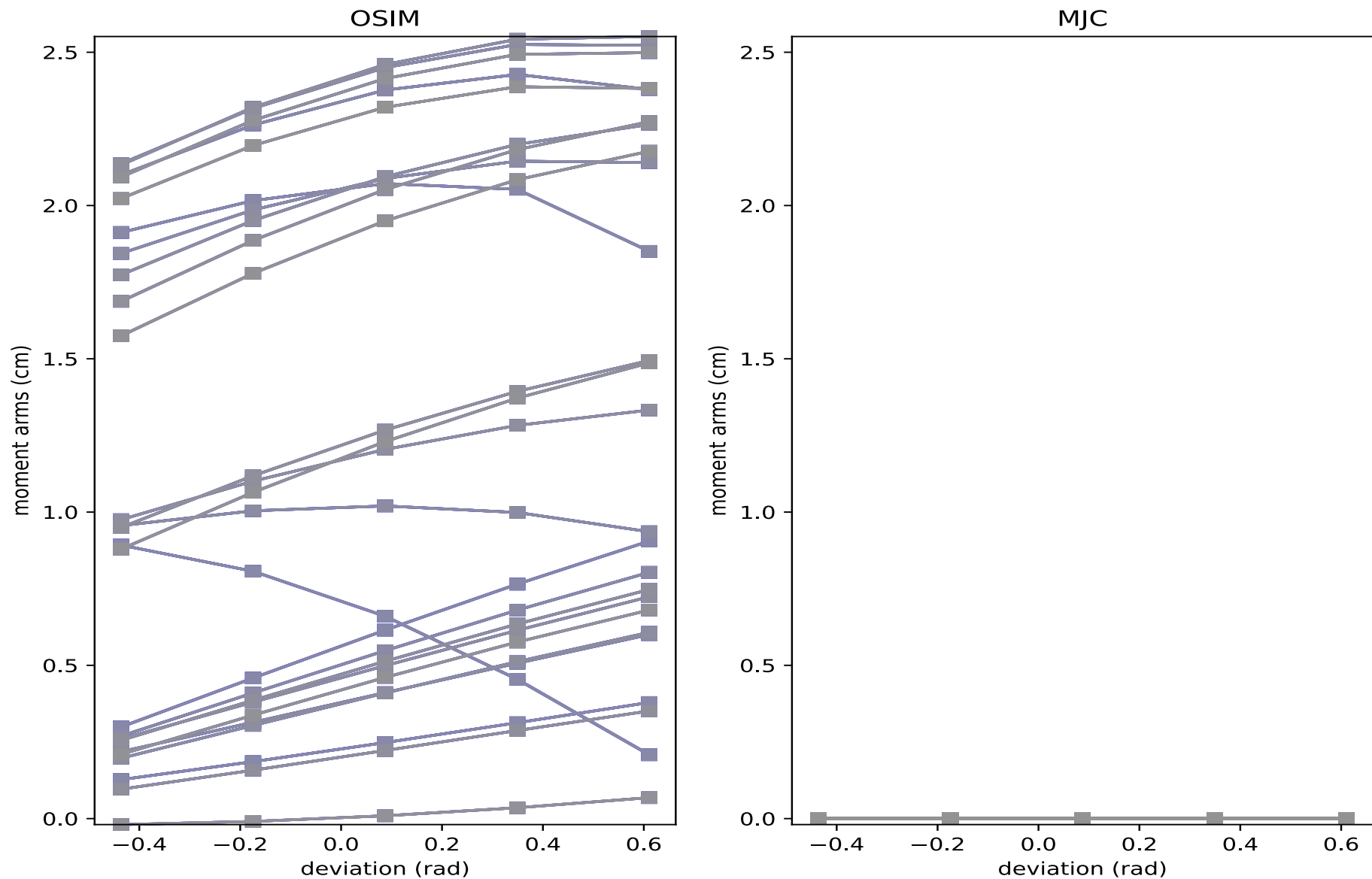
EDCR - elbow\_flexion - pro\_sup





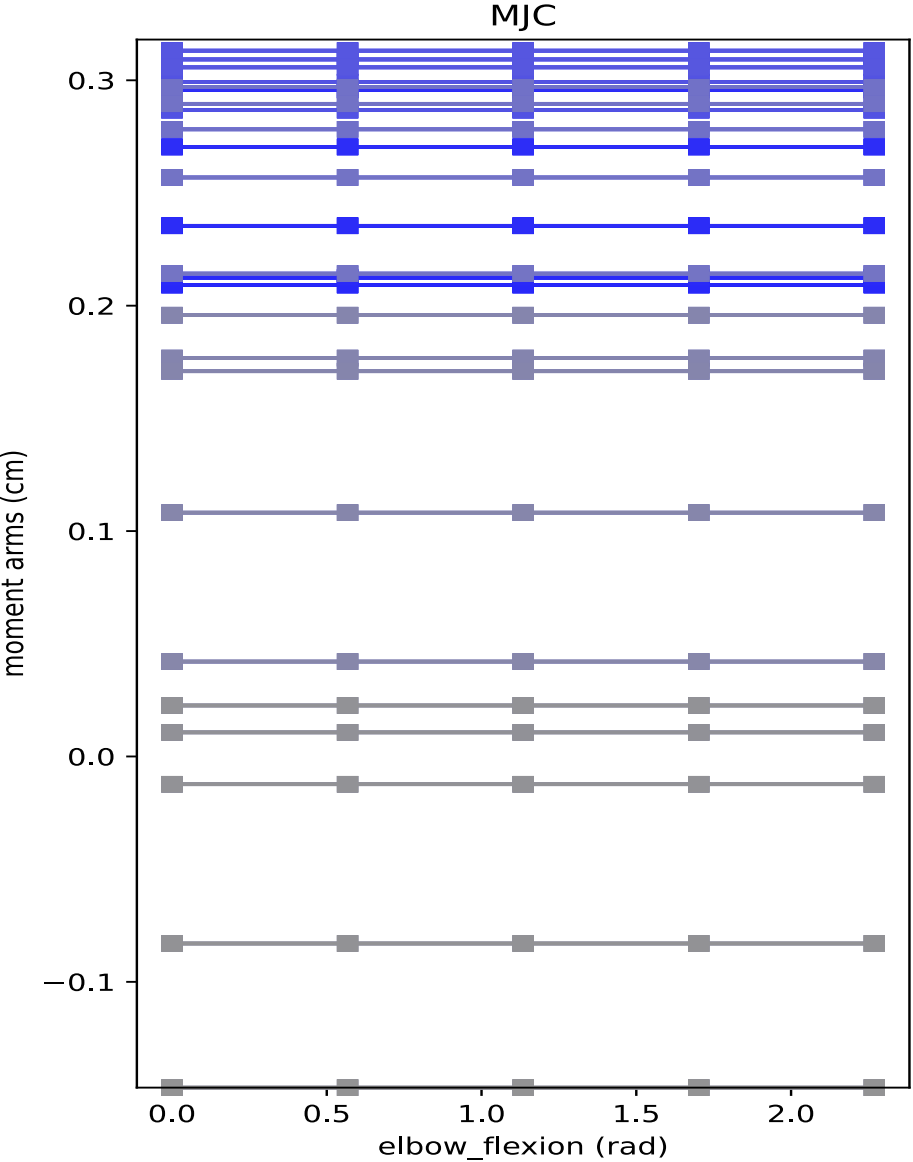
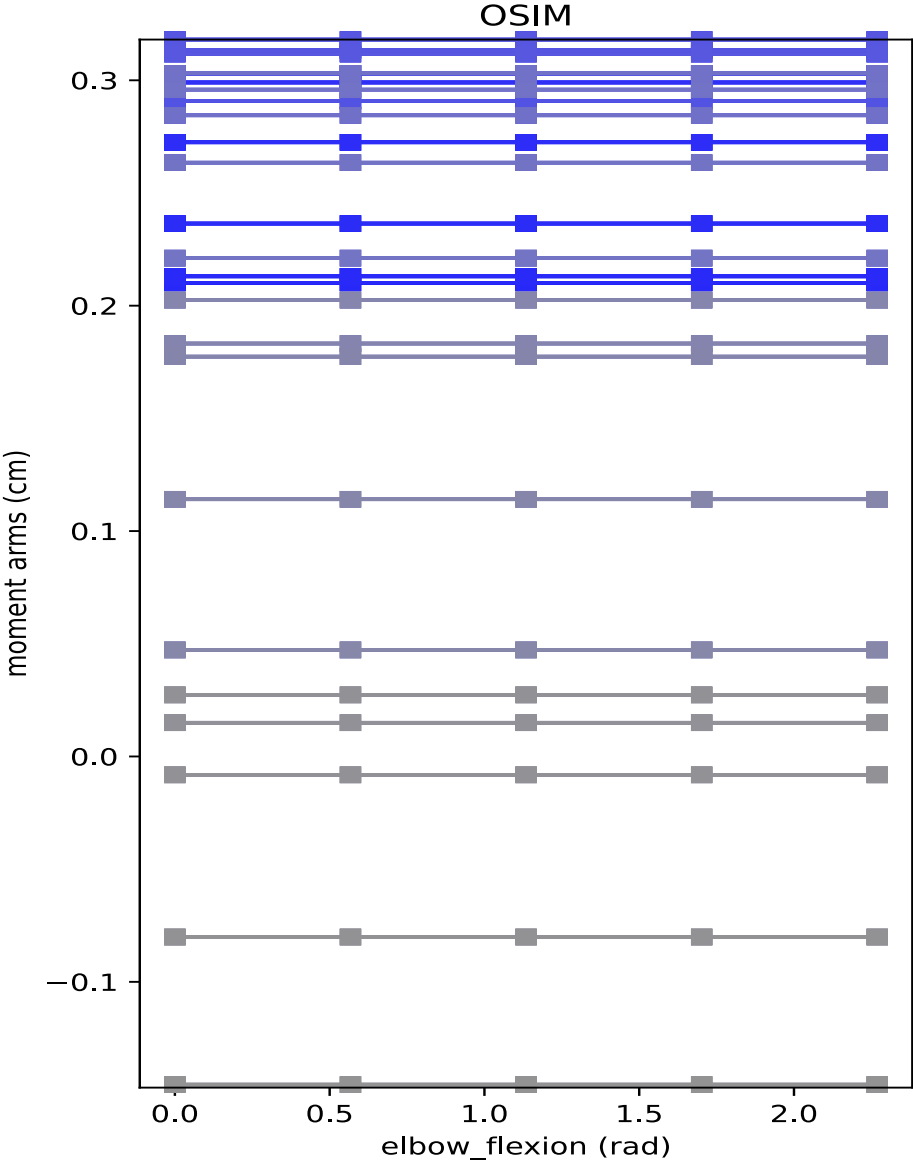
# Muscle specific moment arm comparison with Osim model after optimization

EDM - deviation - elbow\_flexion - flexion - pro\_sup



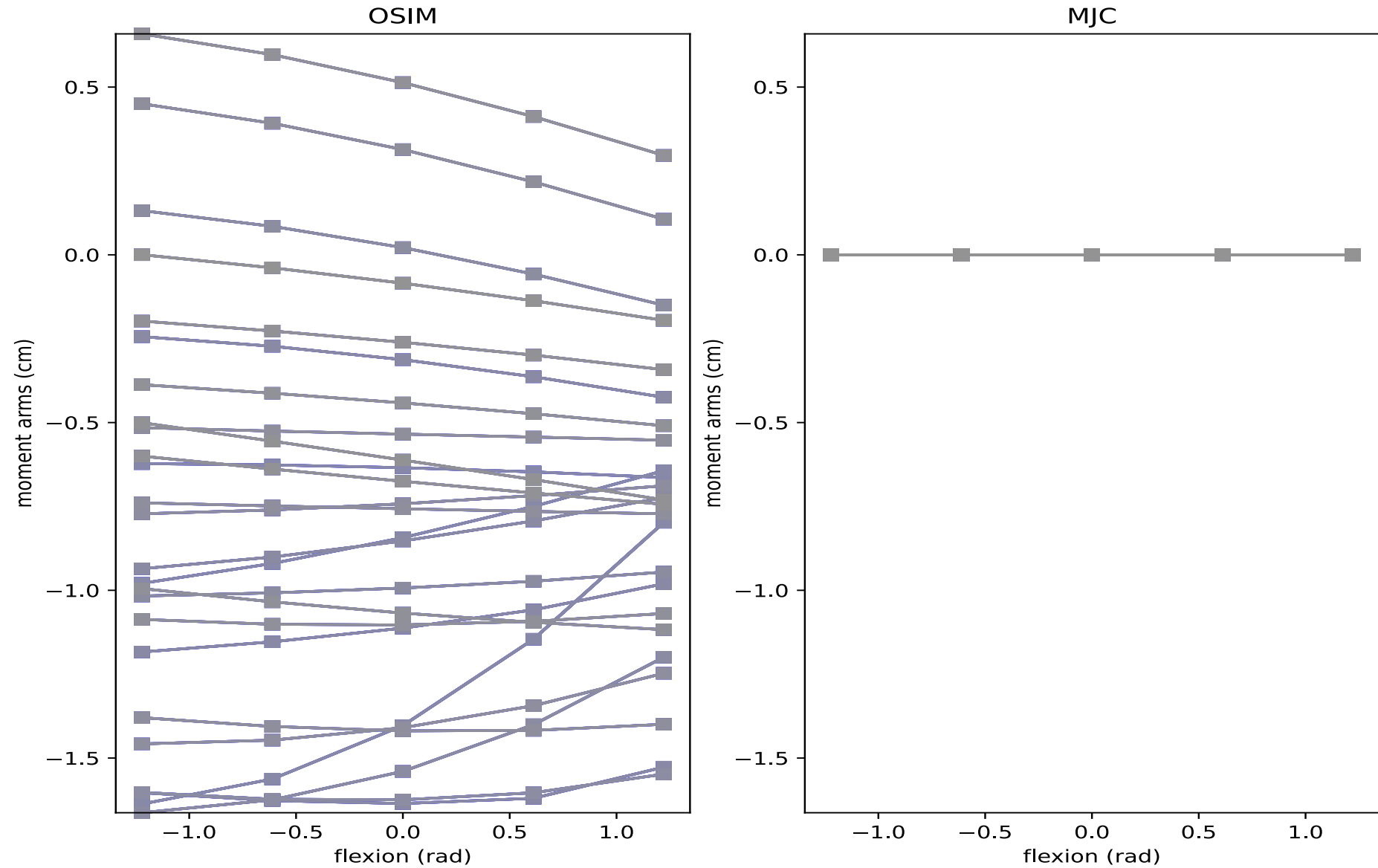
# Muscle specific moment arm comparison with Osim model after optimization

EDM - deviation - elbow\_flexion - flexion - pro\_sup



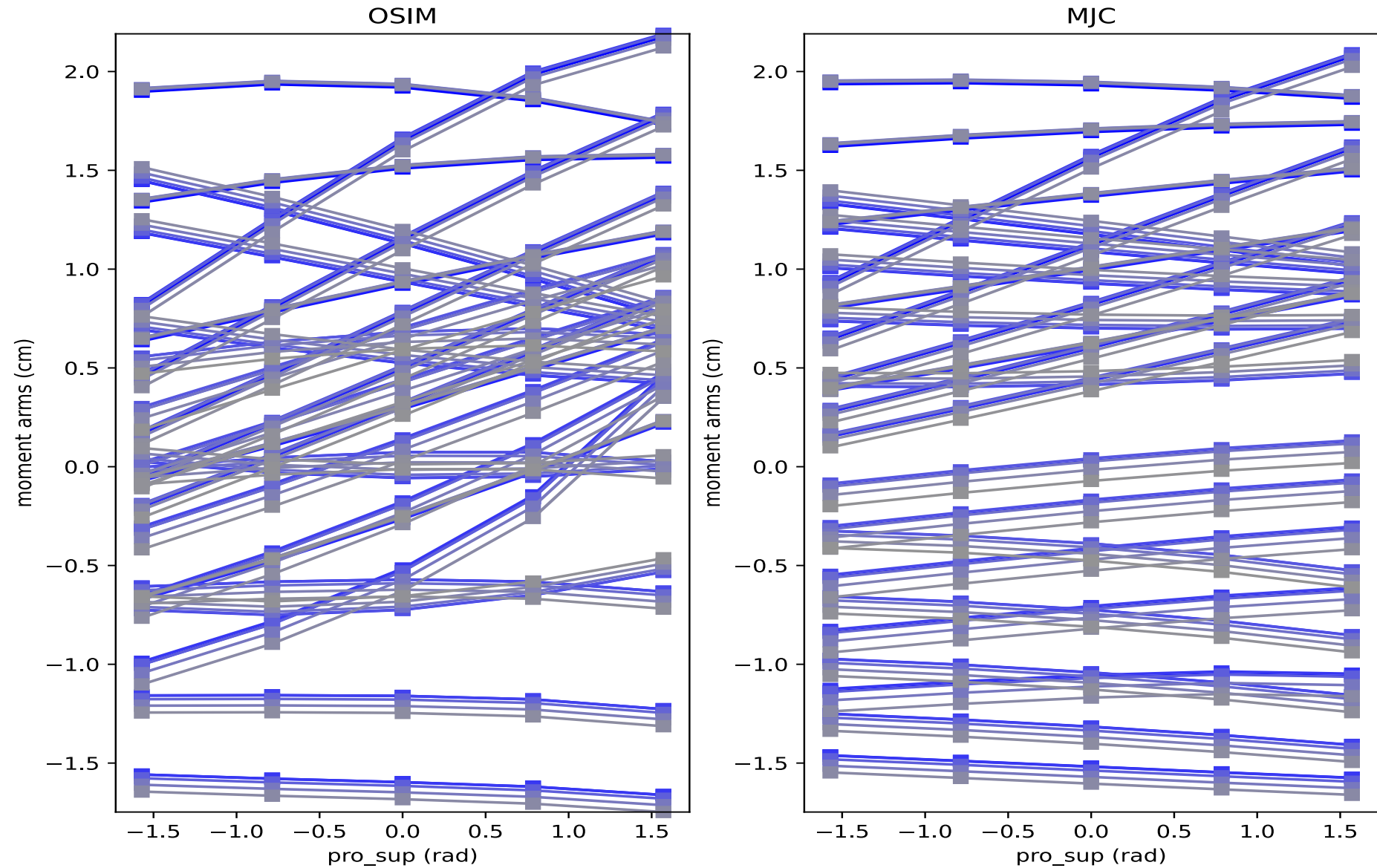
## Muscle specific moment arm comparison with Osim model after optimization

EDM - deviation - elbow\_flexion - flexion - pro\_sup



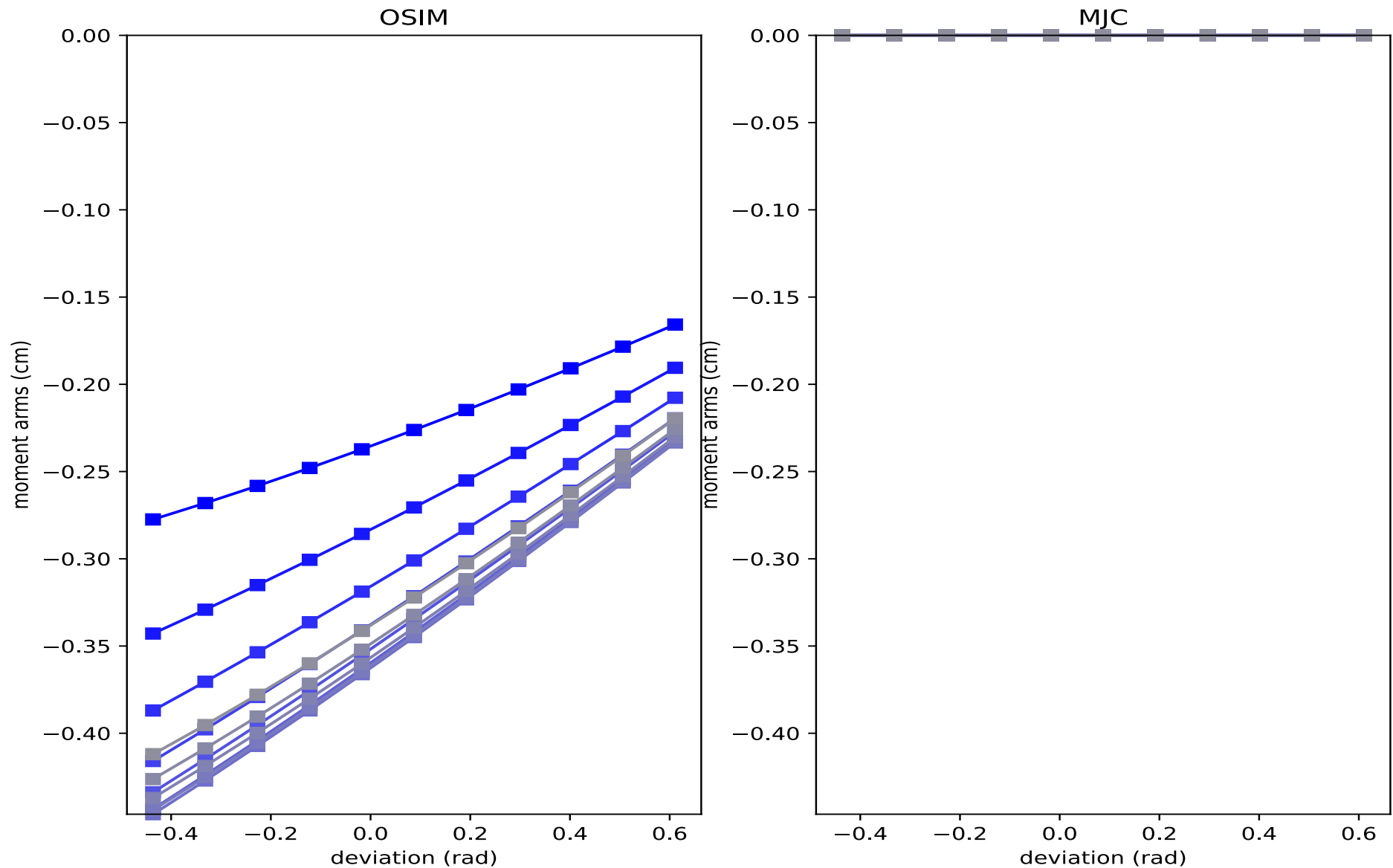
# Muscle specific moment arm comparison with Osim model after optimization

EDM - deviation - elbow\_flexion - flexion - pro\_sup



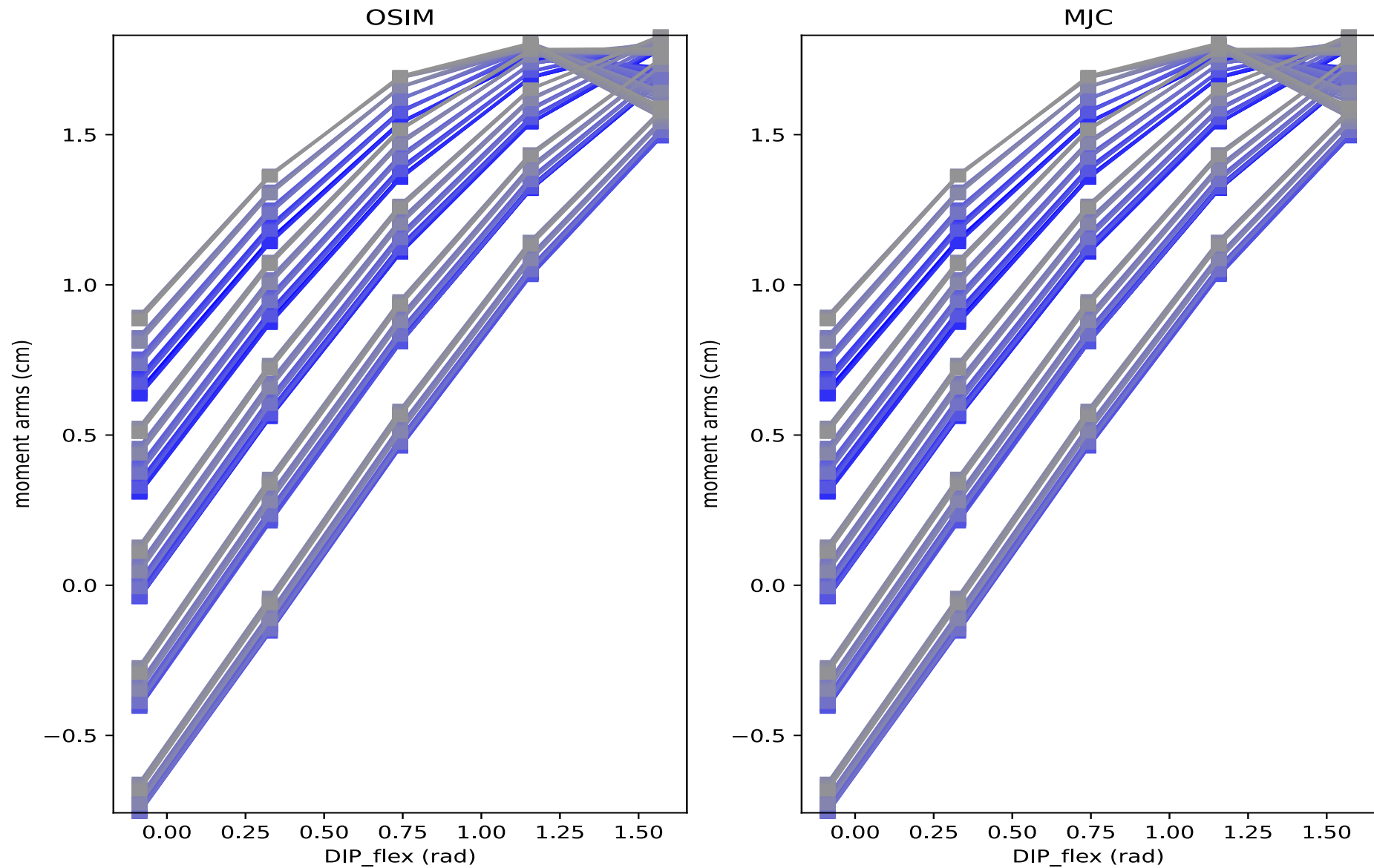
# Muscle specific moment arm comparison with Osim model after optimization

EIP - deviation - flexion



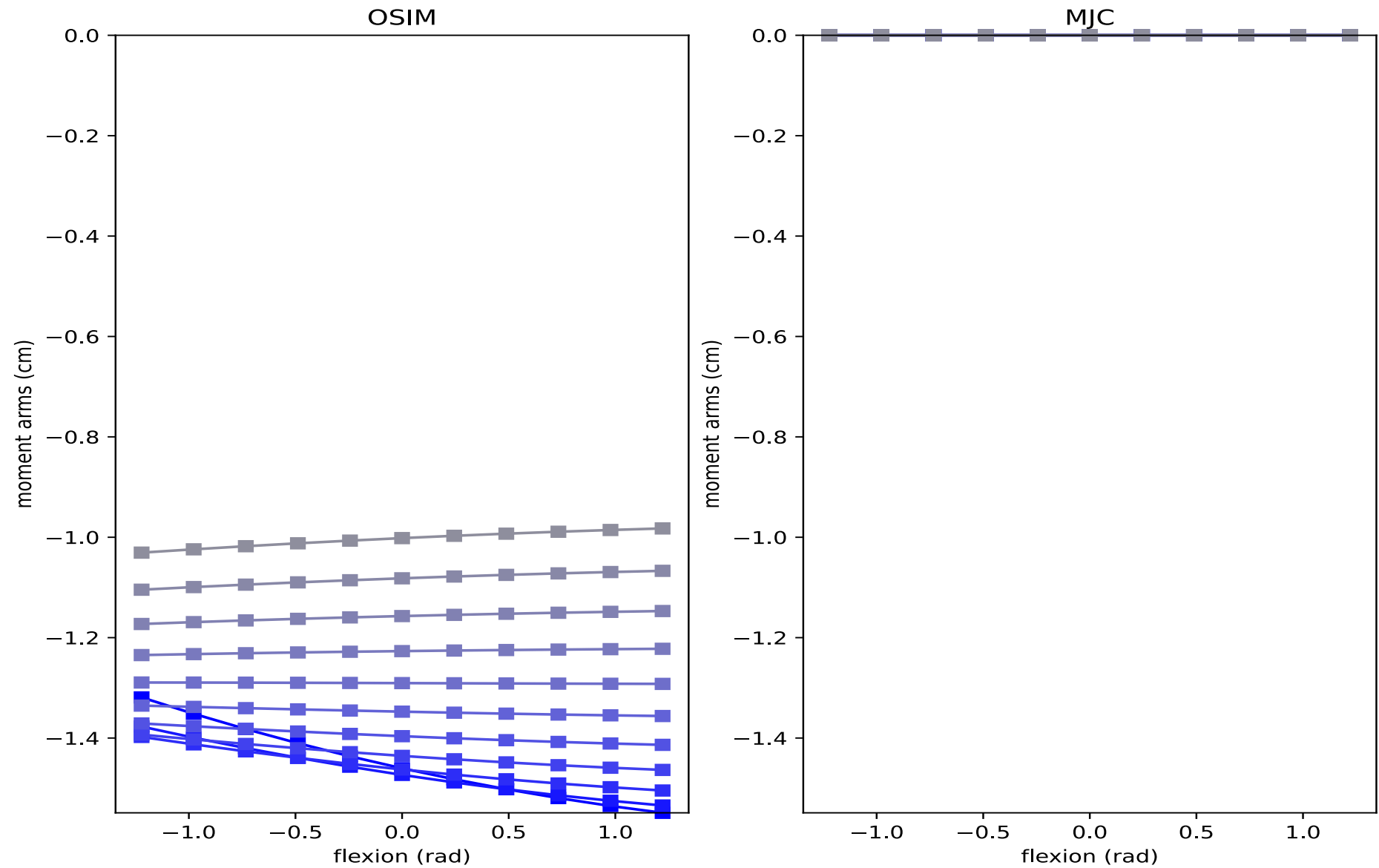
# Muscle specific moment arm comparison with Osim model after optimization

EIP - DIP\_flex - MCP2\_flex - MCP2\_lateral - PIP\_flex



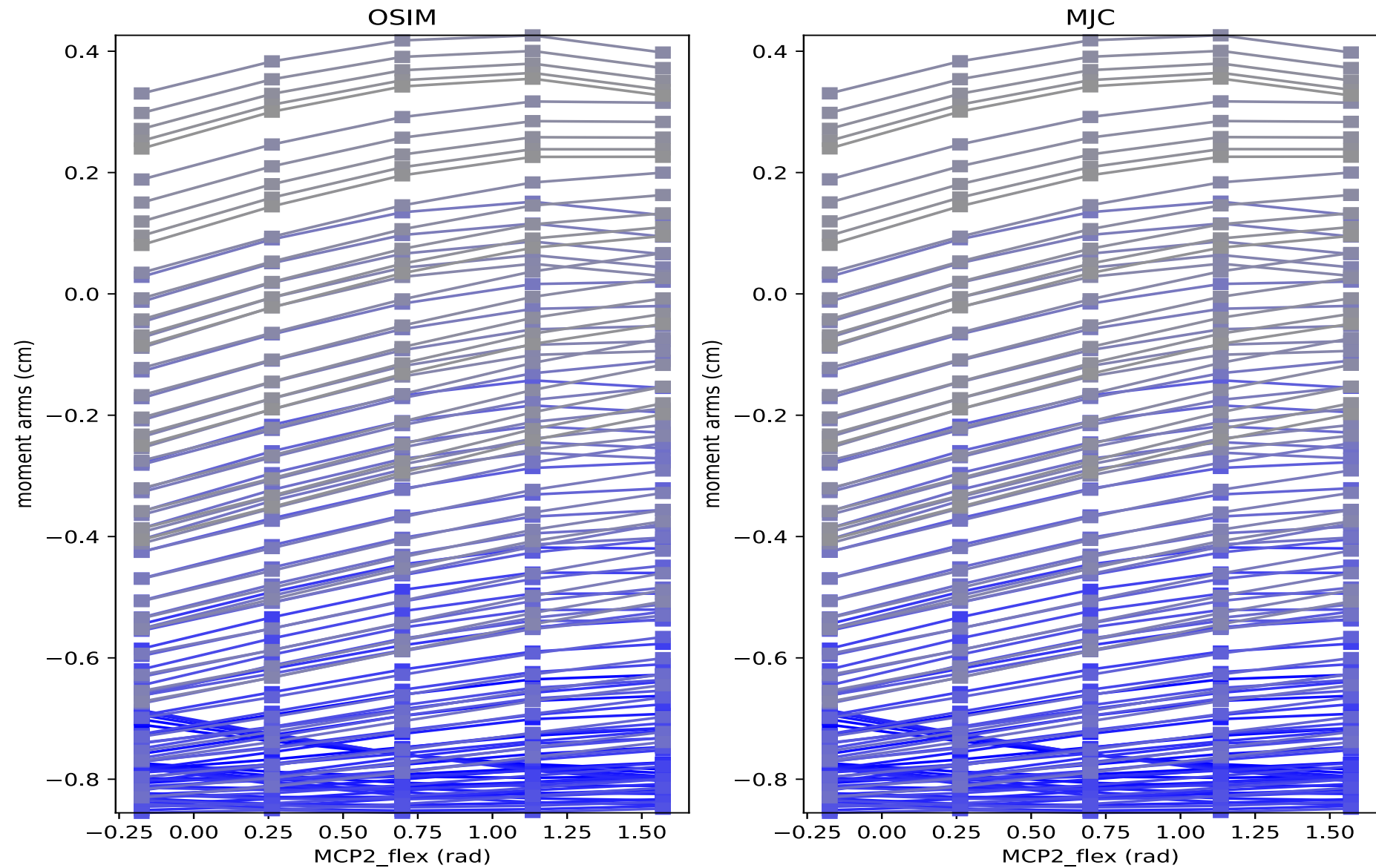
# Muscle specific moment arm comparison with Osim model after optimization

EIP - deviation - flexion



# Muscle specific moment arm comparison with Osim model after optimization

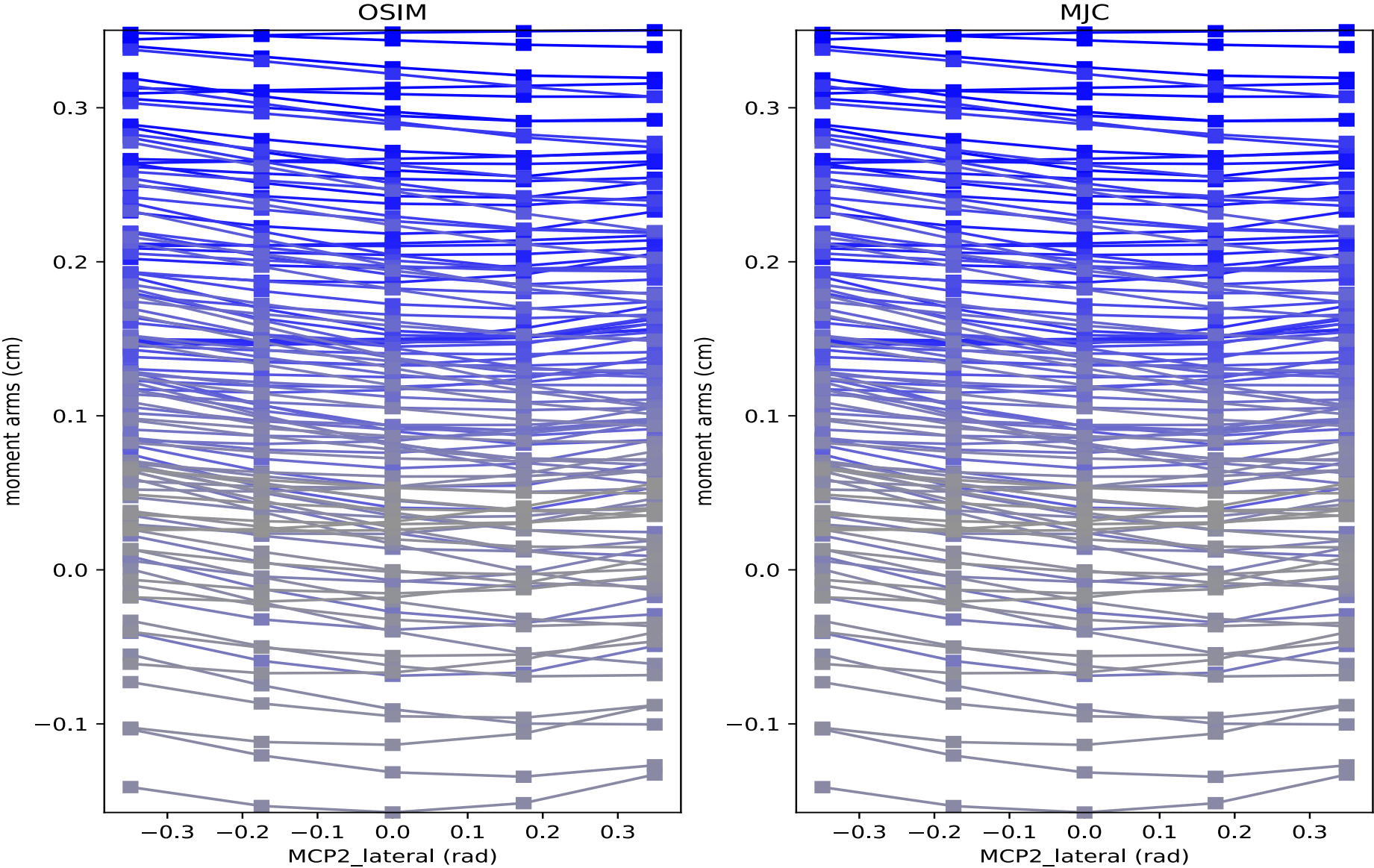
EIP - DIP\_flex - MCP2\_flex - MCP2\_lateral - PIP\_flex





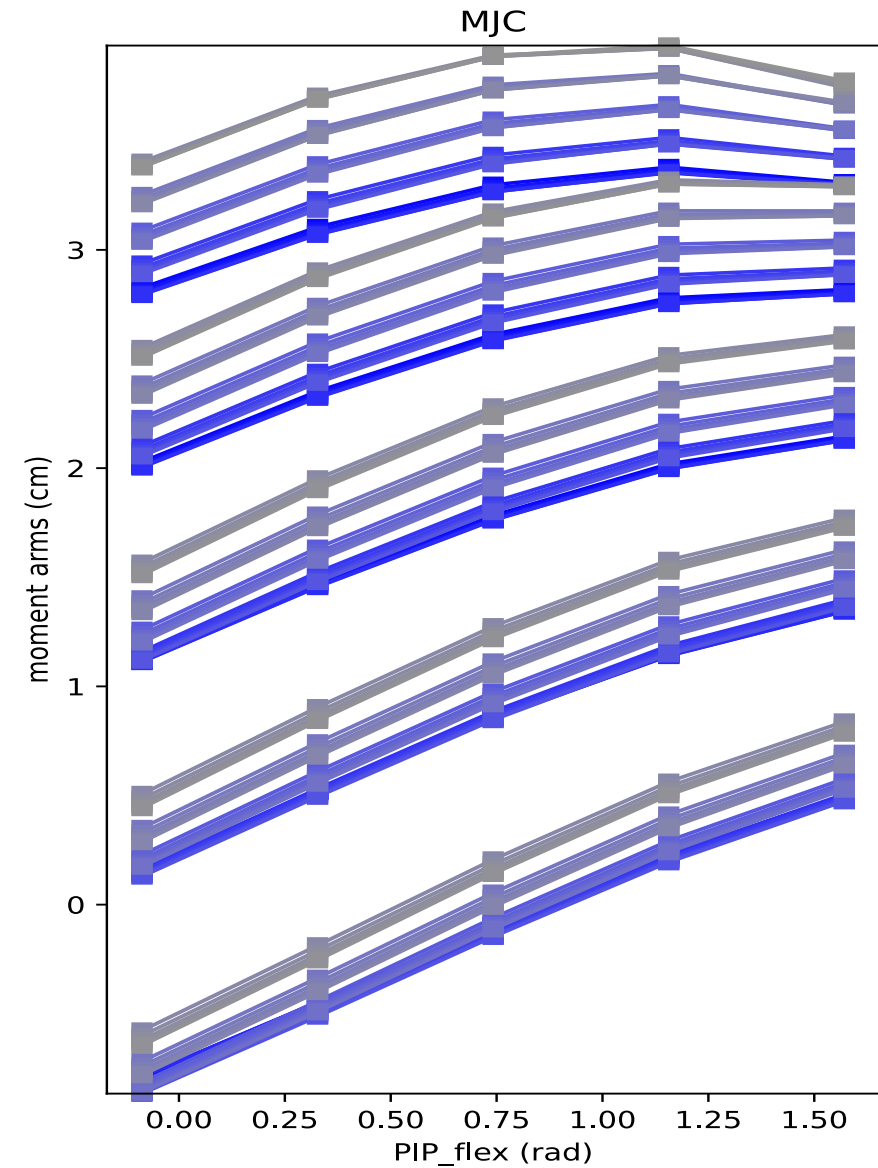
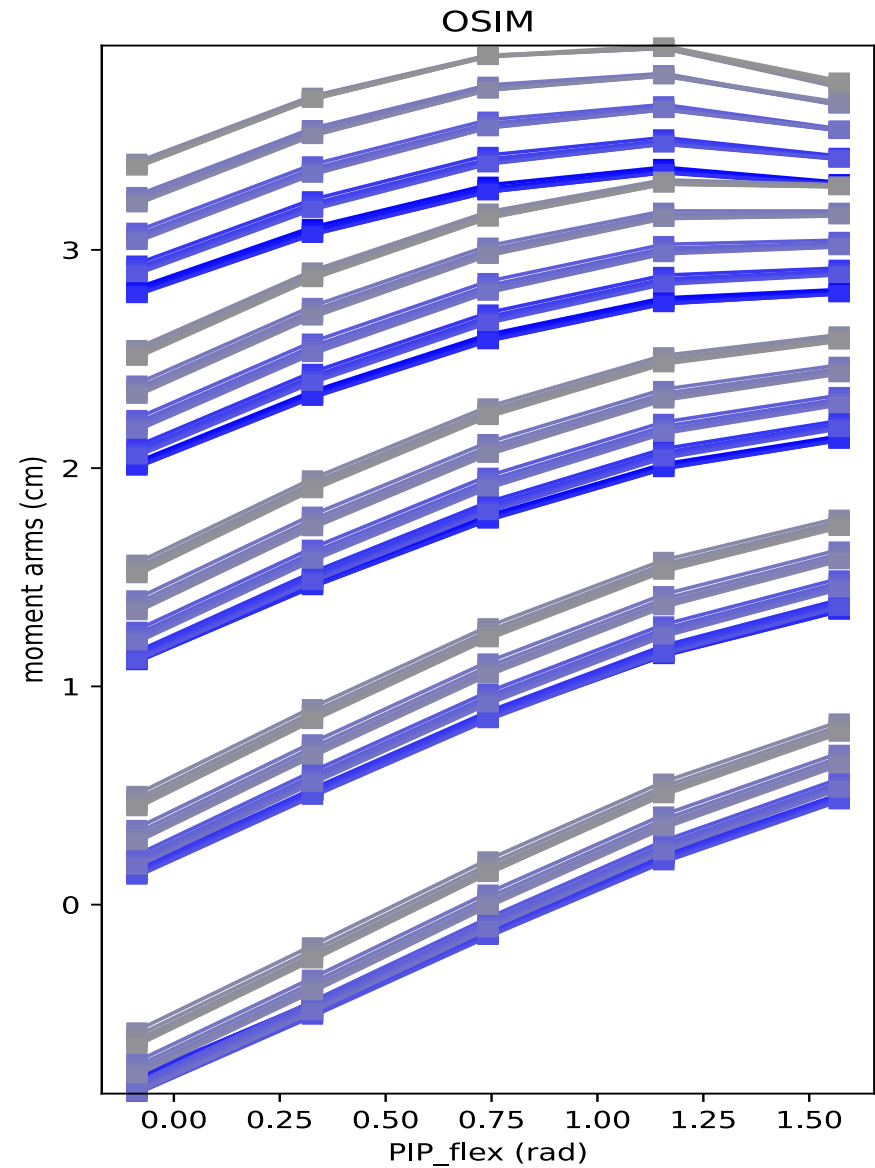
# Muscle specific moment arm comparison with Osim model after optimization

EIP - DIP\_flex - MCP2\_flex - MCP2\_lateral - PIP\_flex



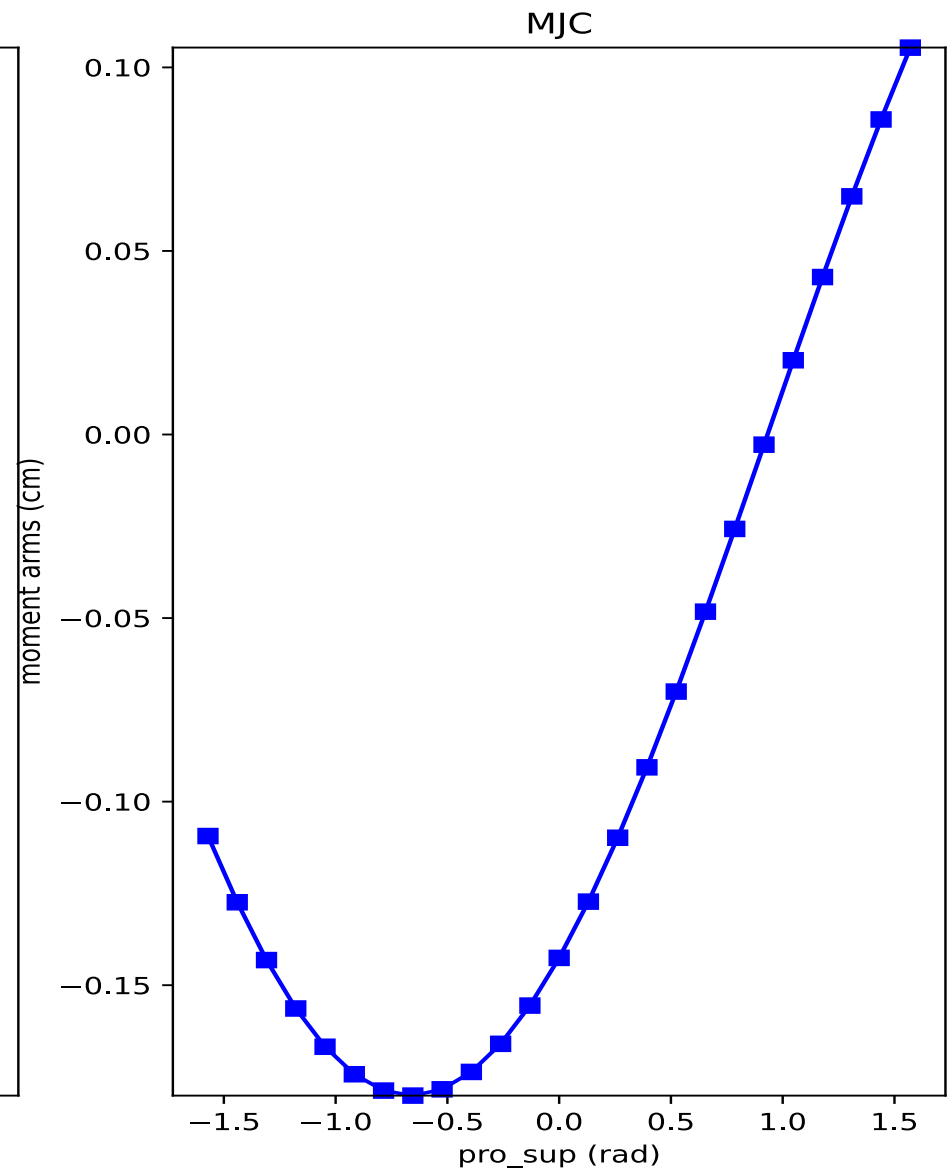
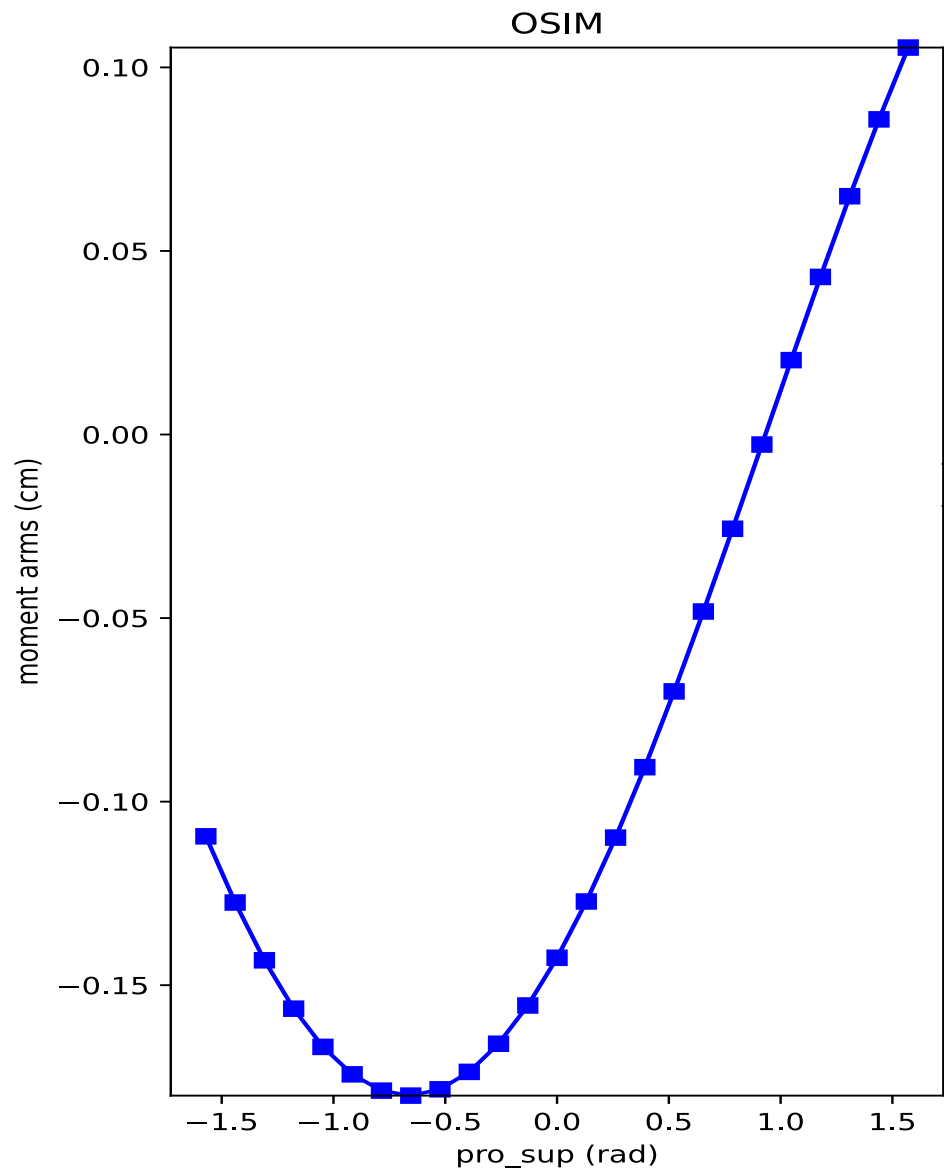
# Muscle specific moment arm comparison with Osim model after optimization

EIP - DIP\_flex - MCP2\_flex - MCP2\_lateral - PIP\_flex



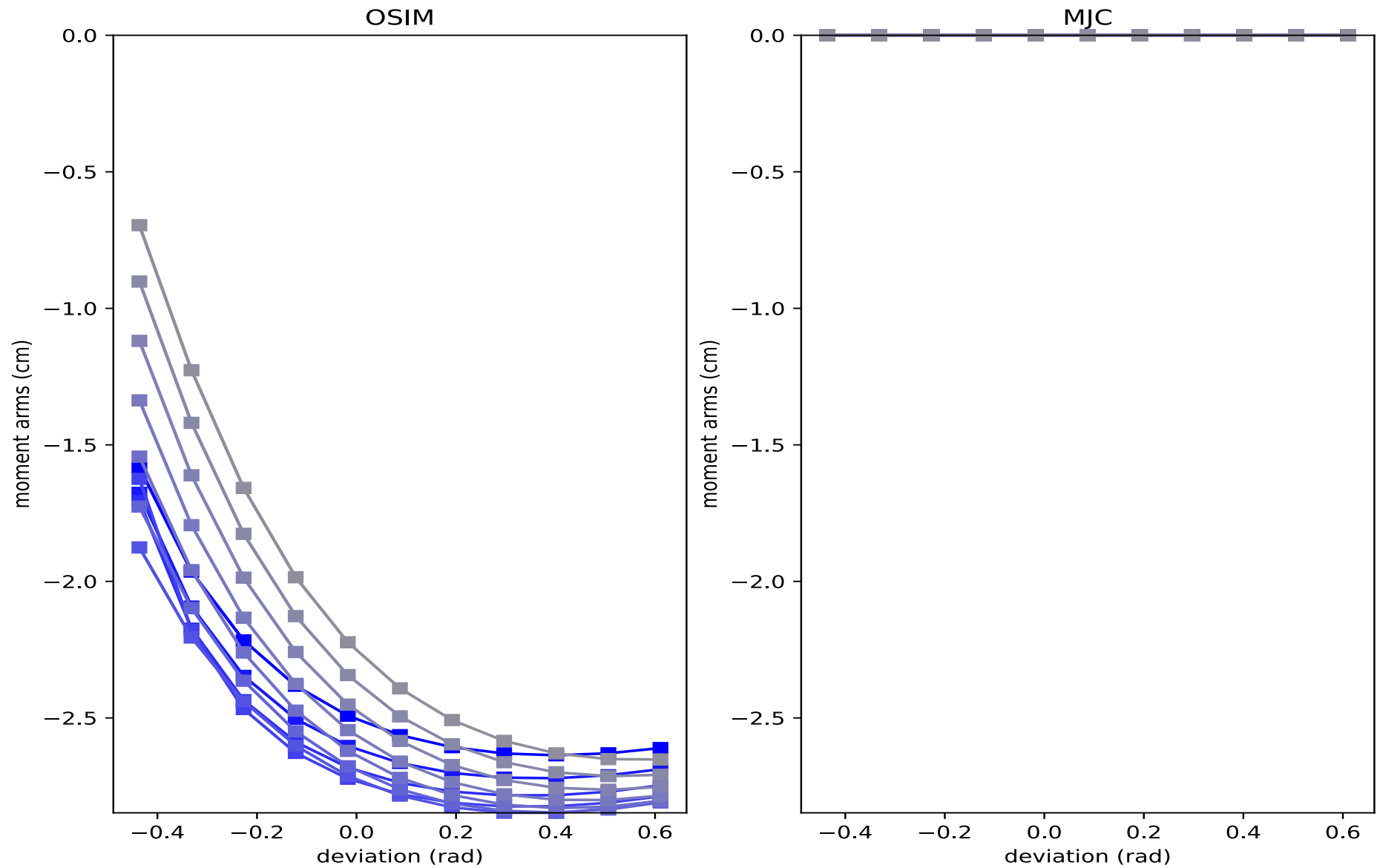
# Muscle specific moment arm comparison with Osim model after optimization

EIP - pro\_sup



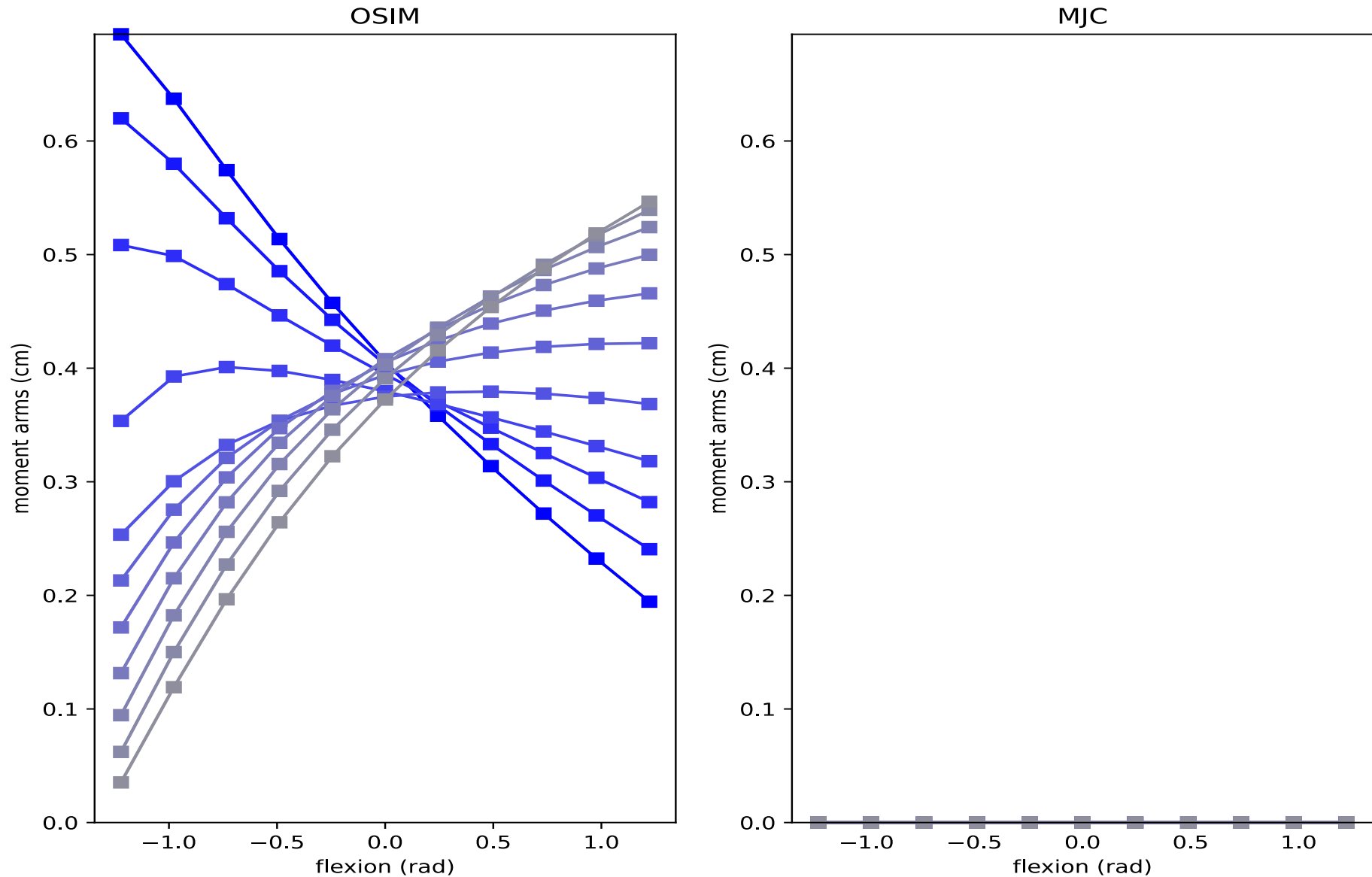
# Muscle specific moment arm comparison with Osim model after optimization

EPB - deviation - flexion



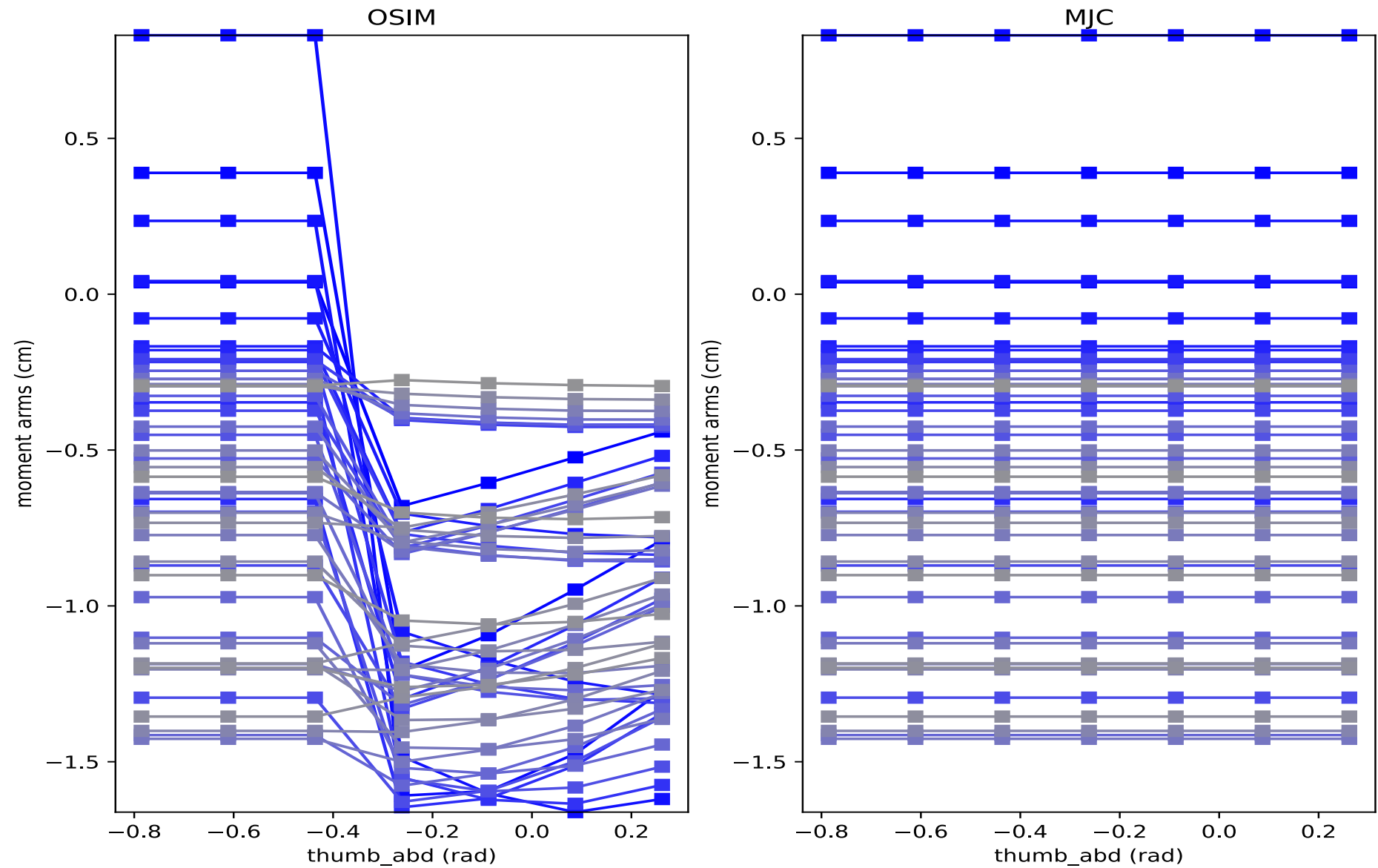
### Muscle specific moment arm comparison with Osim model after optimization

EPB - deviation - flexion



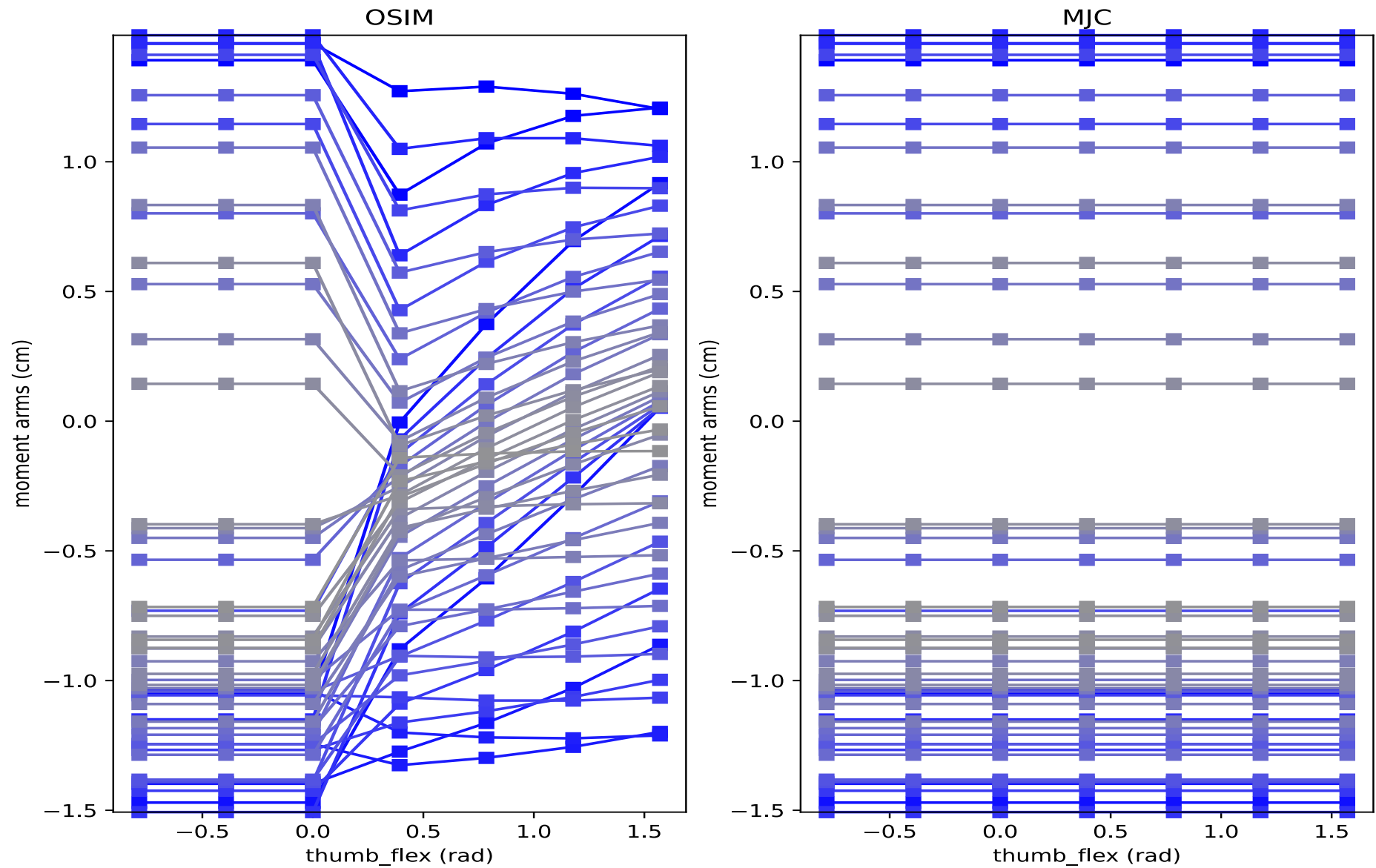
# Muscle specific moment arm comparison with Osim model after optimization

EPB - flexion - thumb\_abd - thumb\_flex



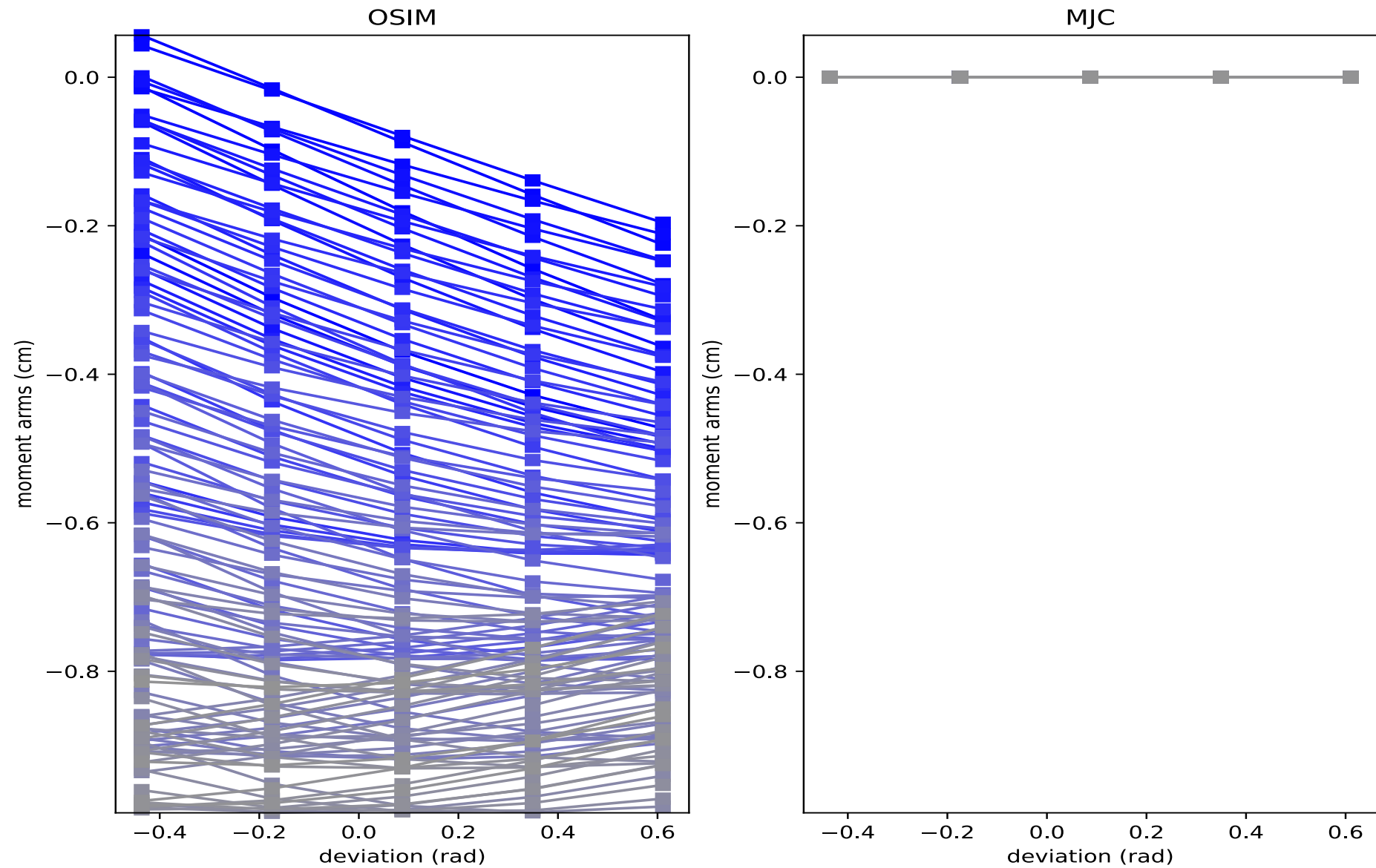
# Muscle specific moment arm comparison with Osim model after optimization

EPB - flexion - thumb\_abd - thumb\_flex



# Muscle specific moment arm comparison with Osim model after optimization

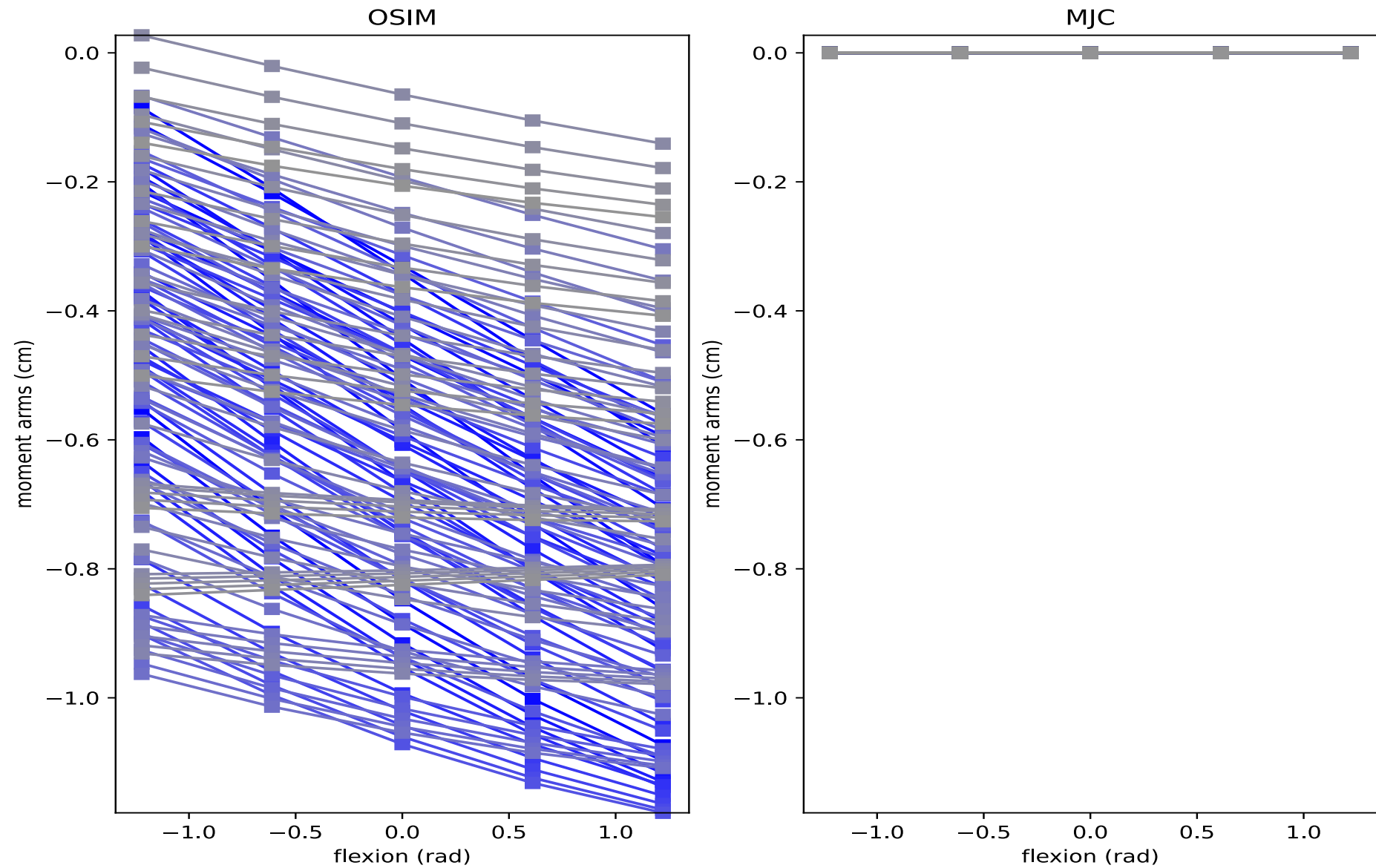
EPL - deviation - flexion - thumb\_abd - thumb\_flex





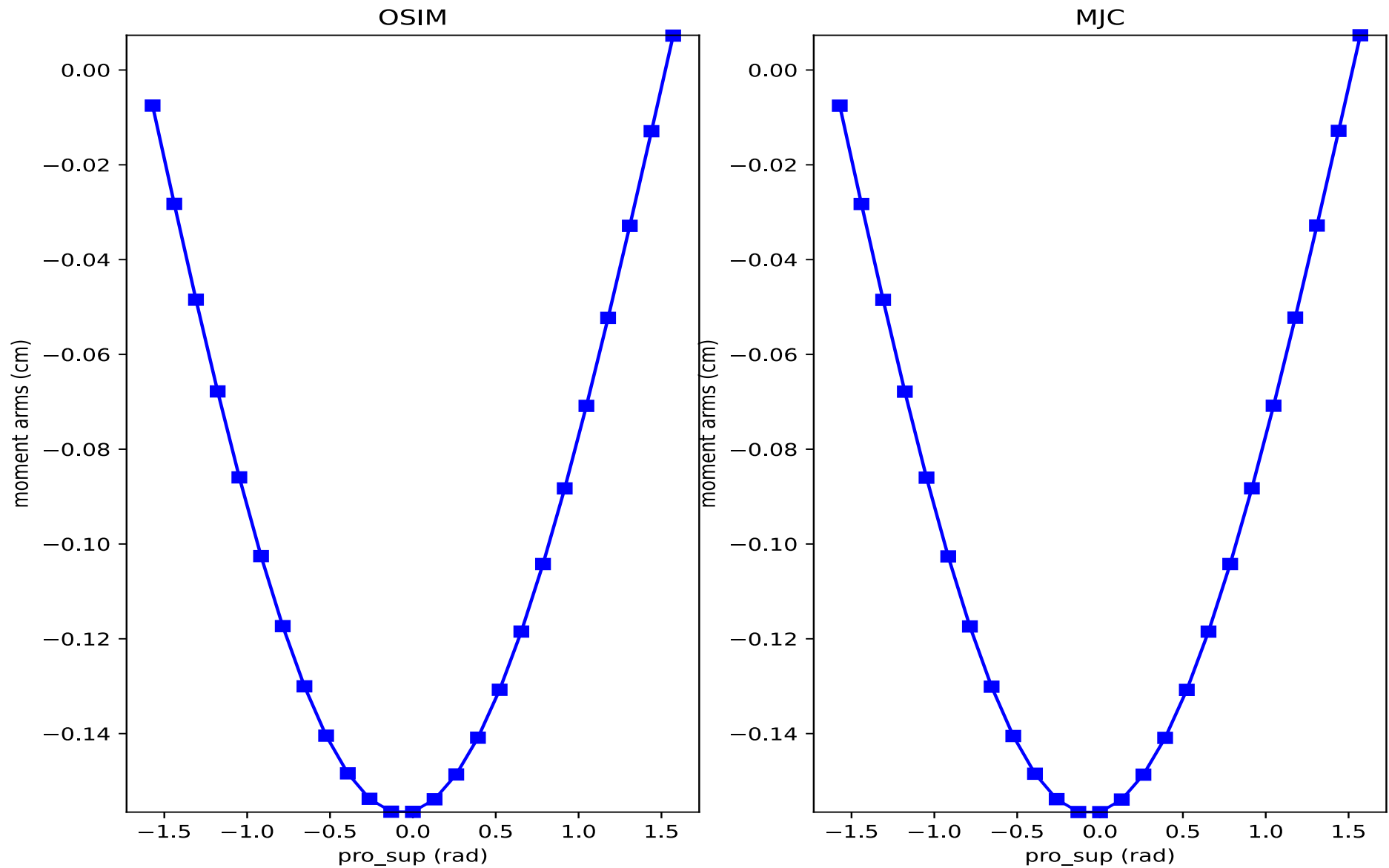
# Muscle specific moment arm comparison with Osim model after optimization

EPL - deviation - flexion - thumb\_abd - thumb\_flex



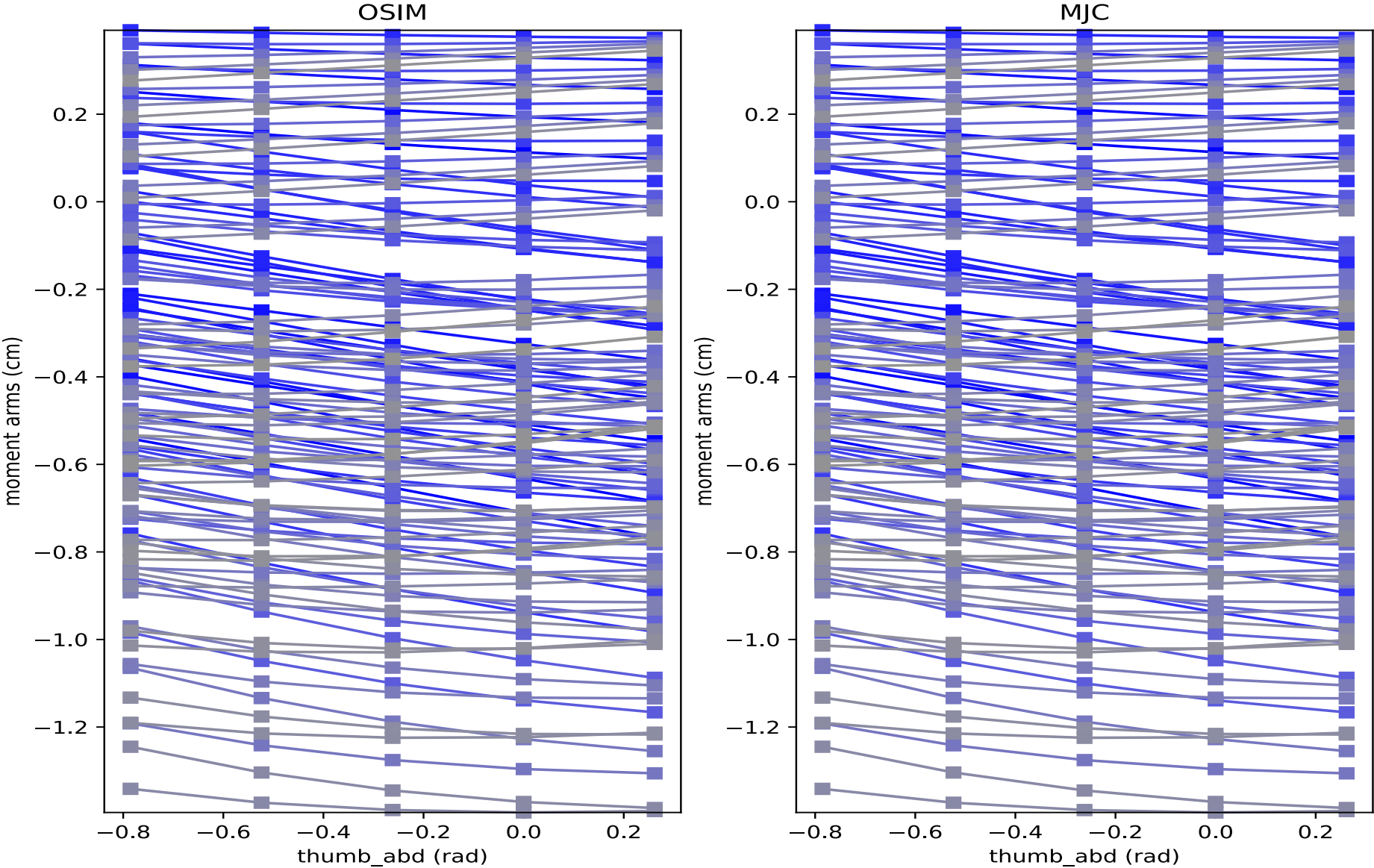
# Muscle specific moment arm comparison with Osim model after optimization

EPL - pro\_sup



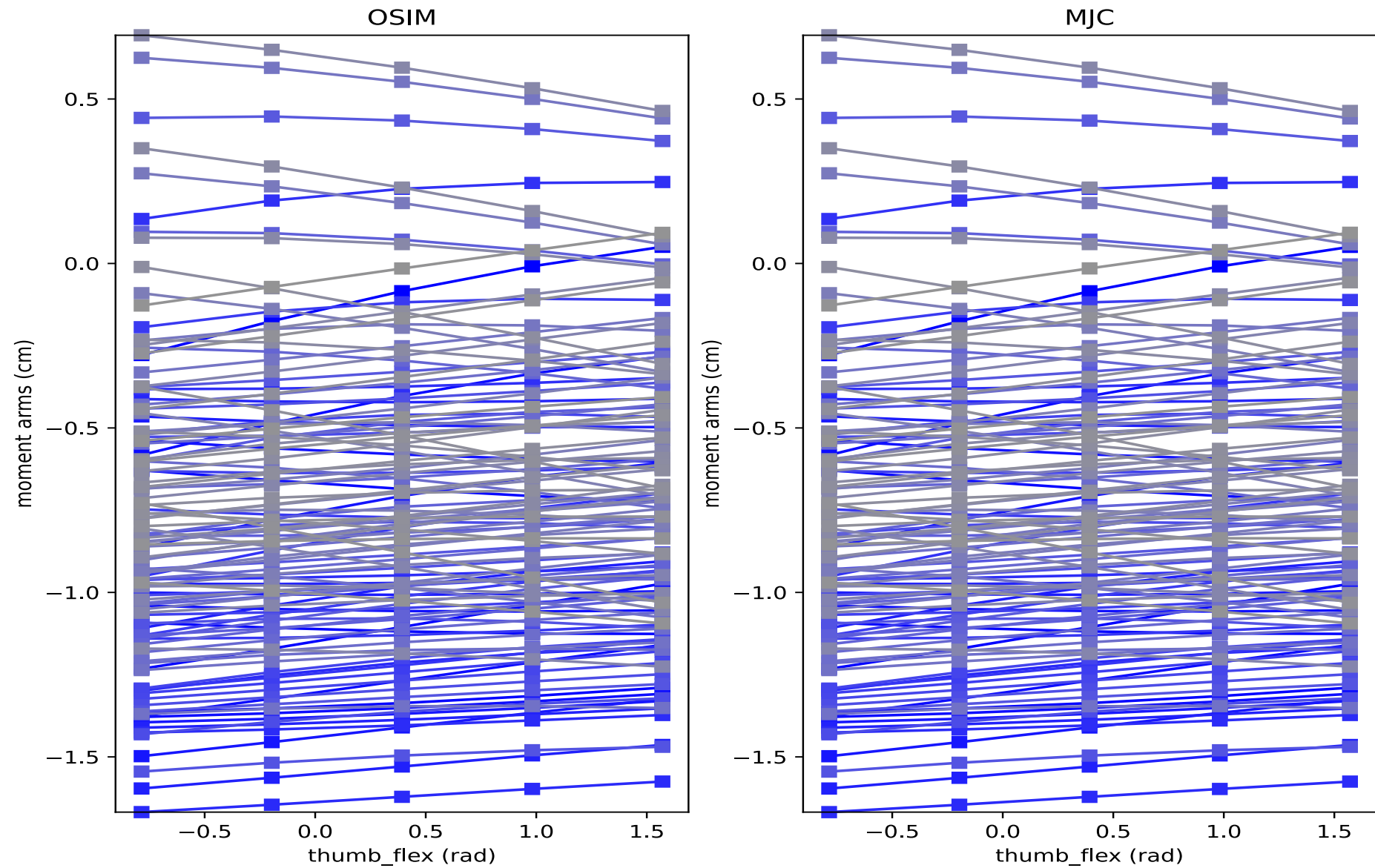
# Muscle specific moment arm comparison with Osim model after optimization

EPL - deviation - flexion - thumb\_abd - thumb\_flex



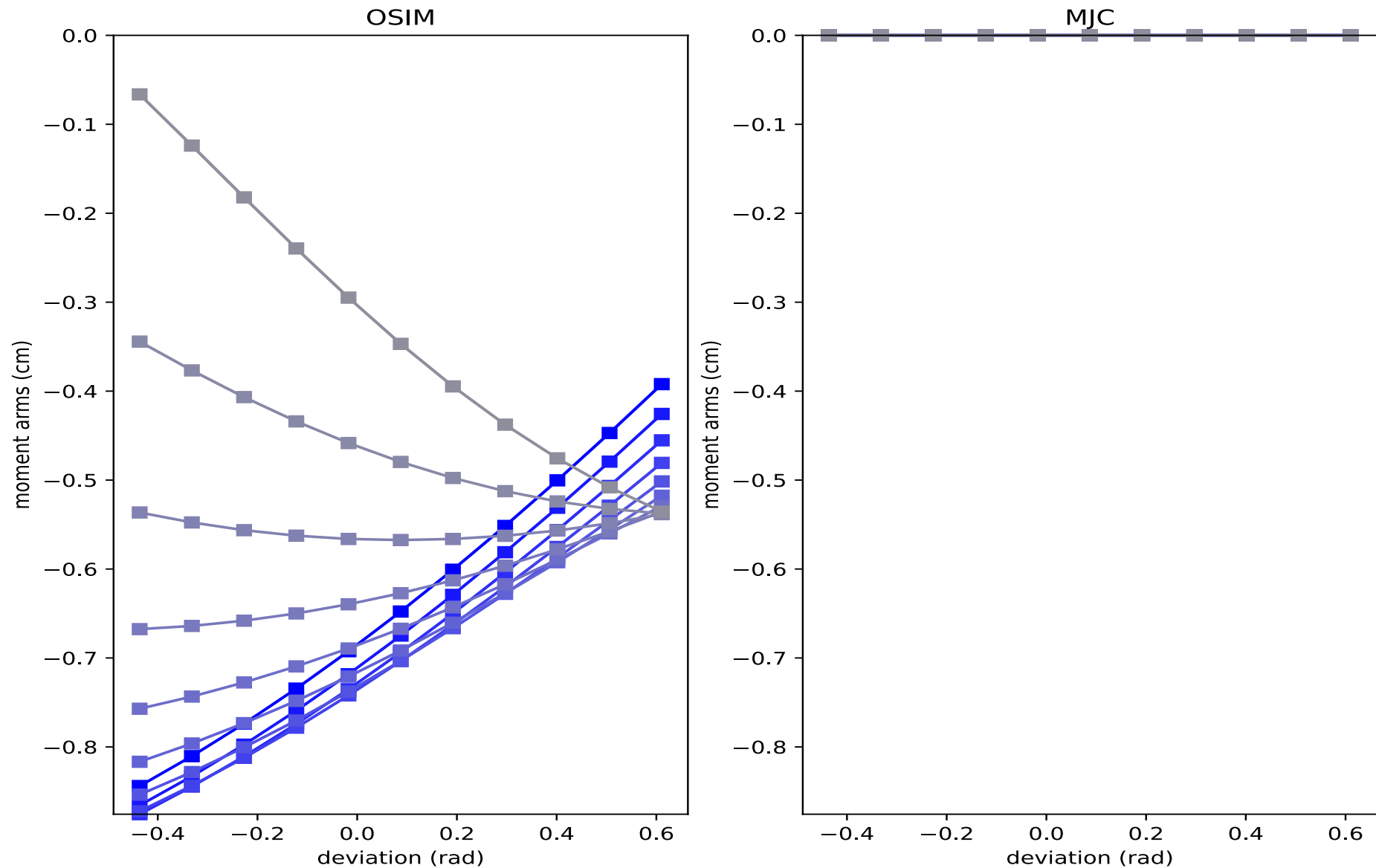
# Muscle specific moment arm comparison with Osim model after optimization

EPL - deviation - flexion - thumb\_abd - thumb\_flex



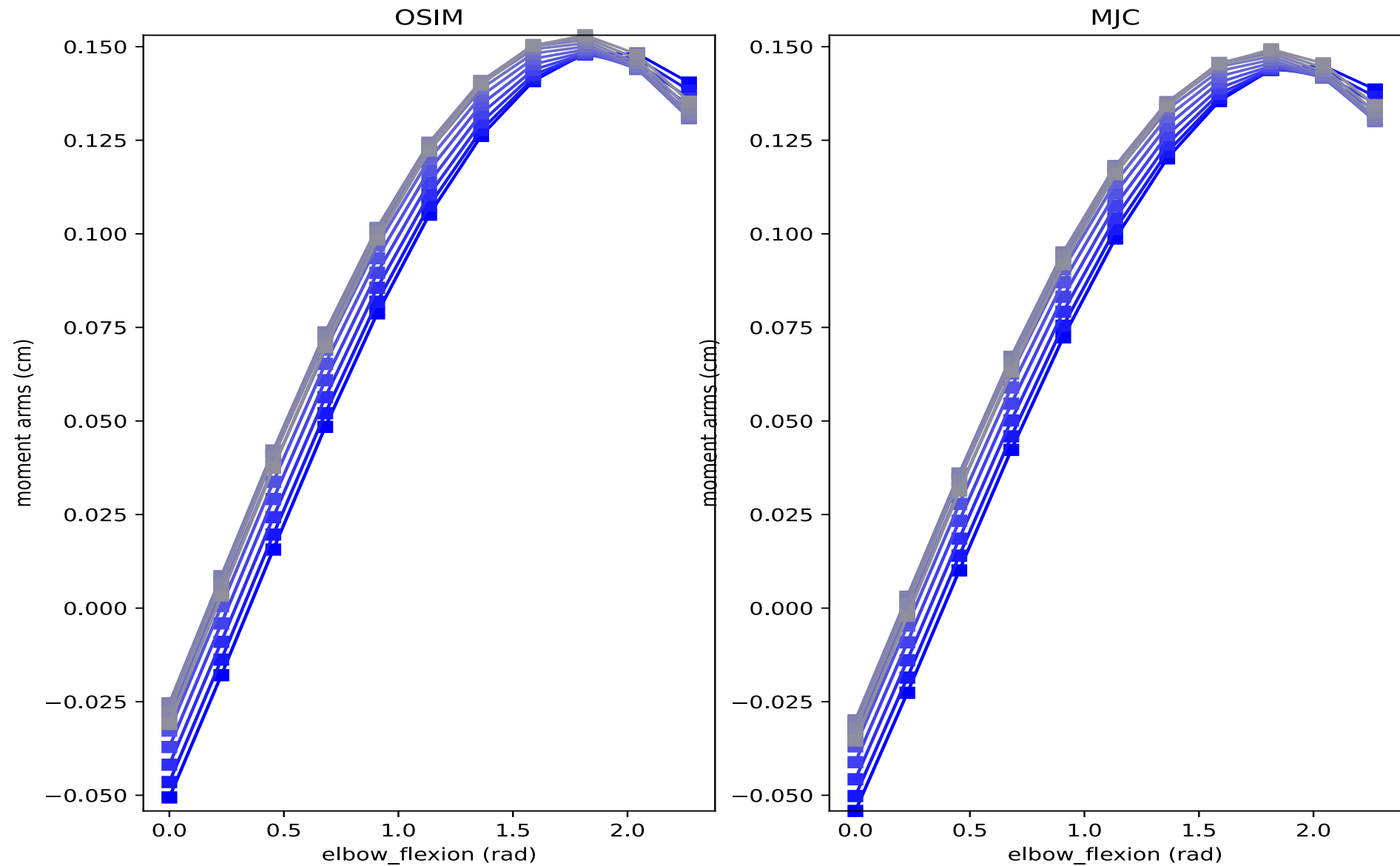
# Muscle specific moment arm comparison with Osim model after optimization

FCR - deviation - flexion



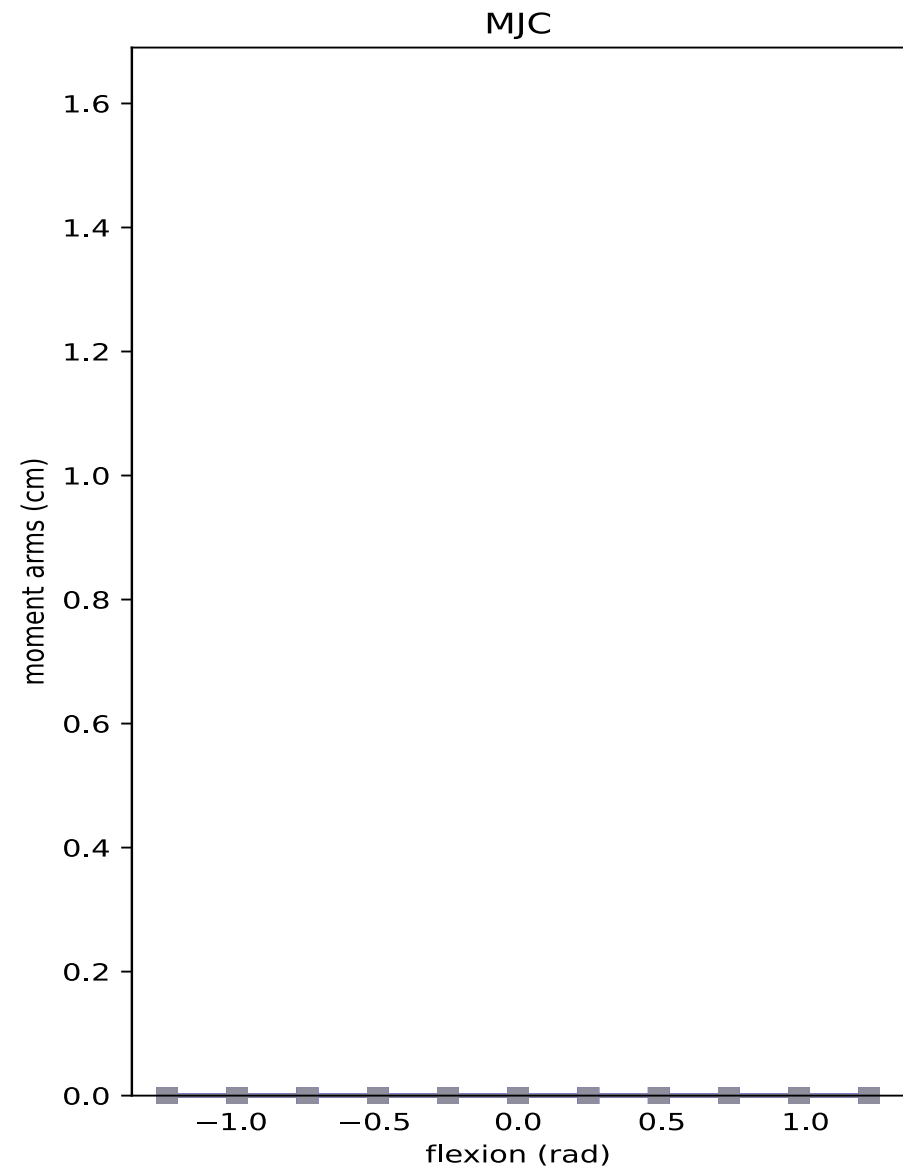
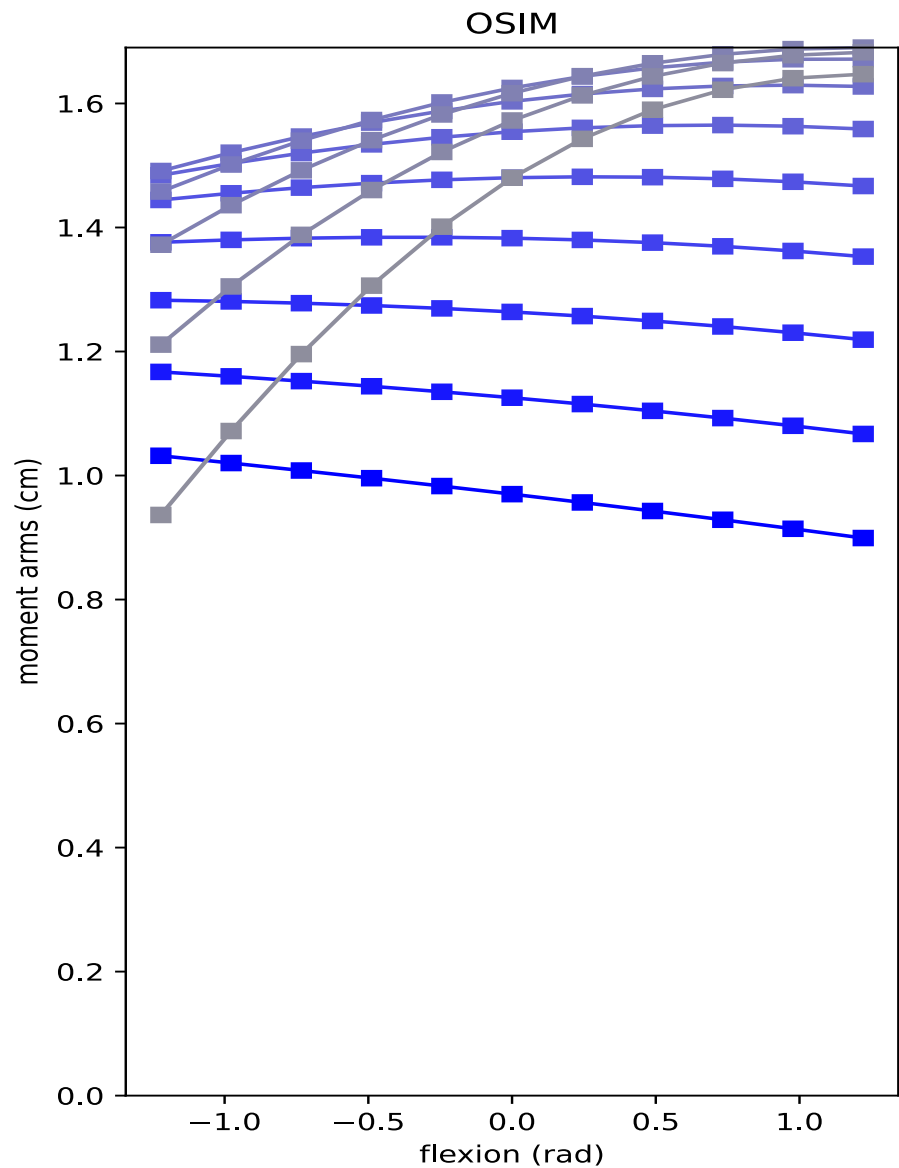
# Muscle specific moment arm comparison with Osim model after optimization

FCR - elbow\_flexion - pro\_sup



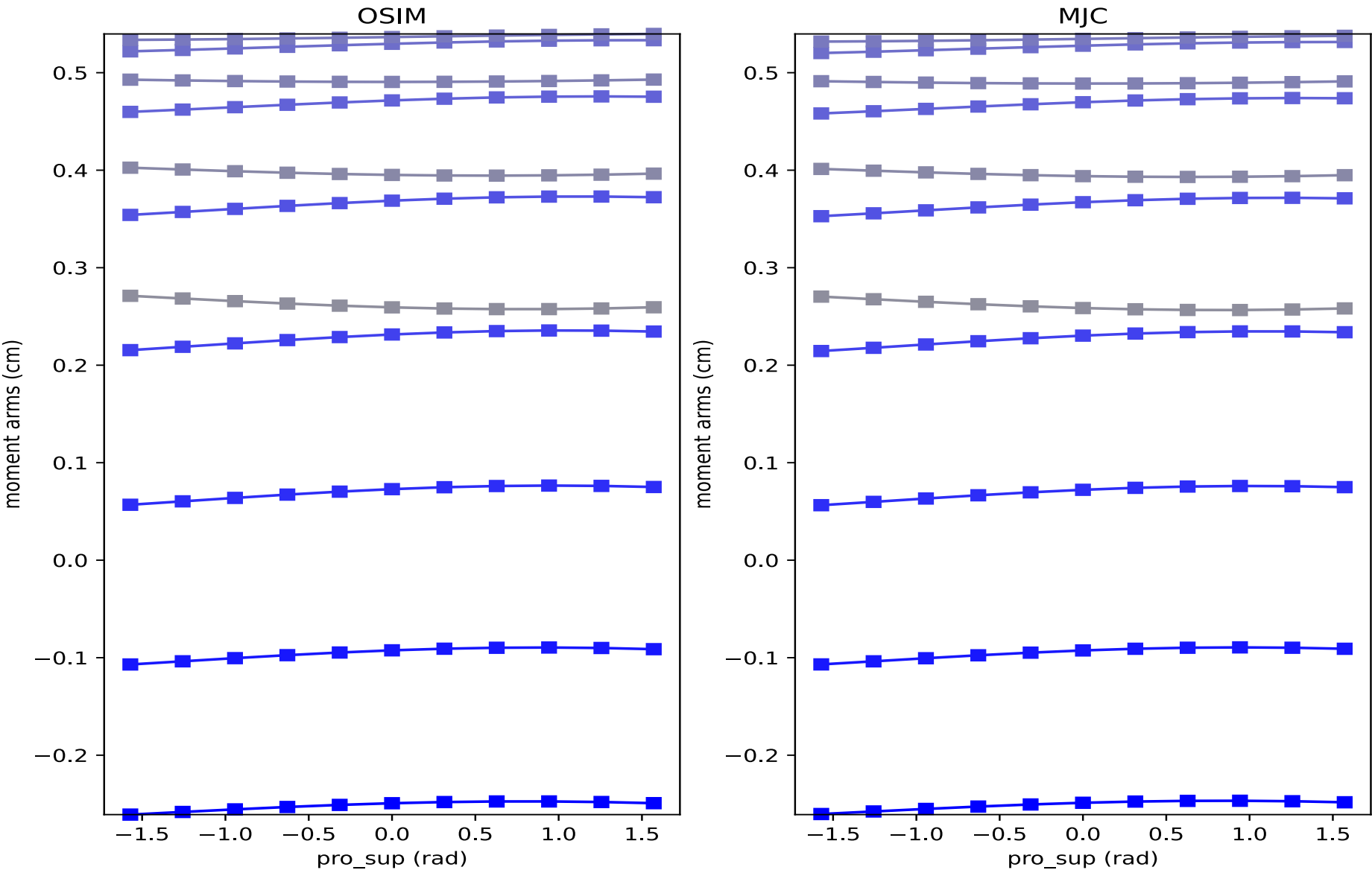
# Muscle specific moment arm comparison with Osim model after optimization

FCR - deviation - flexion



# Muscle specific moment arm comparison with Osim model after optimization

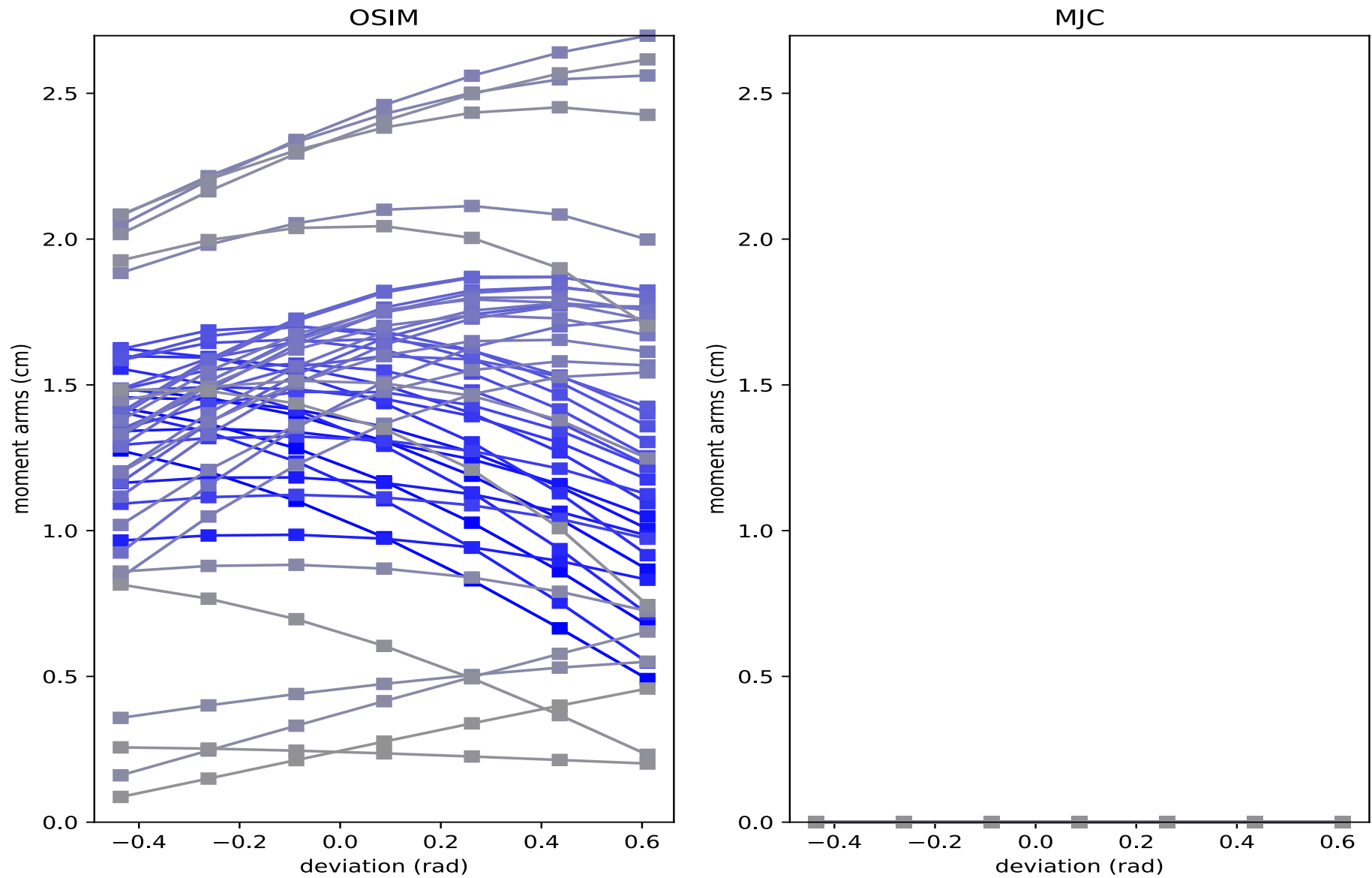
FCR - elbow\_flexion - pro\_sup





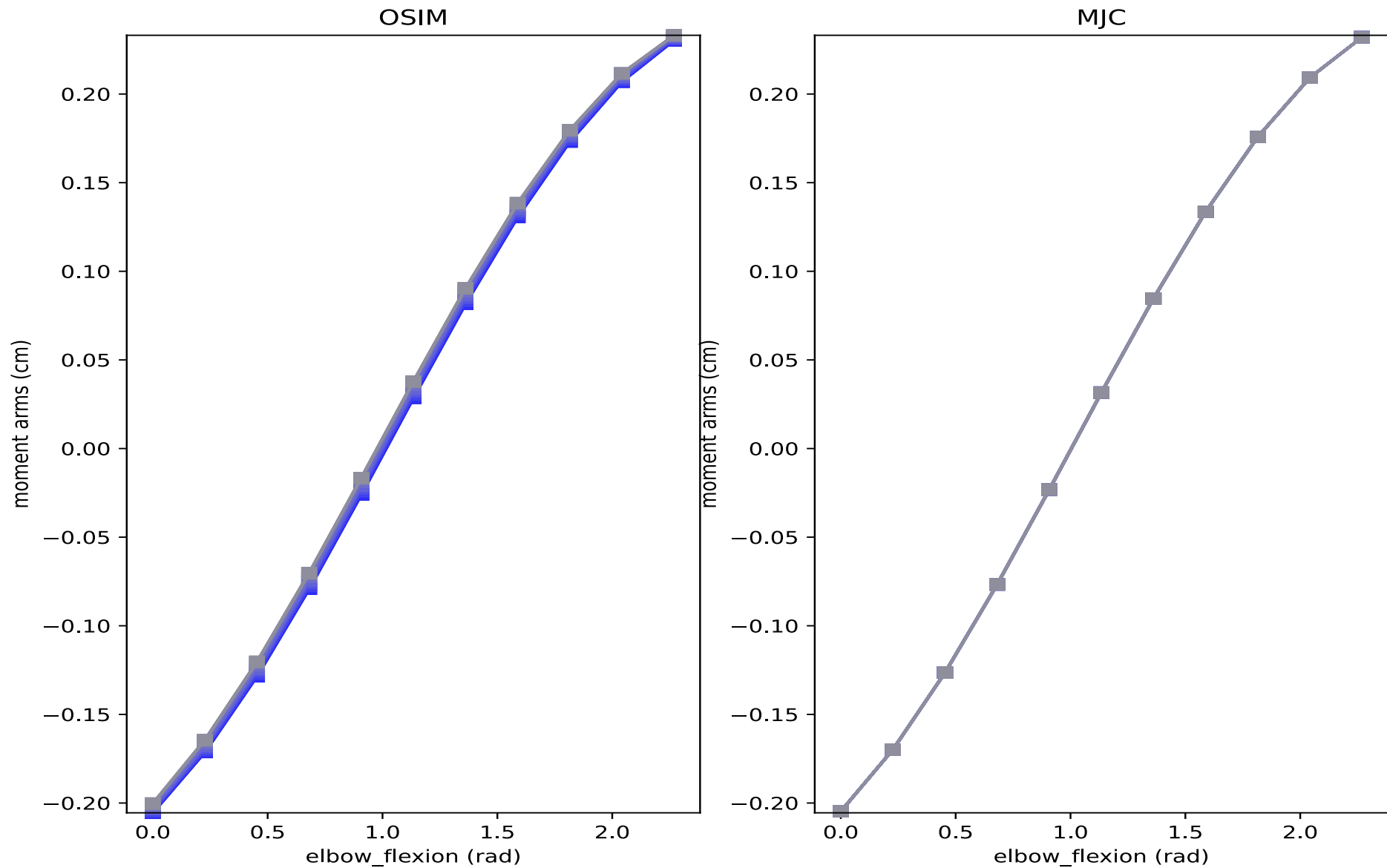
# Muscle specific moment arm comparison with Osim model after optimization

FCU - deviation - flexion - pro\_sup



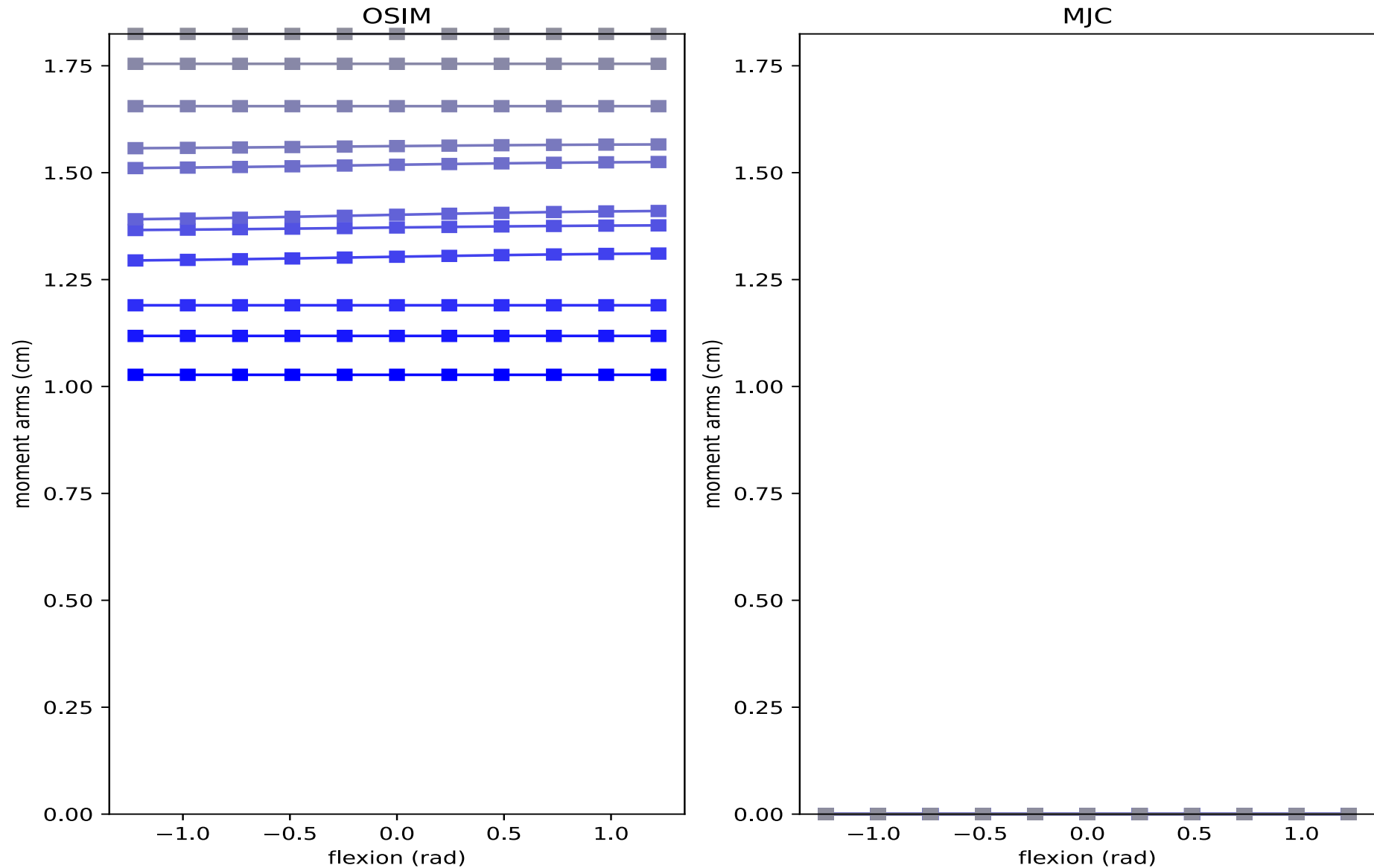
# Muscle specific moment arm comparison with Osim model after optimization

FCU - elbow\_flexion - flexion



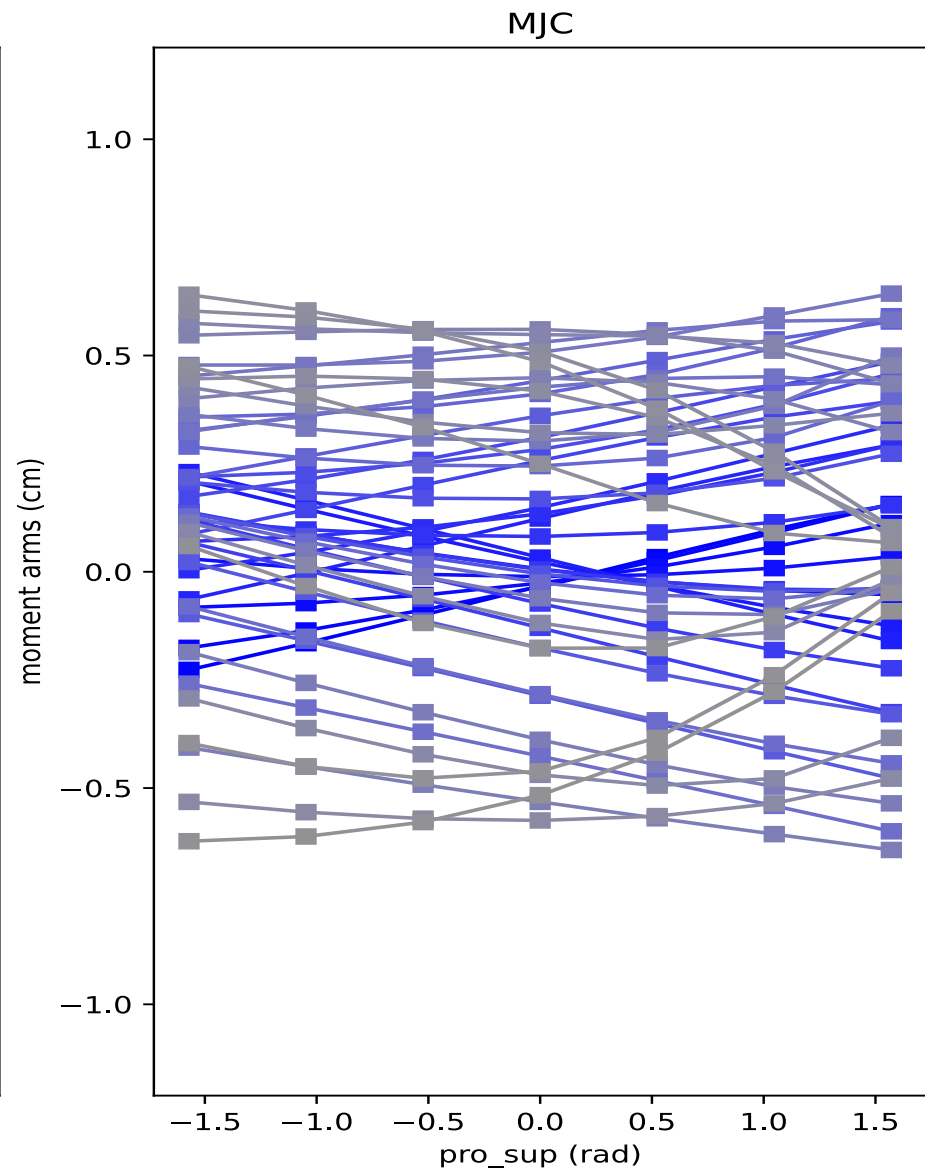
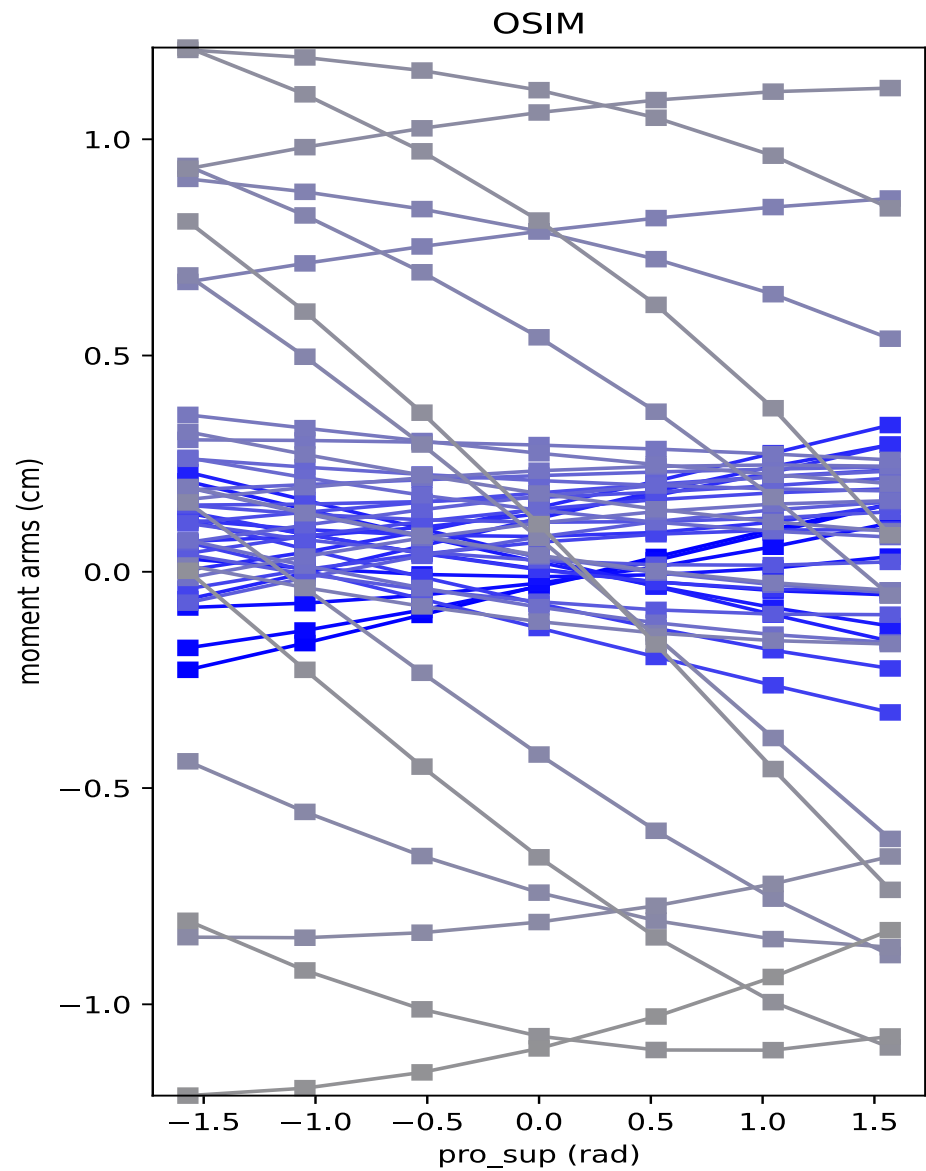
# Muscle specific moment arm comparison with Osim model after optimization

FCU - elbow\_flexion - flexion



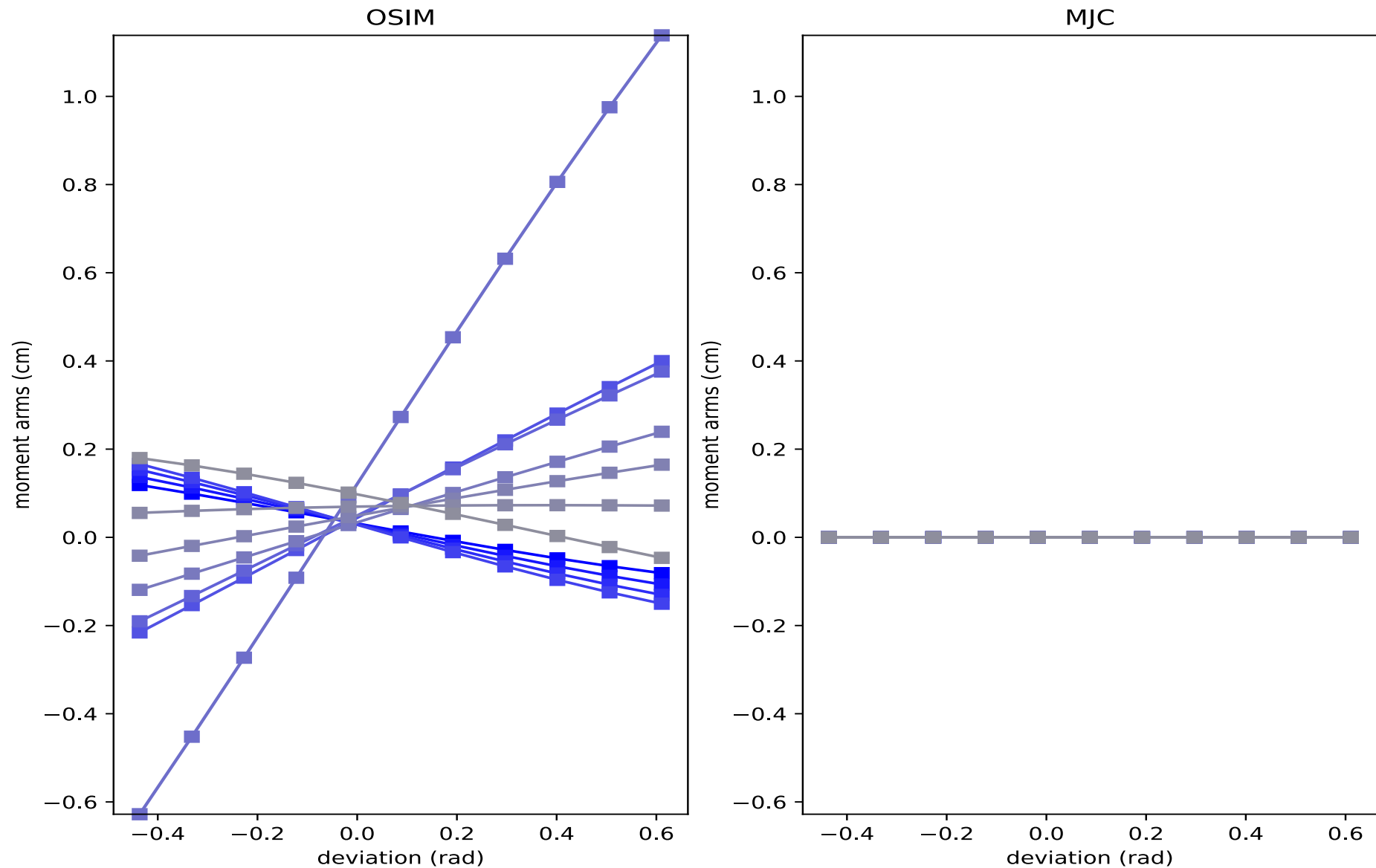
# Muscle specific moment arm comparison with Osim model after optimization

FCU - deviation - flexion - pro\_sup



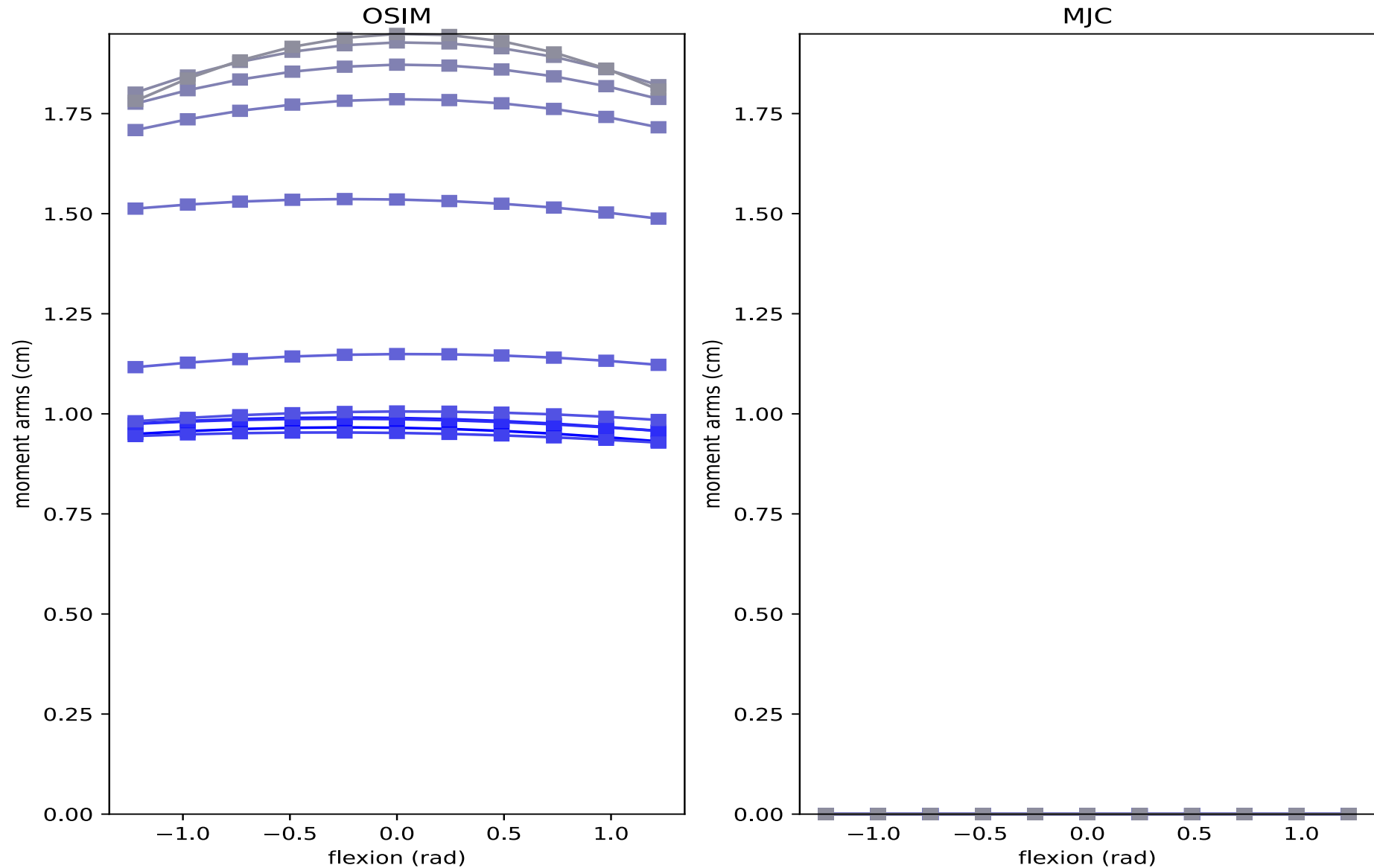
# Muscle specific moment arm comparison with Osim model after optimization

FDPI - deviation - flexion



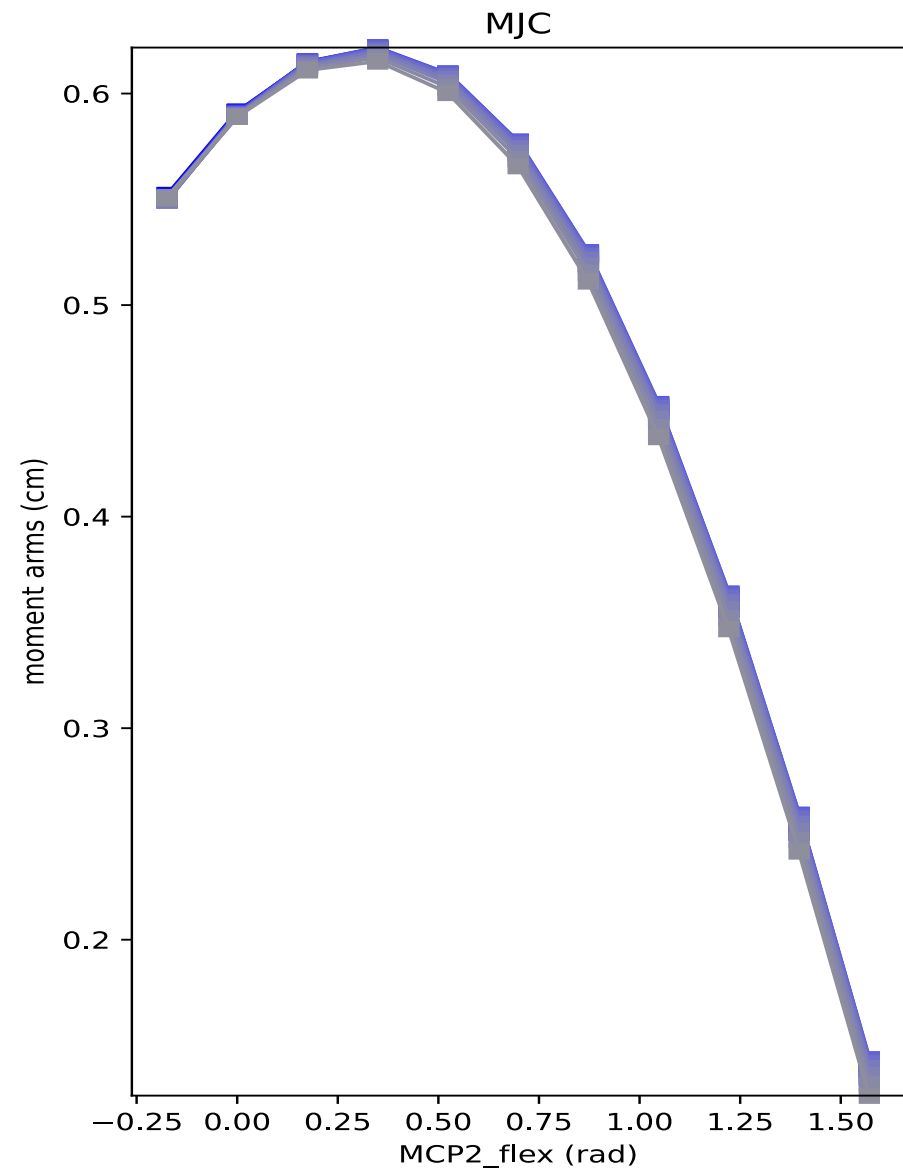
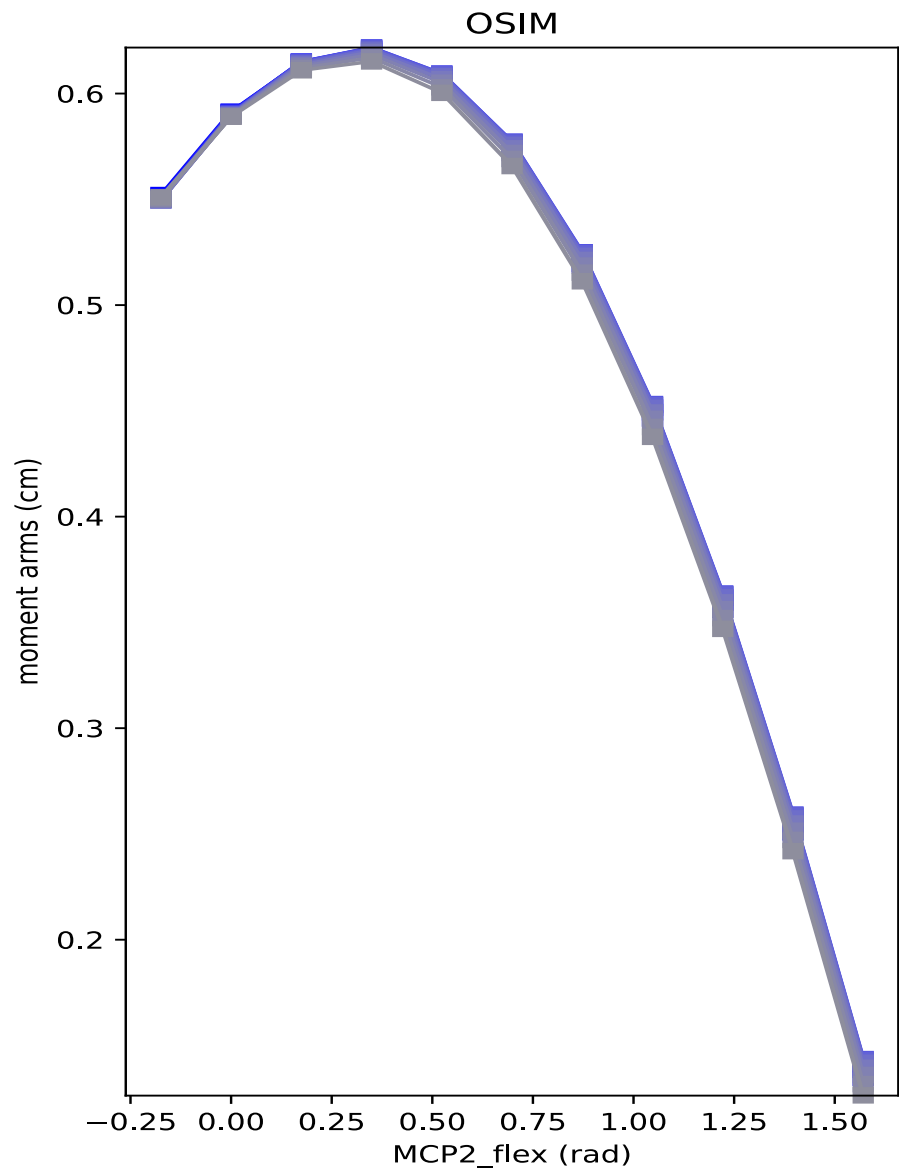
## Muscle specific moment arm comparison with Osim model after optimization

FDPI - deviation - flexion



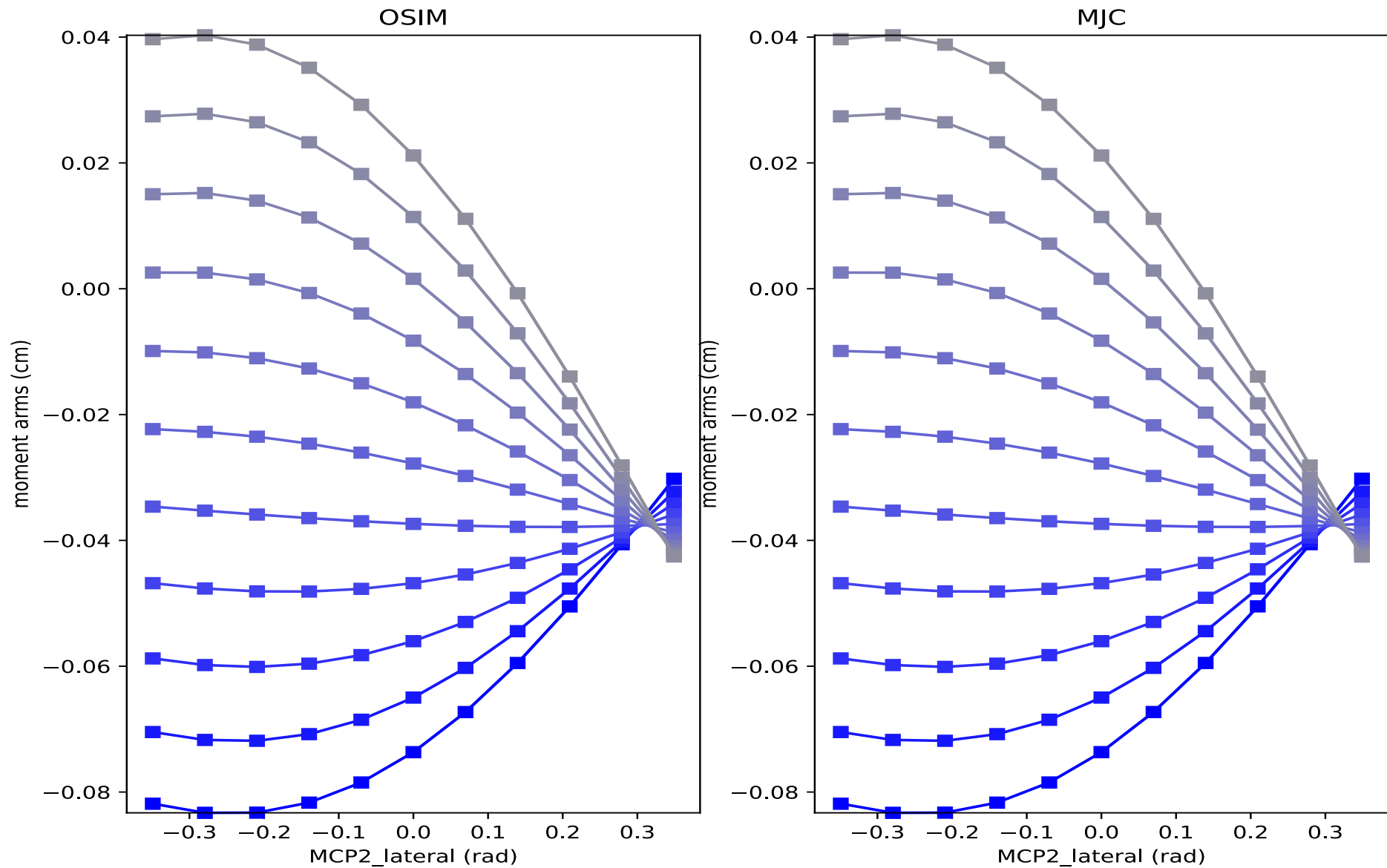
# Muscle specific moment arm comparison with Osim model after optimization

FDPI - MCP2\_flex - MCP2\_lateral



# Muscle specific moment arm comparison with Osim model after optimization

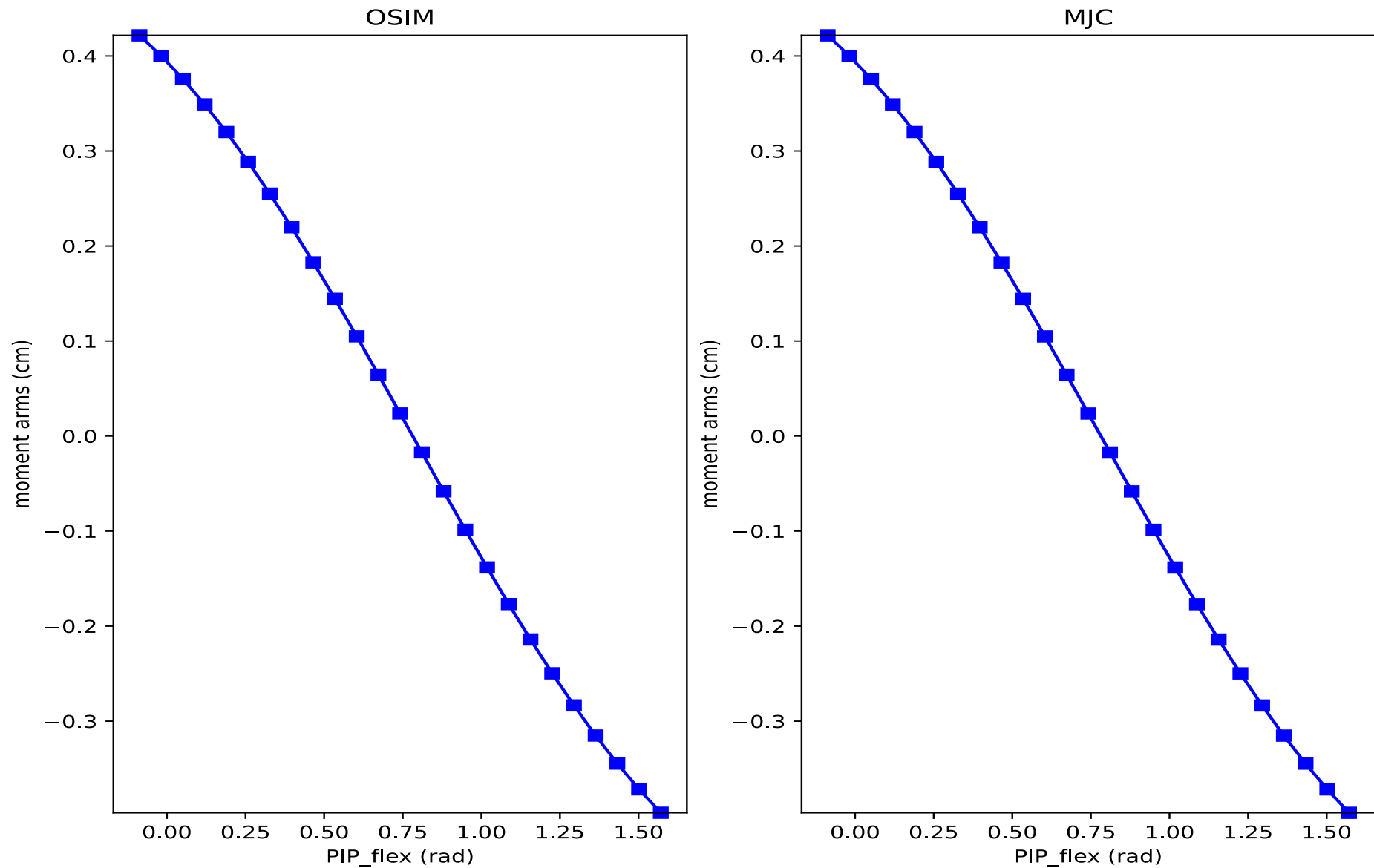
FDPI - MCP2\_flex - MCP2\_lateral





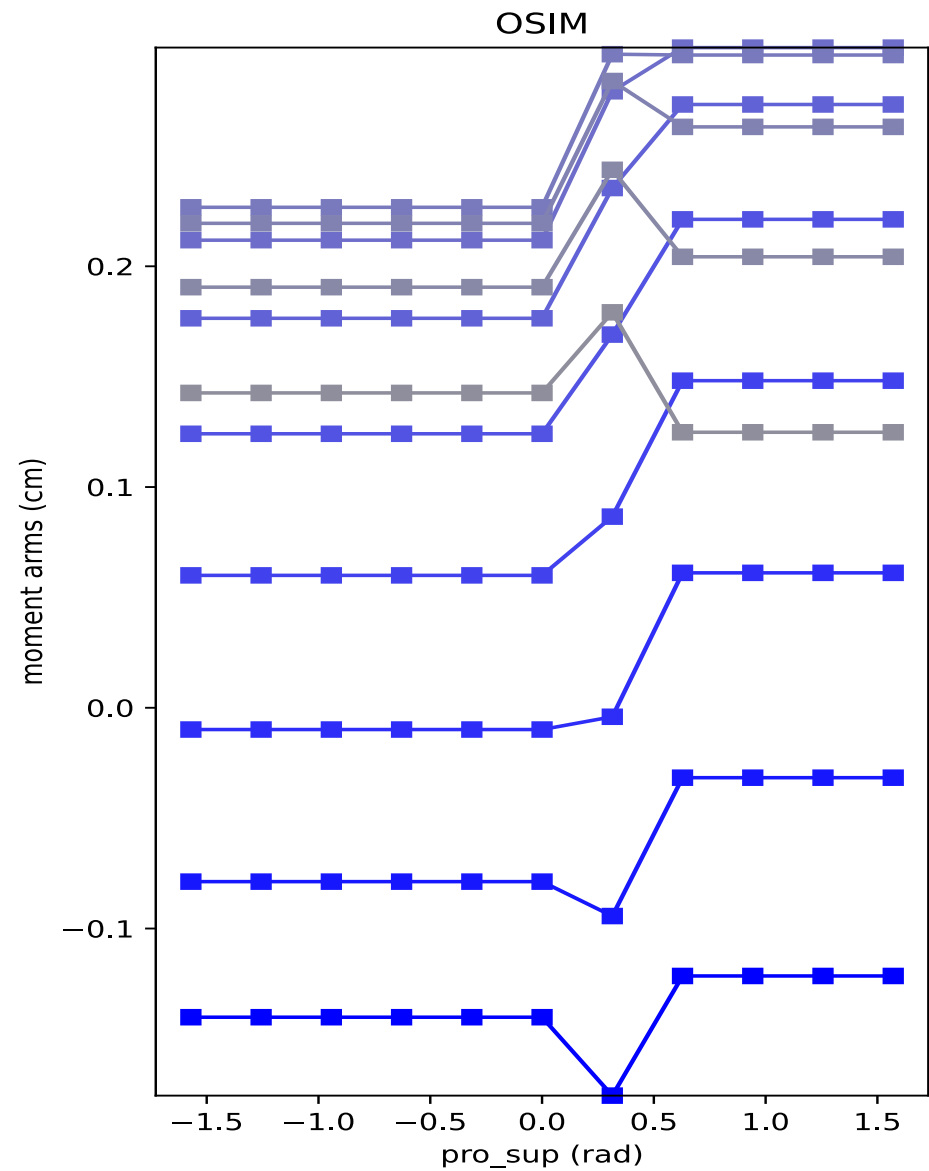
# Muscle specific moment arm comparison with Osim model after optimization

FDPI - PIP\_flex

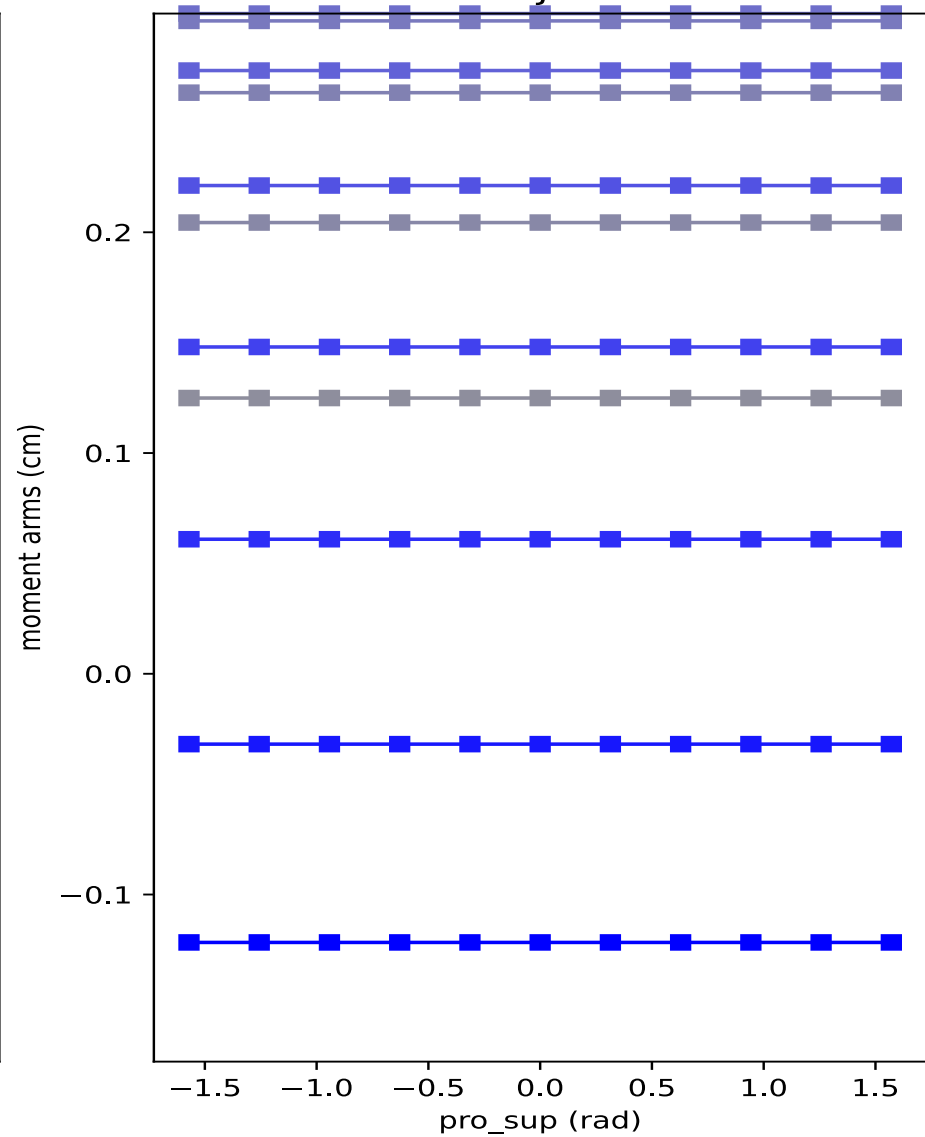


### Muscle specific moment arm comparison with Osim model after optimization

FDPI - flexion - pro\_sup

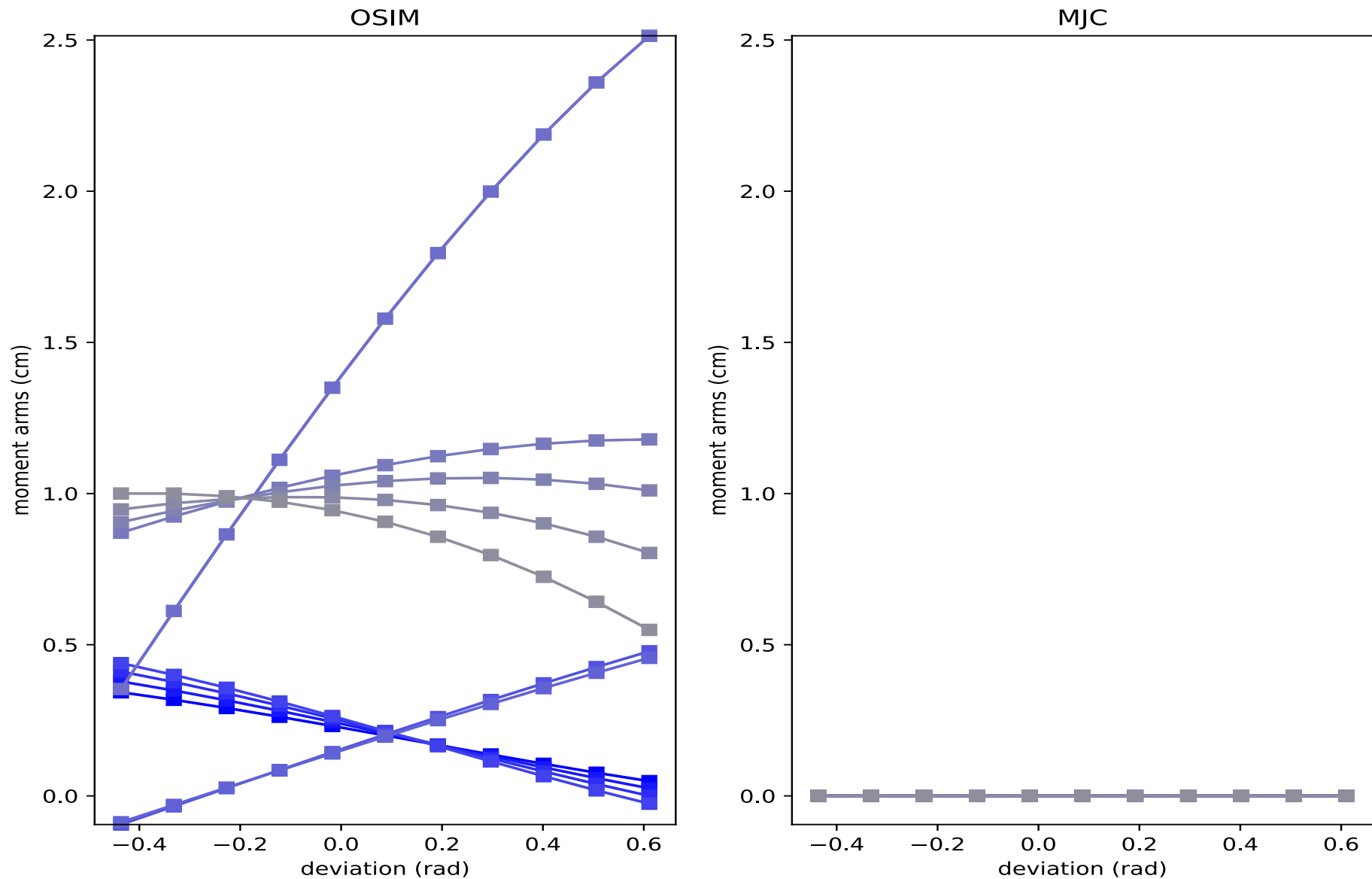


MJC



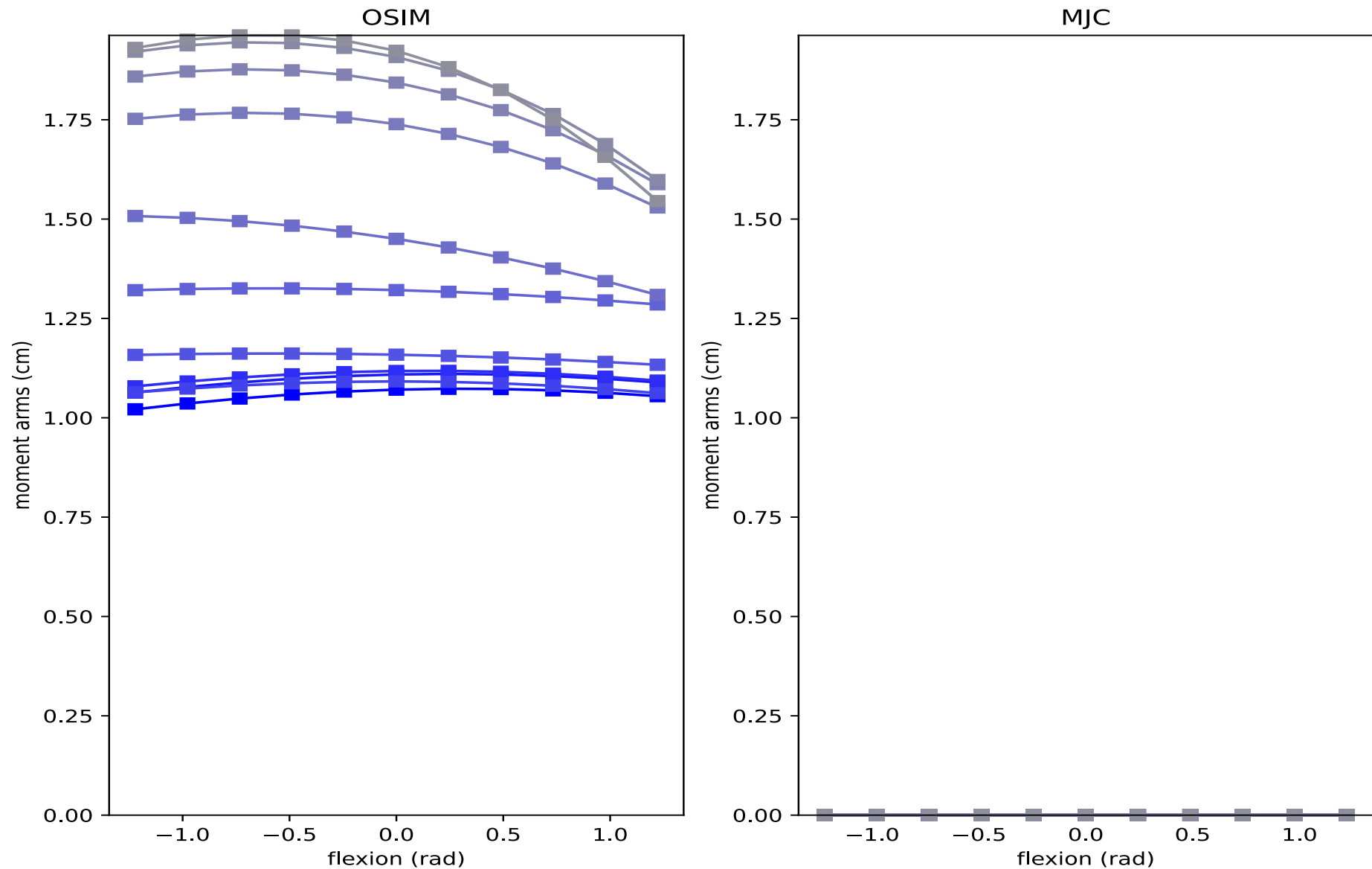
## Muscle specific moment arm comparison with Osim model after optimization

FDPL - deviation - flexion



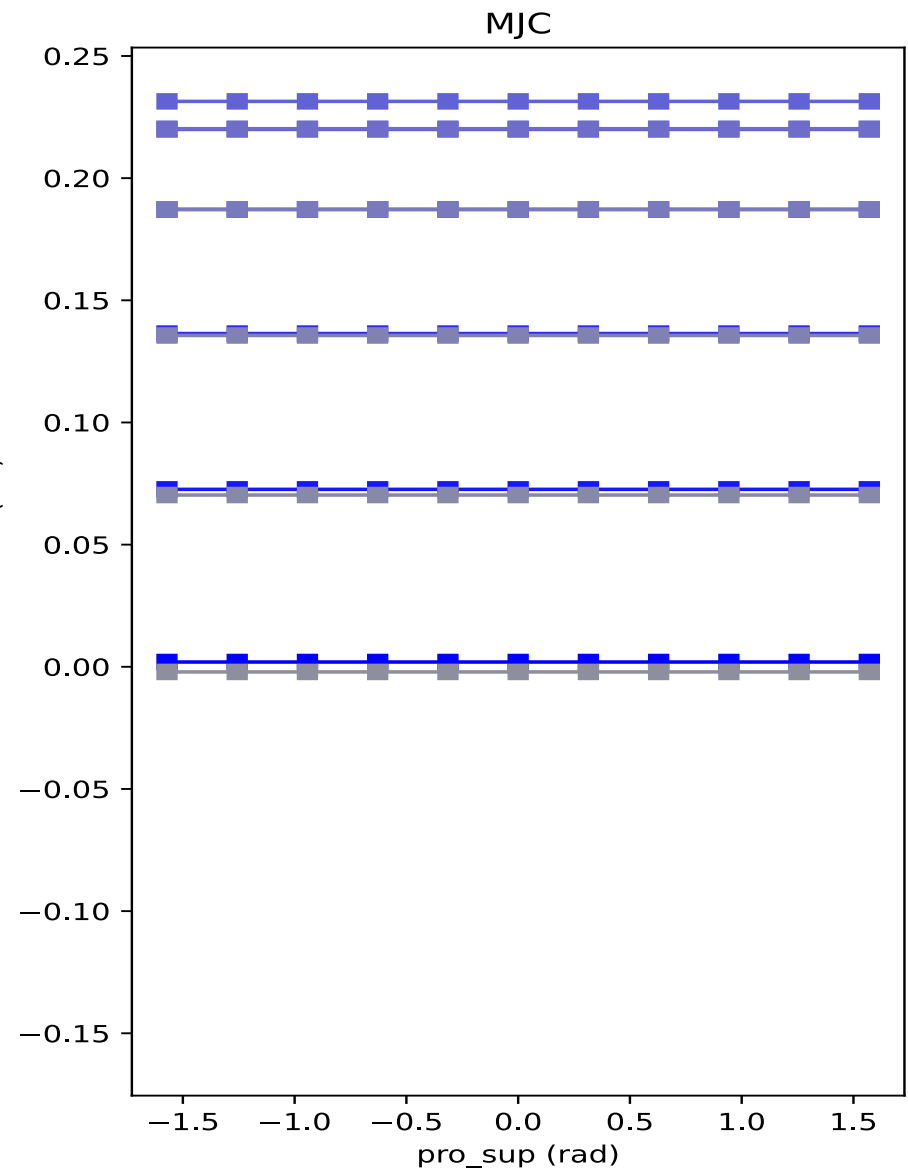
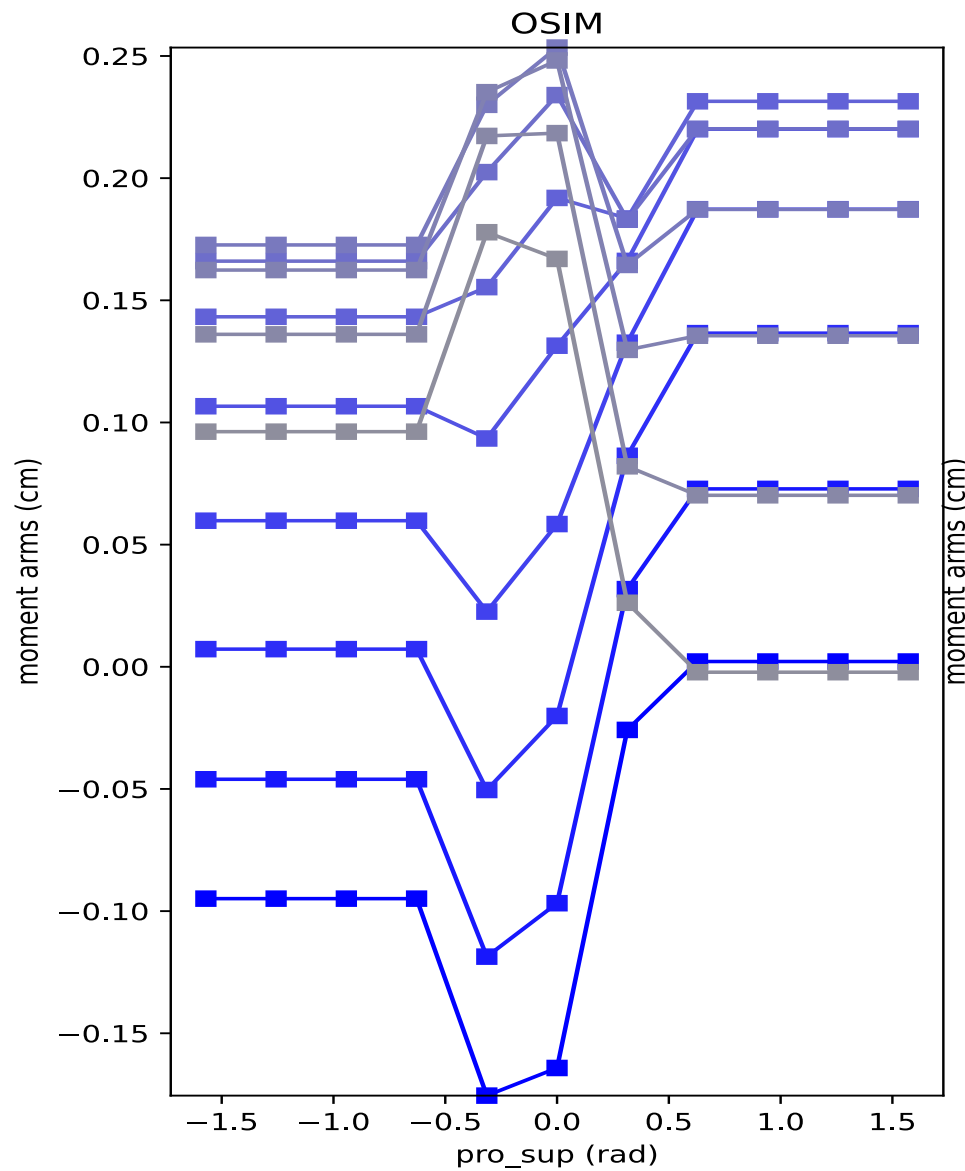
## Muscle specific moment arm comparison with Osim model after optimization

FDPL - deviation - flexion



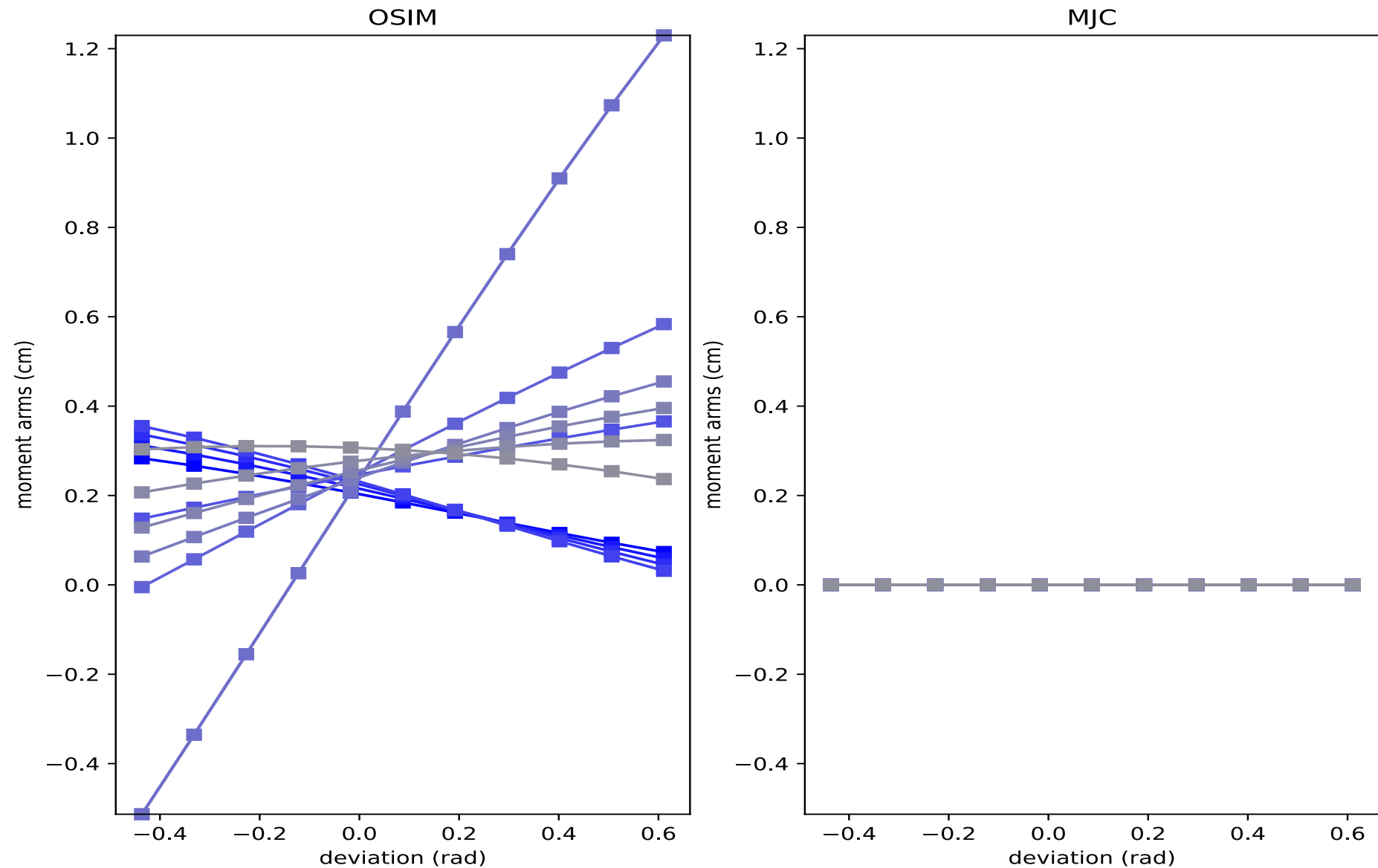
# Muscle specific moment arm comparison with Osim model after optimization

FDPL - flexion - pro\_sup



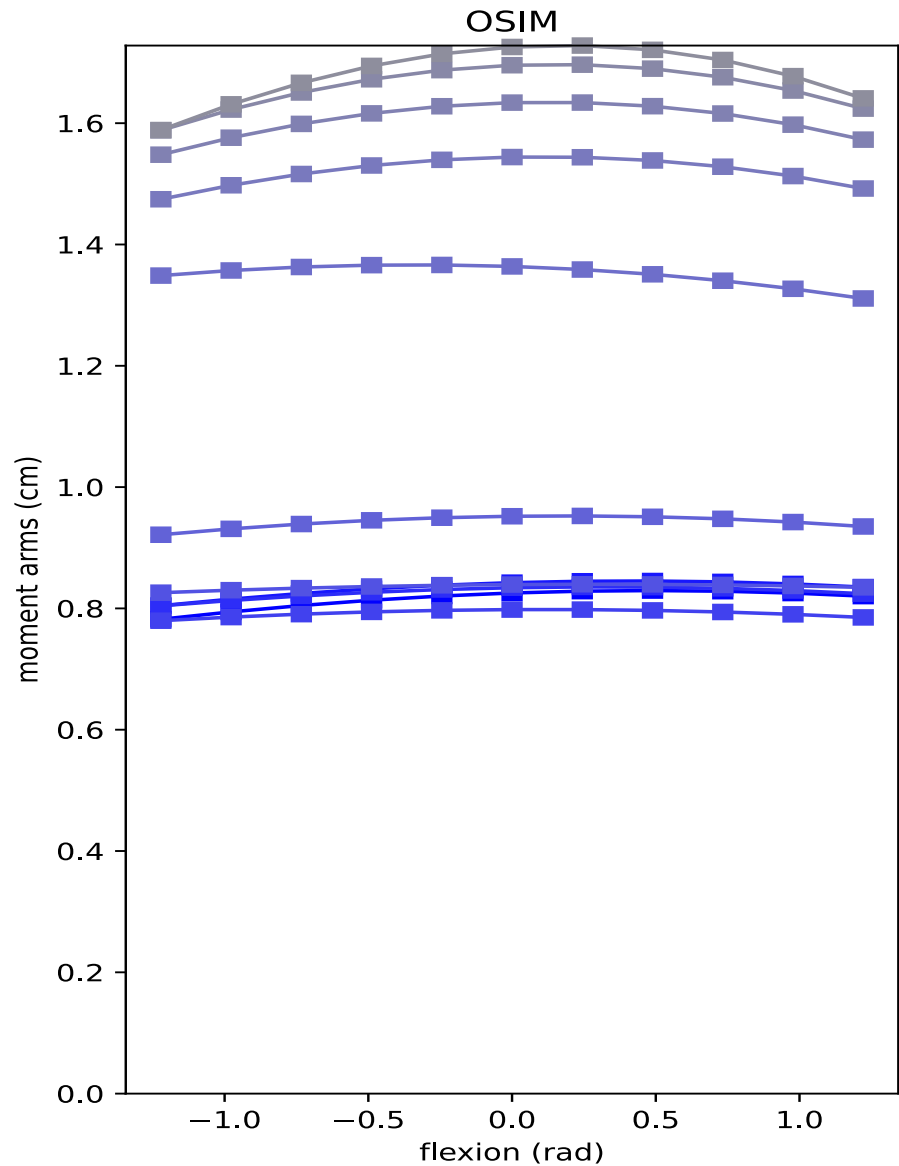
# Muscle specific moment arm comparison with Osim model after optimization

FDPM - deviation - flexion

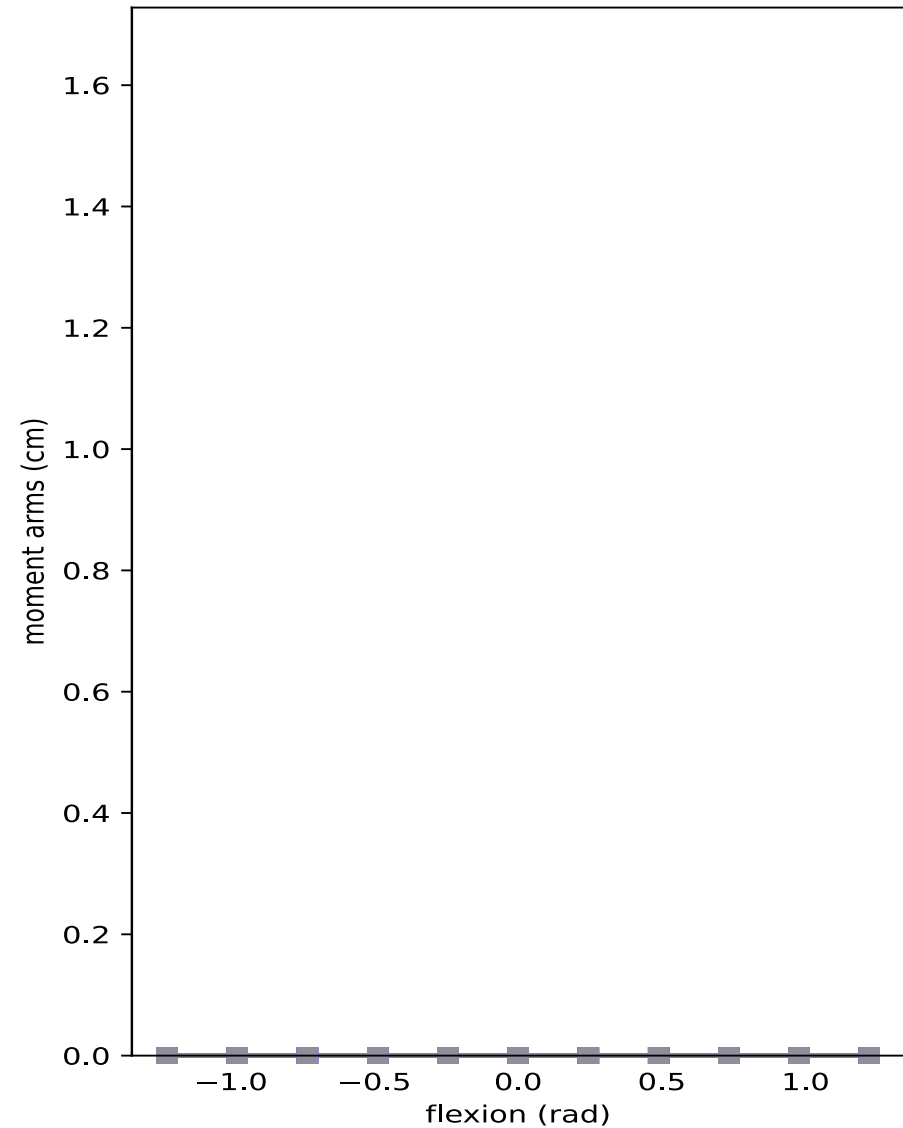


## Muscle specific moment arm comparison with Osim model after optimization

FDPM - deviation - flexion

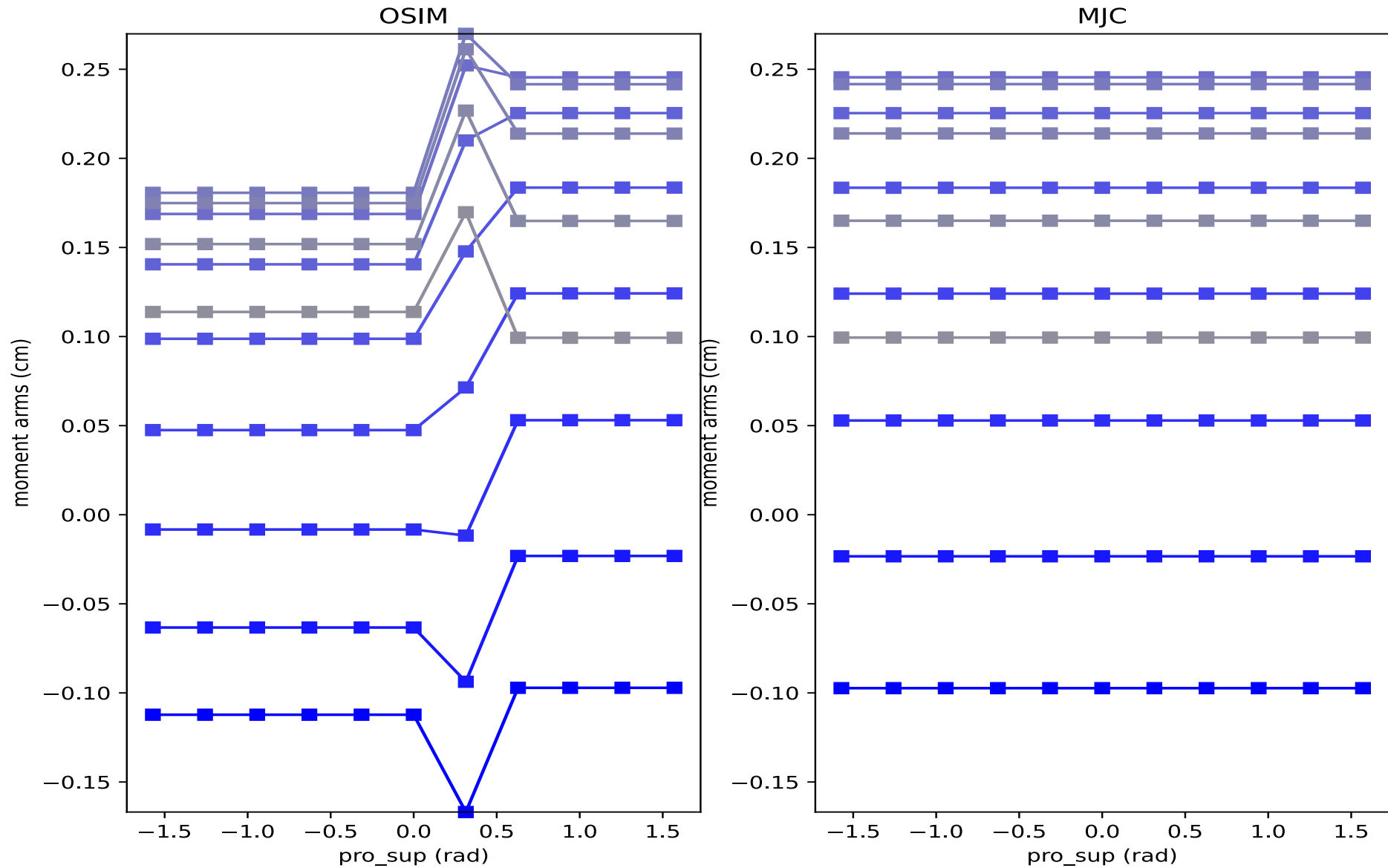


MJC



# Muscle specific moment arm comparison with Osim model after optimization

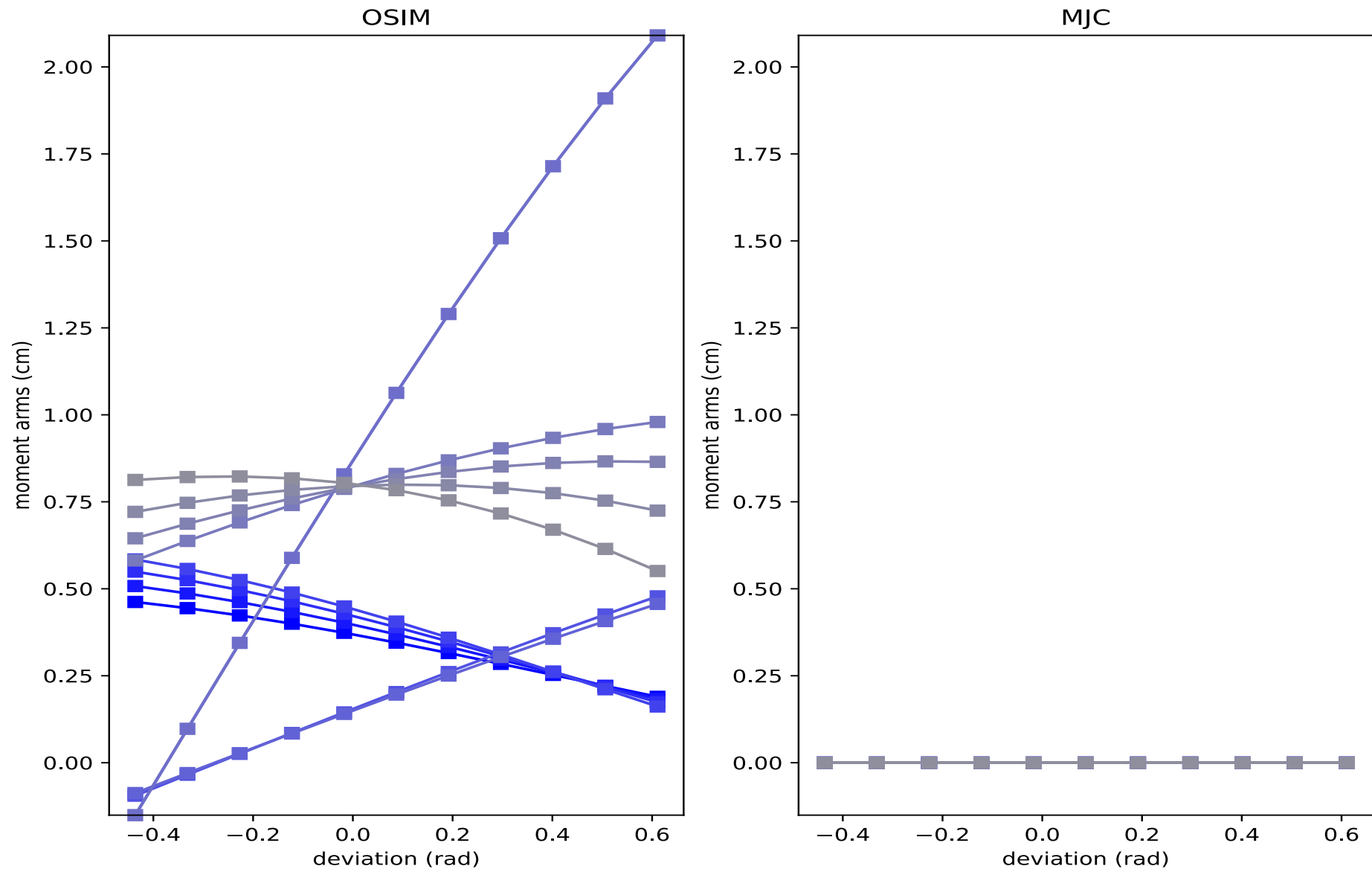
FDPM - flexion - pro\_sup





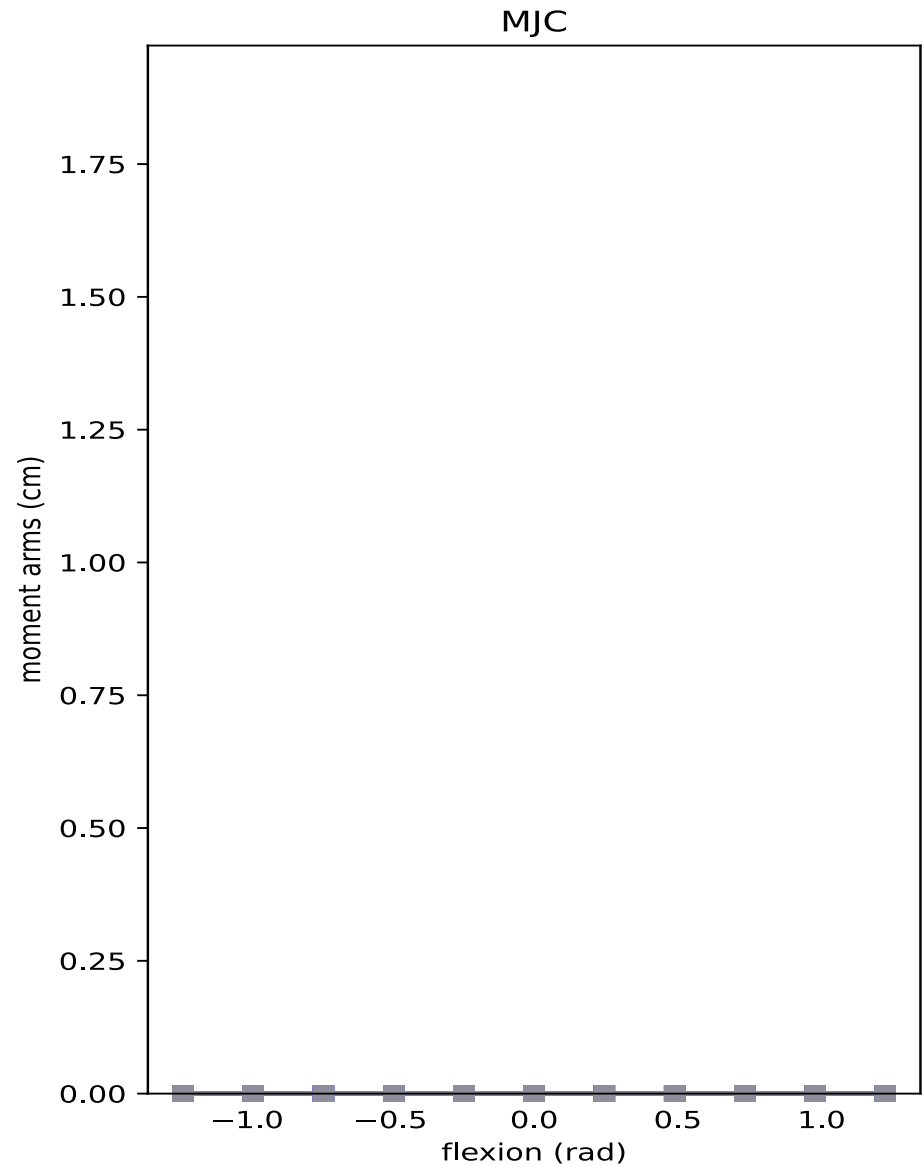
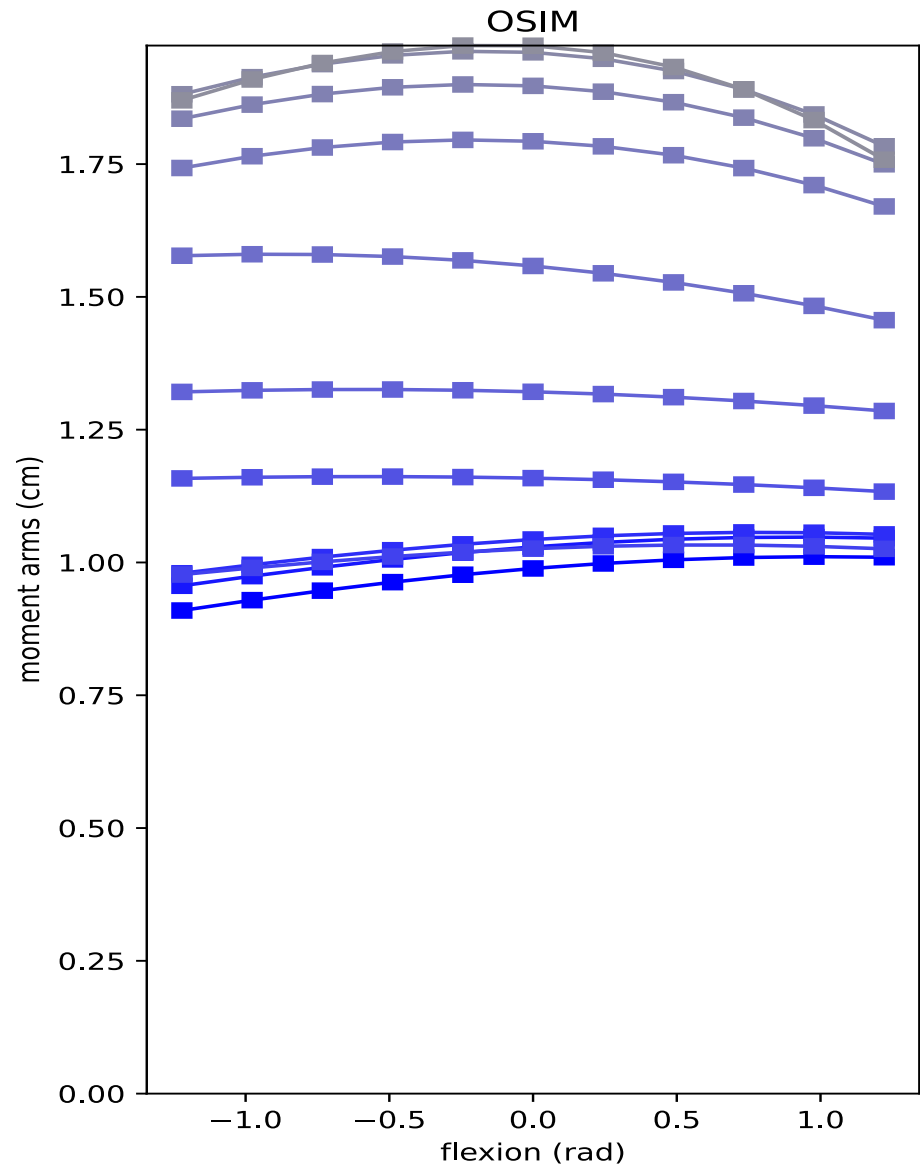
# Muscle specific moment arm comparison with Osim model after optimization

FDPR - deviation - flexion



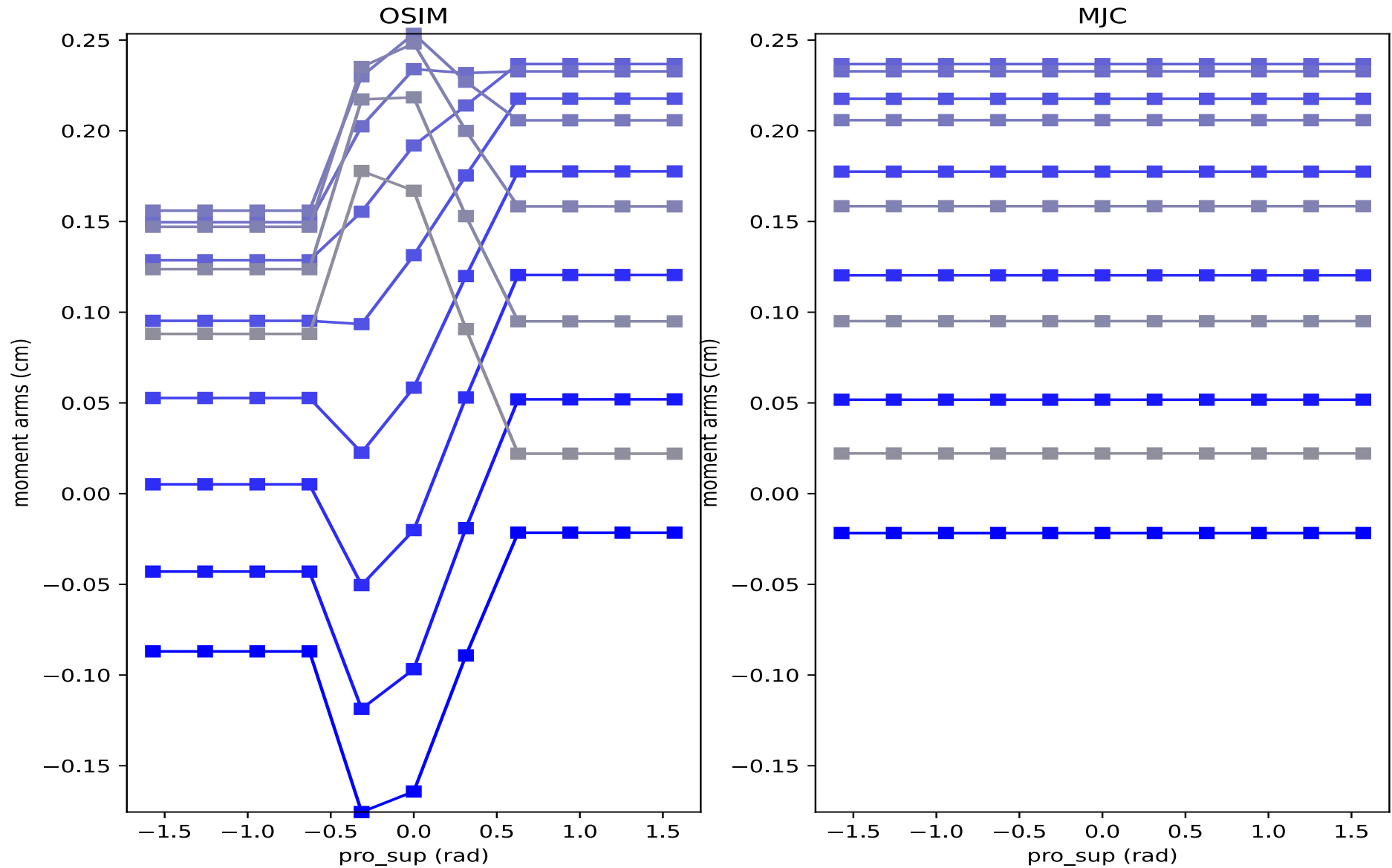
## Muscle specific moment arm comparison with Osim model after optimization

FDPR - deviation - flexion



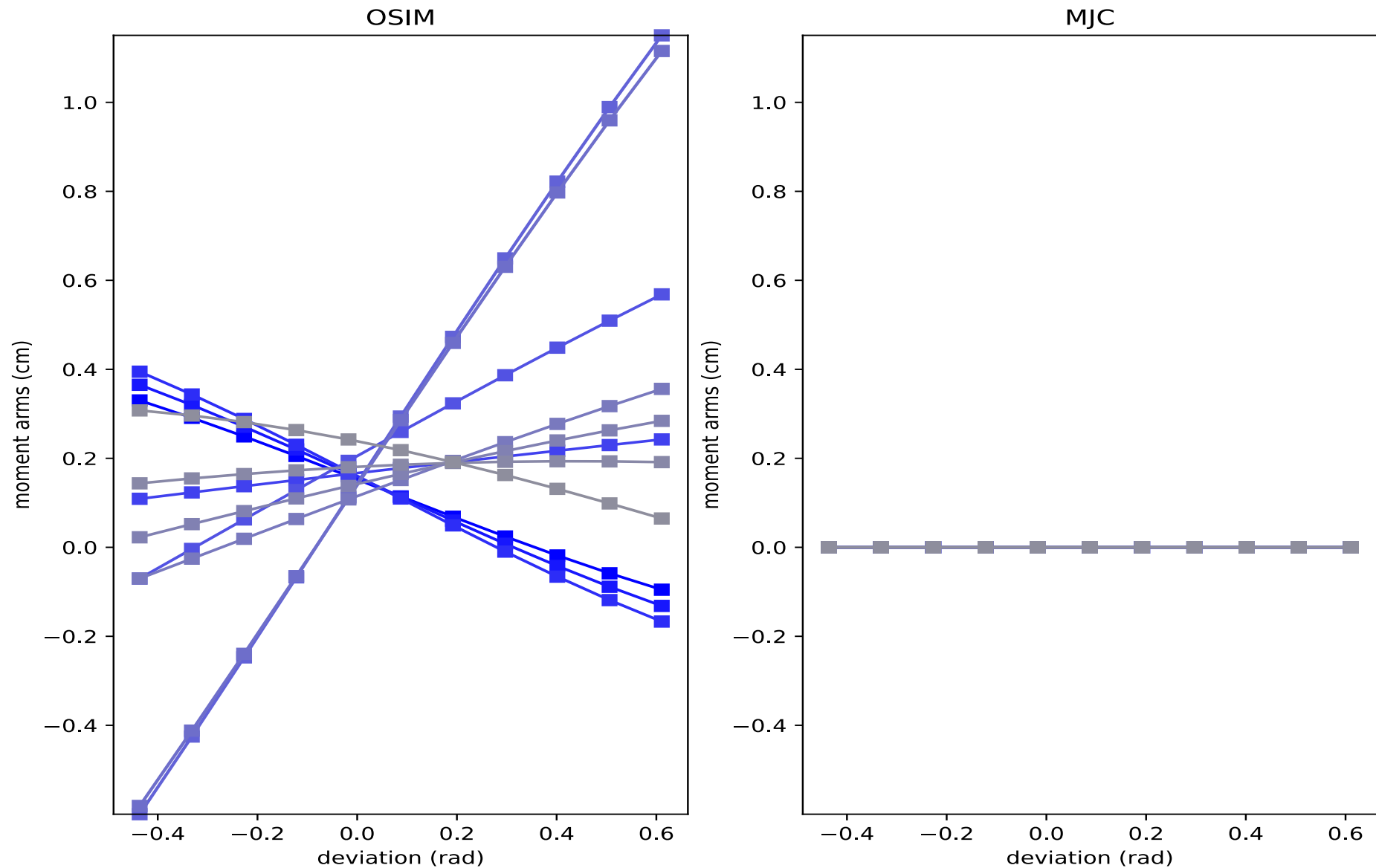
# Muscle specific moment arm comparison with Osim model after optimization

FDPR - flexion - pro\_sup



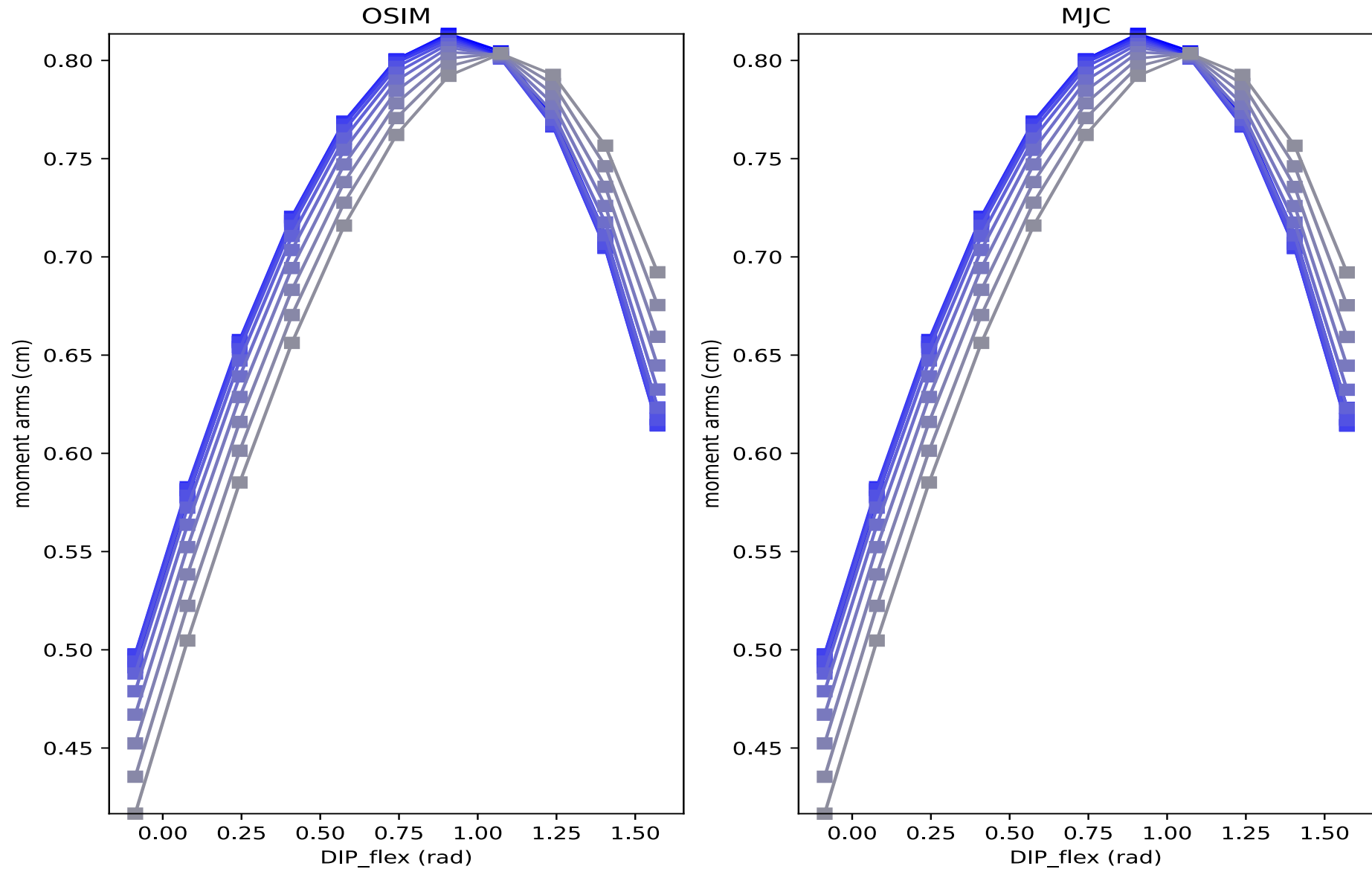
# Muscle specific moment arm comparison with Osim model after optimization

FDSI - deviation - flexion



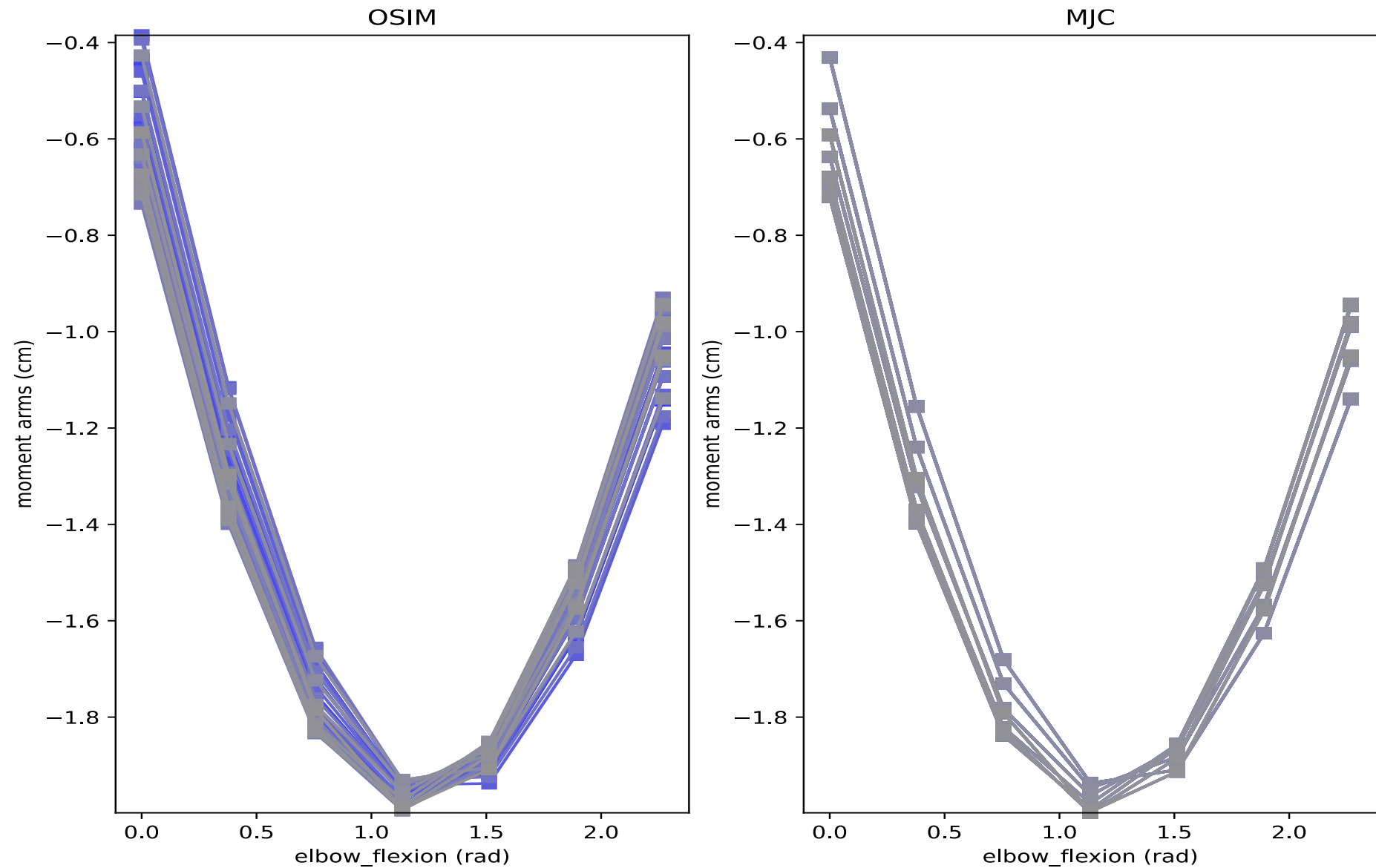
# Muscle specific moment arm comparison with Osim model after optimization

FDSI - DIP\_flex - PIP\_flex



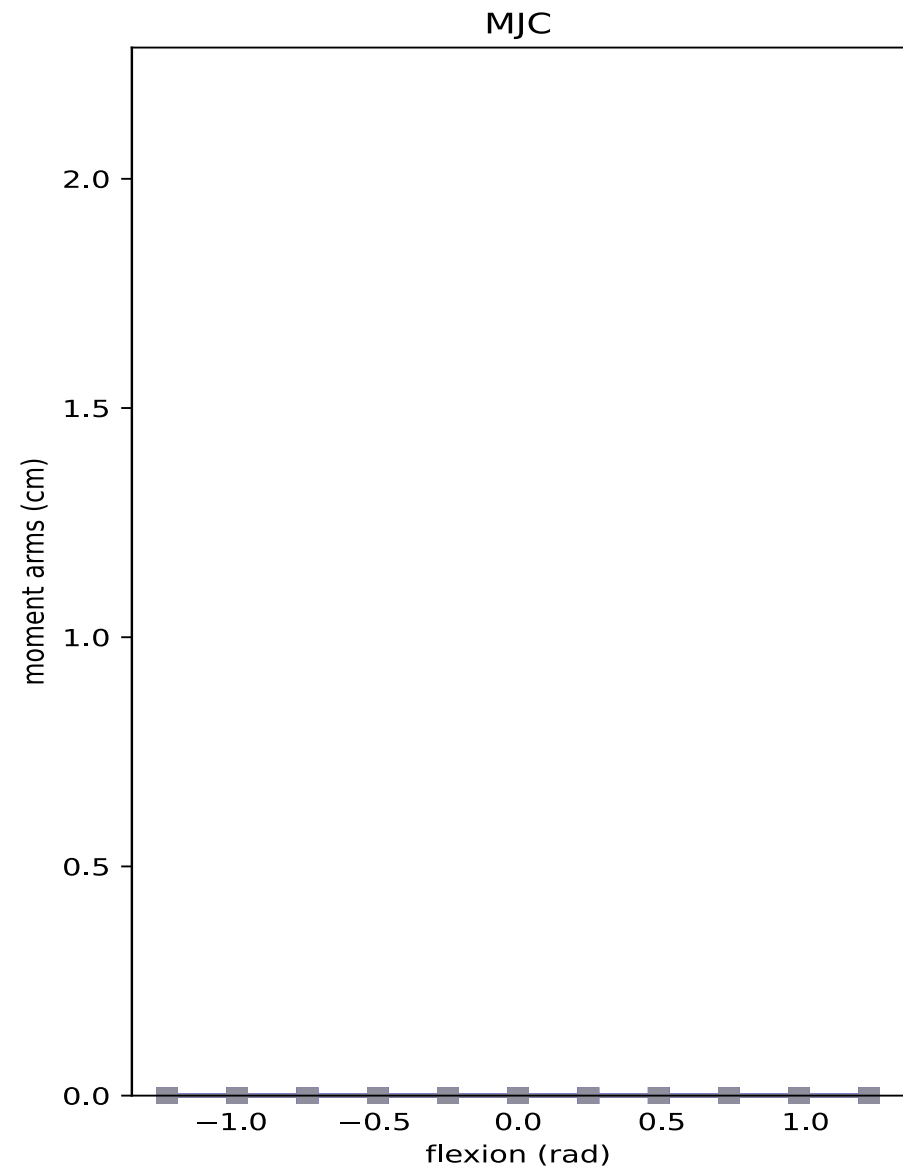
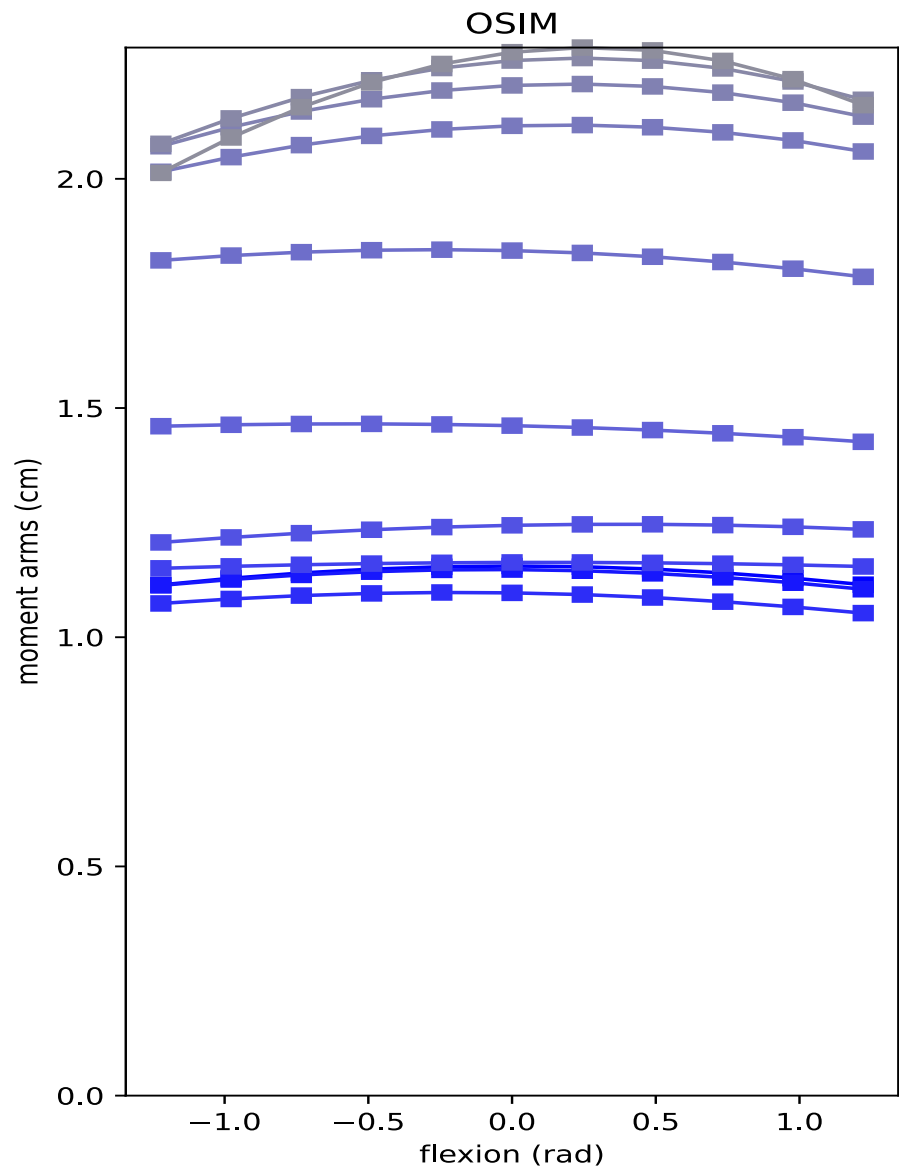
# Muscle specific moment arm comparison with Osim model after optimization

FDSI - elbow\_flexion - flexion - pro\_sup



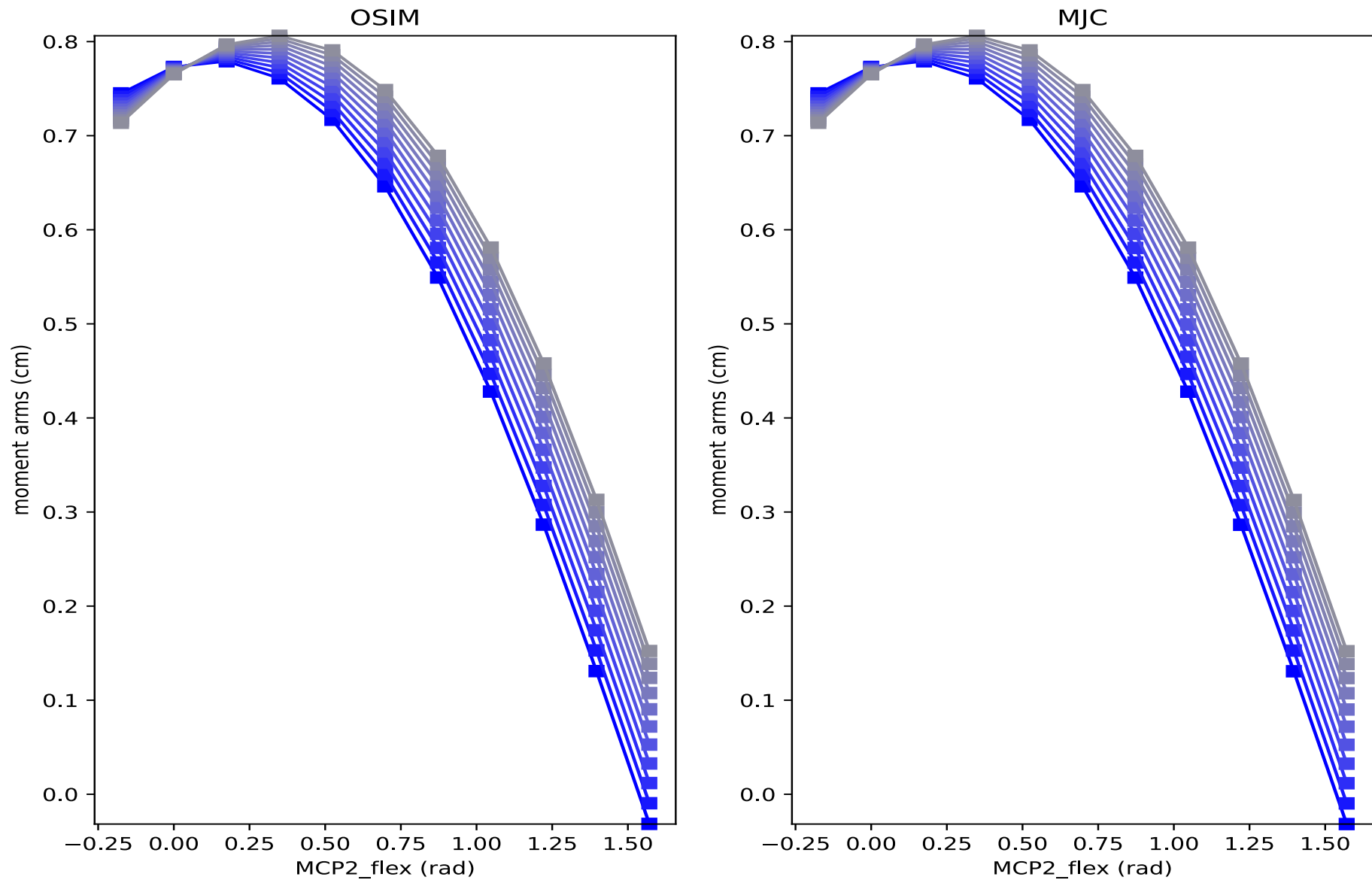
### Muscle specific moment arm comparison with Osim model after optimization

FDSI - deviation - flexion



# Muscle specific moment arm comparison with Osim model after optimization

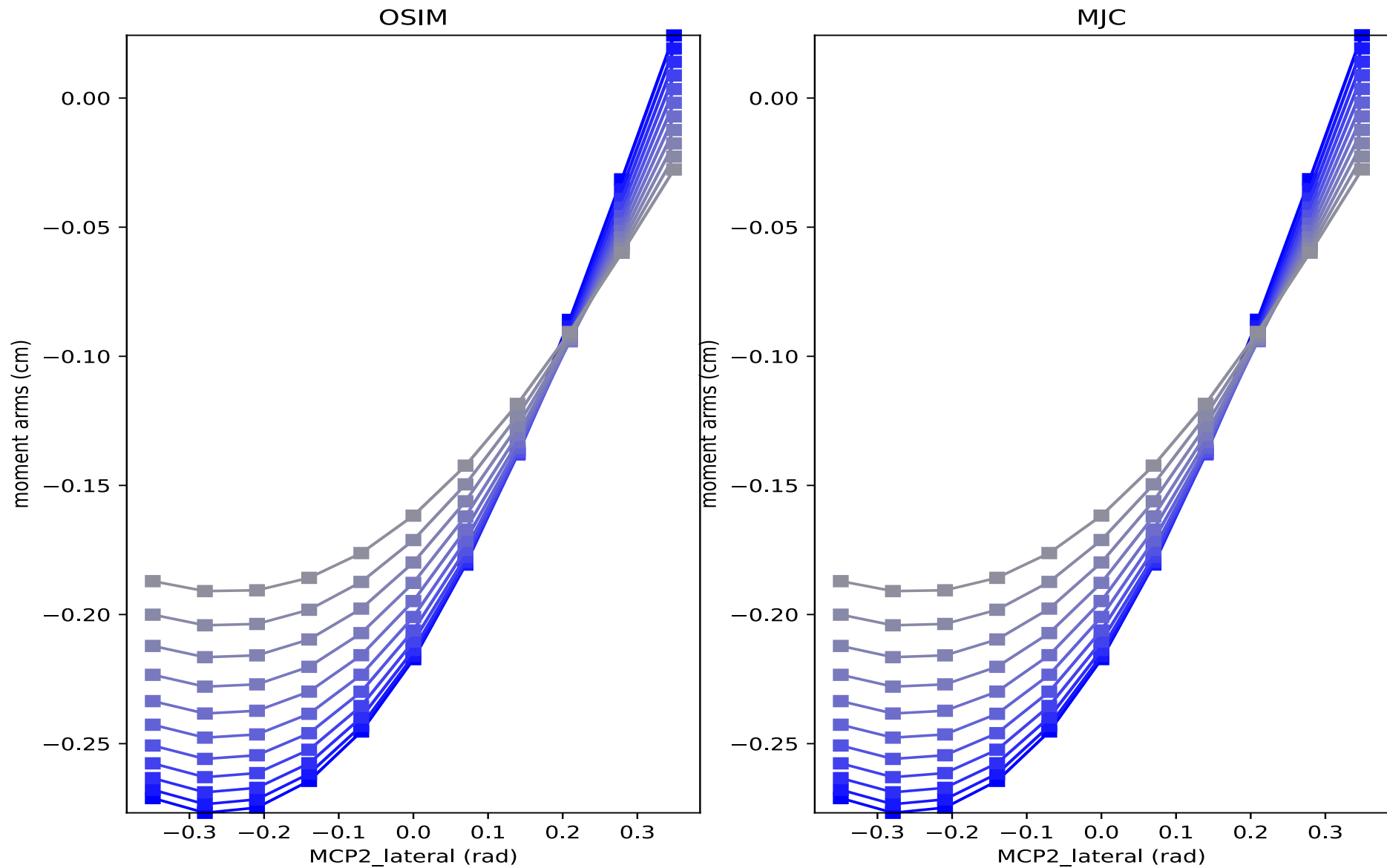
FDSI - MCP2\_flex - MCP2\_lateral





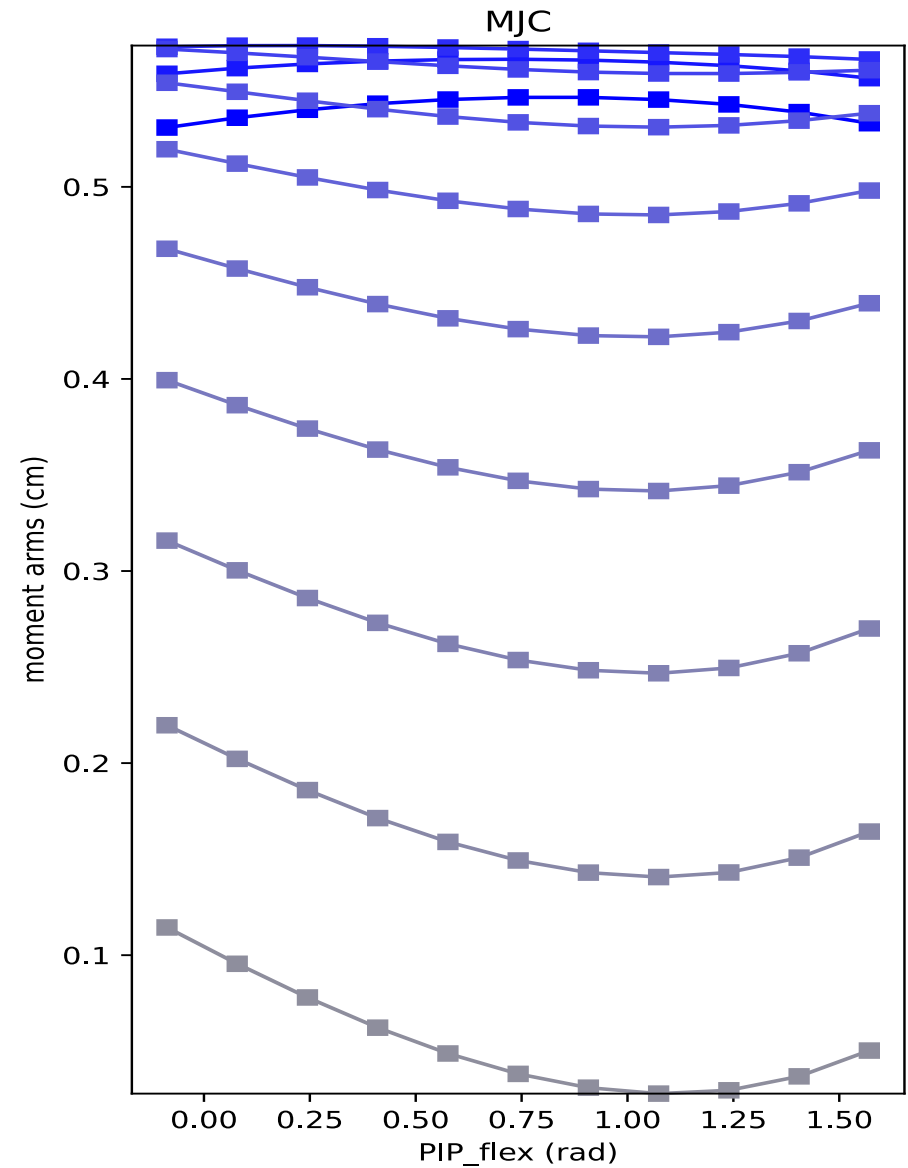
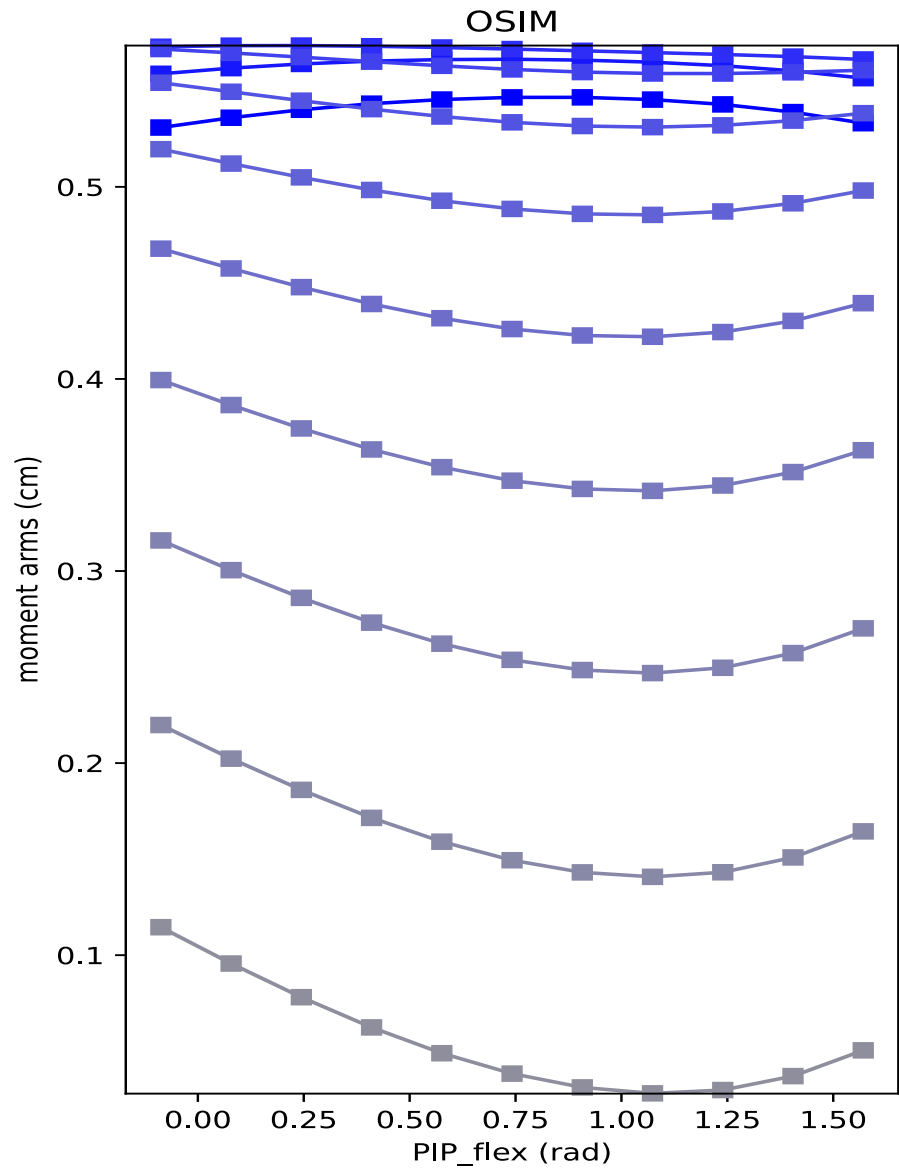
# Muscle specific moment arm comparison with Osim model after optimization

FDSI - MCP2\_flex - MCP2\_lateral



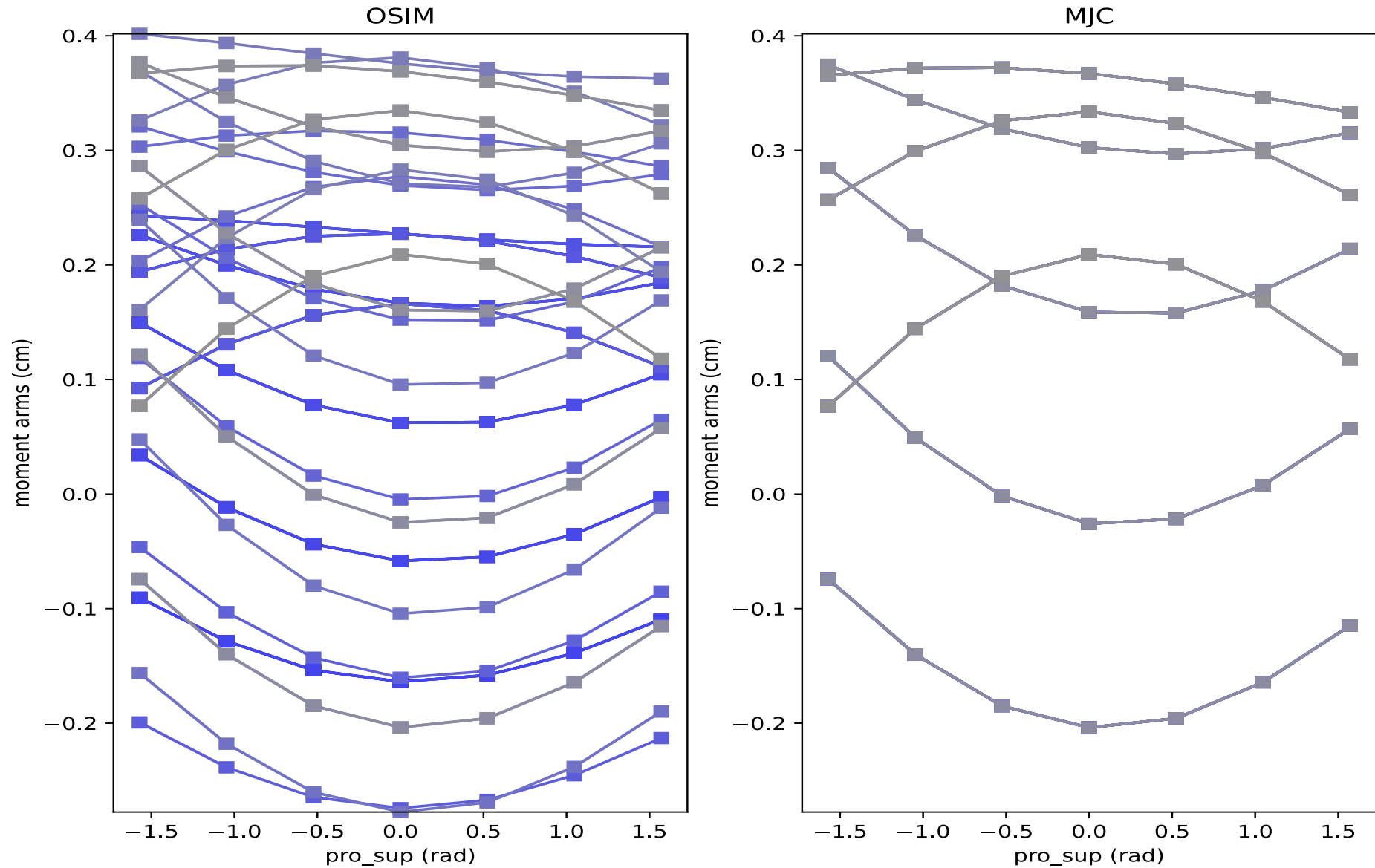
# Muscle specific moment arm comparison with Osim model after optimization

FDSI - DIP\_flex - PIP\_flex



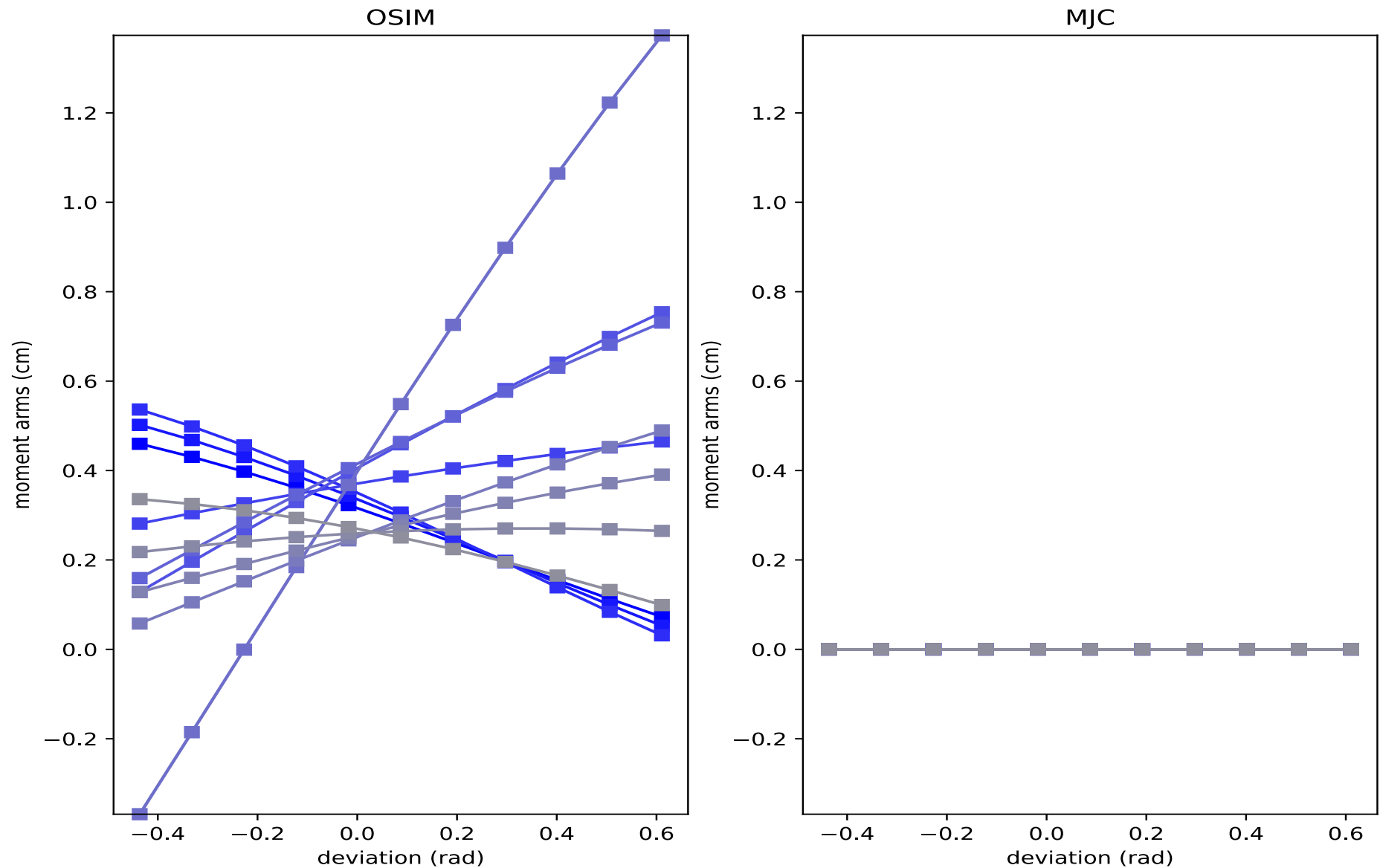
# Muscle specific moment arm comparison with Osim model after optimization

FDSI - elbow\_flexion - flexion - pro\_sup



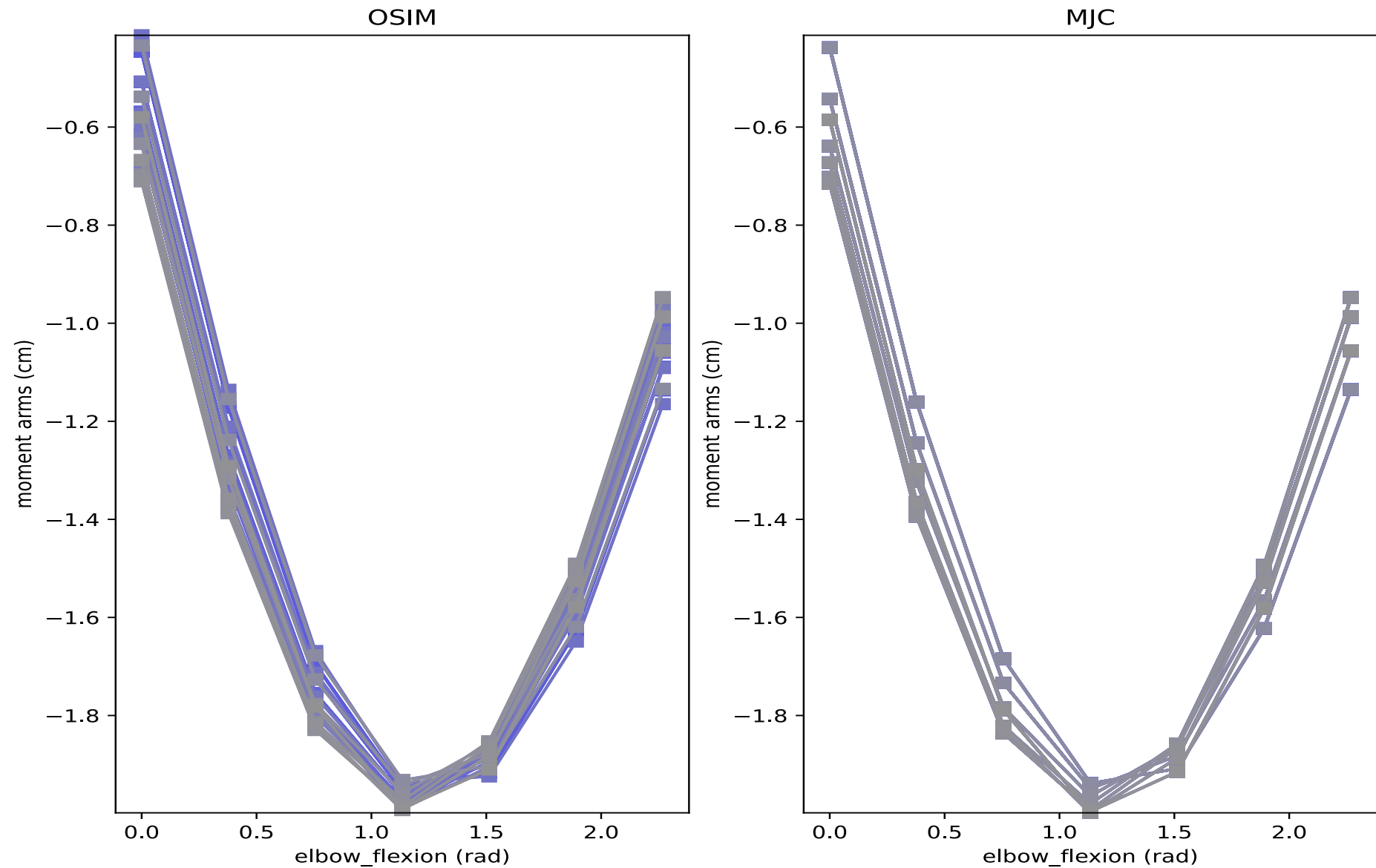
# Muscle specific moment arm comparison with Osim model after optimization

FDSL - deviation - flexion



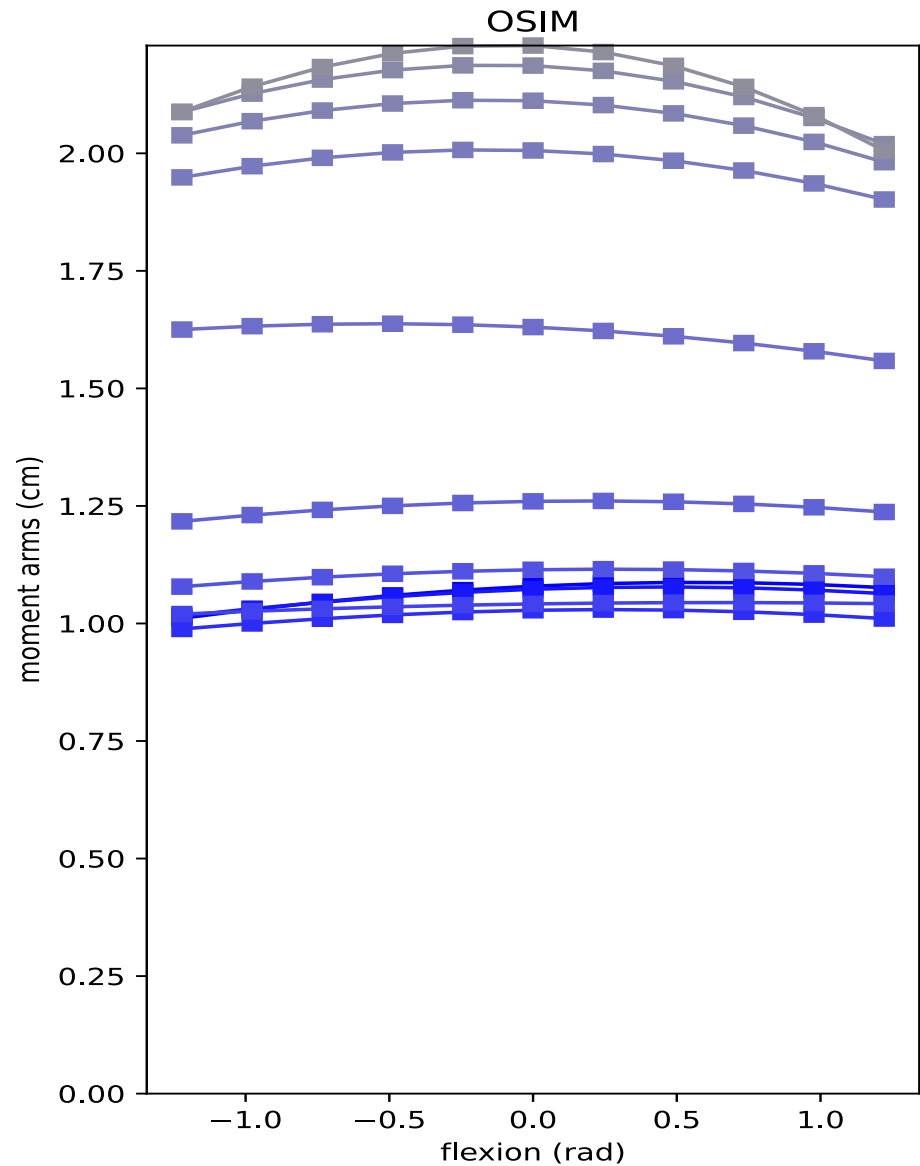
# Muscle specific moment arm comparison with Osim model after optimization

FDSL - elbow\_flexion - flexion - pro\_sup

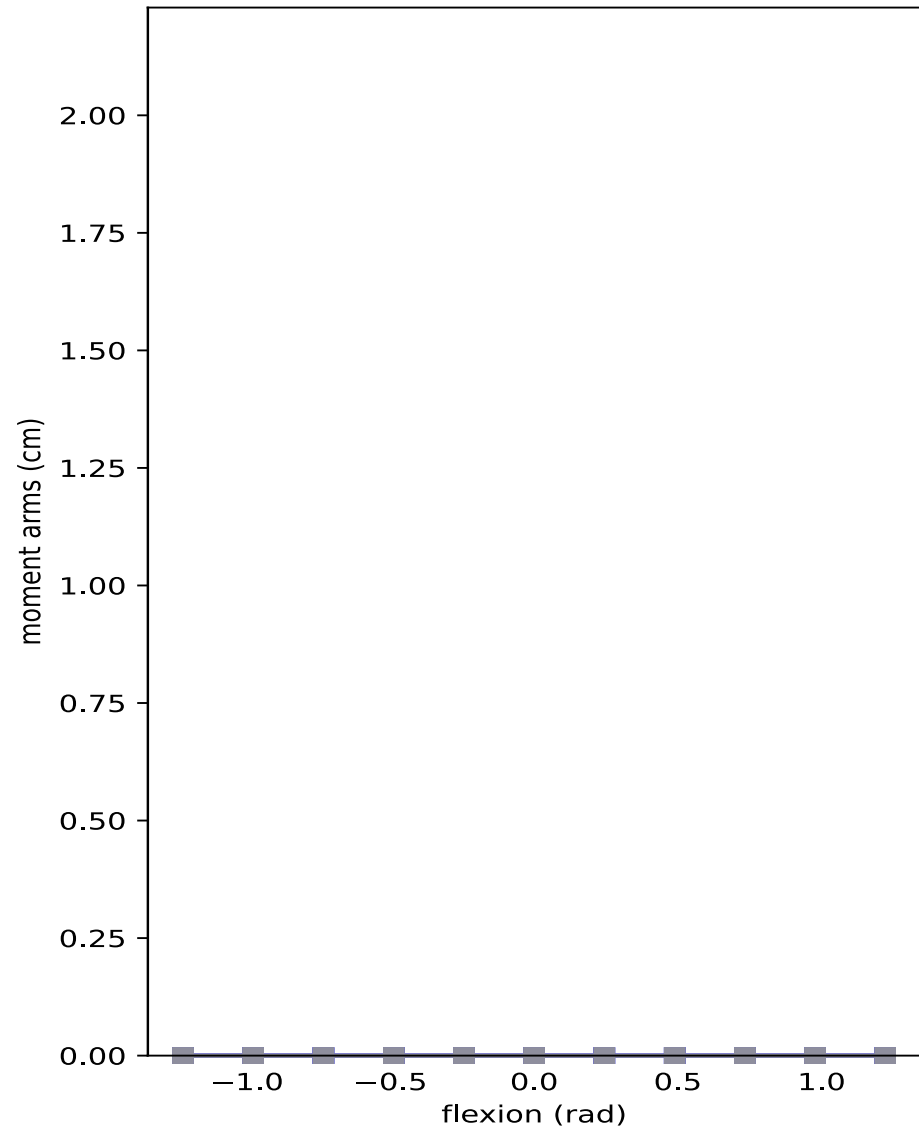


# Muscle specific moment arm comparison with Osim model after optimization

FDSL - deviation - flexion

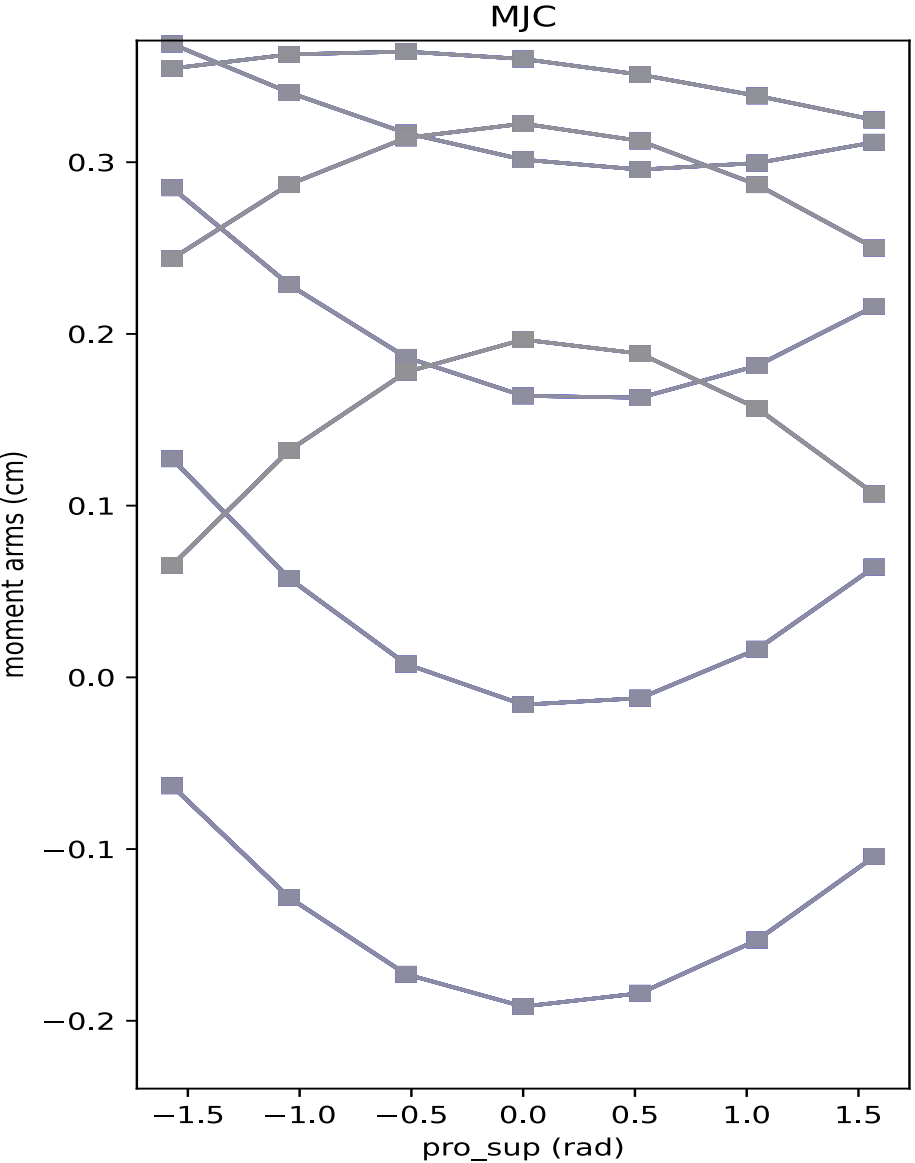
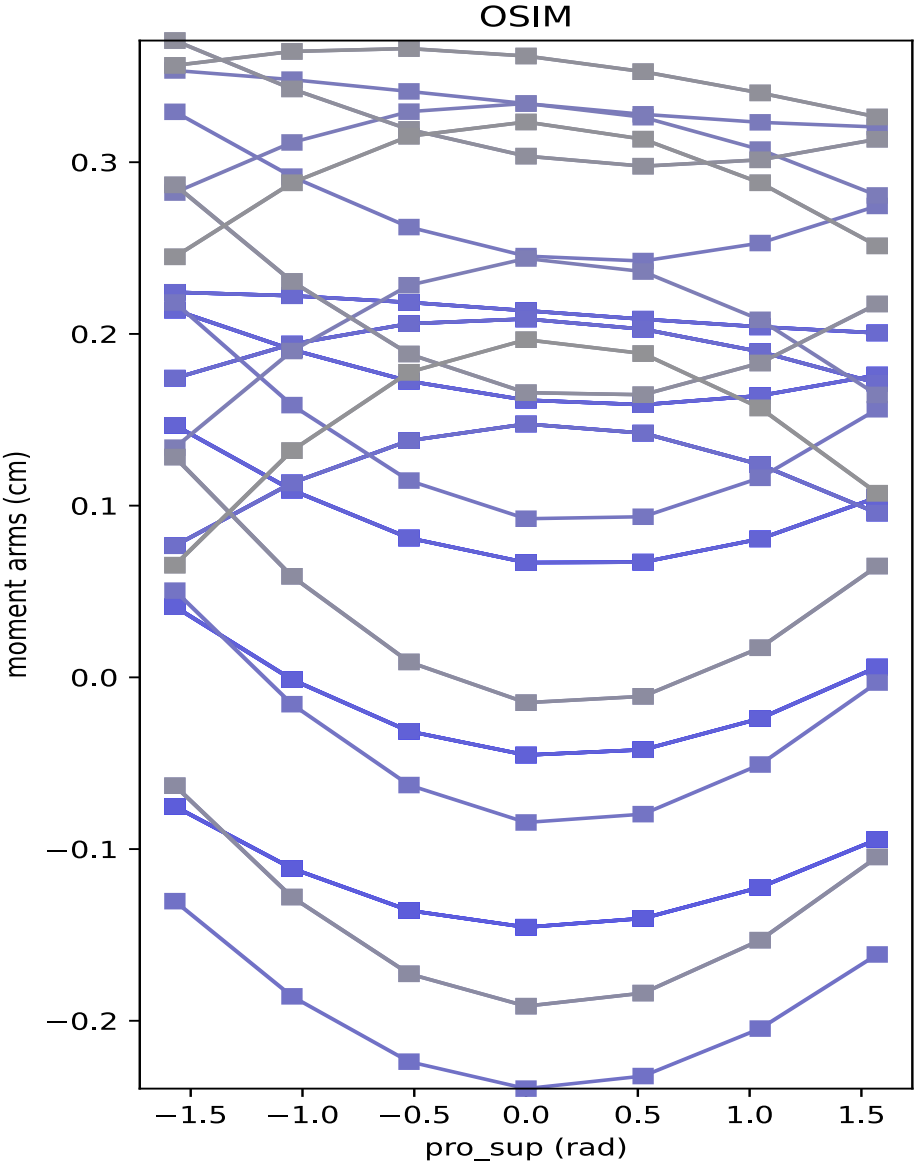


MJC



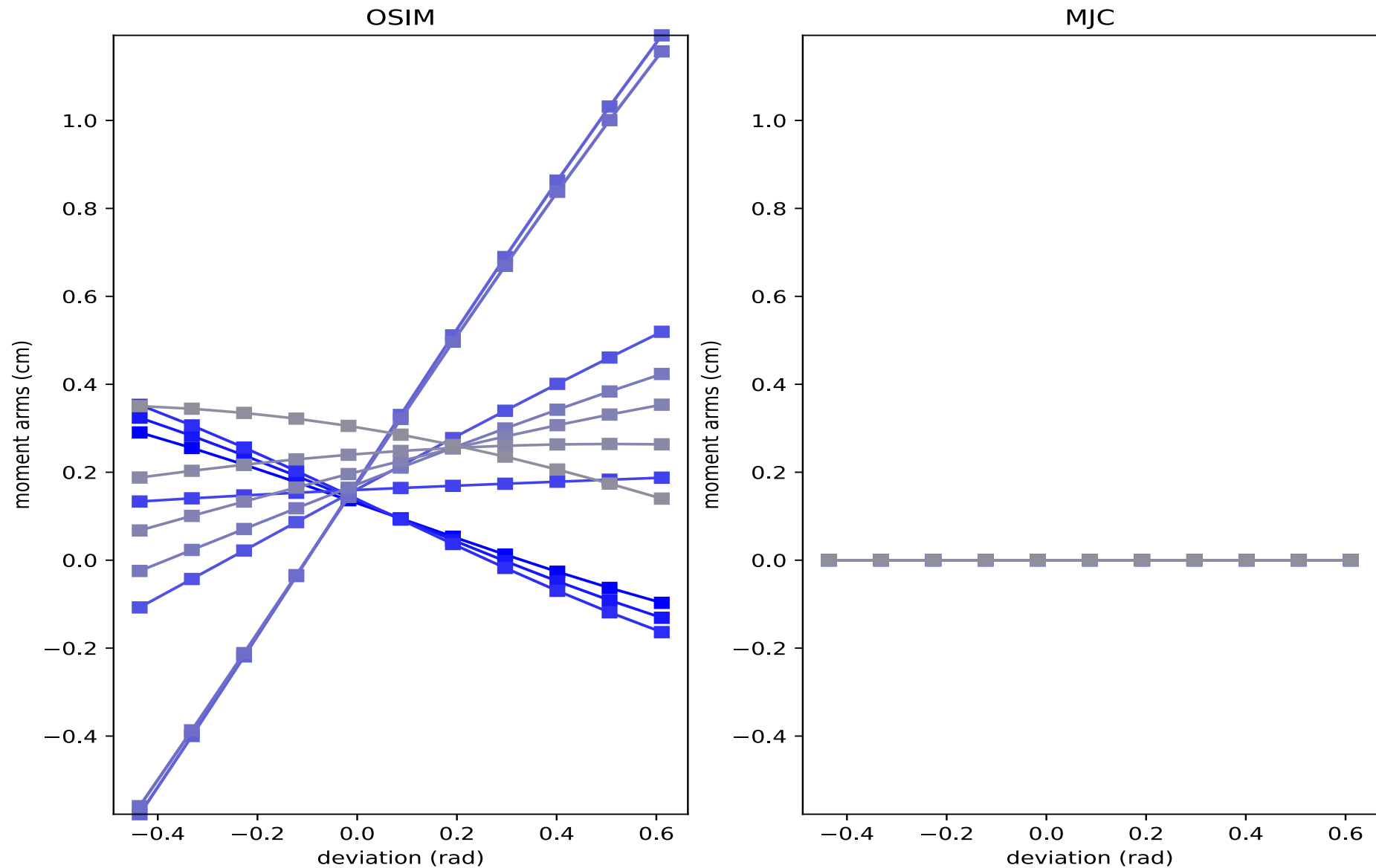
# Muscle specific moment arm comparison with Osim model after optimization

FDSL - elbow\_flexion - flexion - pro\_sup



# Muscle specific moment arm comparison with Osim model after optimization

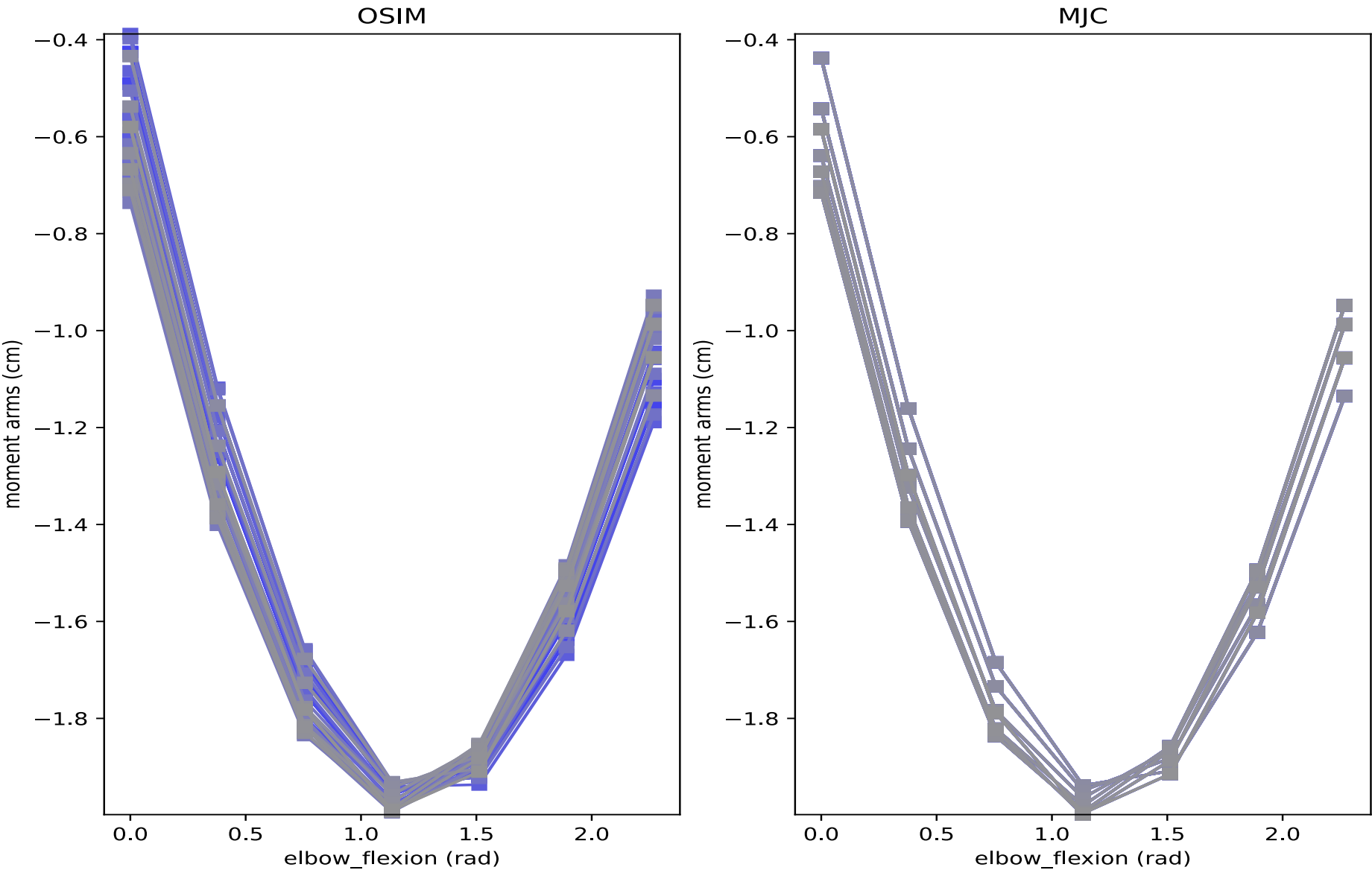
FDSM - deviation - flexion





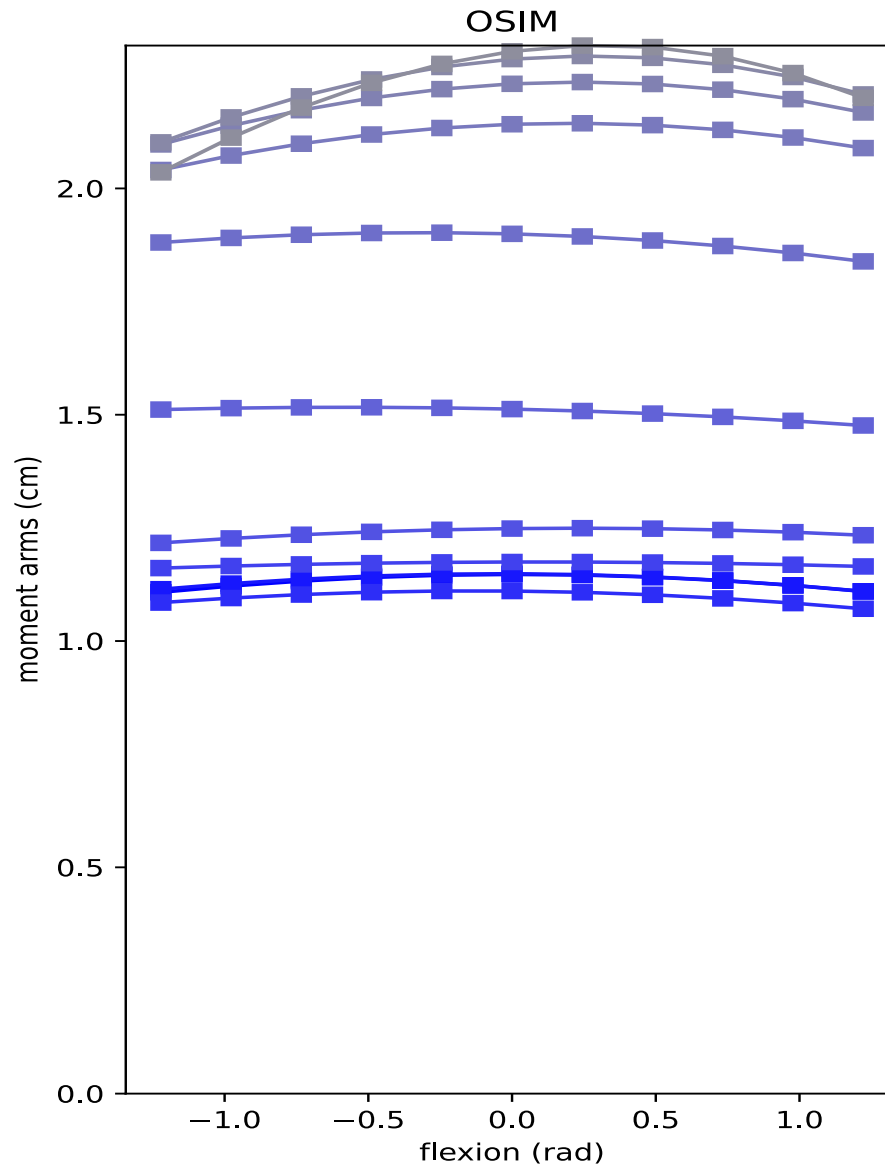
# Muscle specific moment arm comparison with Osim model after optimization

FDSM - elbow\_flexion - flexion - pro\_sup

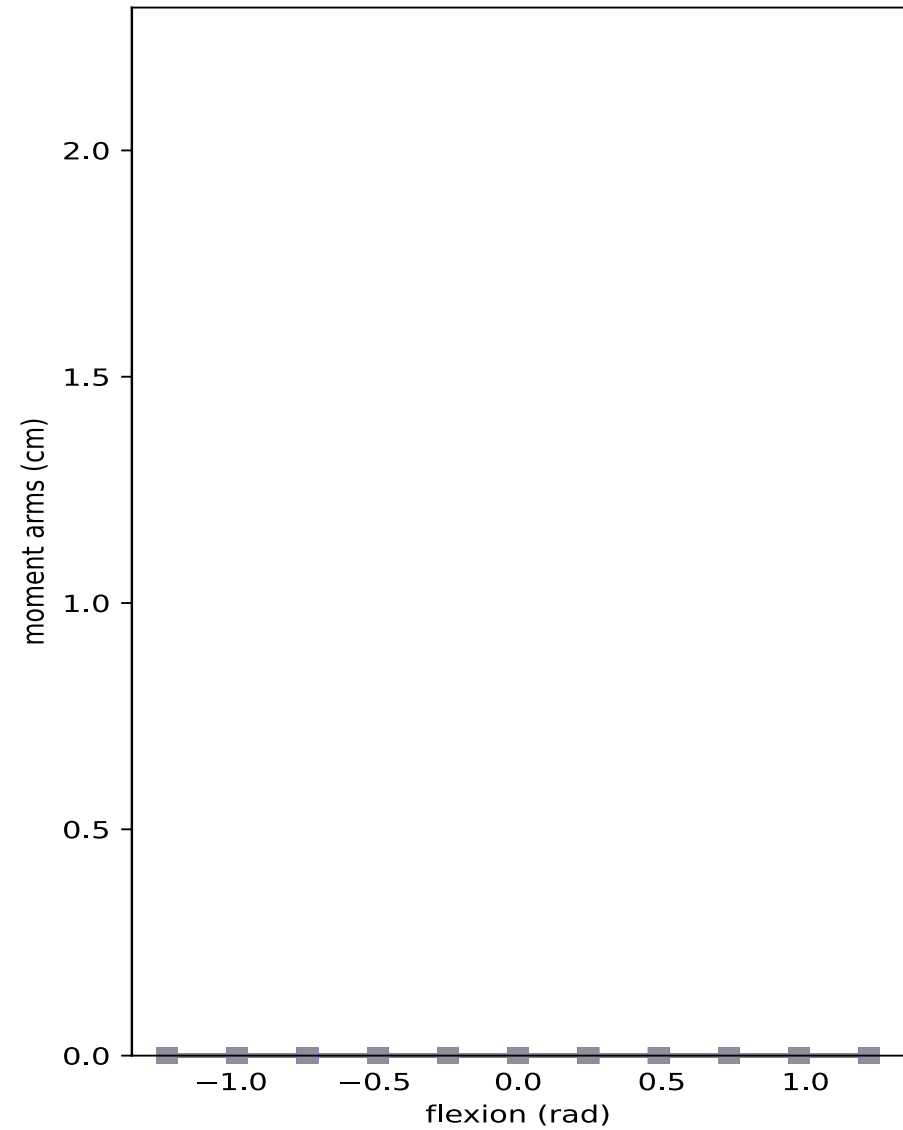


## Muscle specific moment arm comparison with Osim model after optimization

FDSM - deviation - flexion

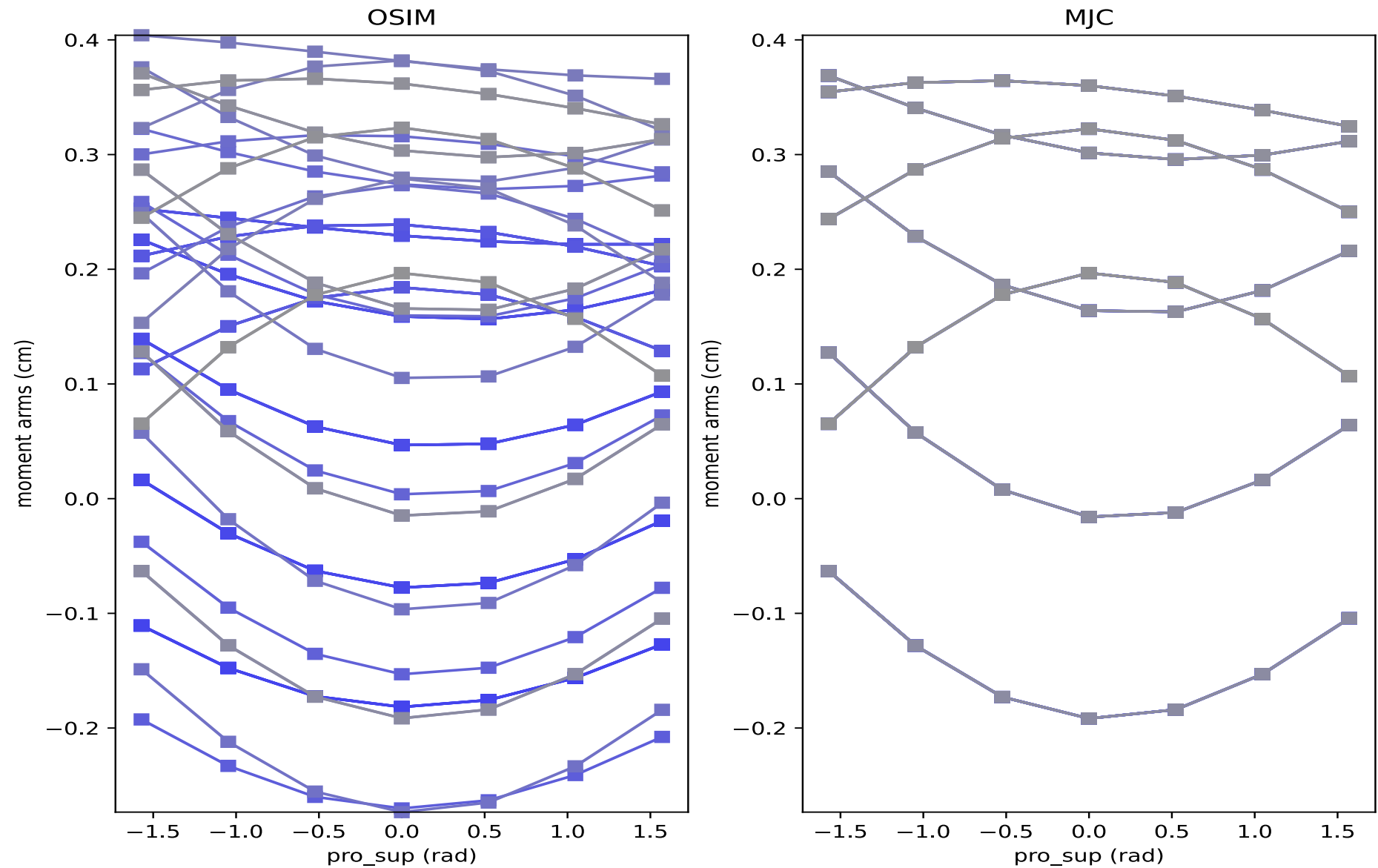


MJC



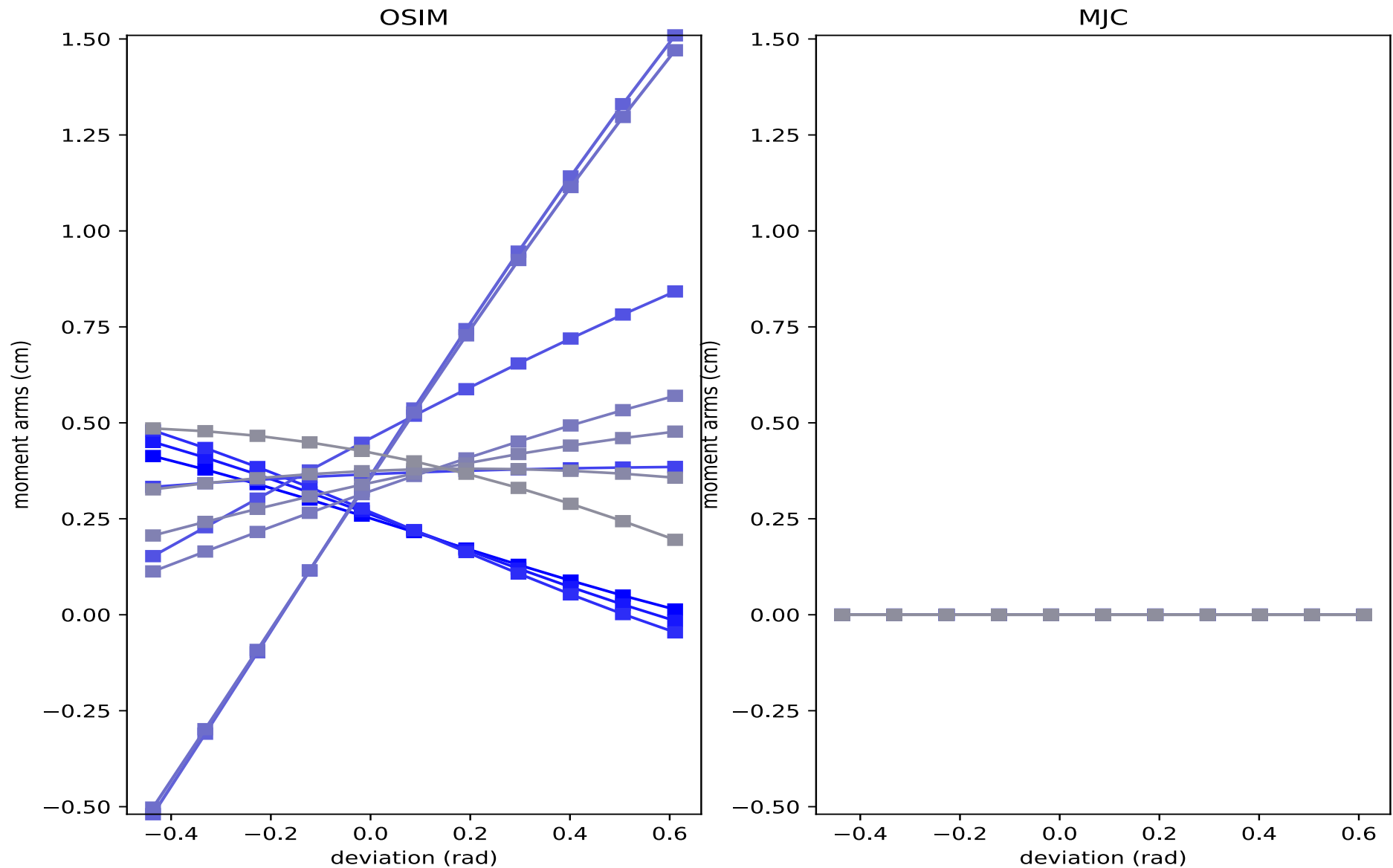
# Muscle specific moment arm comparison with Osim model after optimization

FDSM - elbow\_flexion - flexion - pro\_sup



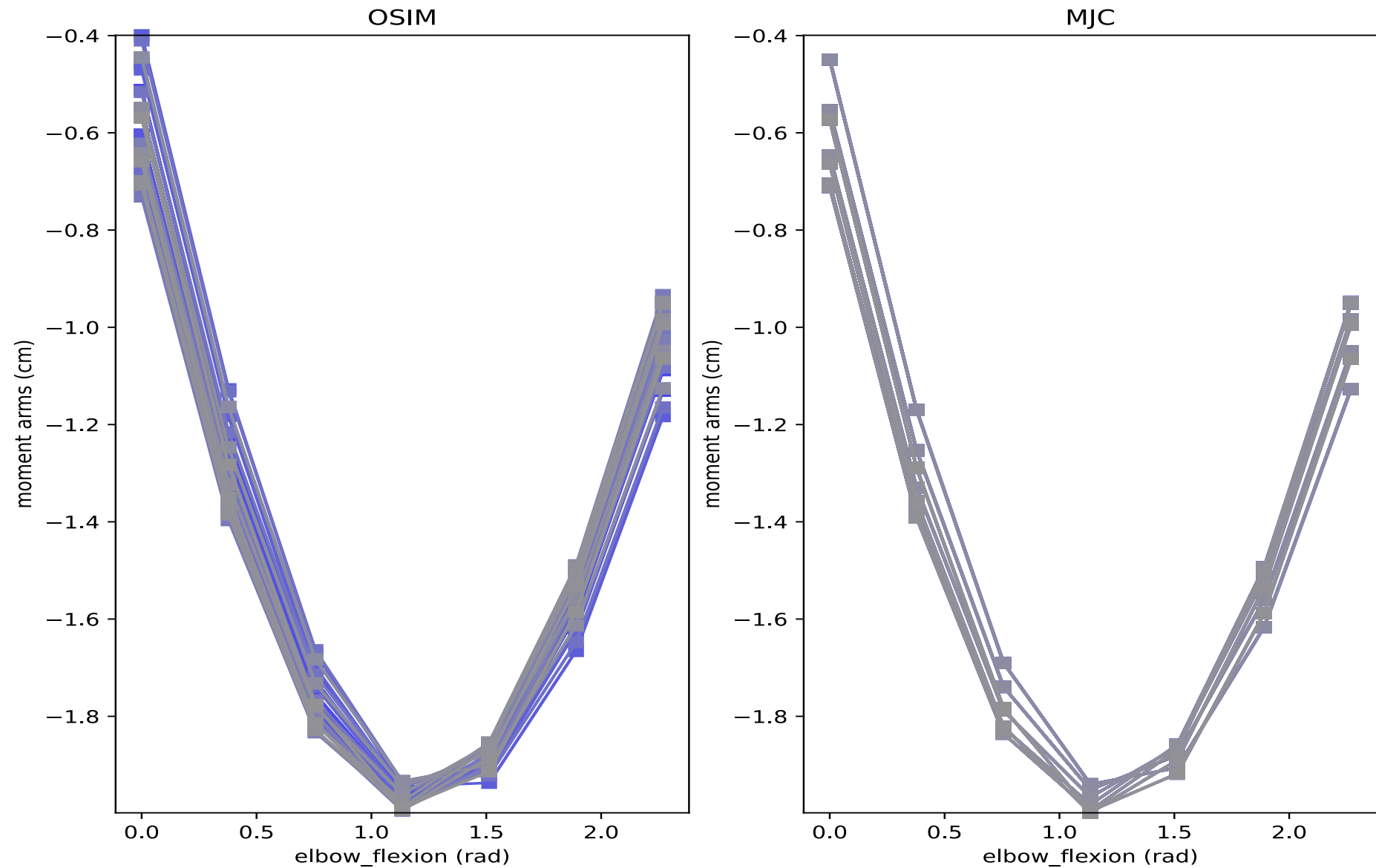
# Muscle specific moment arm comparison with Osim model after optimization

FDSR - deviation - flexion



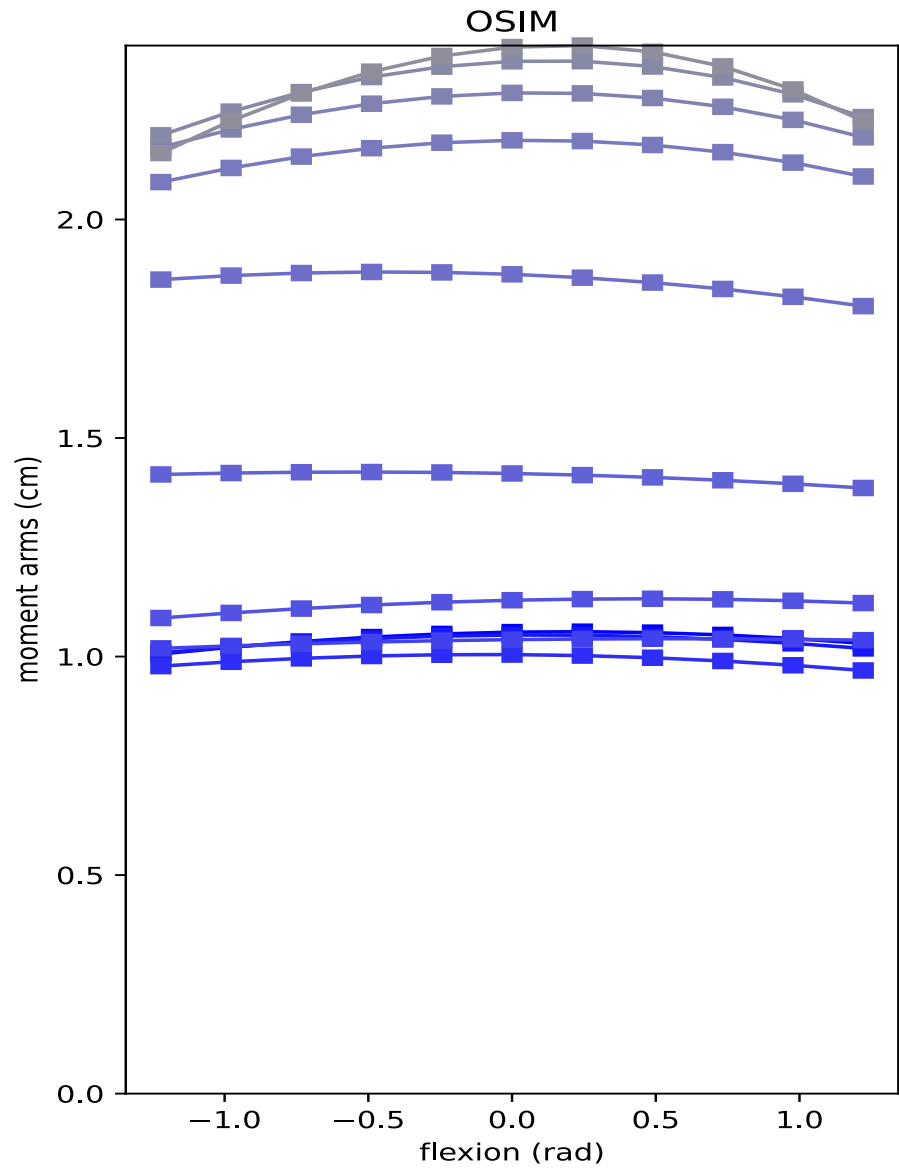
# Muscle specific moment arm comparison with Osim model after optimization

FDSR - elbow\_flexion - flexion - pro\_sup

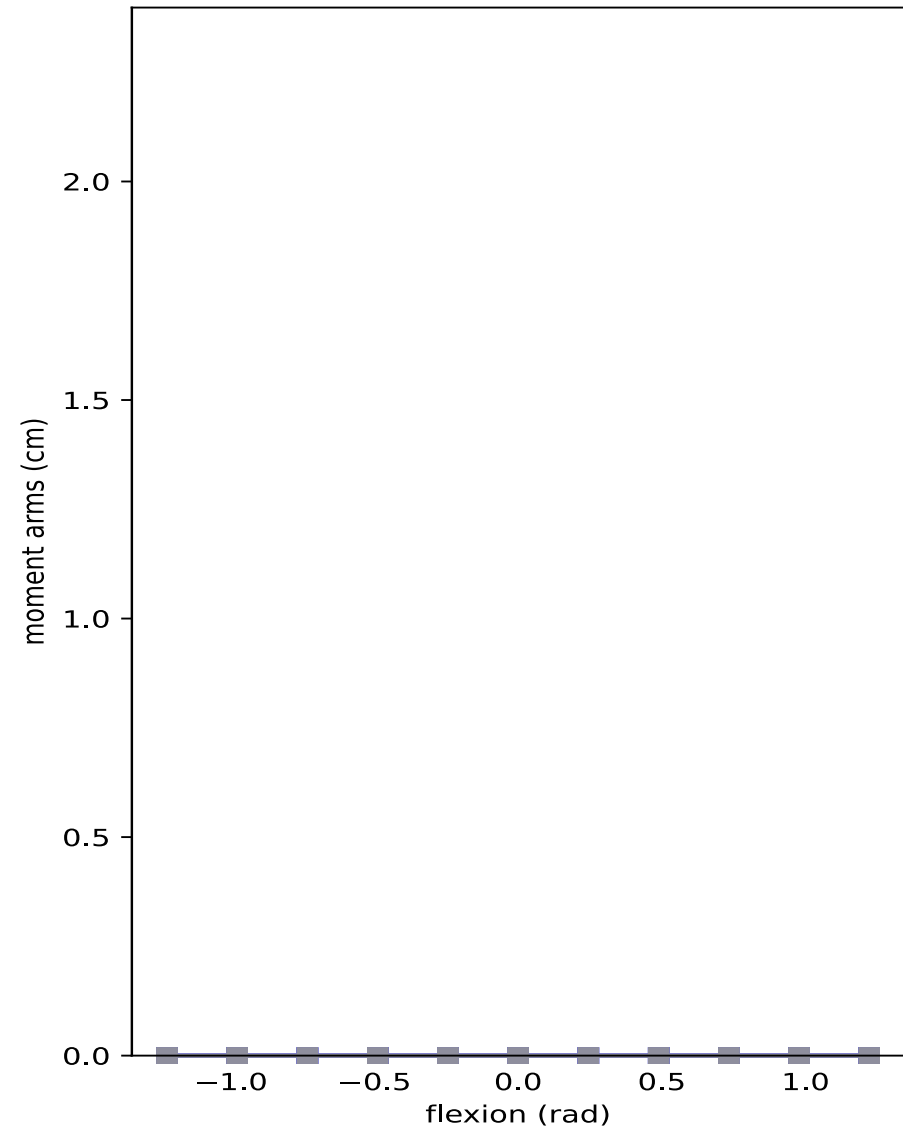


### Muscle specific moment arm comparison with Osim model after optimization

FDSR - deviation - flexion

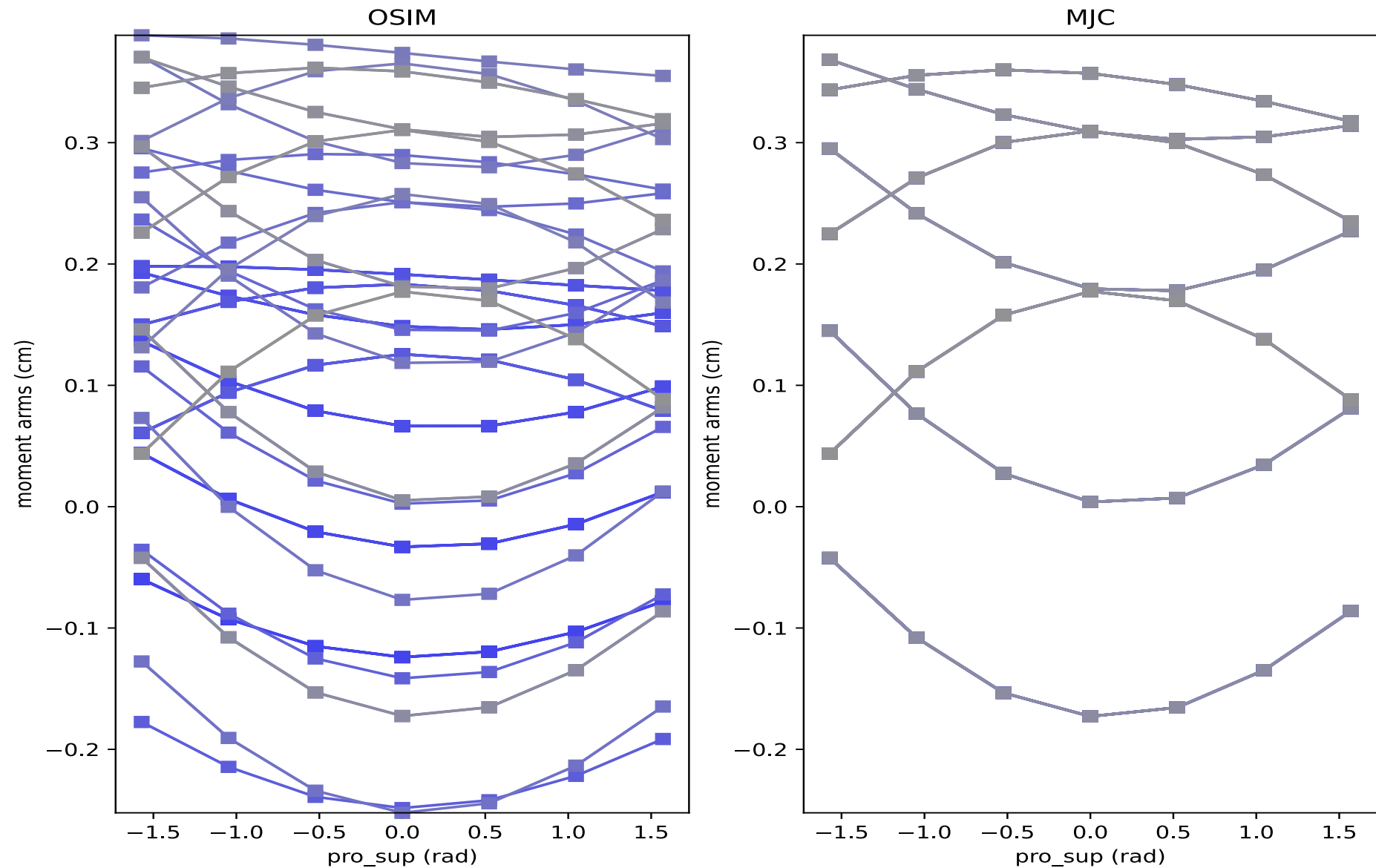


MJC



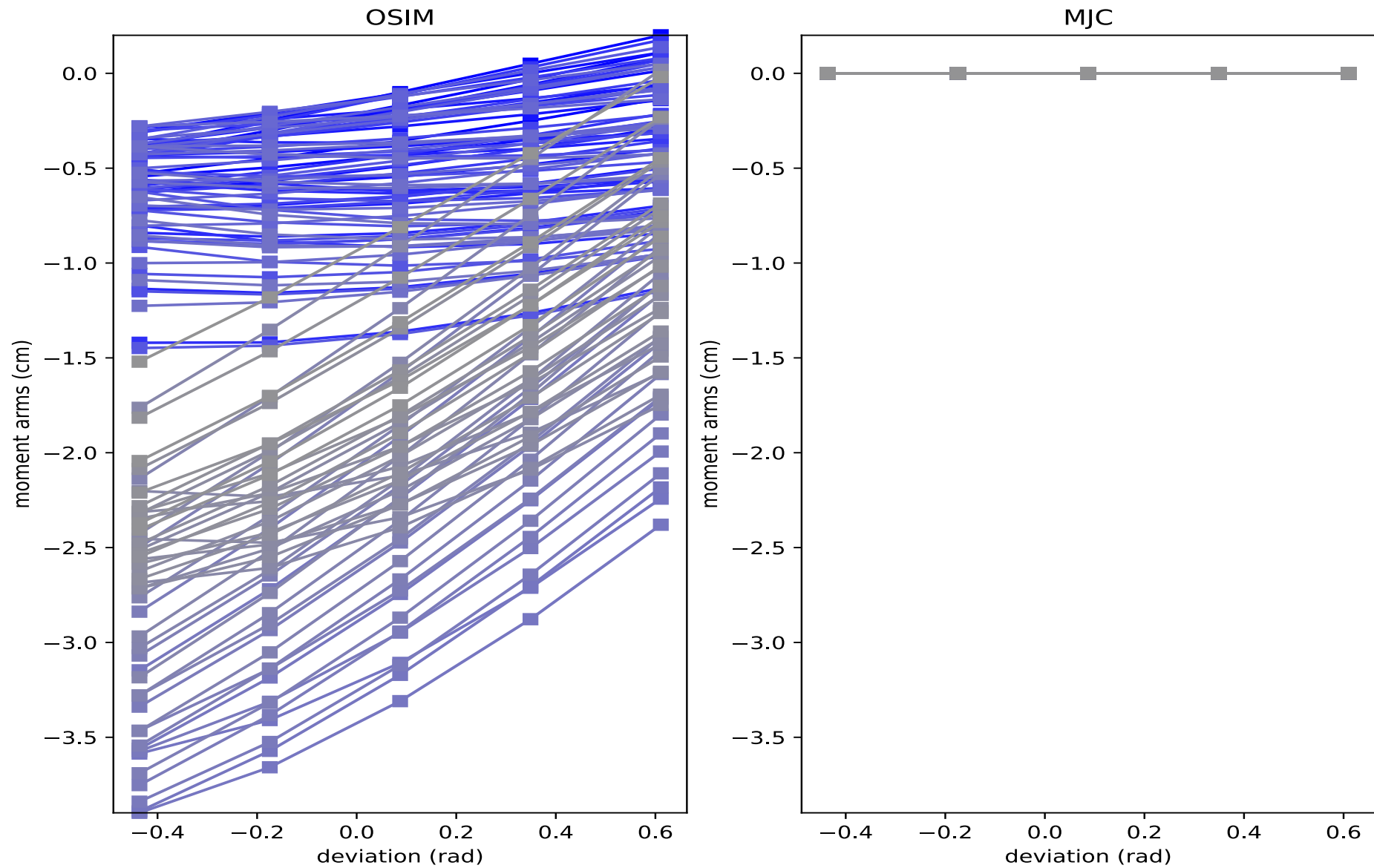
# Muscle specific moment arm comparison with Osim model after optimization

FDSR - elbow\_flexion - flexion - pro\_sup



# Muscle specific moment arm comparison with Osim model after optimization

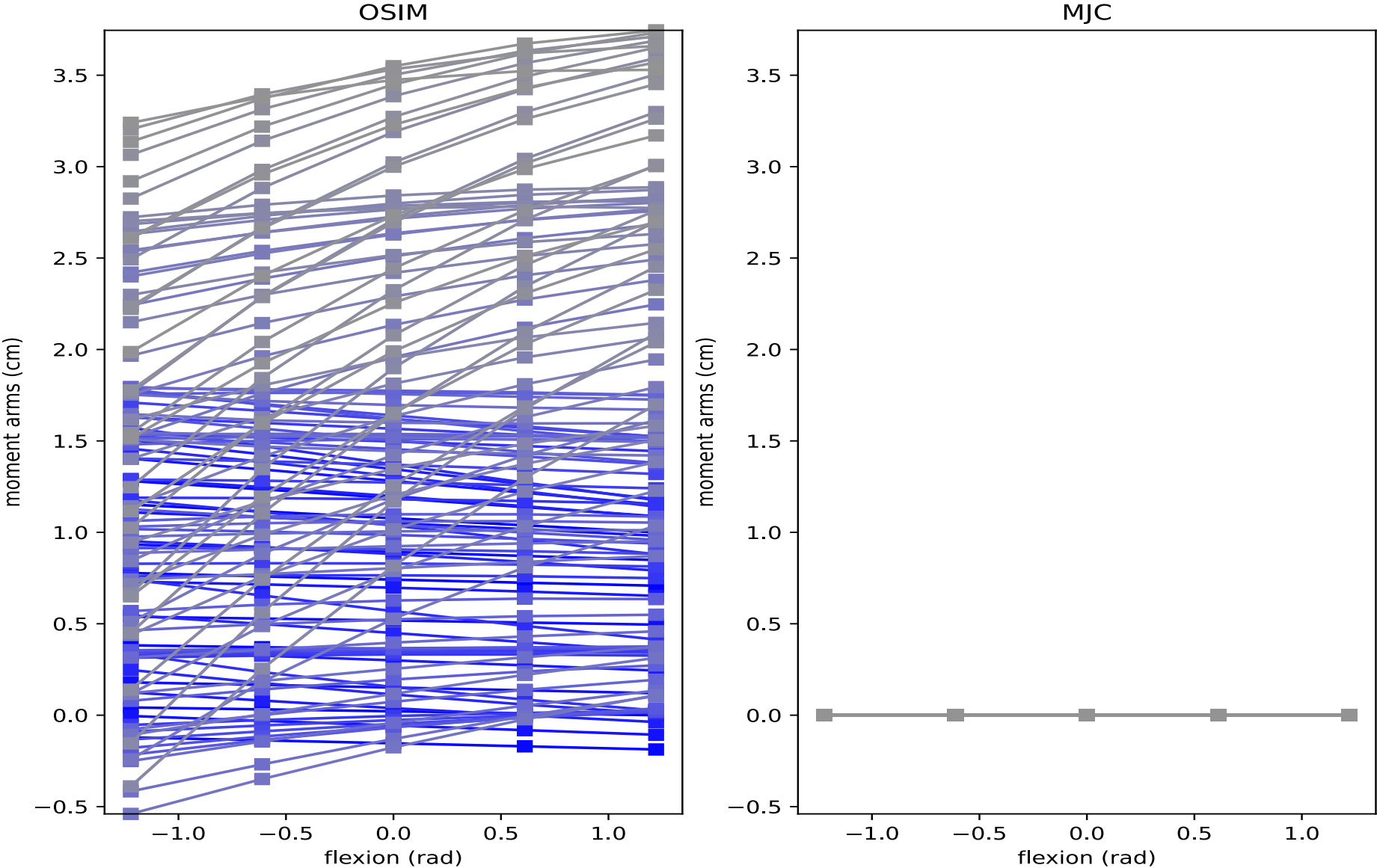
FPL - deviation - flexion - thumb\_abd - thumb\_flex





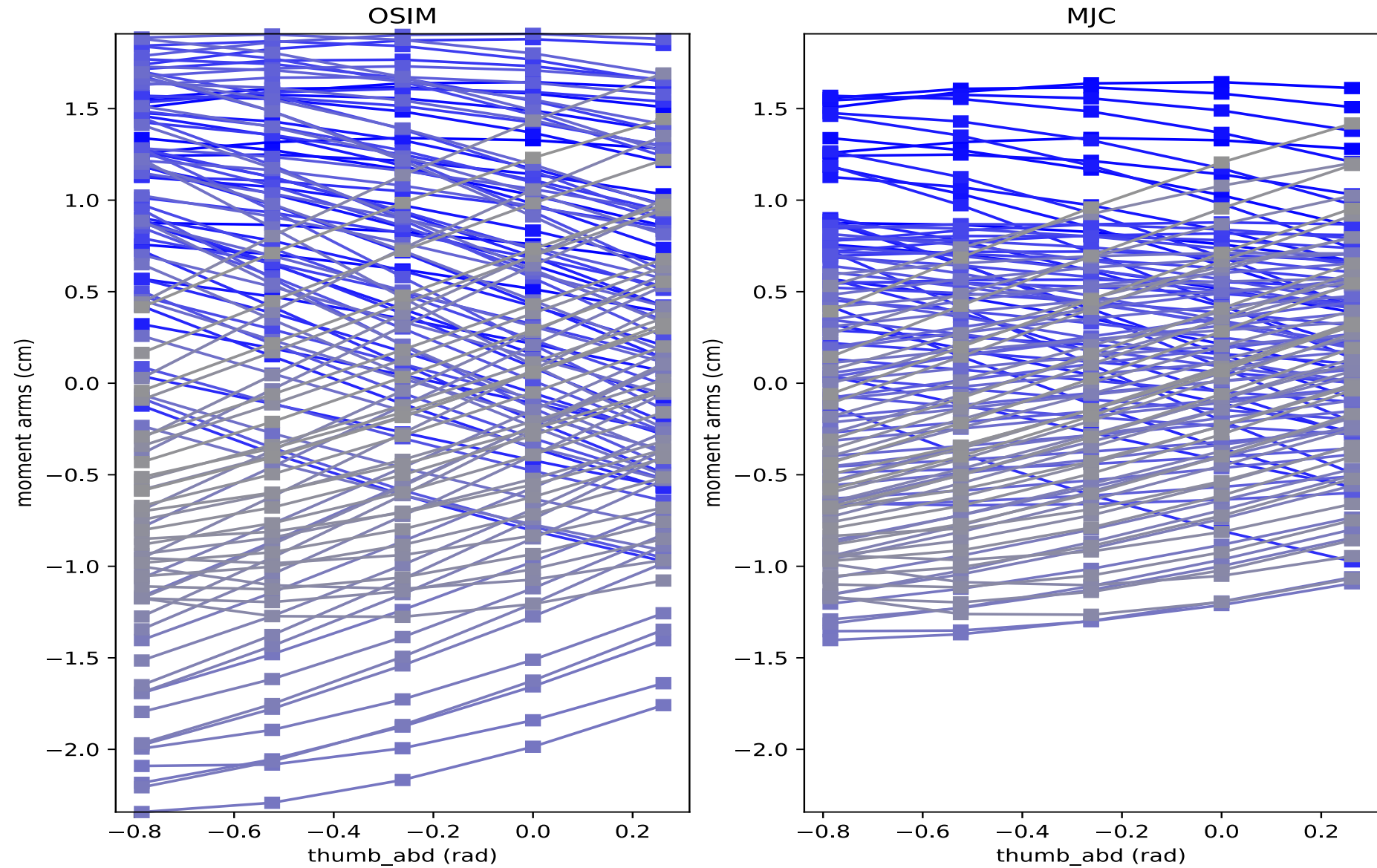
# Muscle specific moment arm comparison with Osim model after optimization

FPL - deviation - flexion - thumb\_abd - thumb\_flex



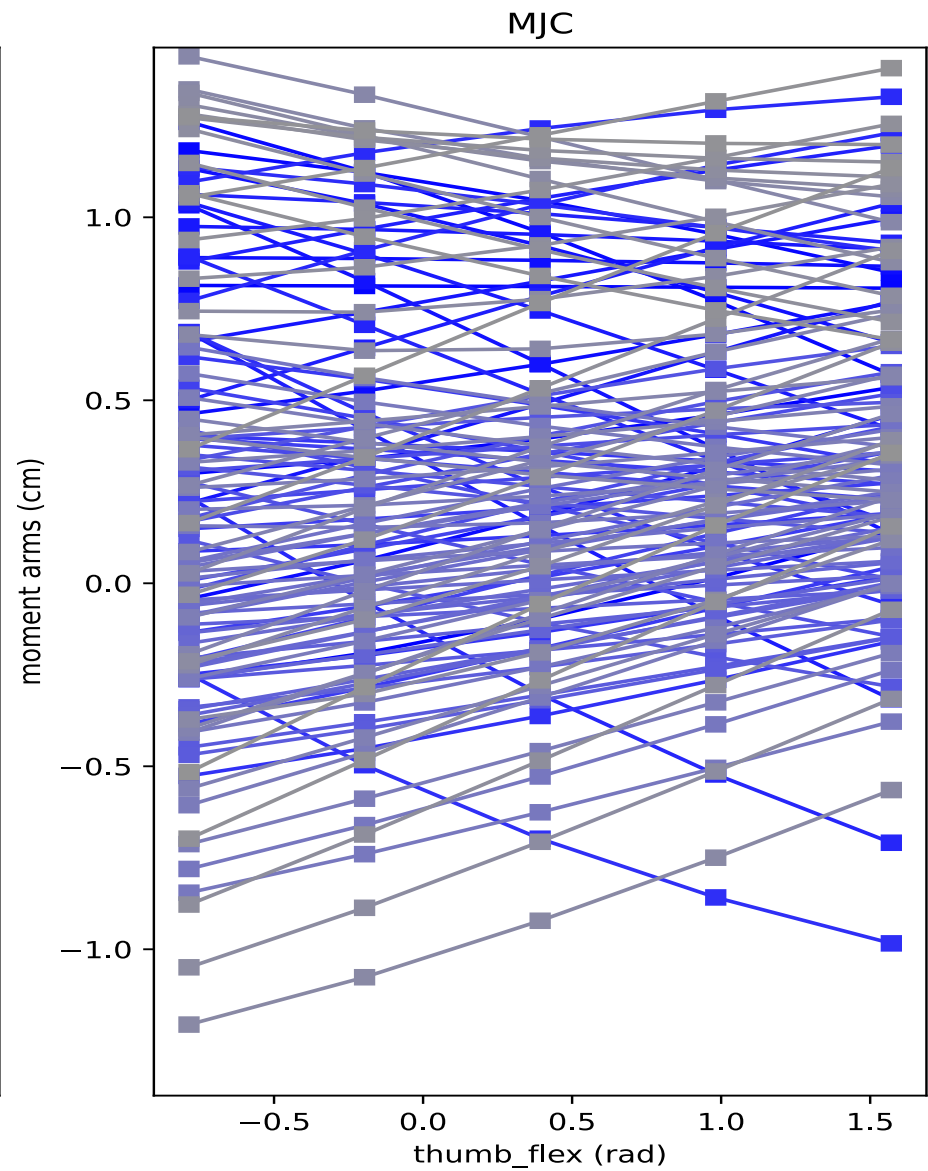
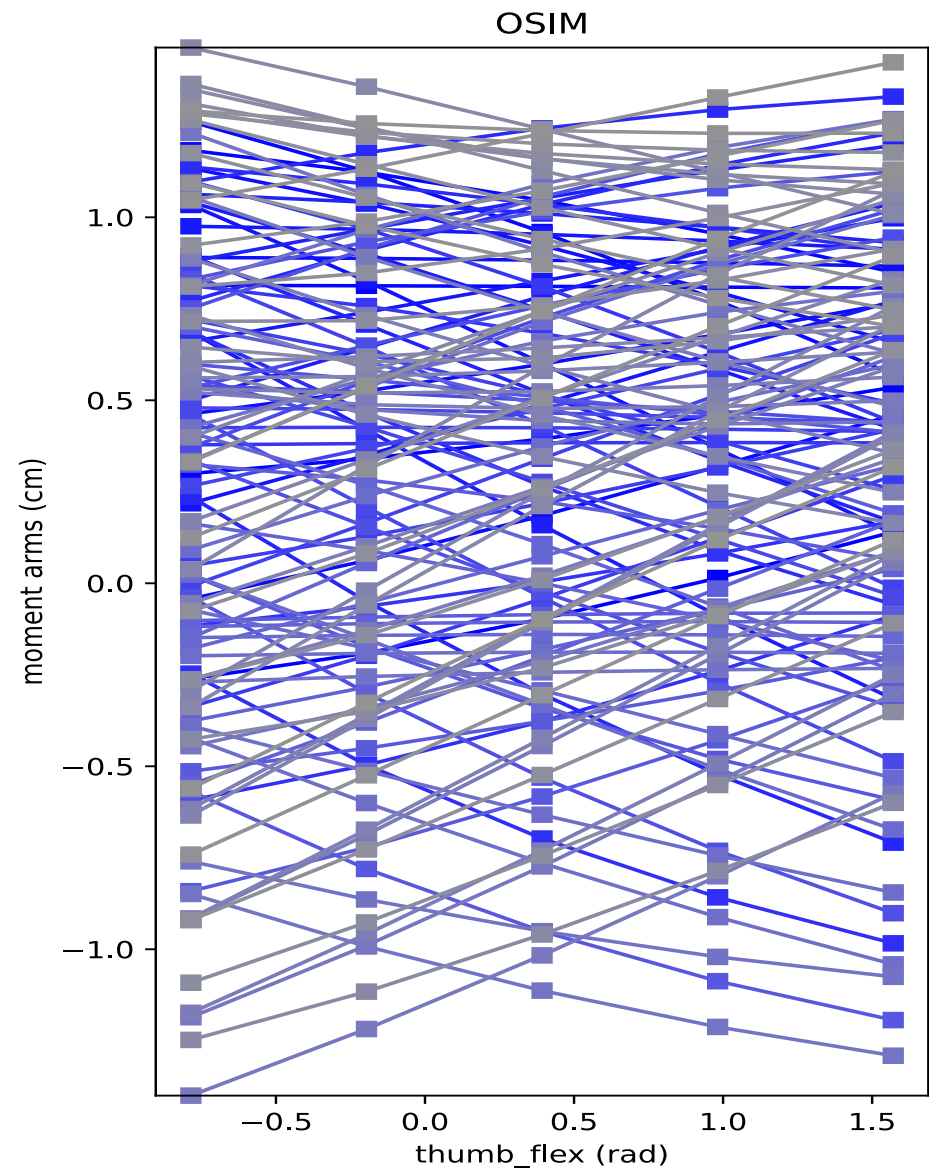
# Muscle specific moment arm comparison with Osim model after optimization

FPL - deviation - flexion - thumb\_abd - thumb\_flex



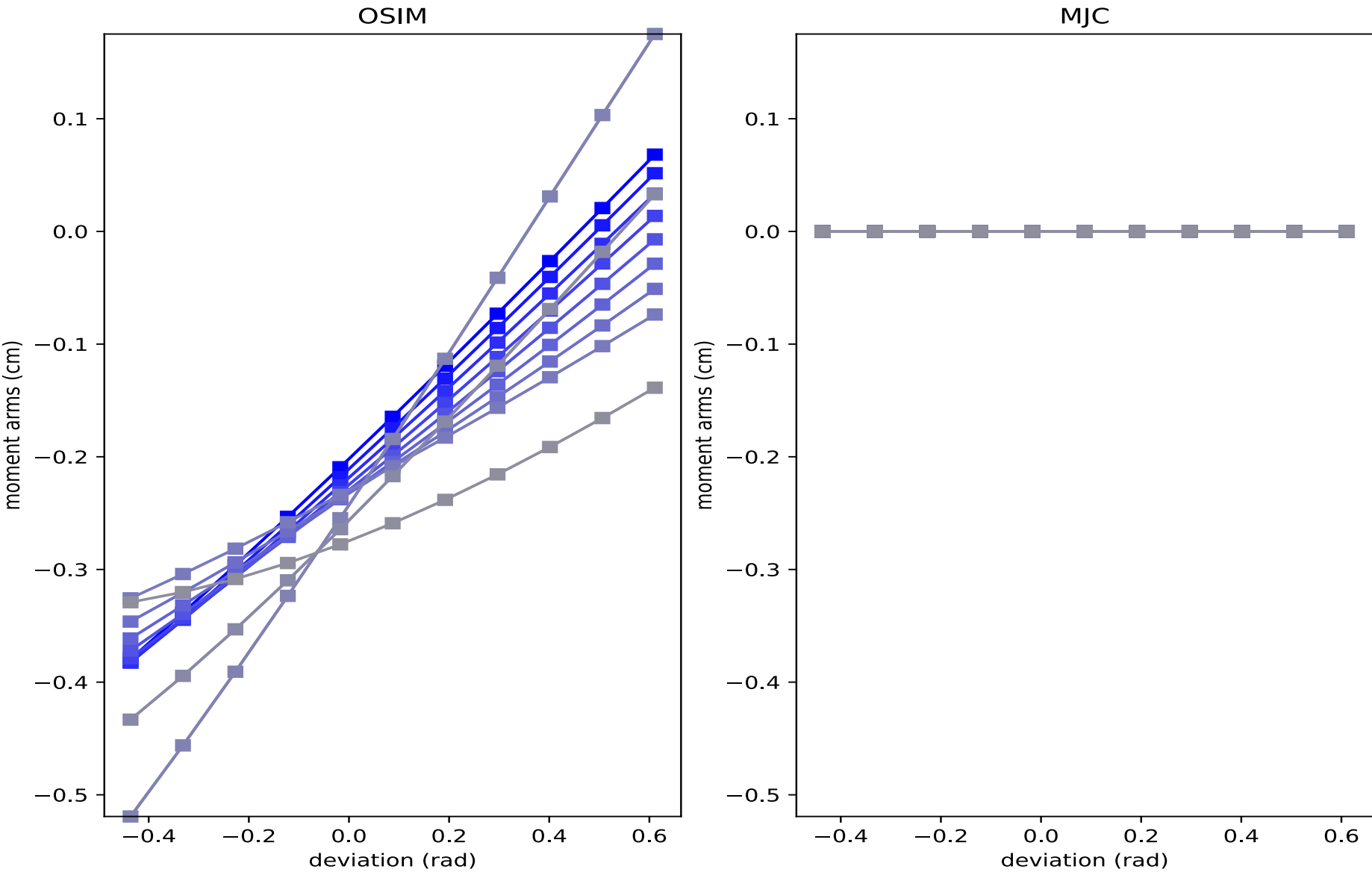
# Muscle specific moment arm comparison with Osim model after optimization

FPL - deviation - flexion - thumb\_abd - thumb\_flex



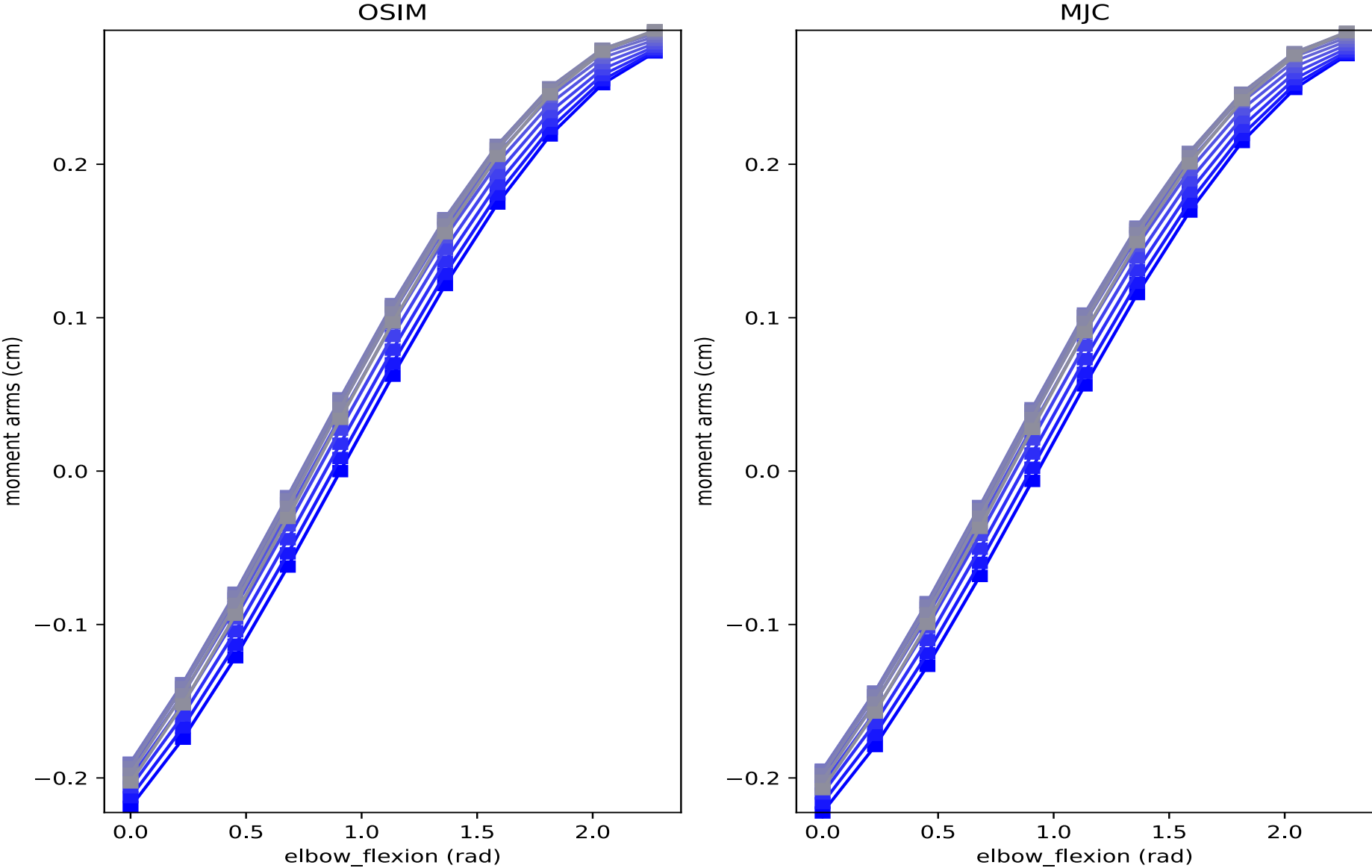
# Muscle specific moment arm comparison with Osim model after optimization

PL - deviation - flexion



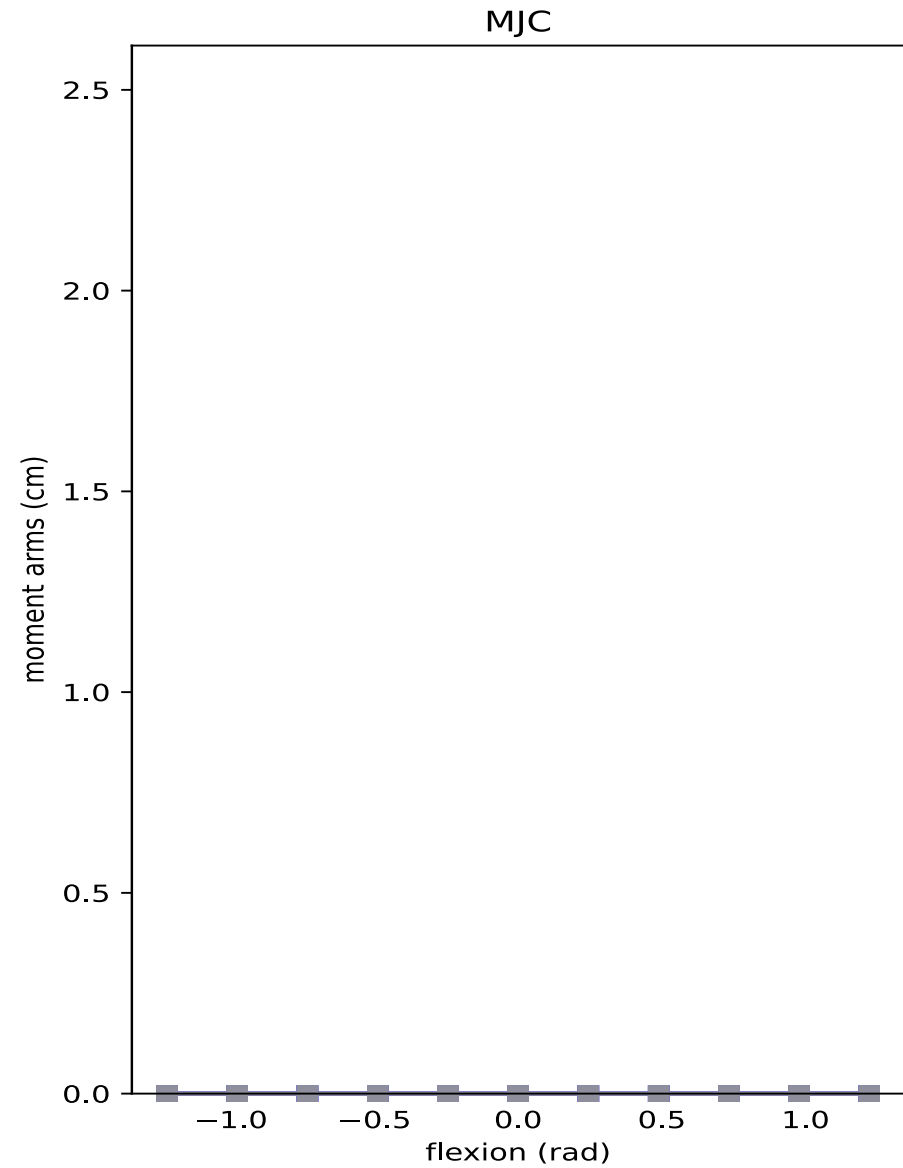
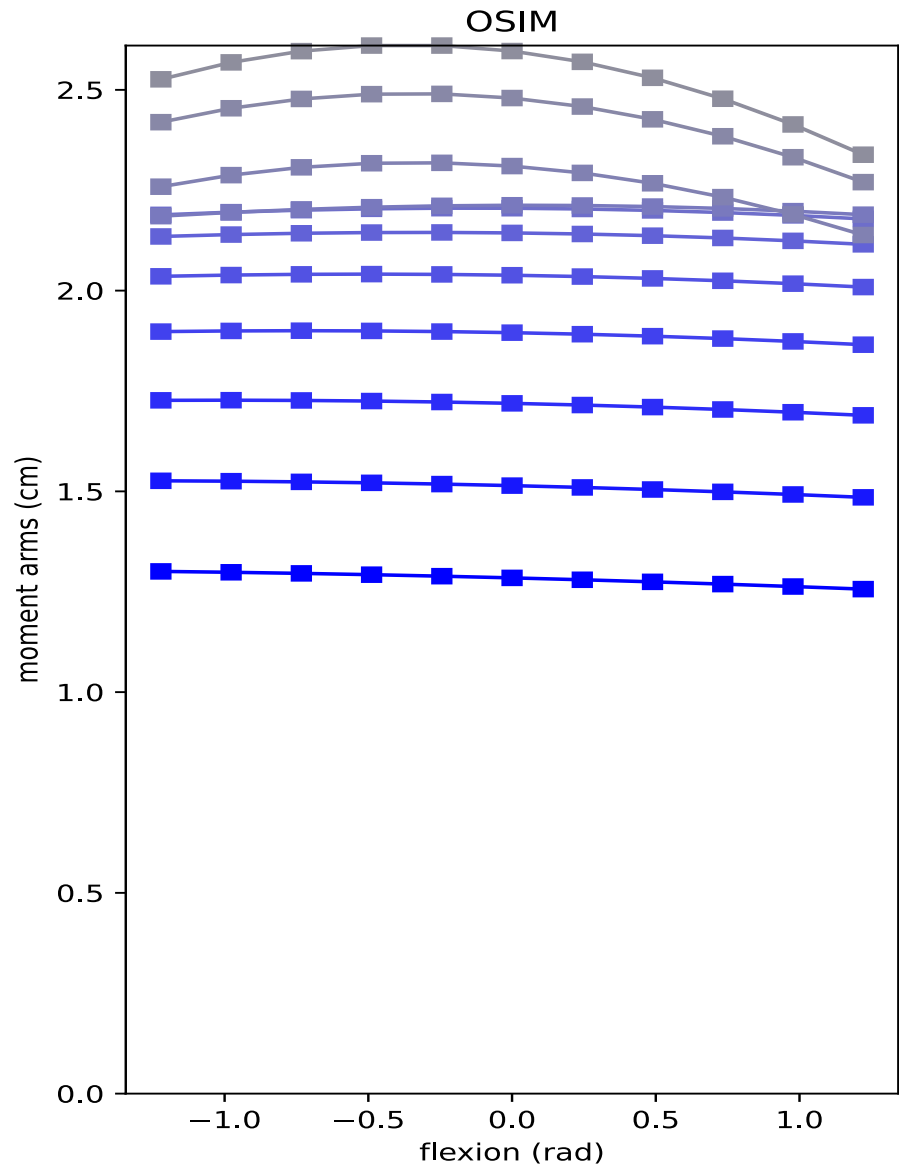
# Muscle specific moment arm comparison with Osim model after optimization

PL - elbow\_flexion - pro\_sup



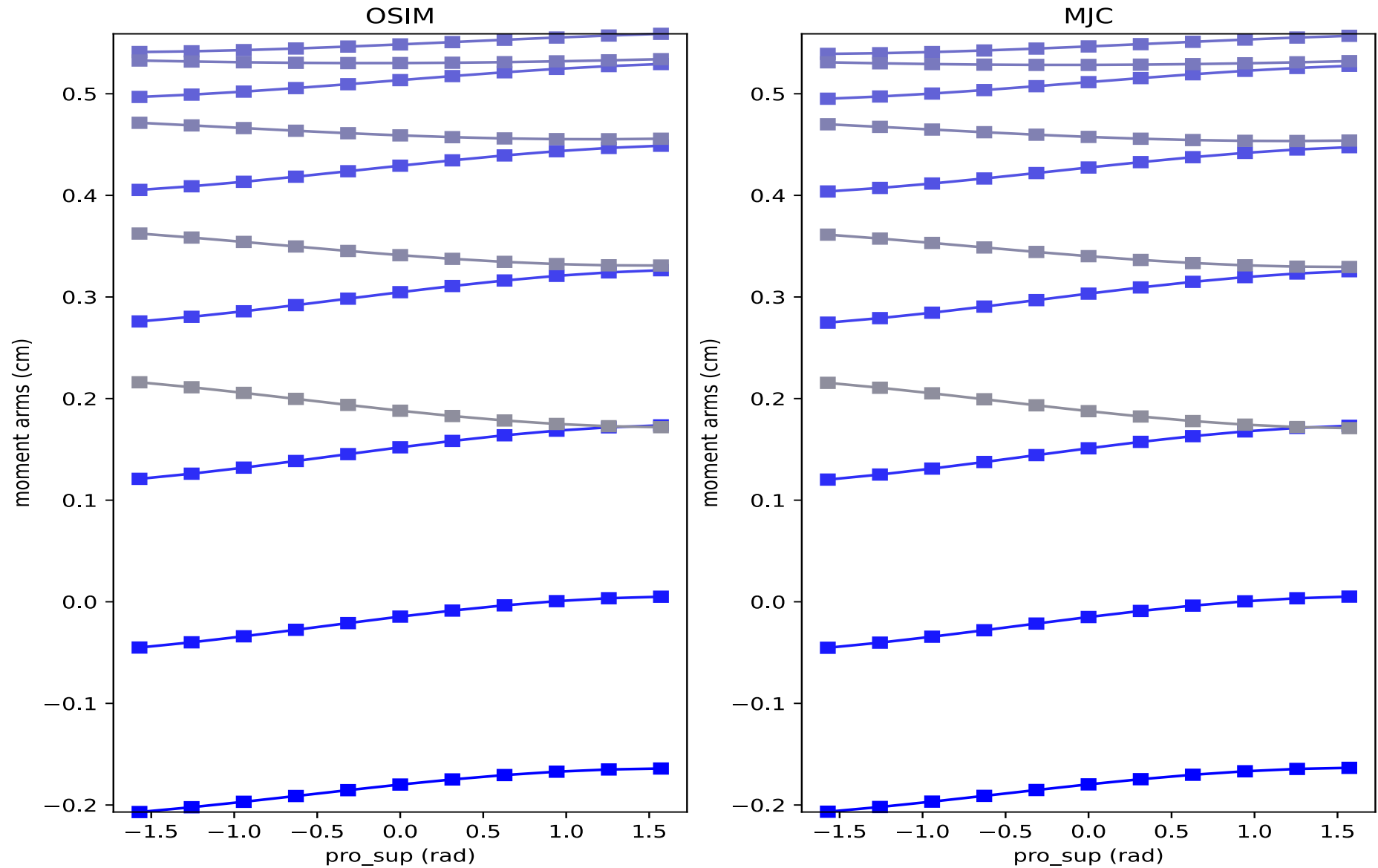
### Muscle specific moment arm comparison with Osim model after optimization

PL - deviation - flexion



# Muscle specific moment arm comparison with Osim model after optimization

PL - elbow\_flexion - pro\_sup



## Step 3: Muscle Kinetic Validation

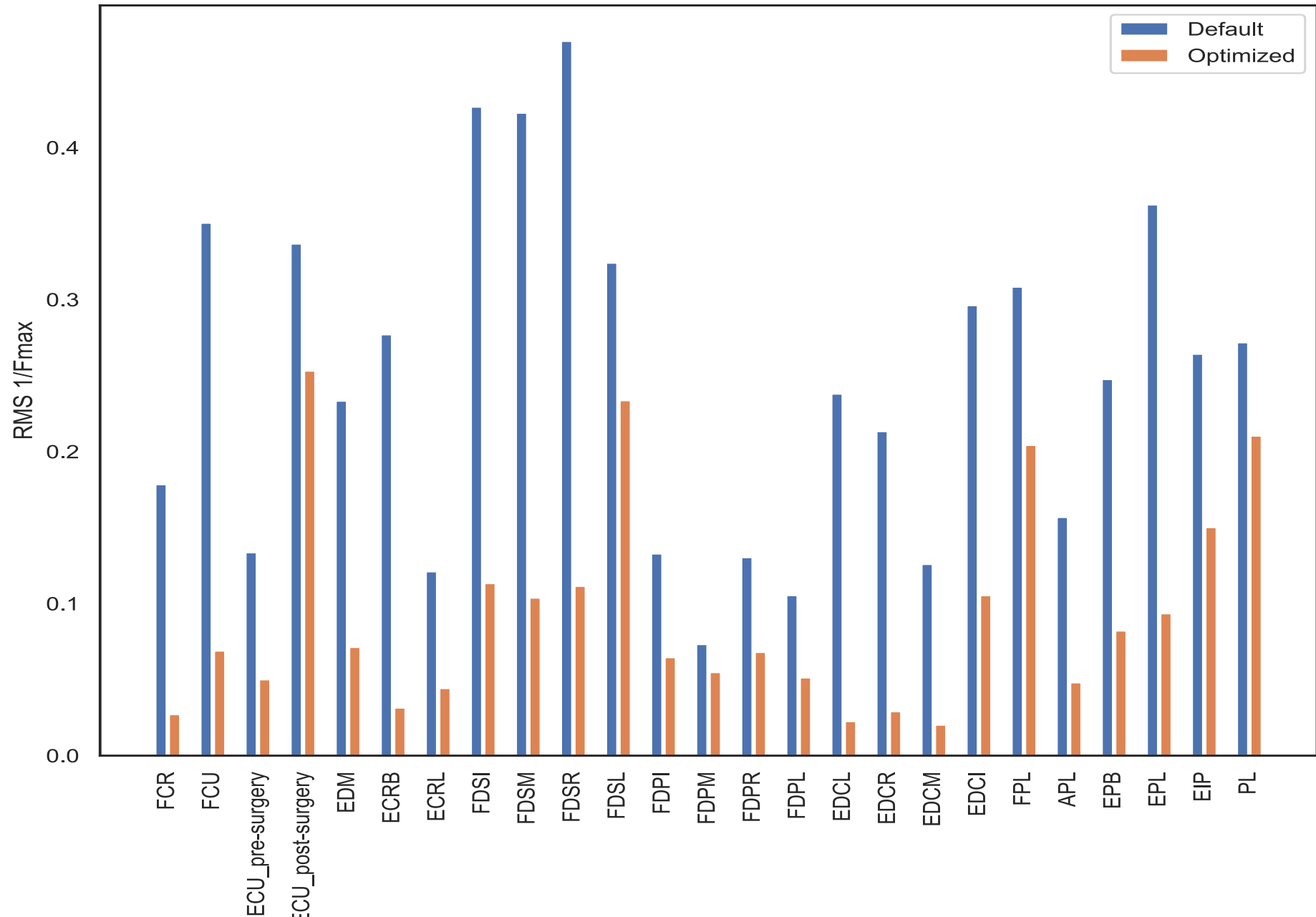
Muscle force-length property are compared between Osim and Mjc models. This force-length property only depends on muscle-fiber-tendon unit lengths. We made it isolated with the moment arm, so that the change in moment arms will not affect the muscle force properties. The muscle-fiber-tendon unit lengths were roughly even extracted (from shorest to longest) with all possible body postures. A bar plot of the froce errors of all muscle before and after optimization is included. Then the detail force-length curve comparsion plot of each muscle is included.

How to interpret the plot:

Global title indicate the muscle name. X axis indicate the muscle-fiber-tendon unit length. Y axis is the muscle force (unnormalized)

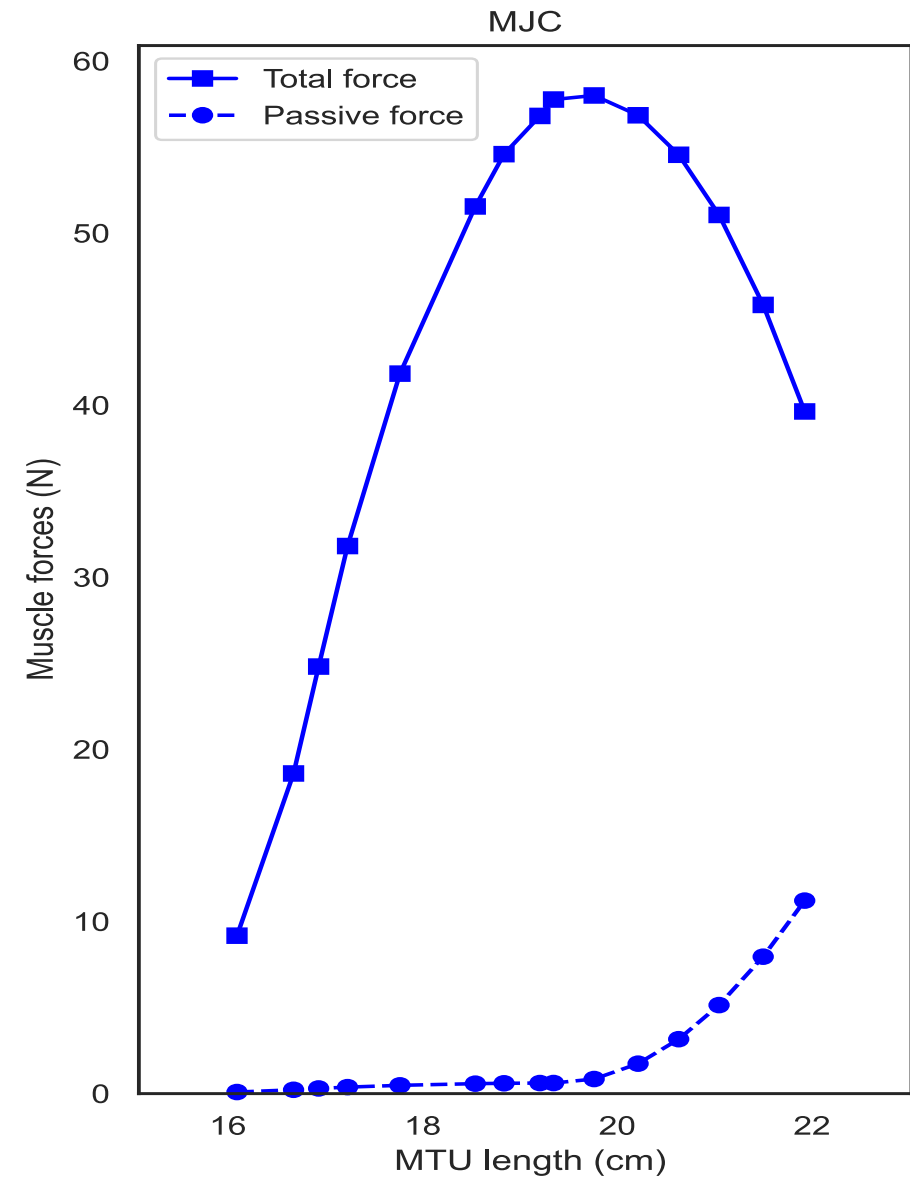
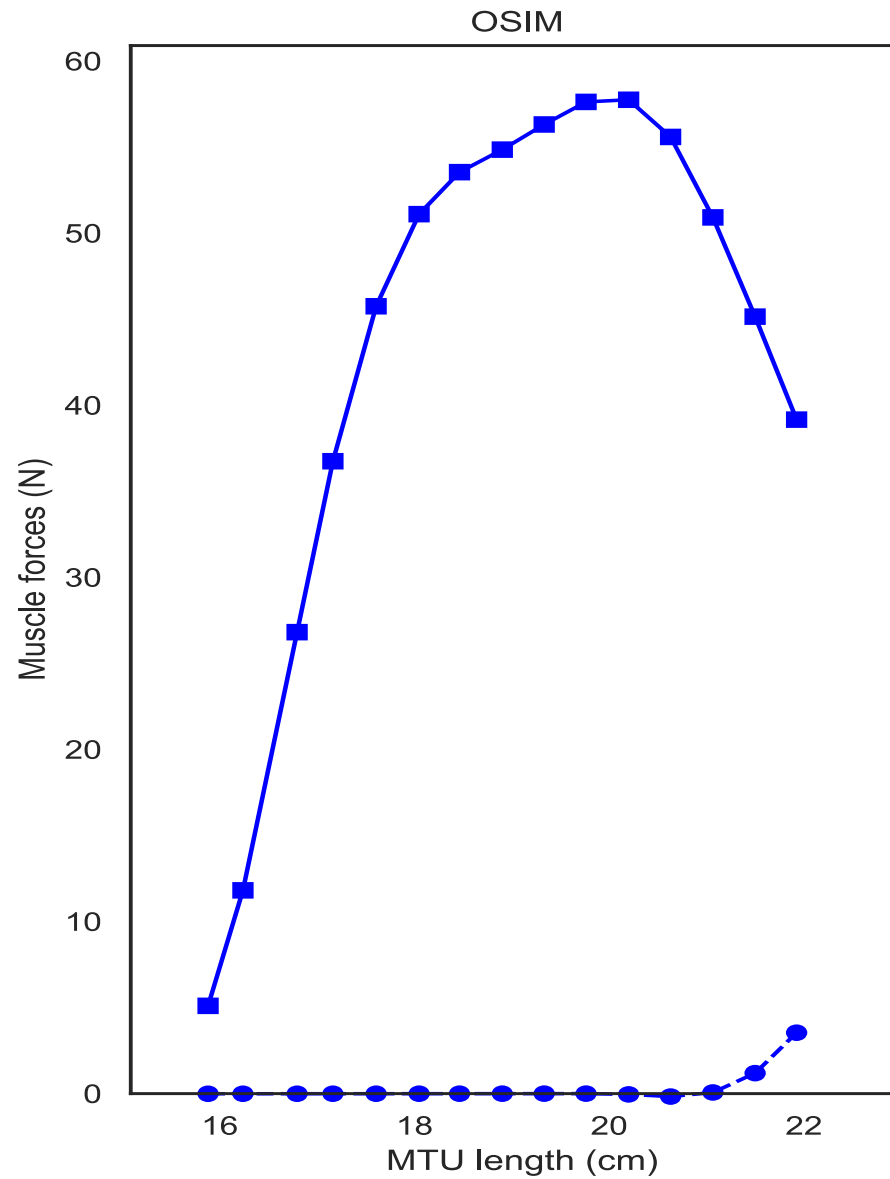


# Overall comparison of muscle force-length relationship before/after optimization



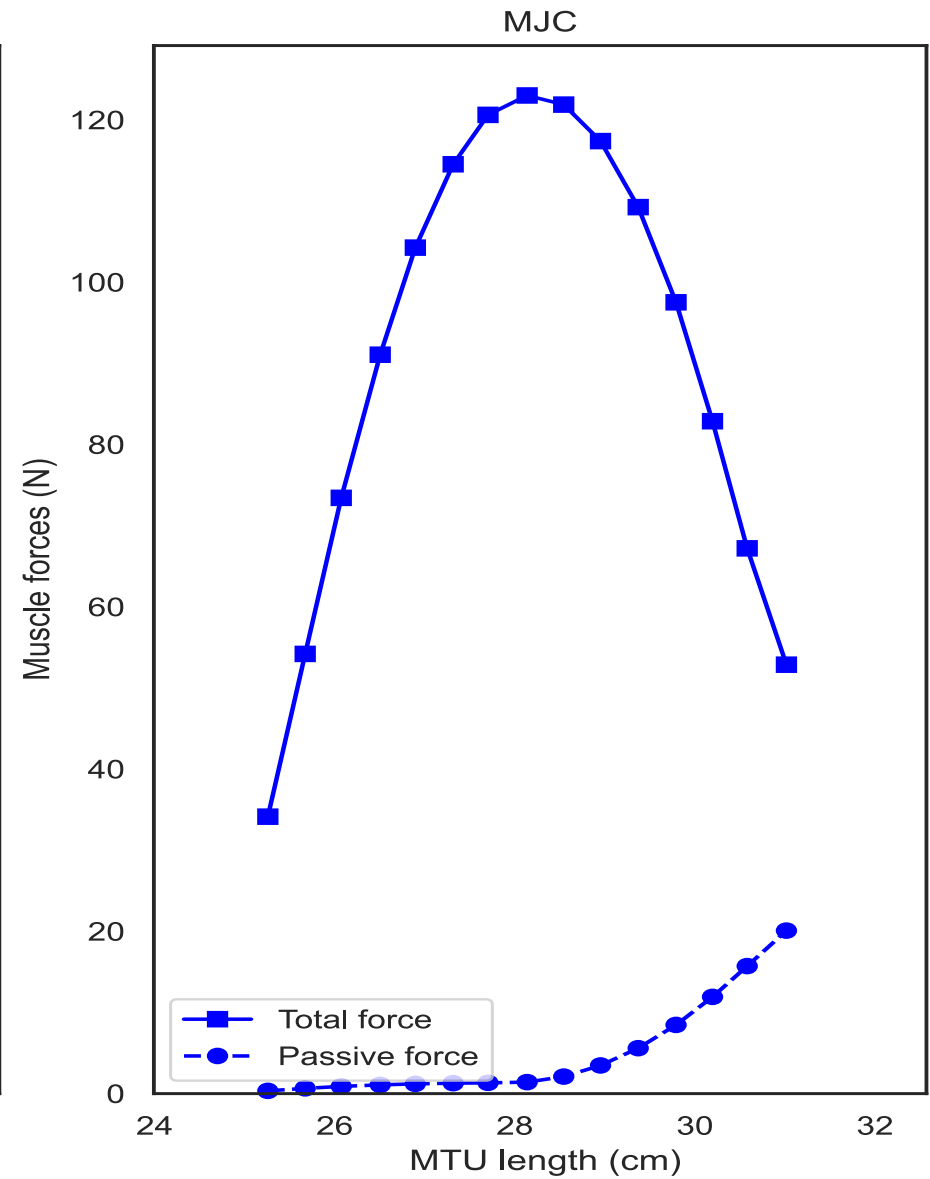
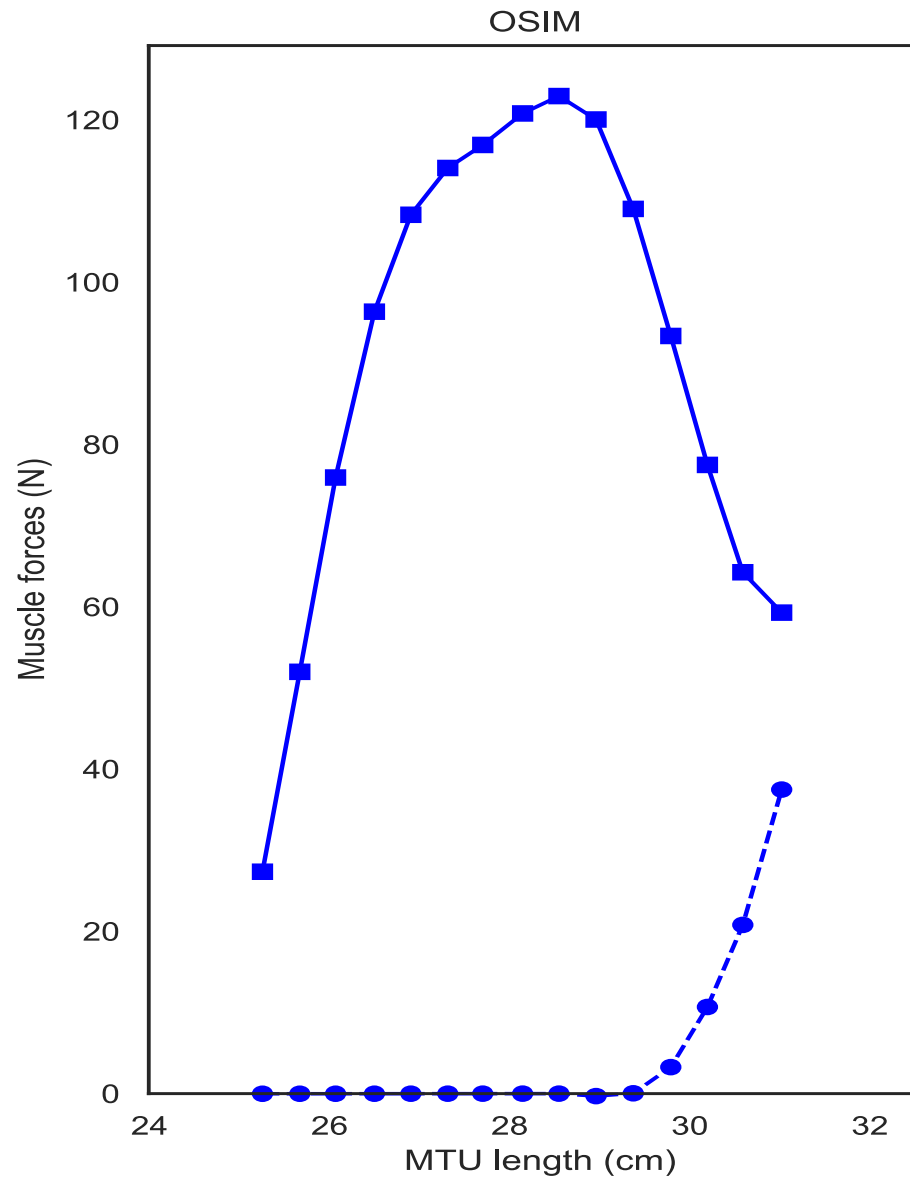
# Muscle specific force comparison before/after optimization

APL



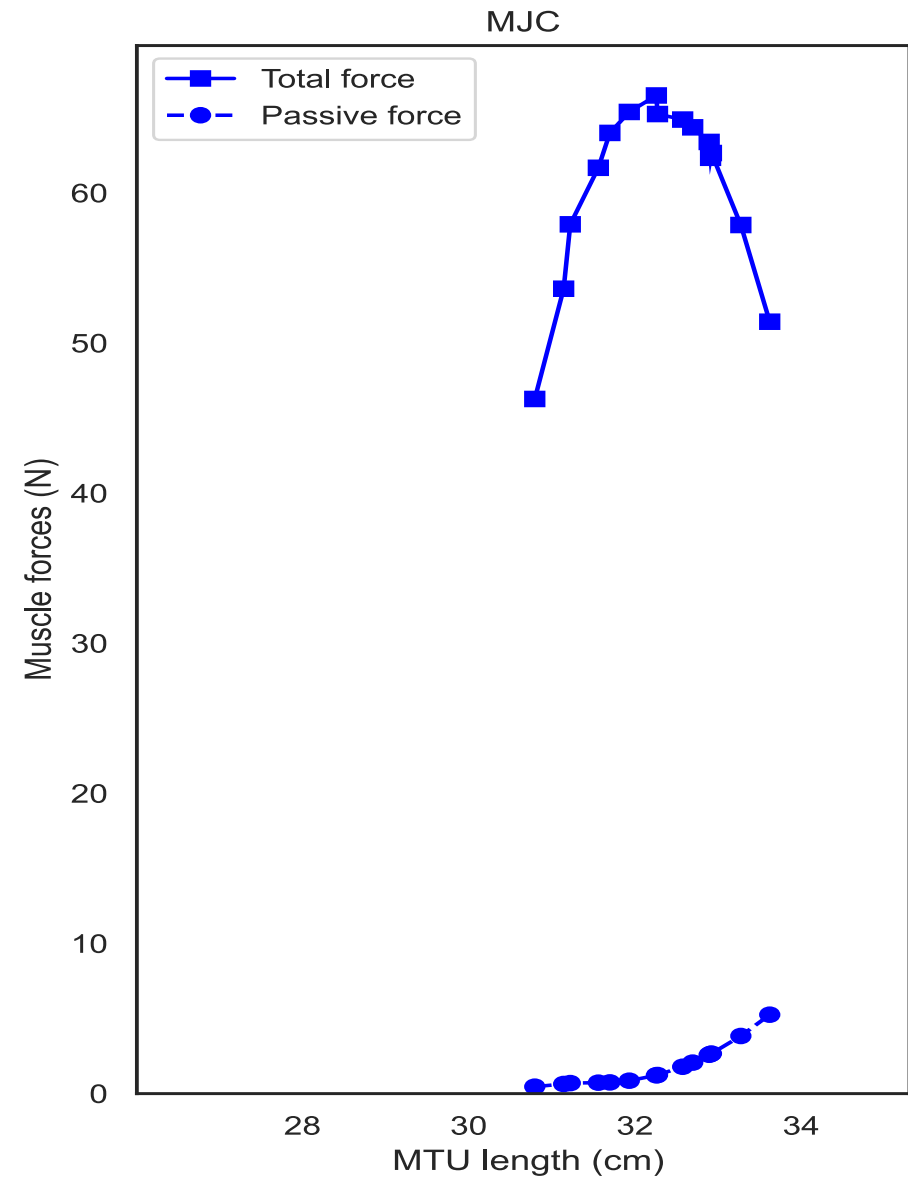
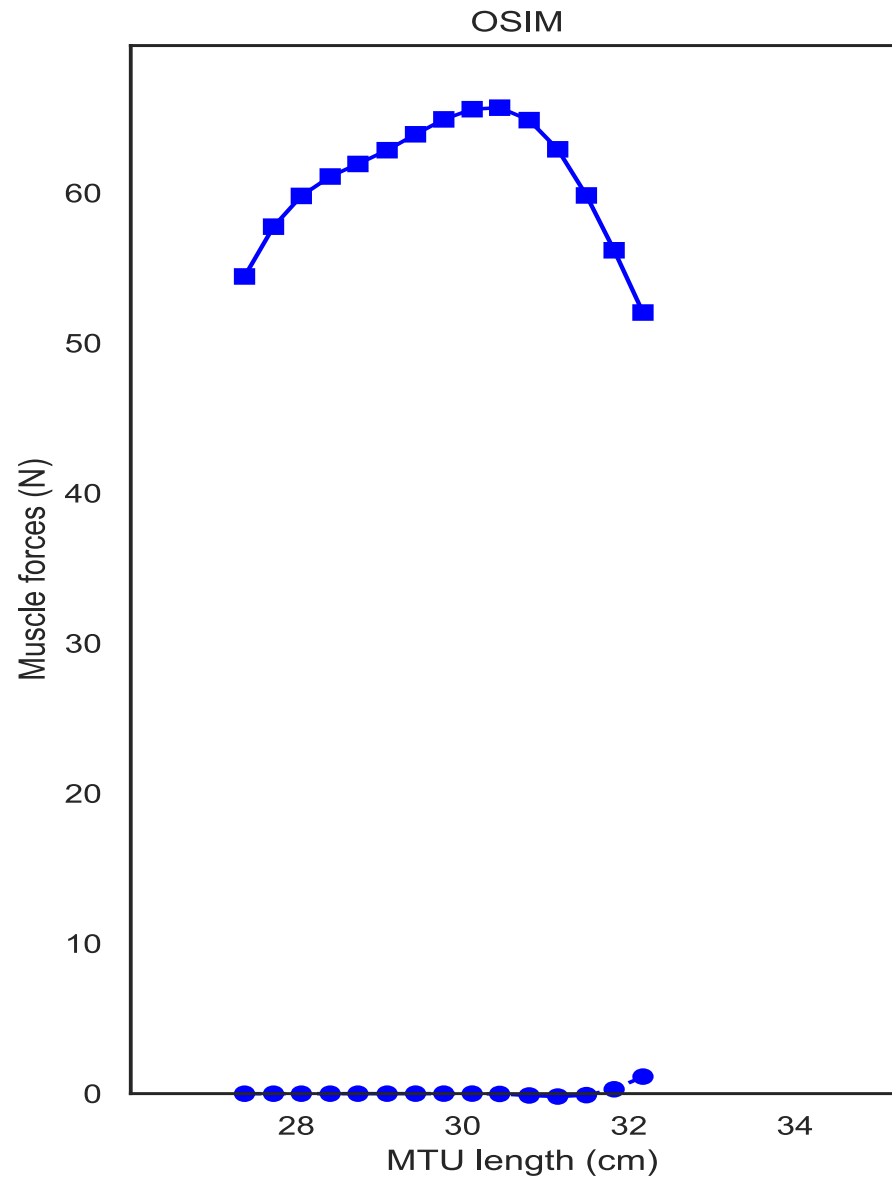
# Muscle specific force comparison before/after optimization

ECRB



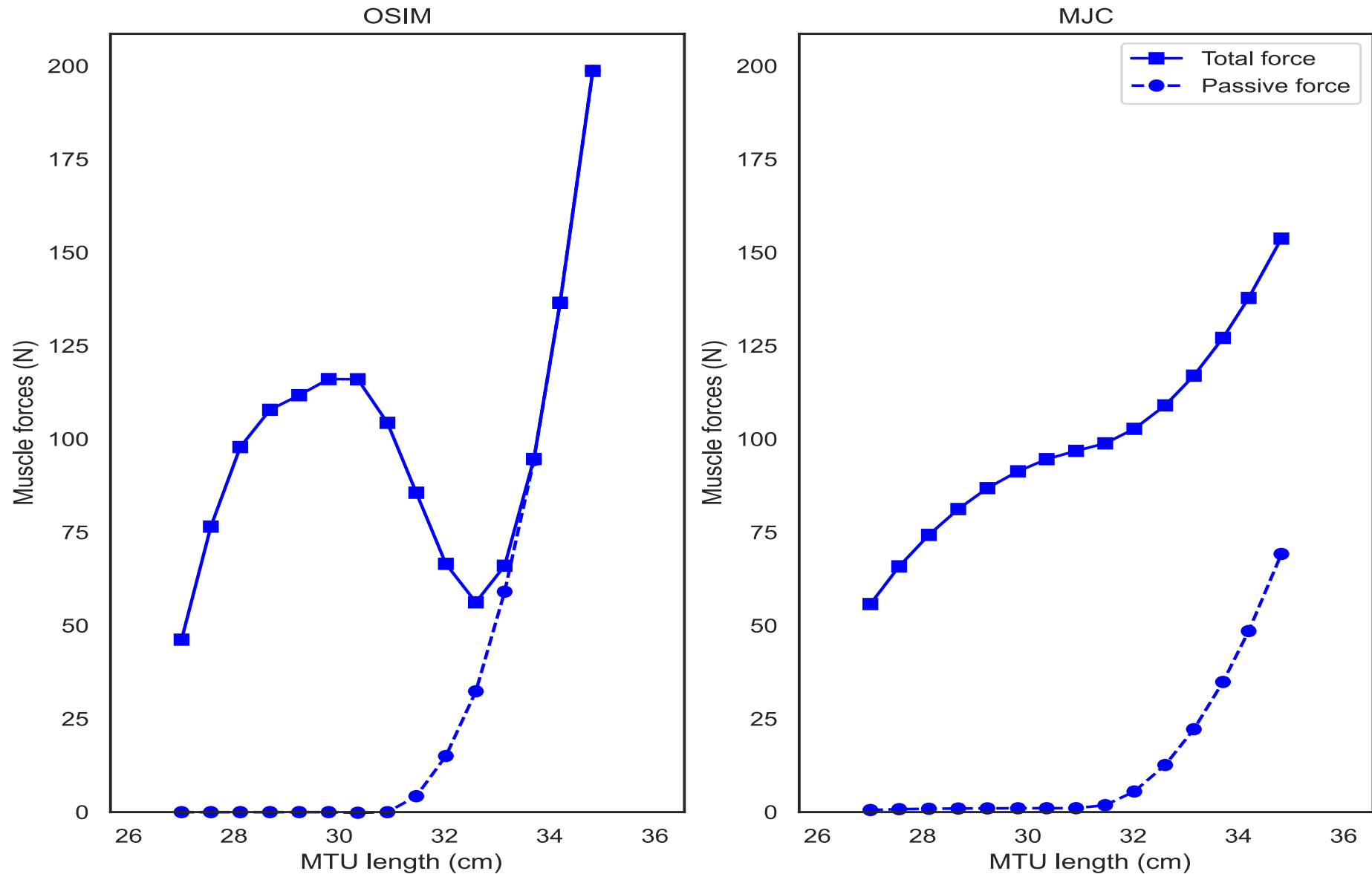
# Muscle specific force comparison before/after optimization

ECRL



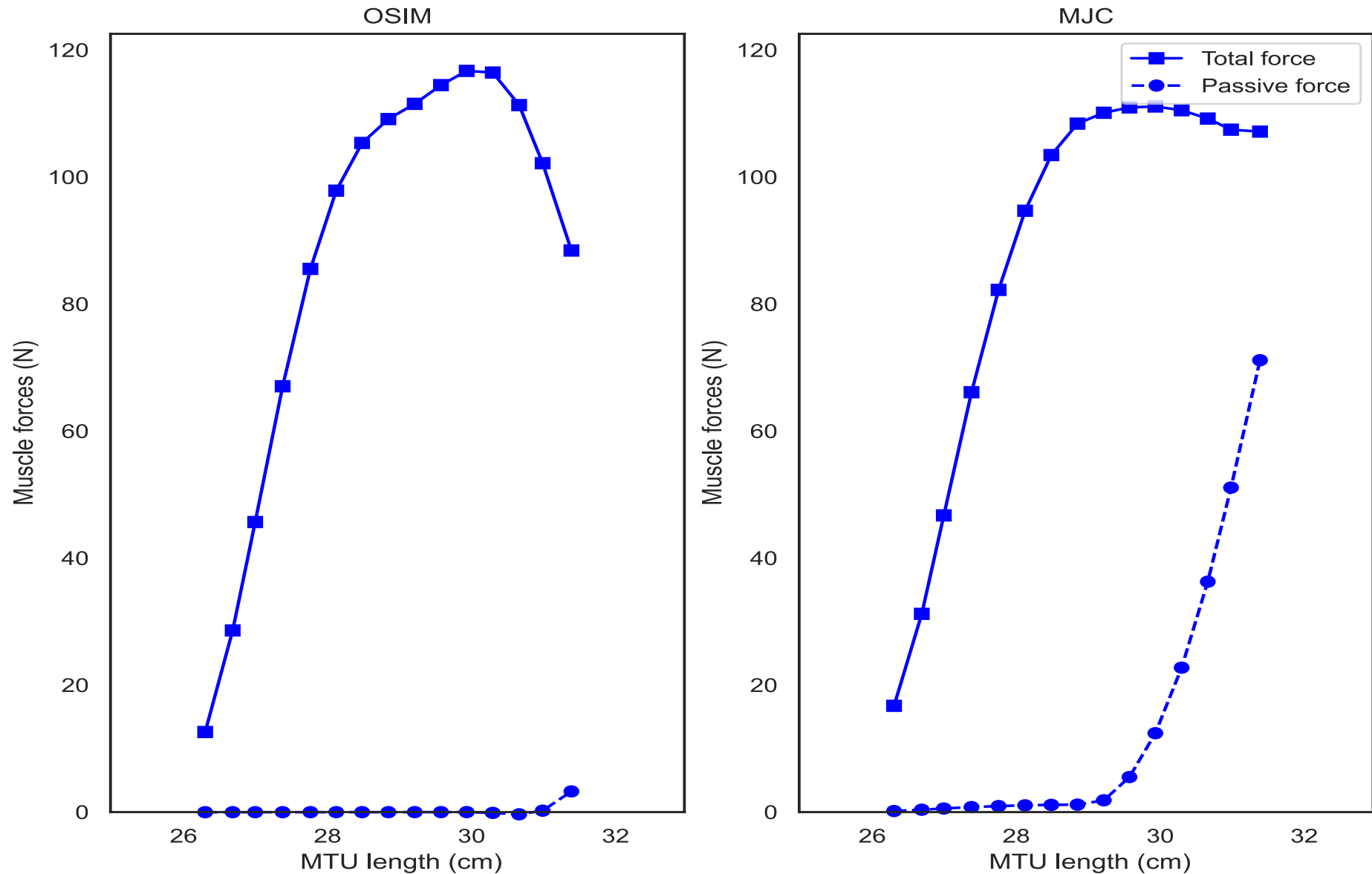
# Muscle specific force comparison before/after optimization

ECU\_post-surgery



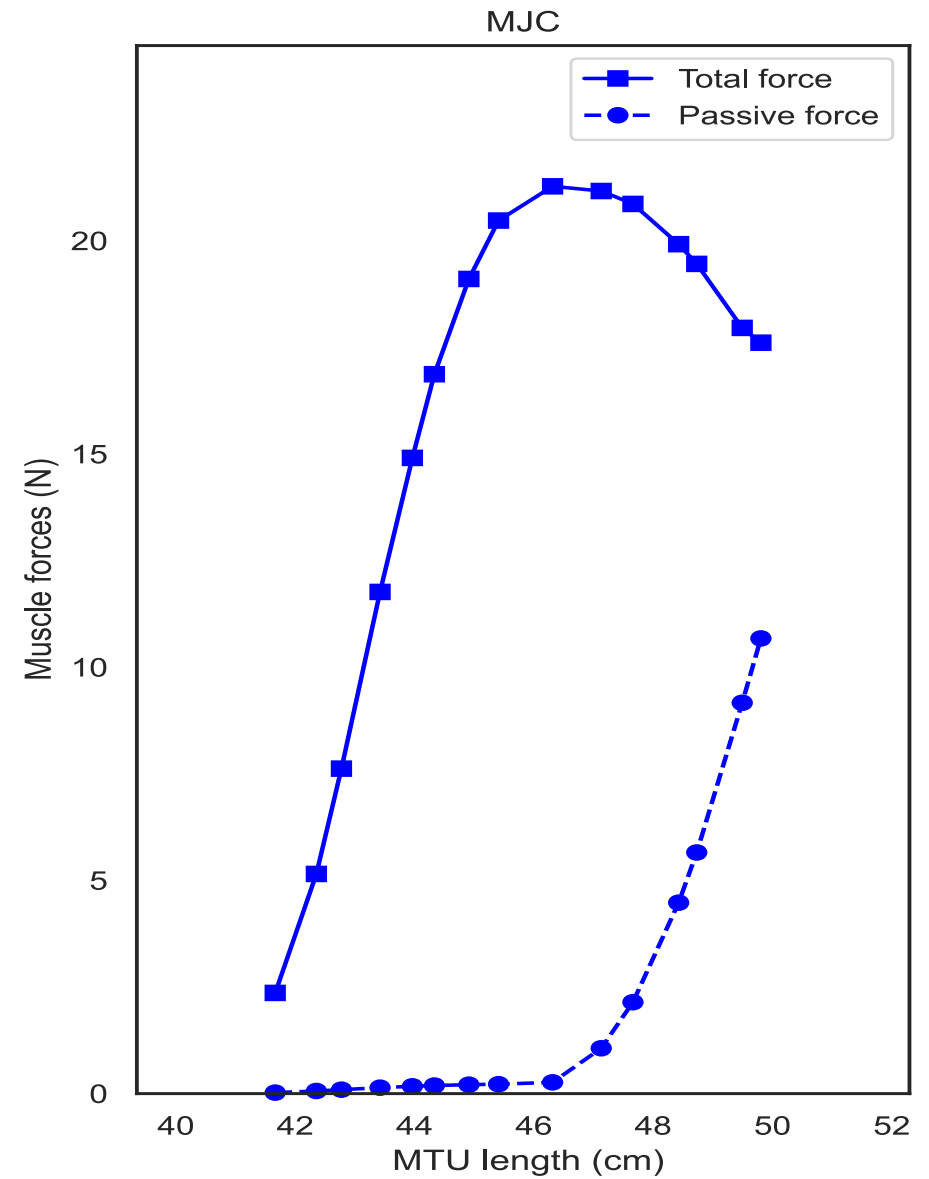
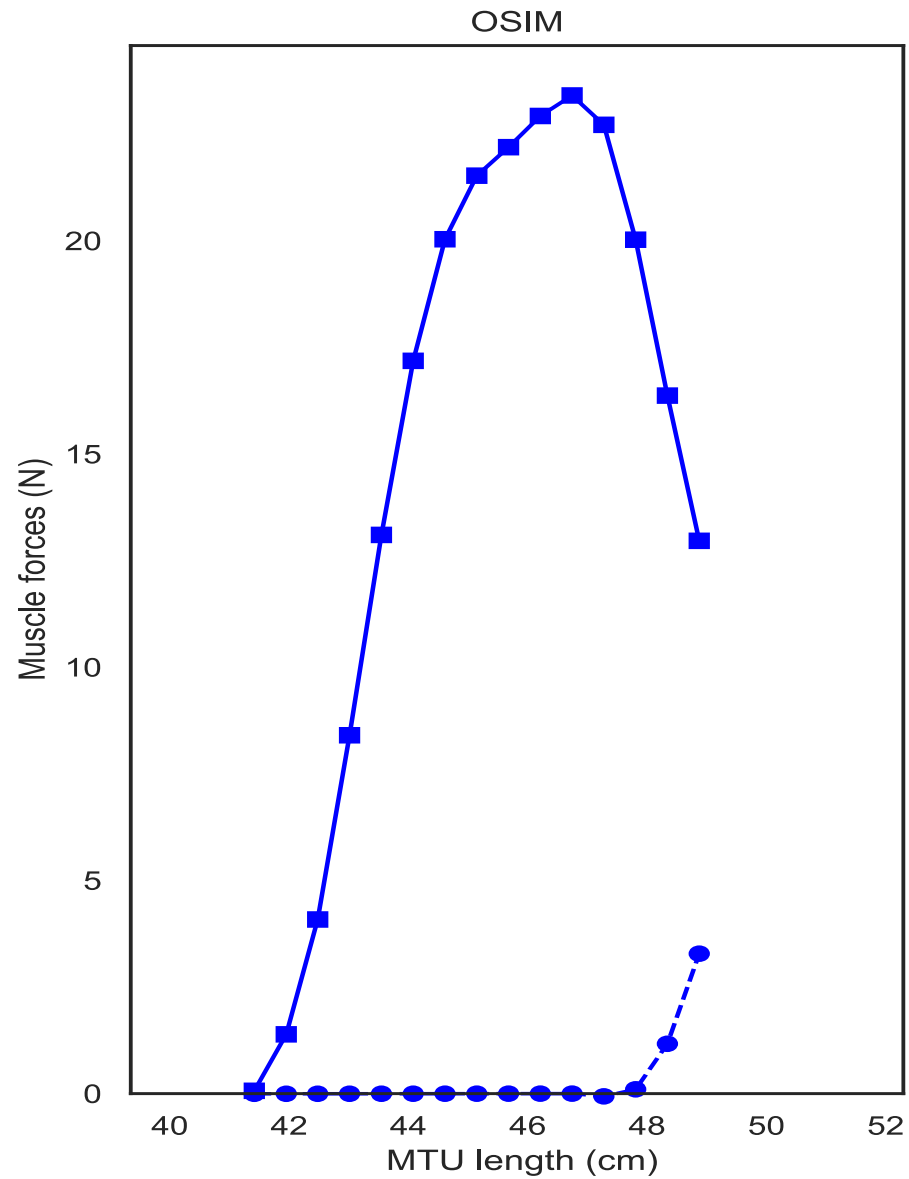
# Muscle specific force comparison before/after optimization

ECU\_pre-surgery



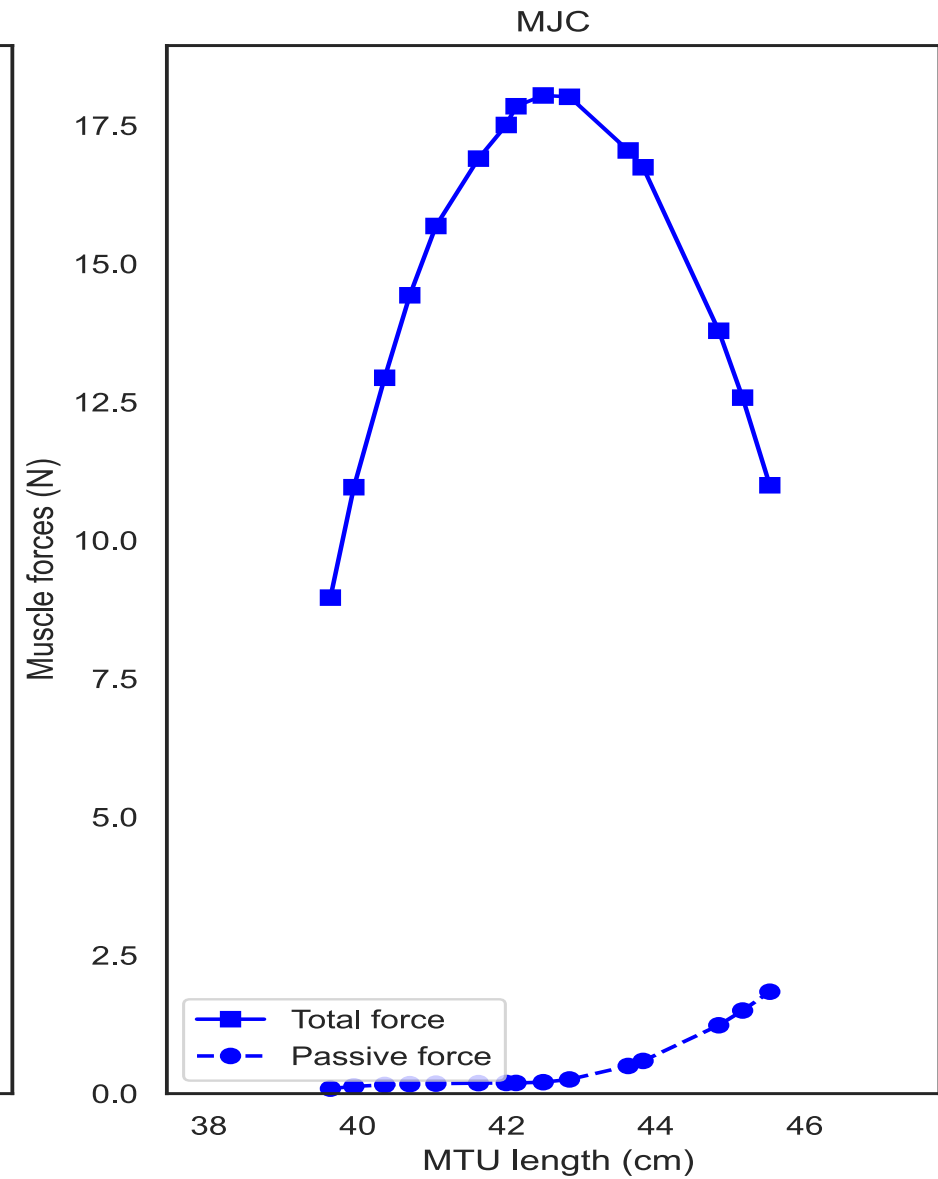
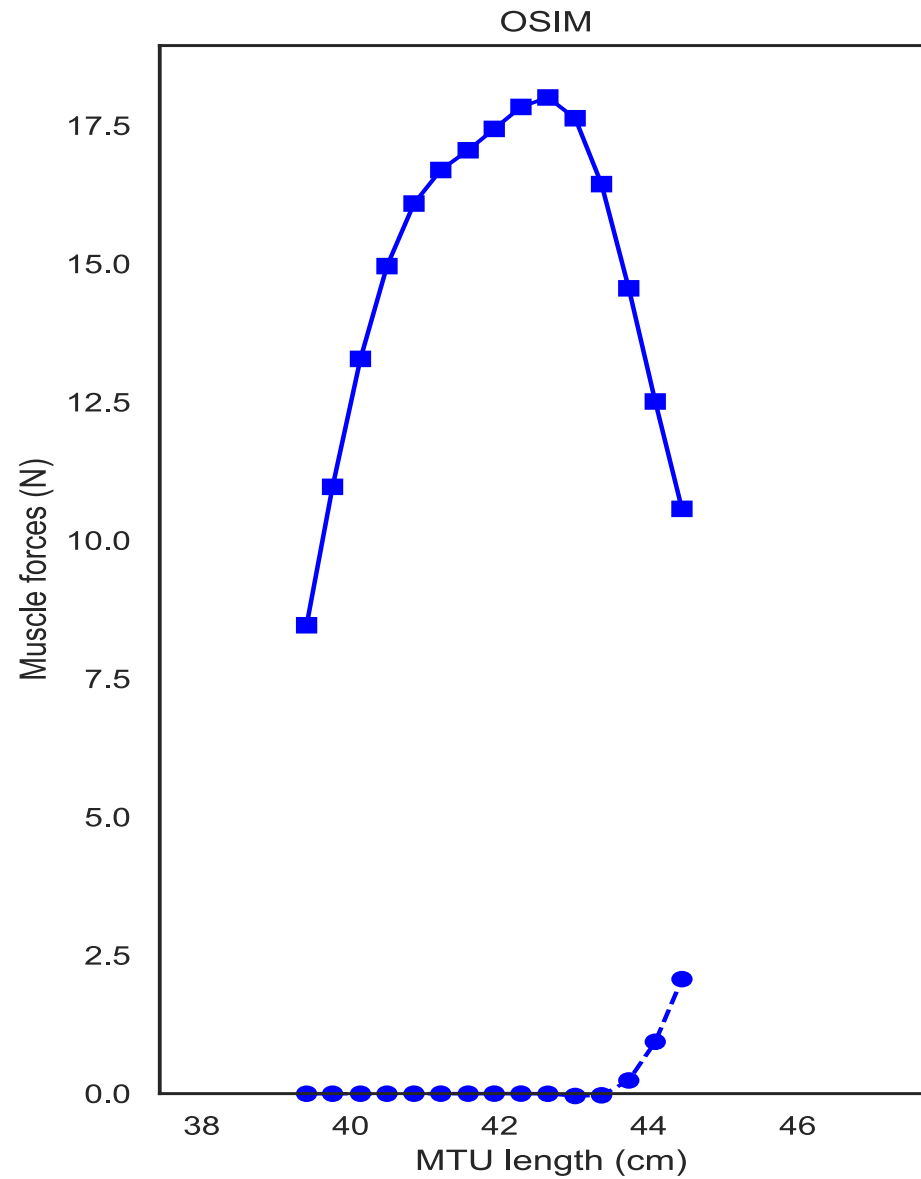
# Muscle specific force comparison before/after optimization

EDCI



# Muscle specific force comparison before/after optimization

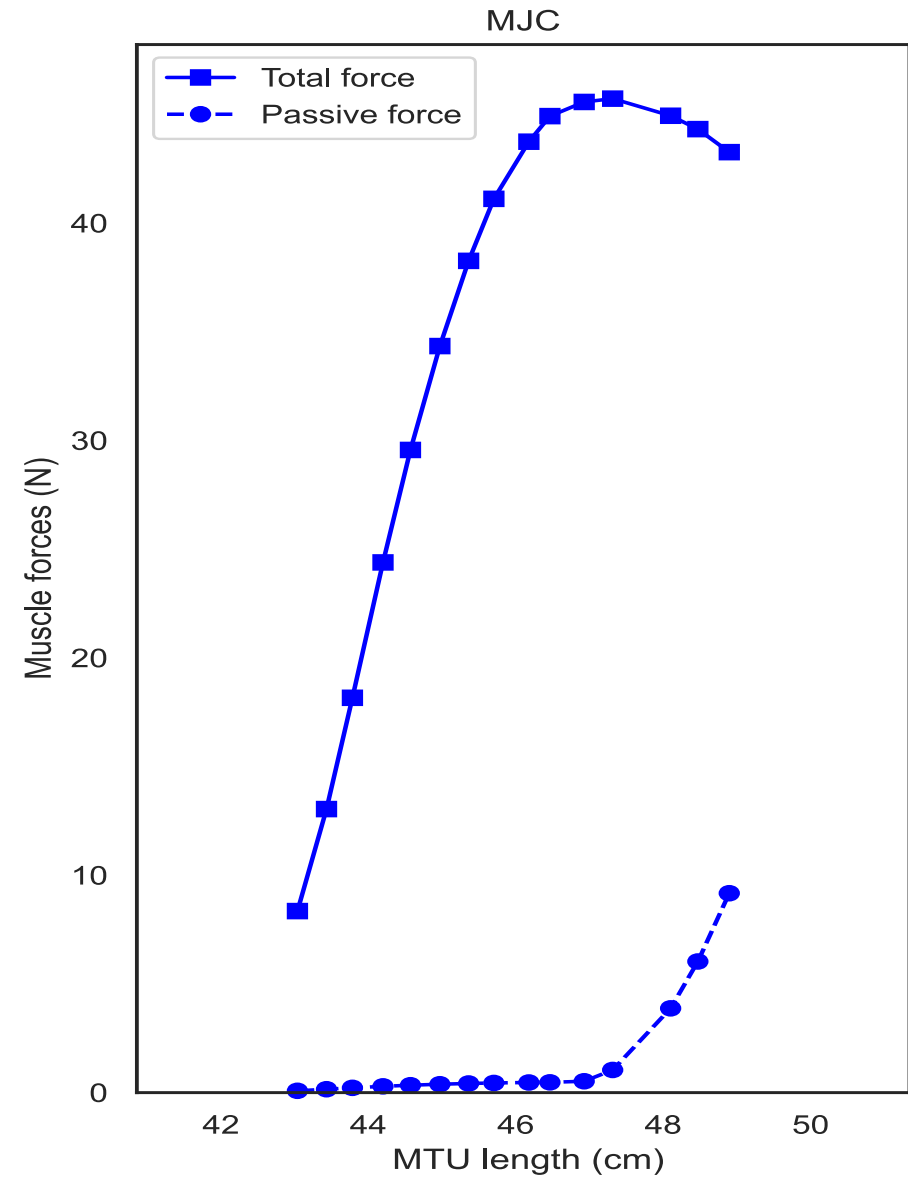
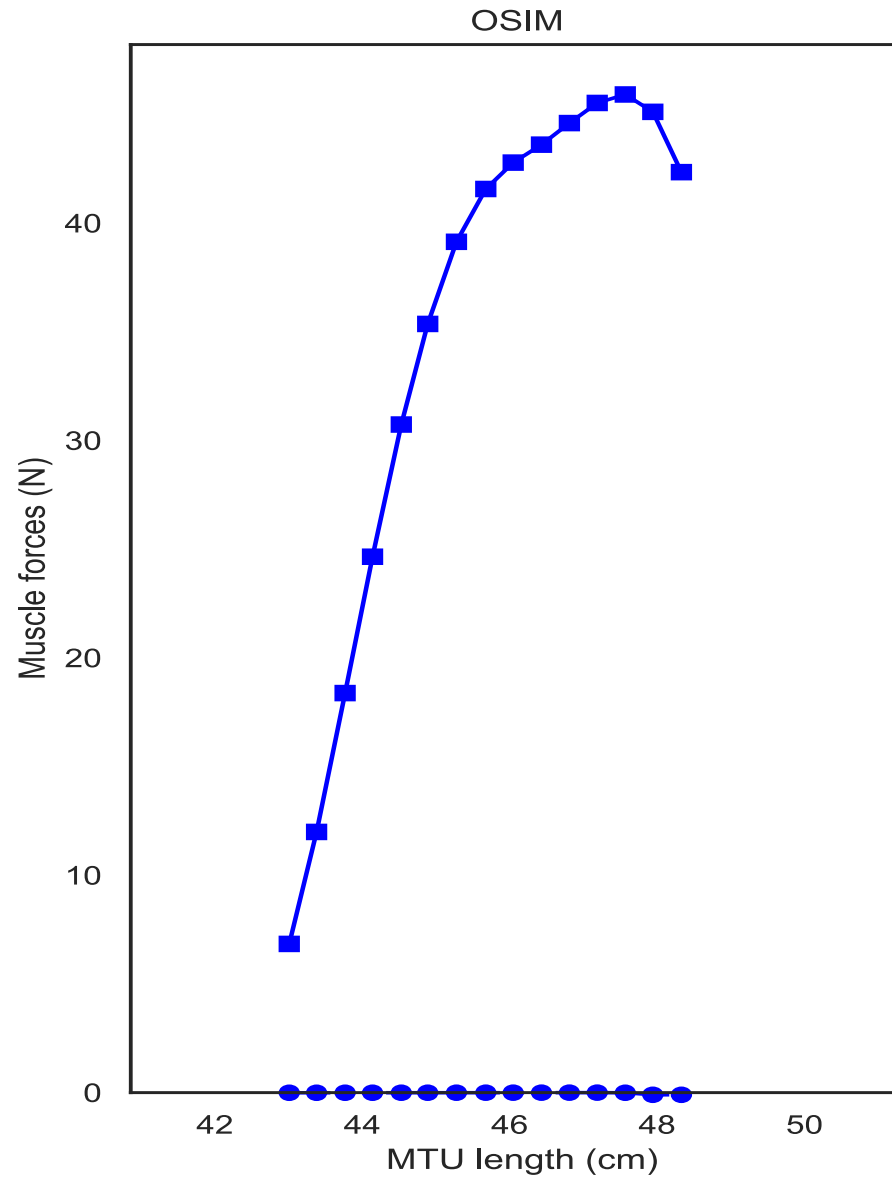
EDCL





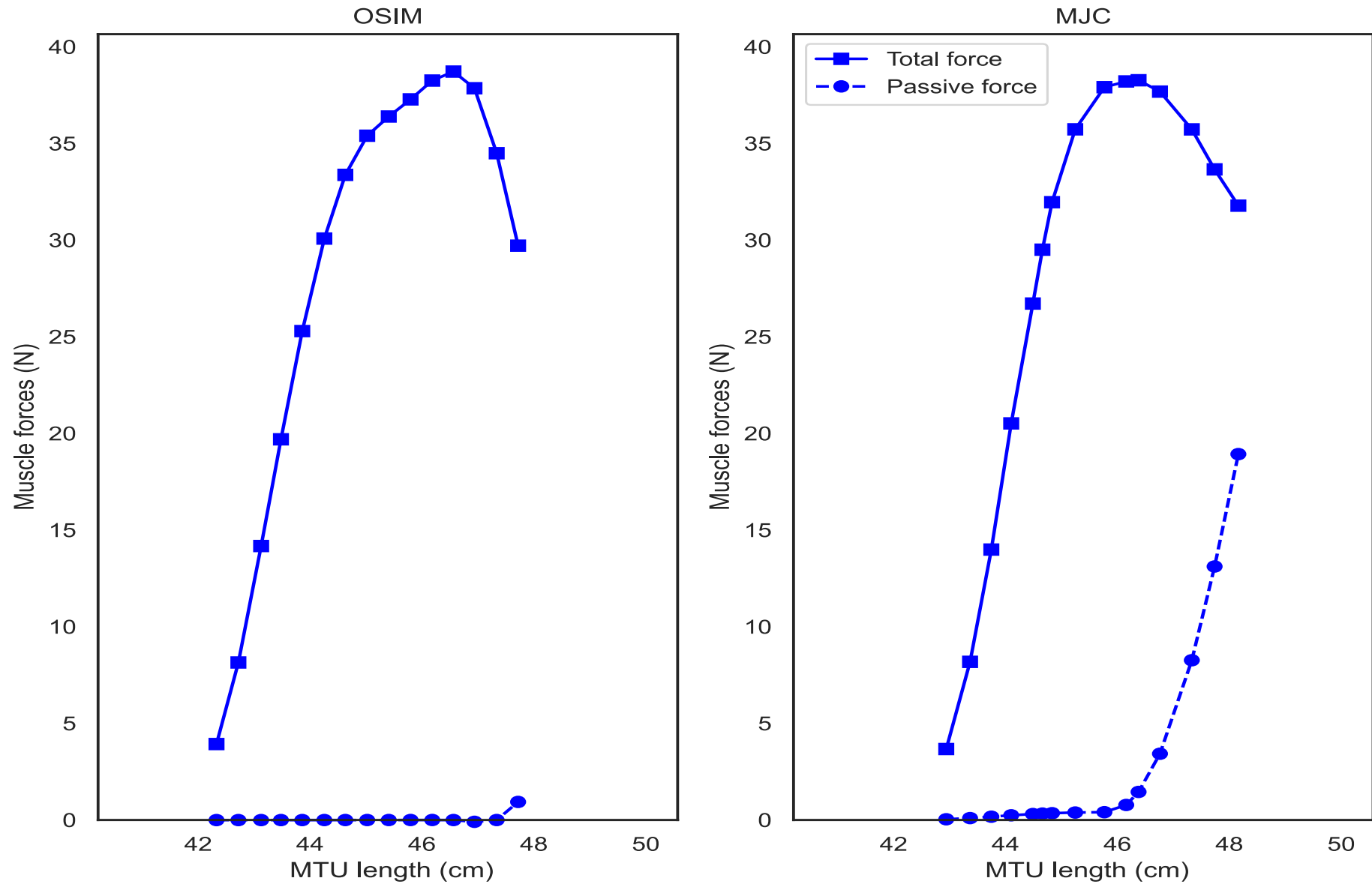
# Muscle specific force comparison before/after optimization

EDCM



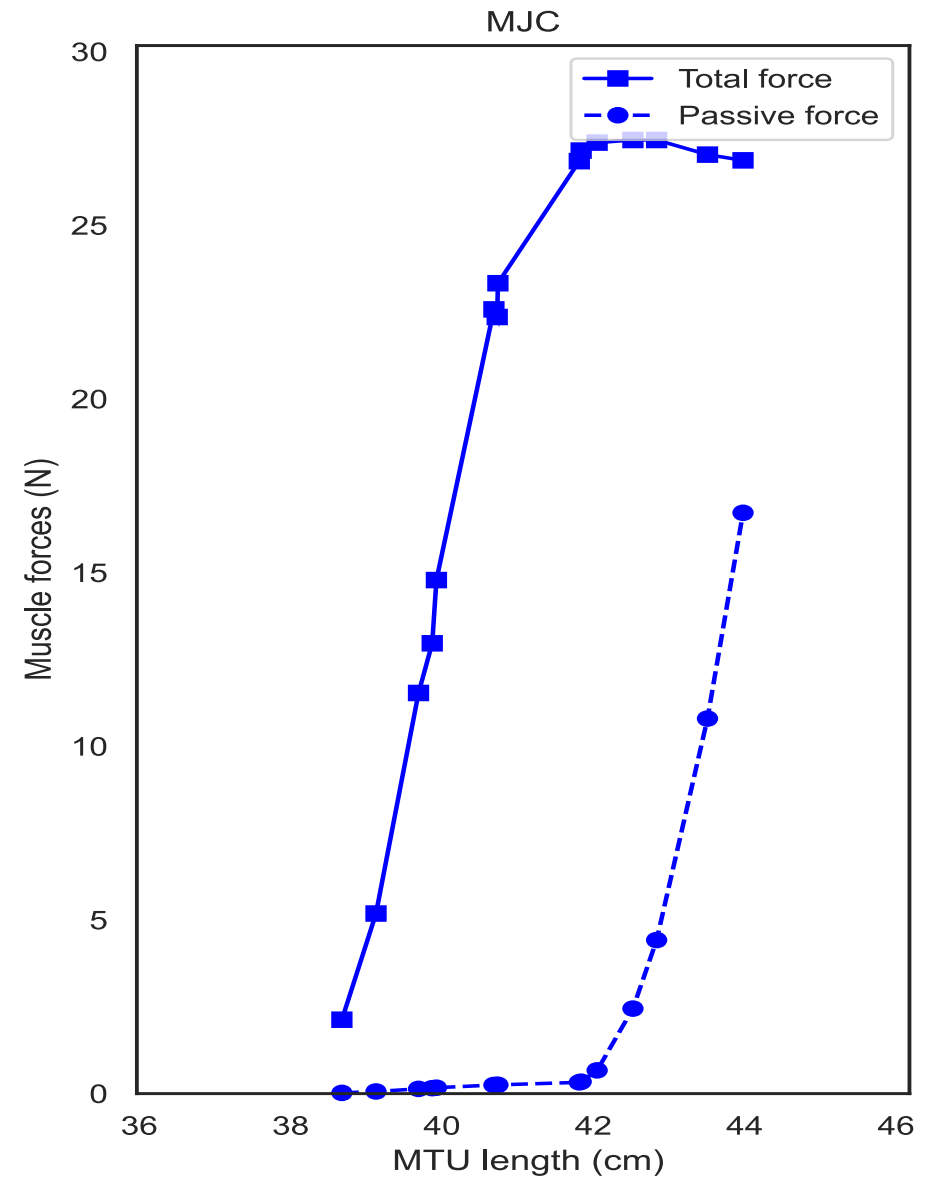
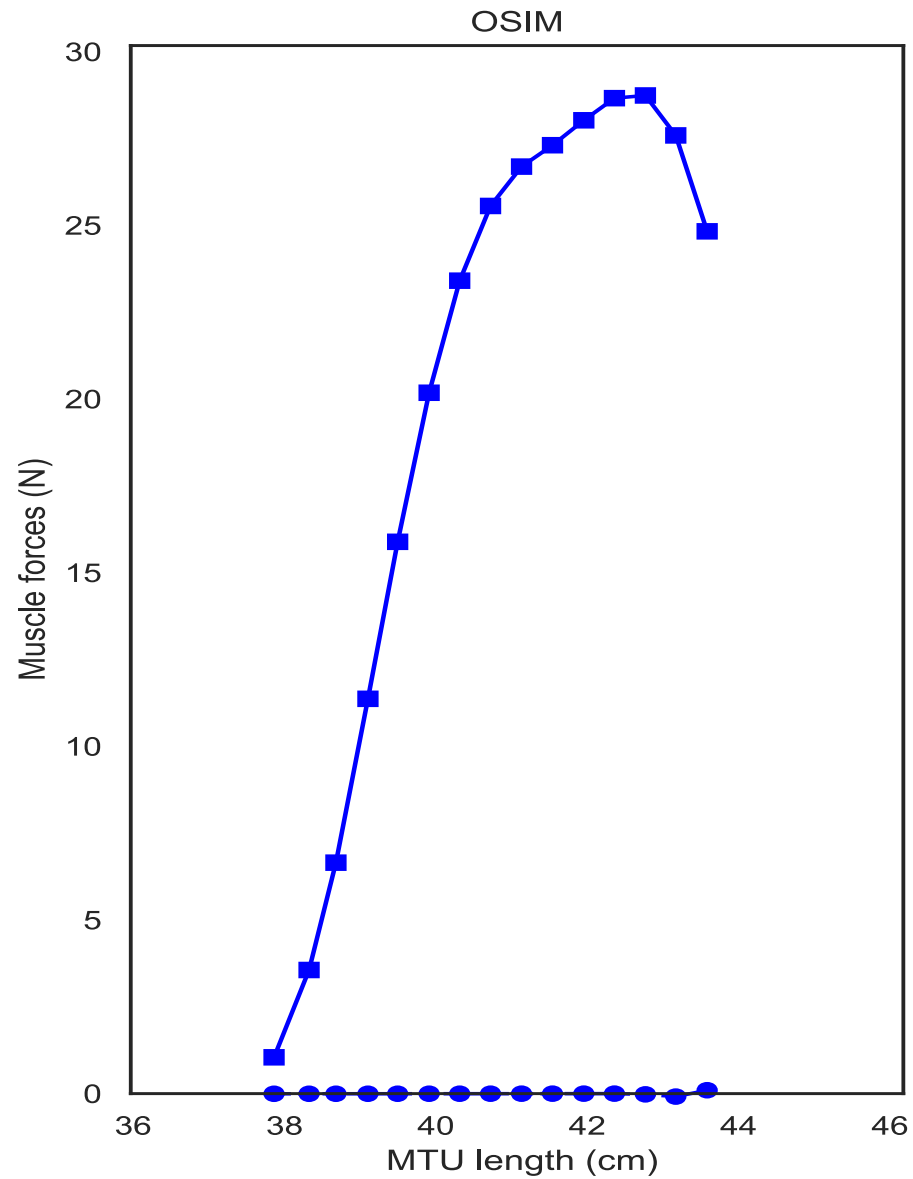
# Muscle specific force comparison before/after optimization

EDCR



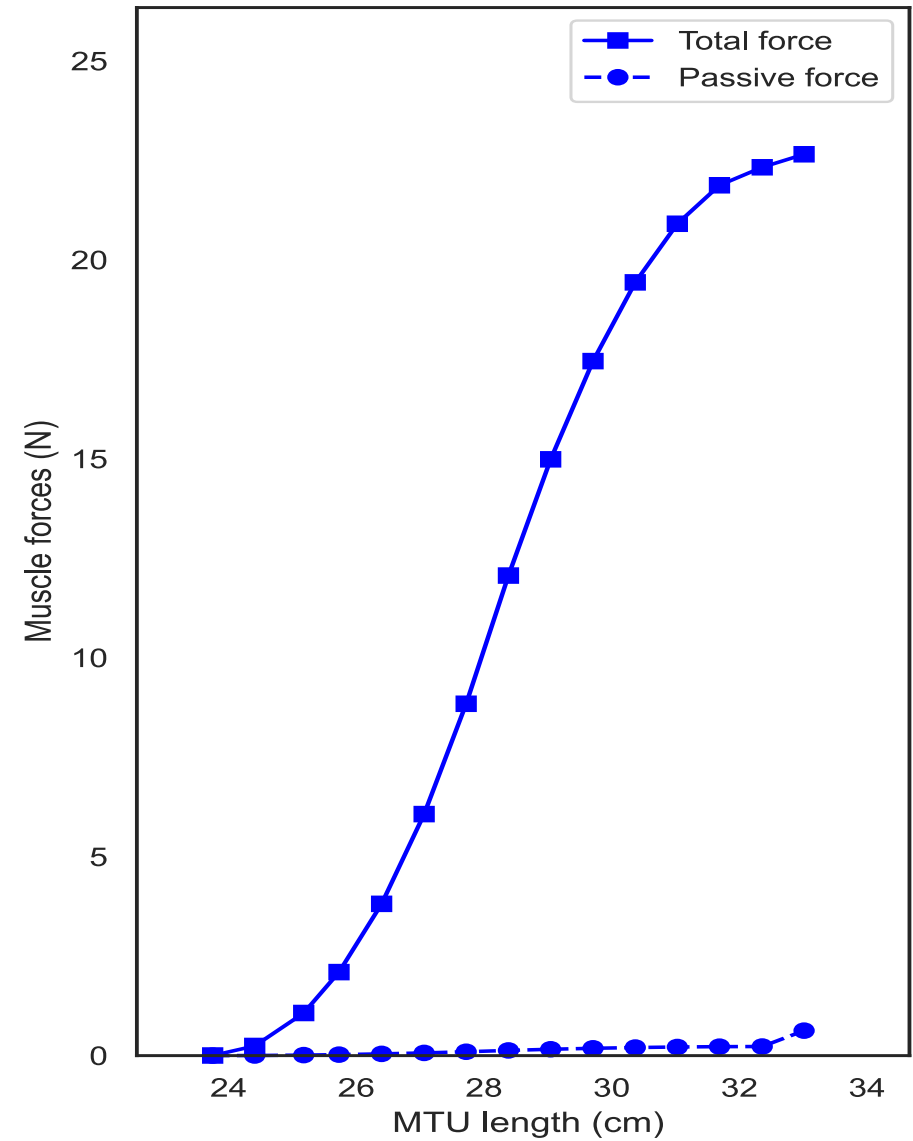
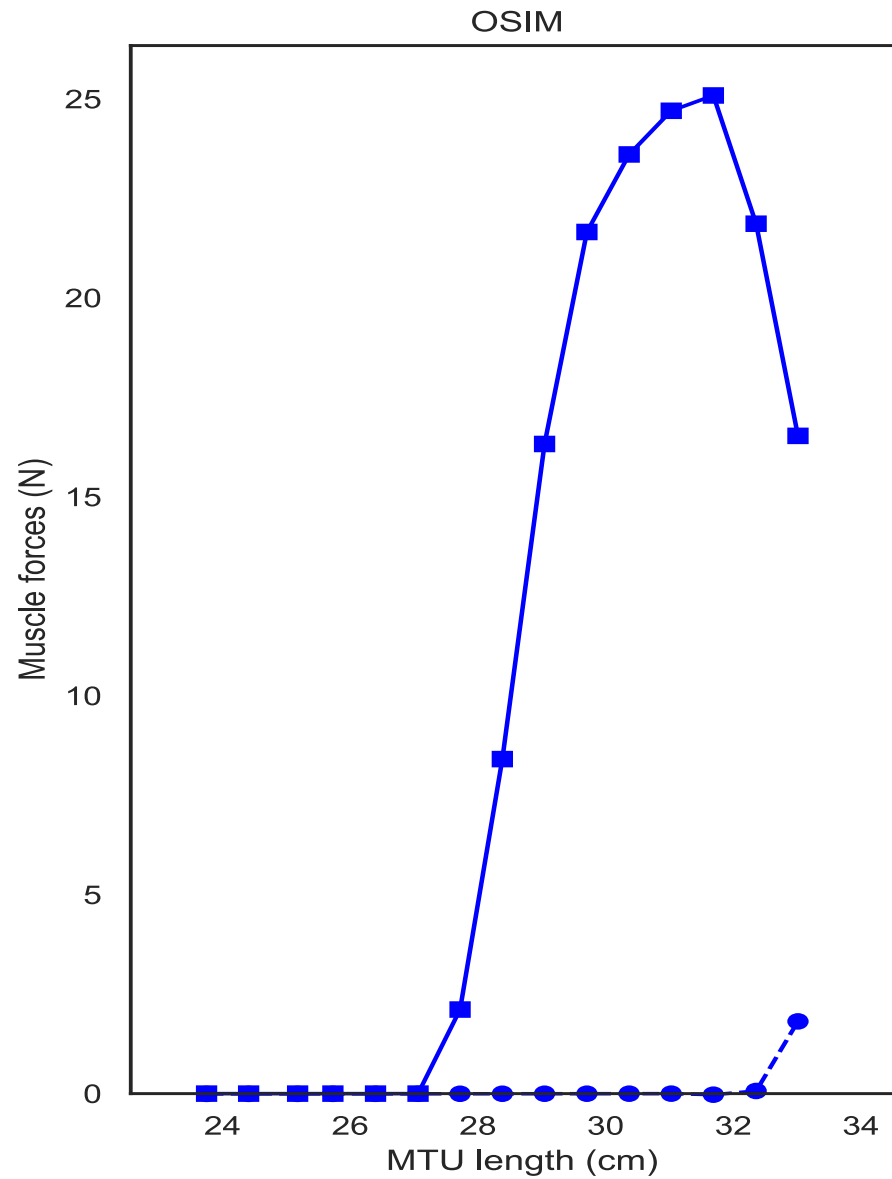
# Muscle specific force comparison before/after optimization

EDM



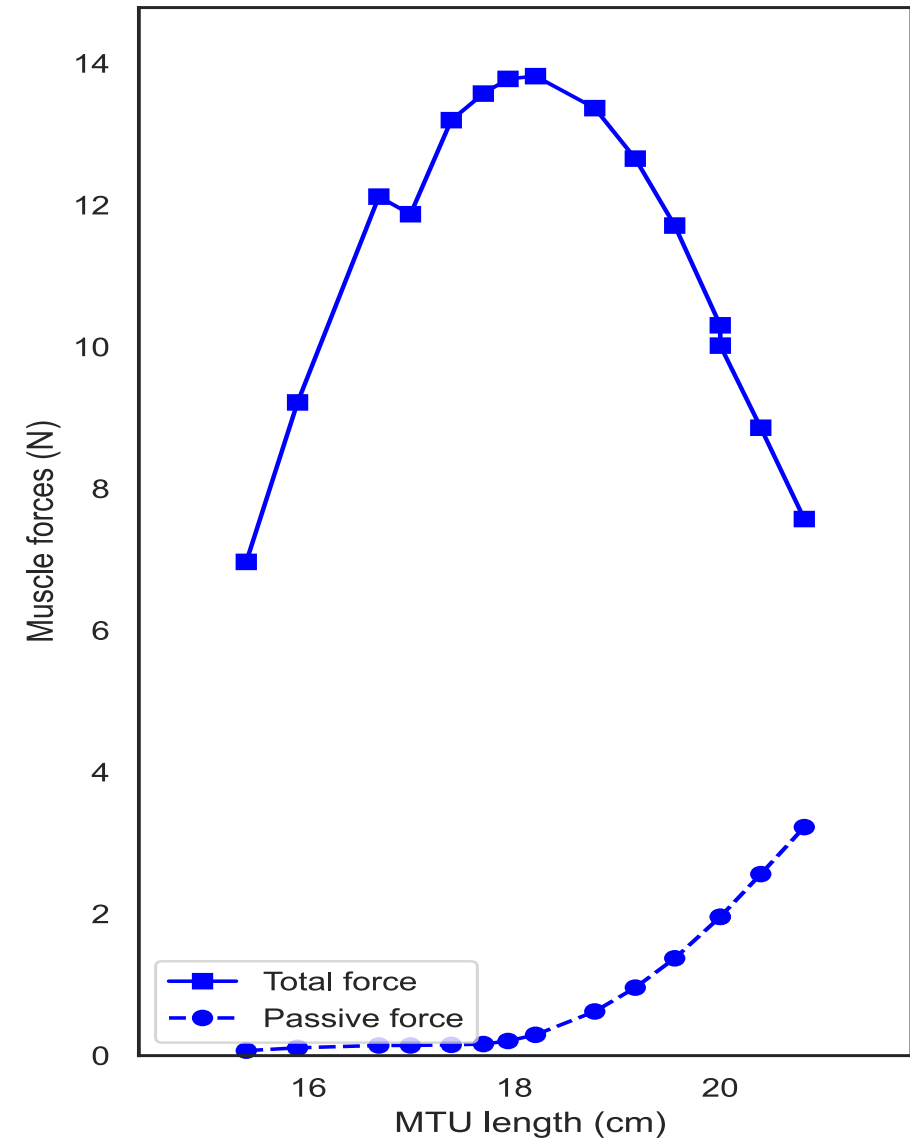
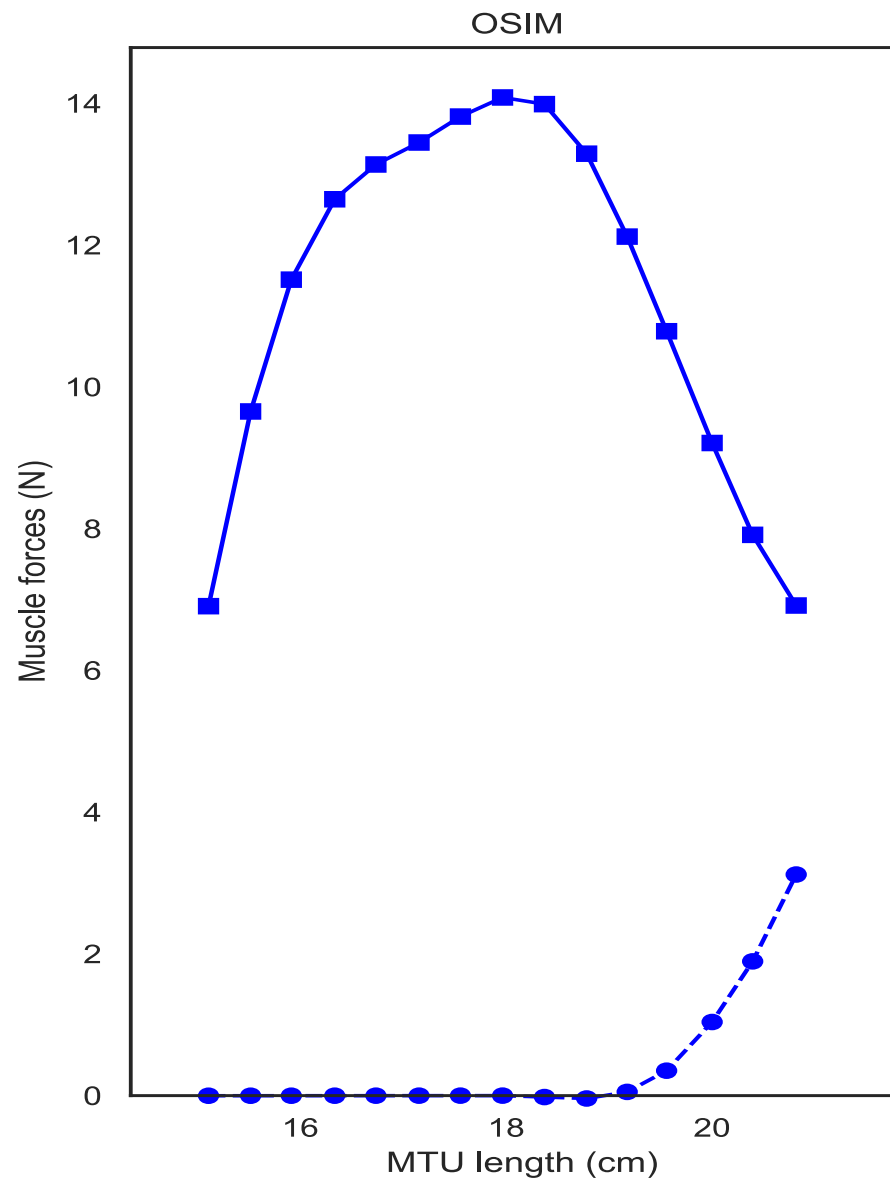
# Muscle specific force comparison before/after optimization

EIP



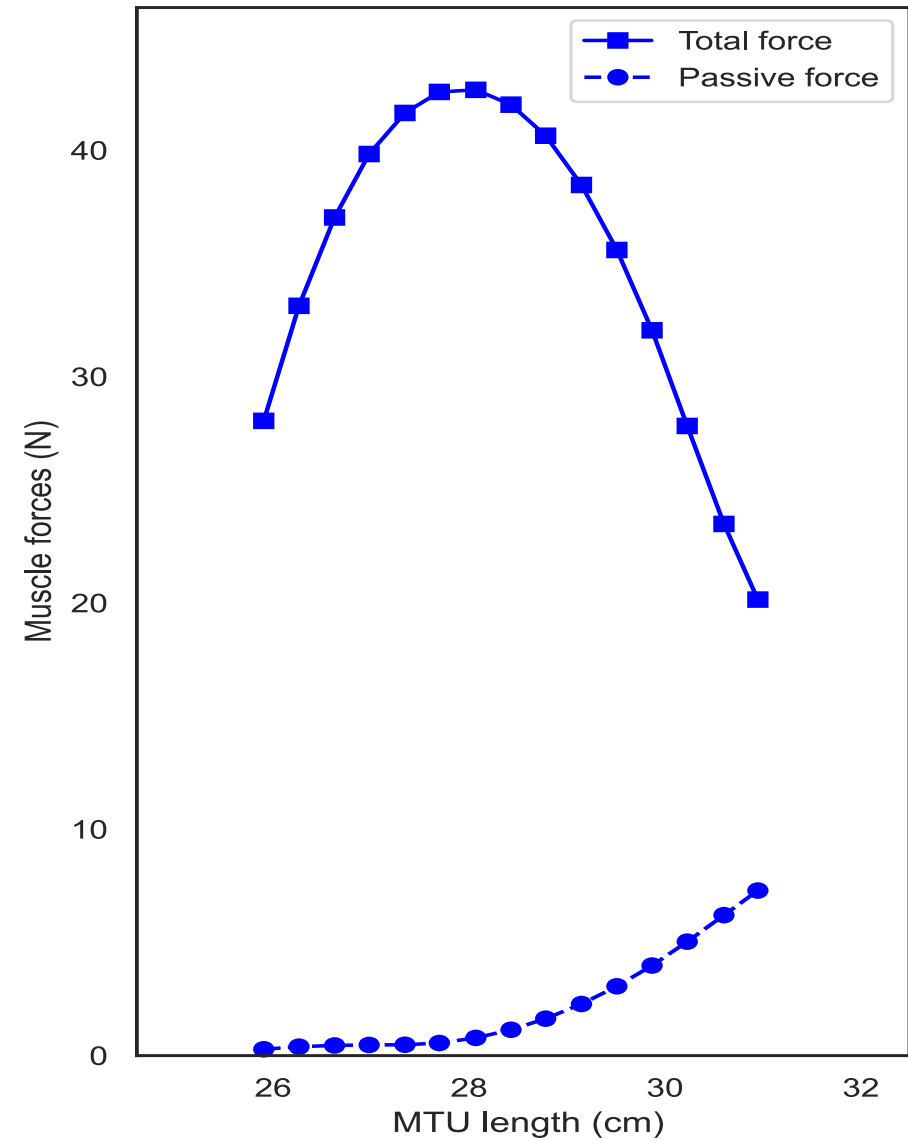
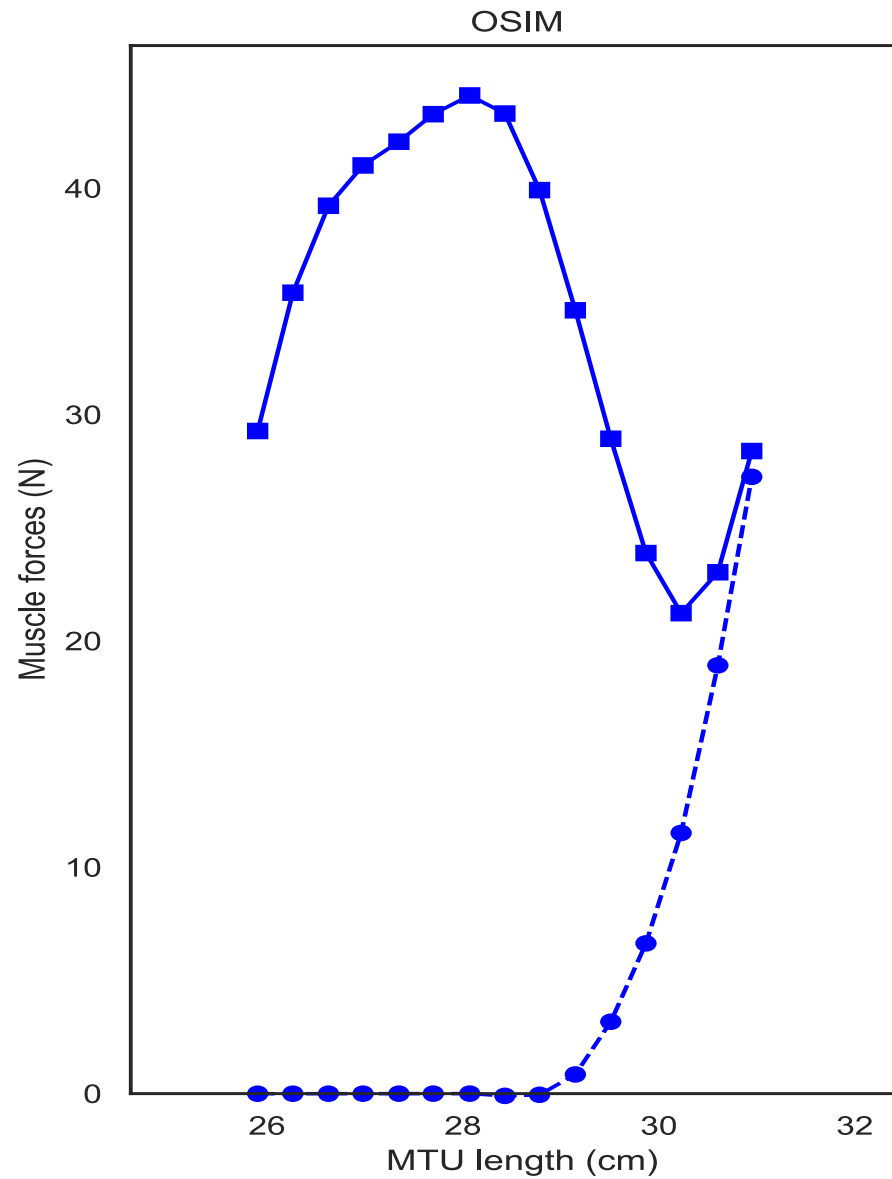
# Muscle specific force comparison before/after optimization

EPB



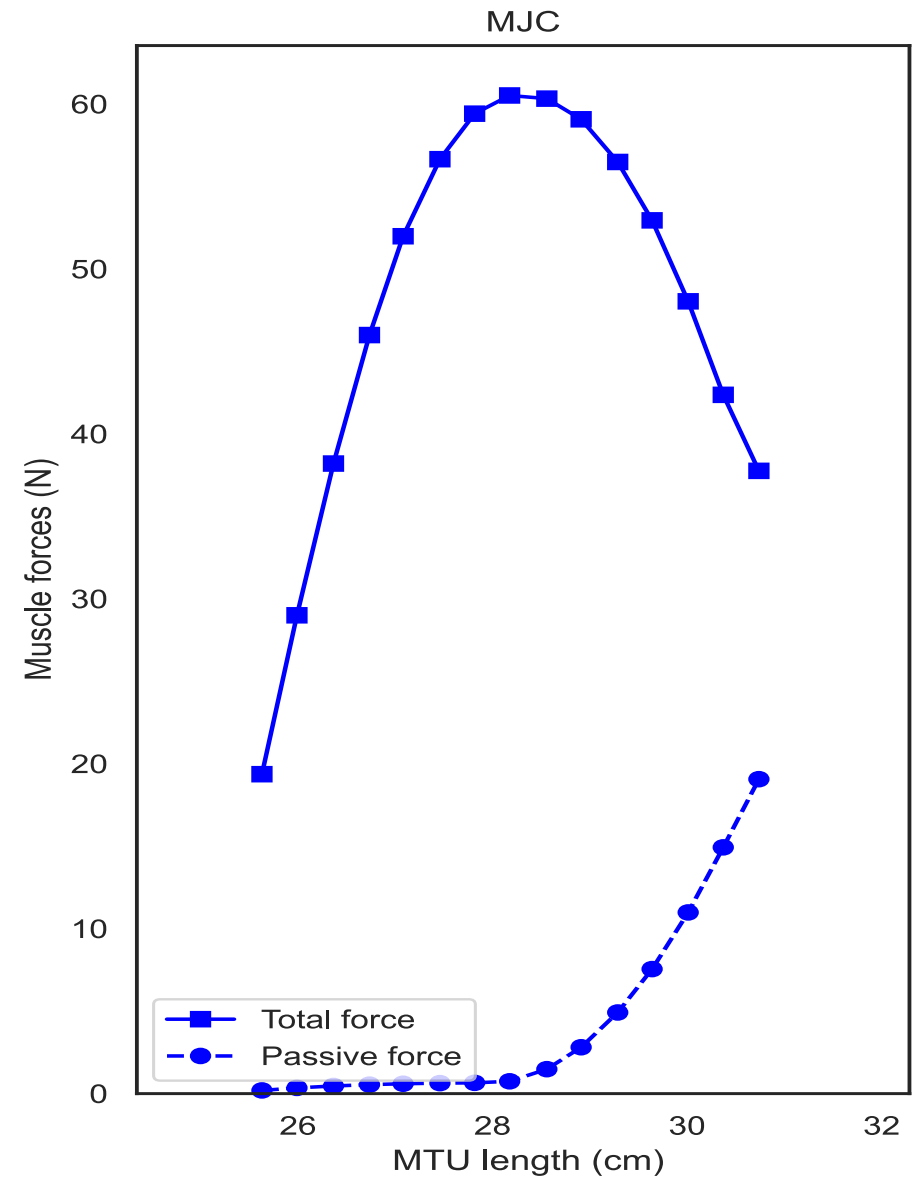
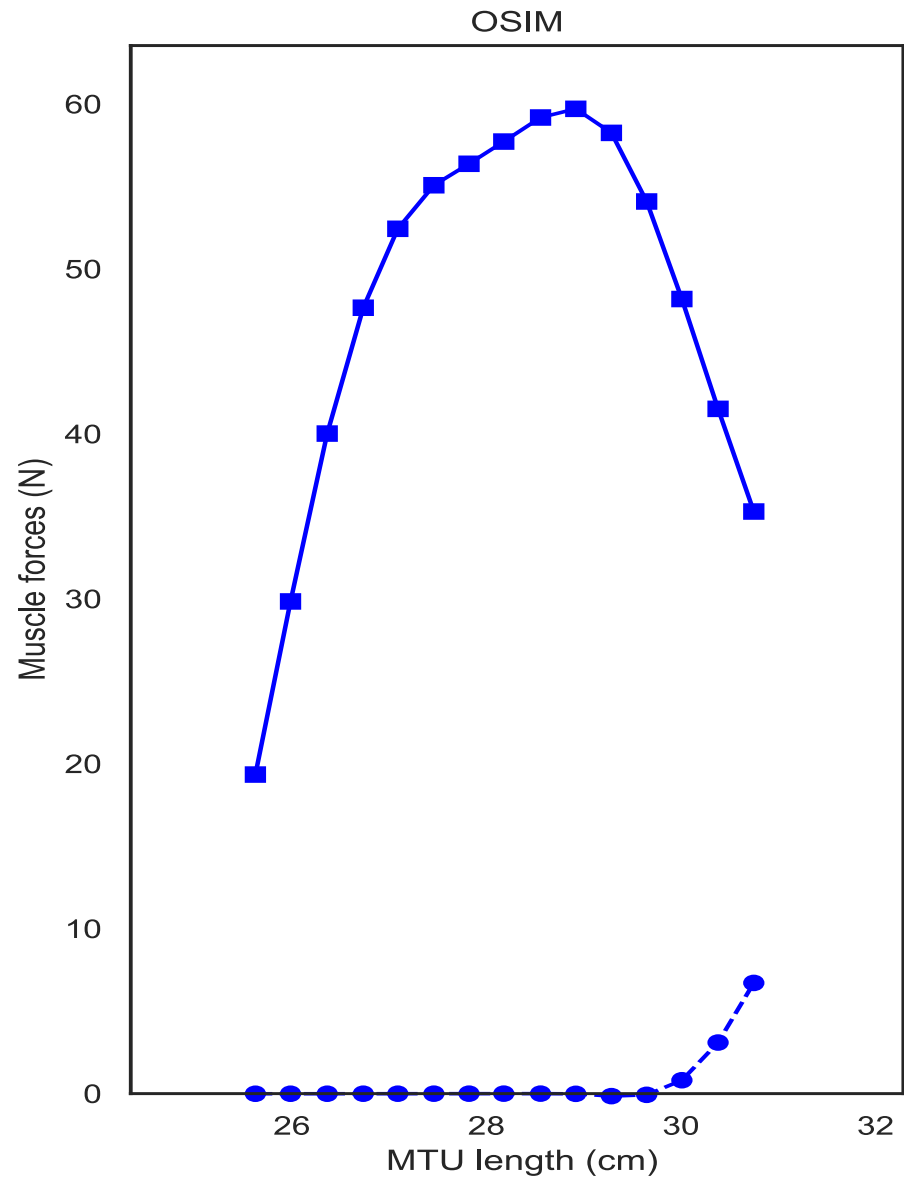
# Muscle specific force comparison before/after optimization

EPL



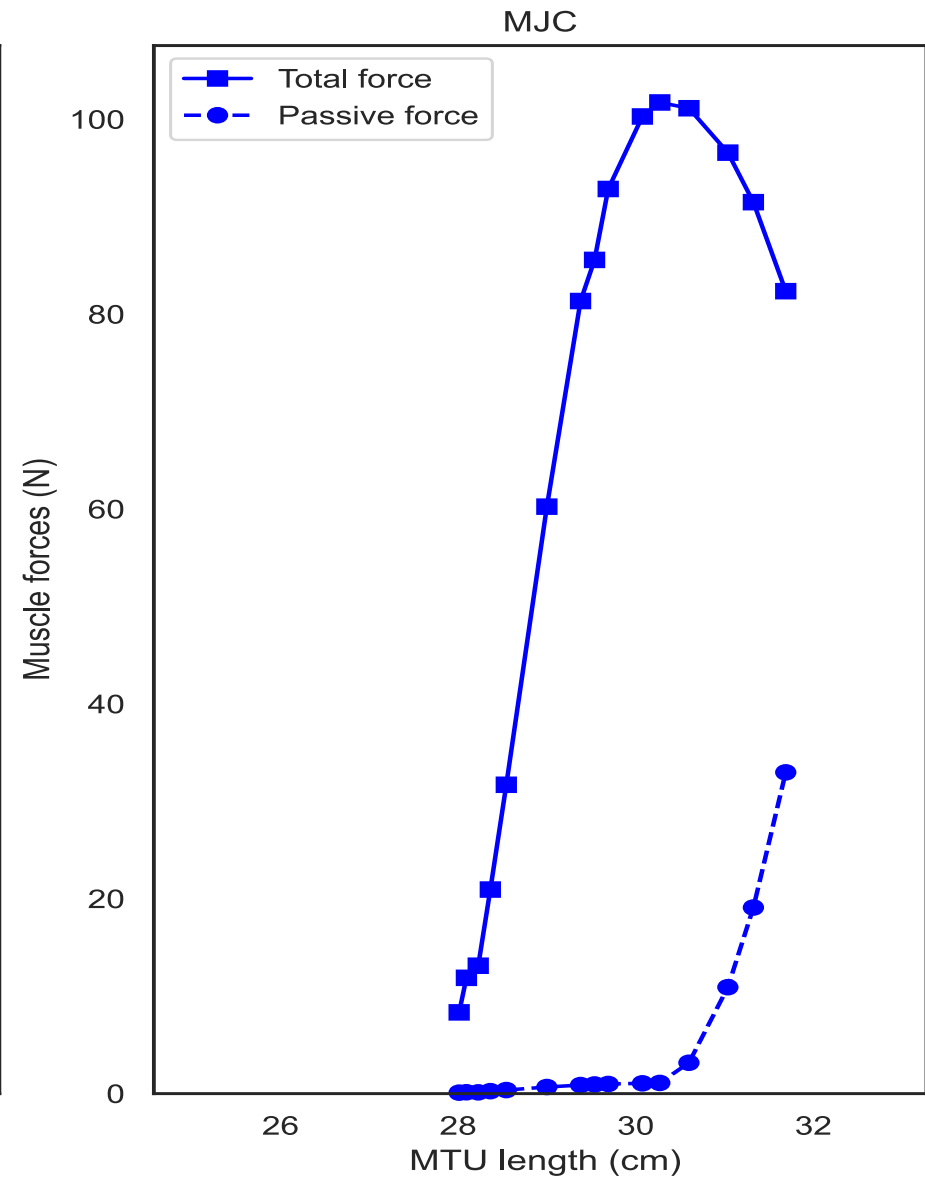
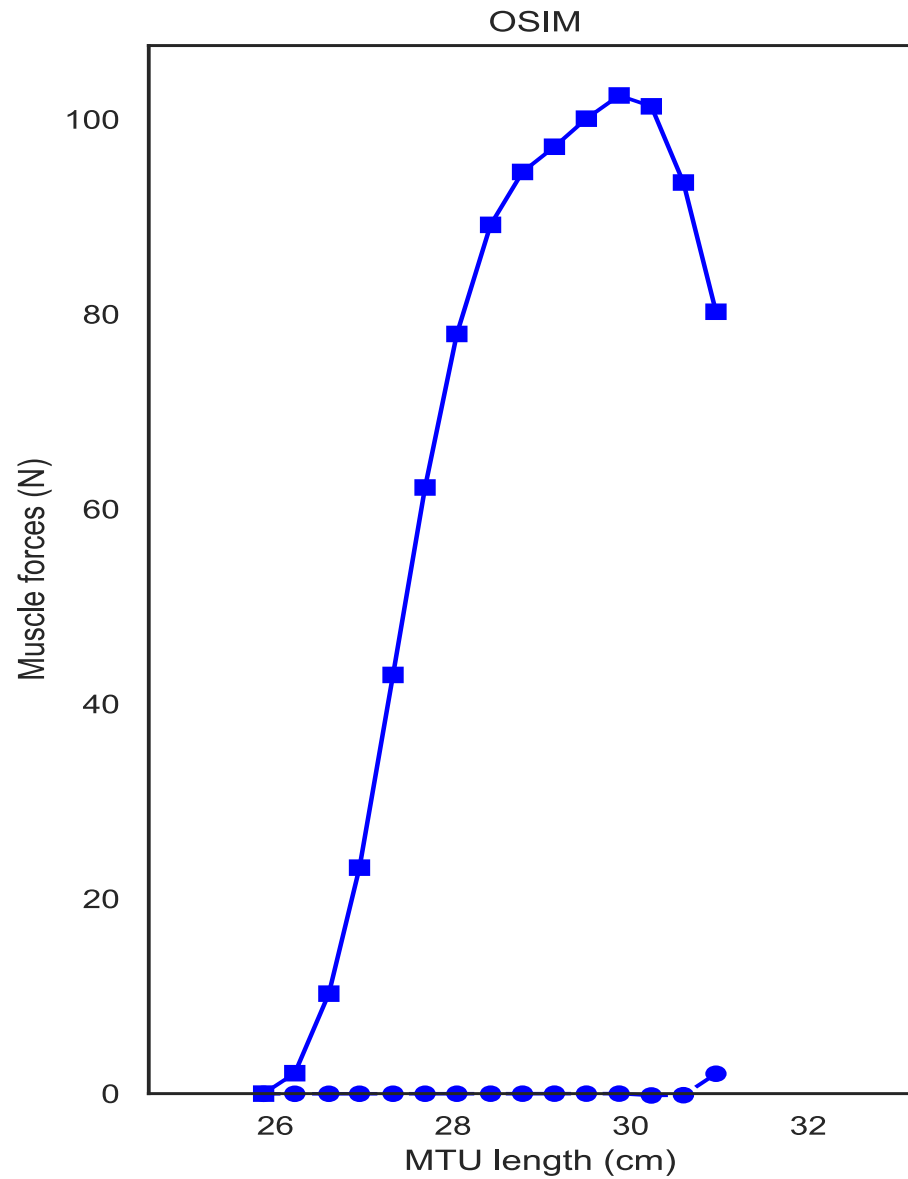
# Muscle specific force comparison before/after optimization

FCR



# Muscle specific force comparison before/after optimization

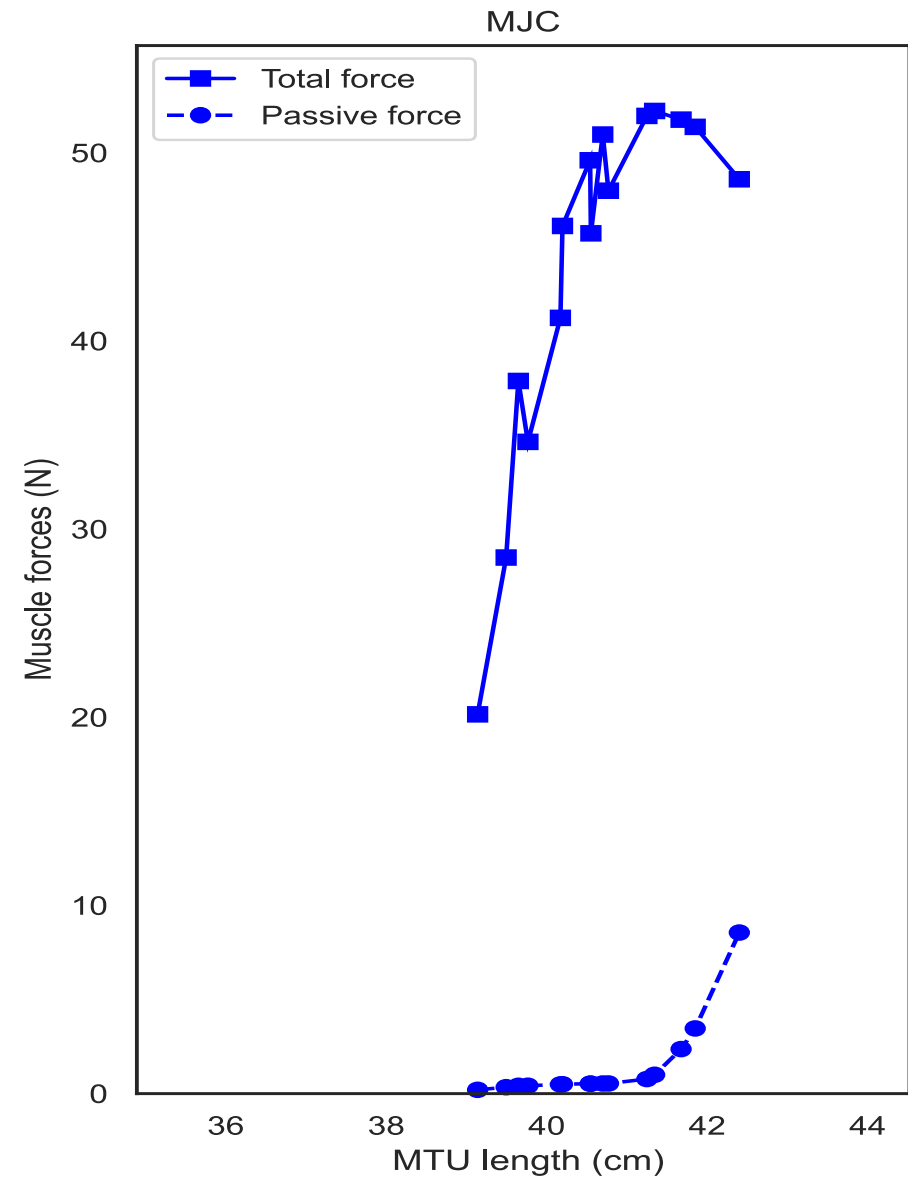
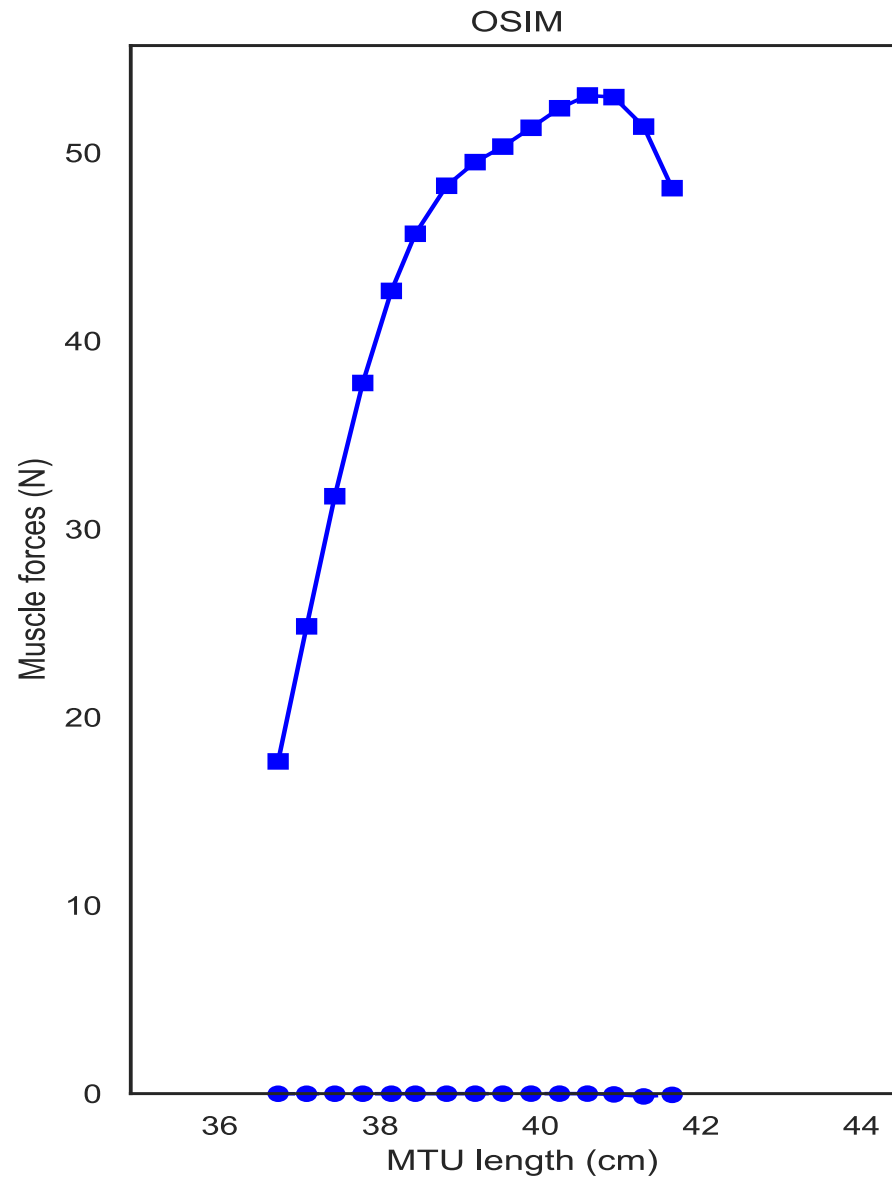
FCU





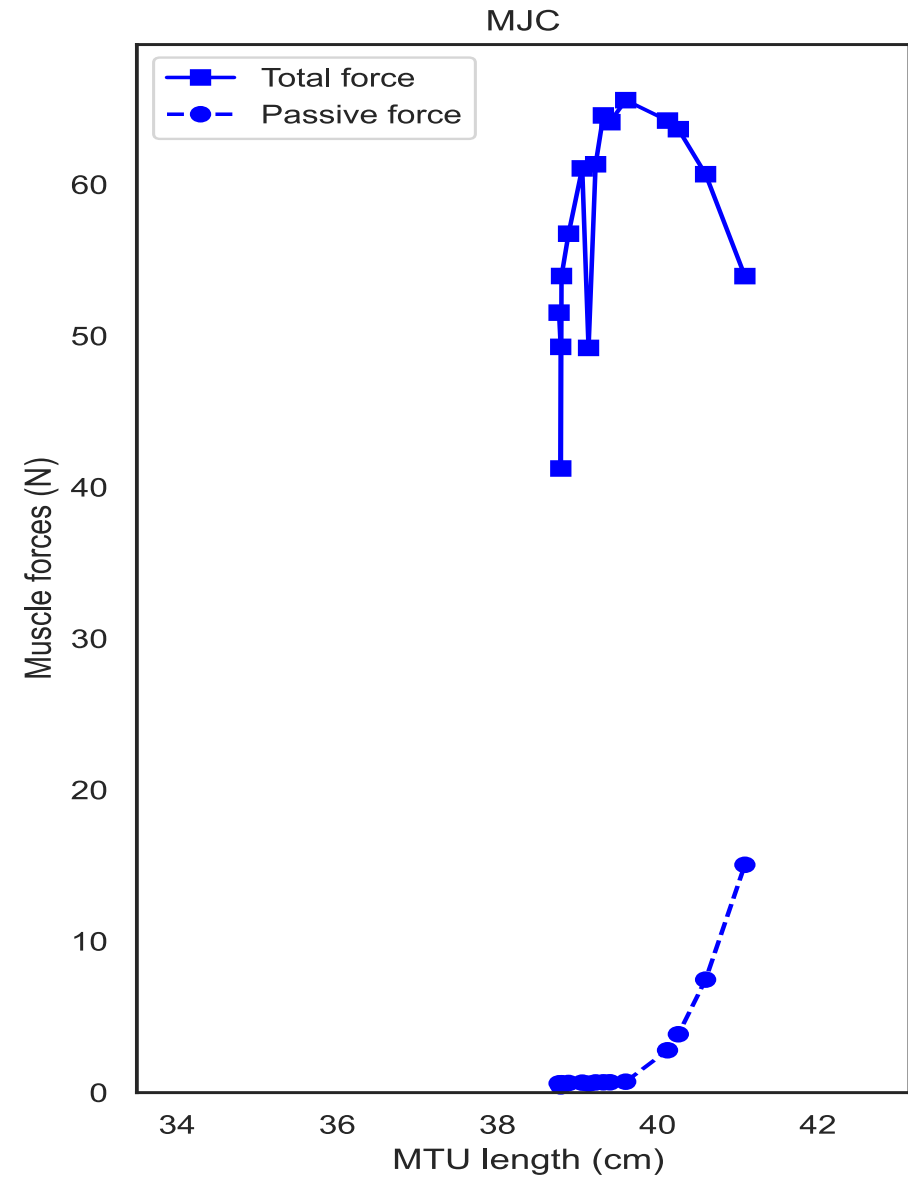
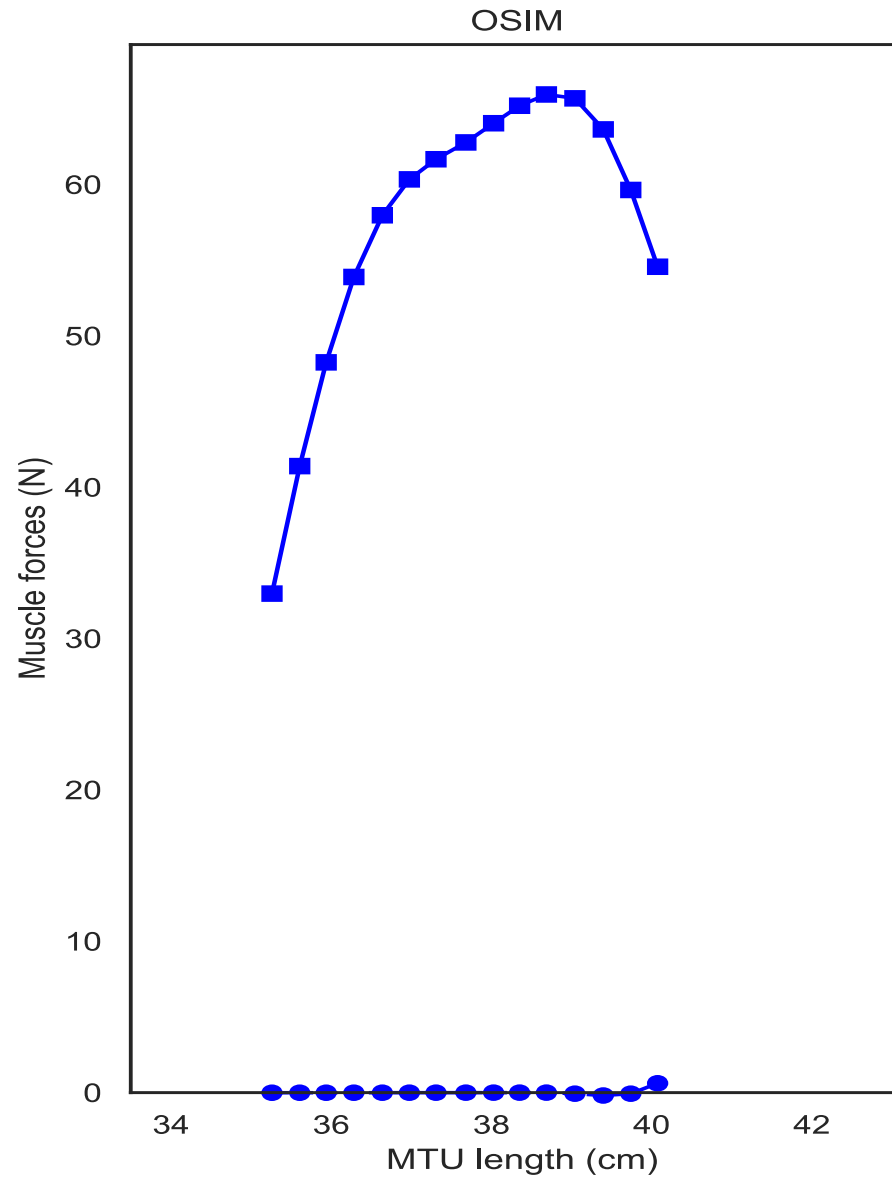
# Muscle specific force comparison before/after optimization

FDPI



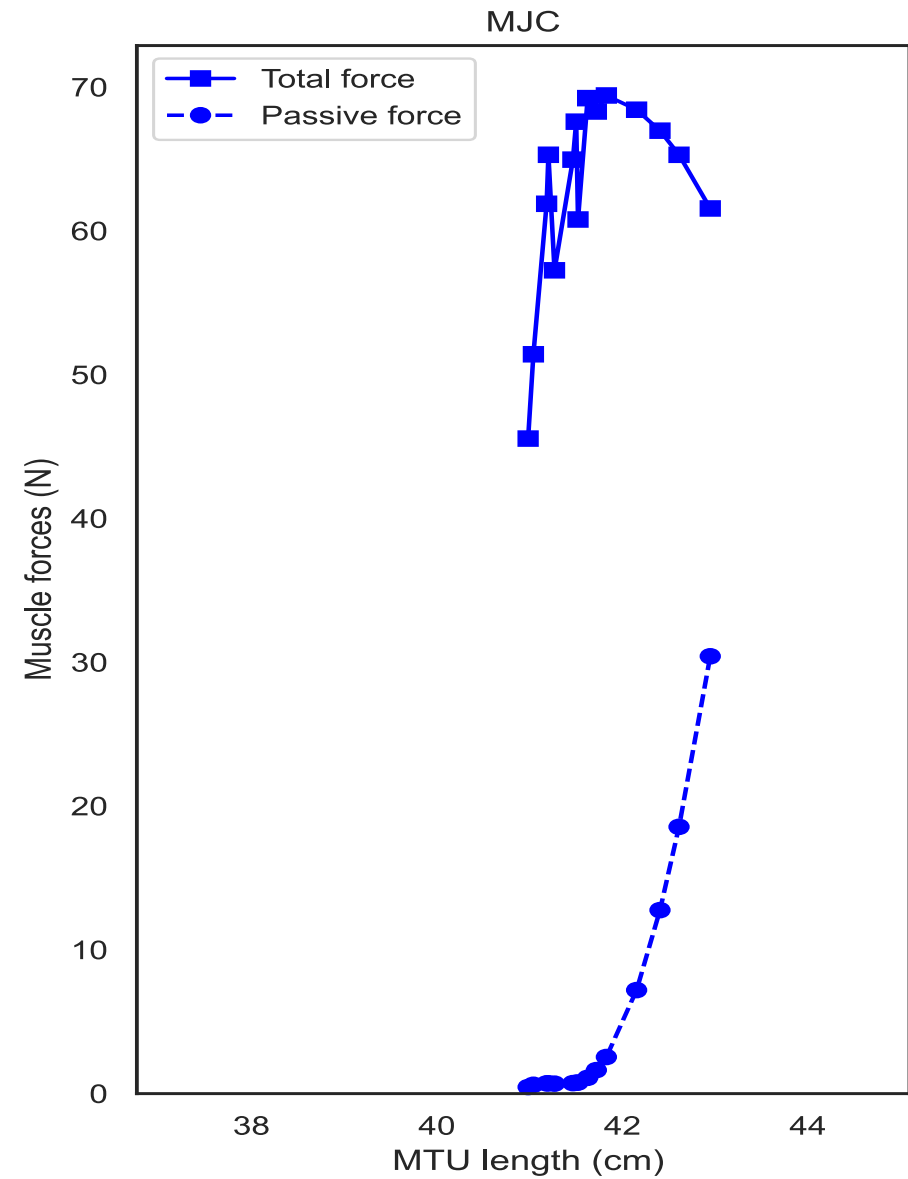
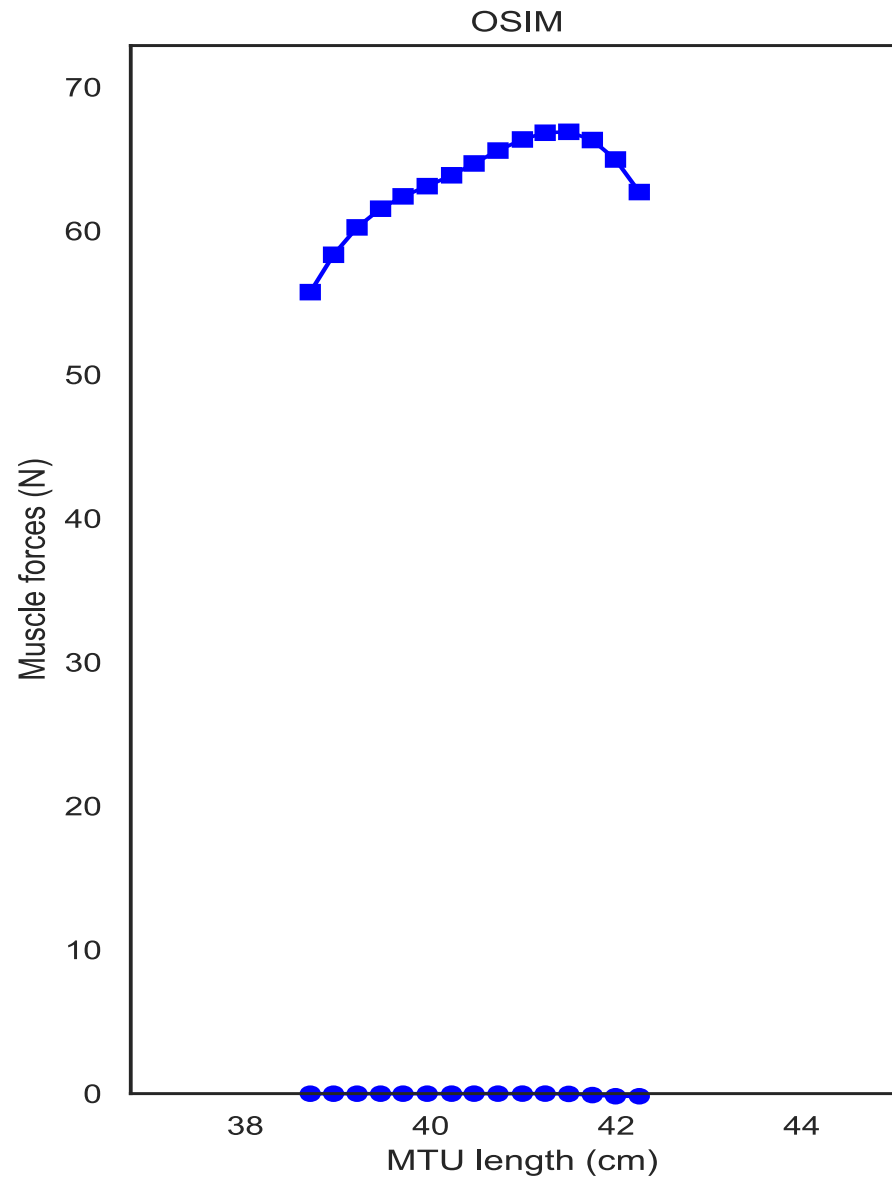
# Muscle specific force comparison before/after optimization

FDPL



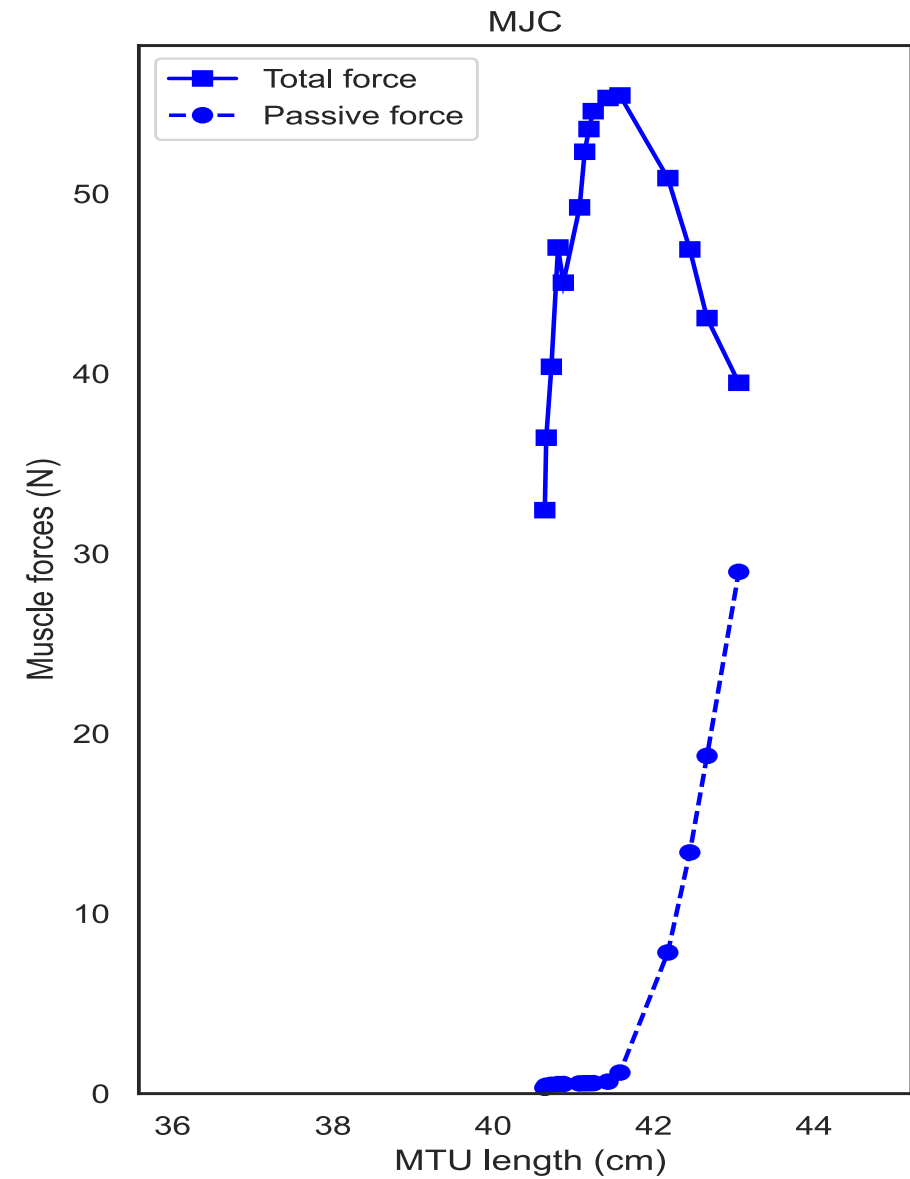
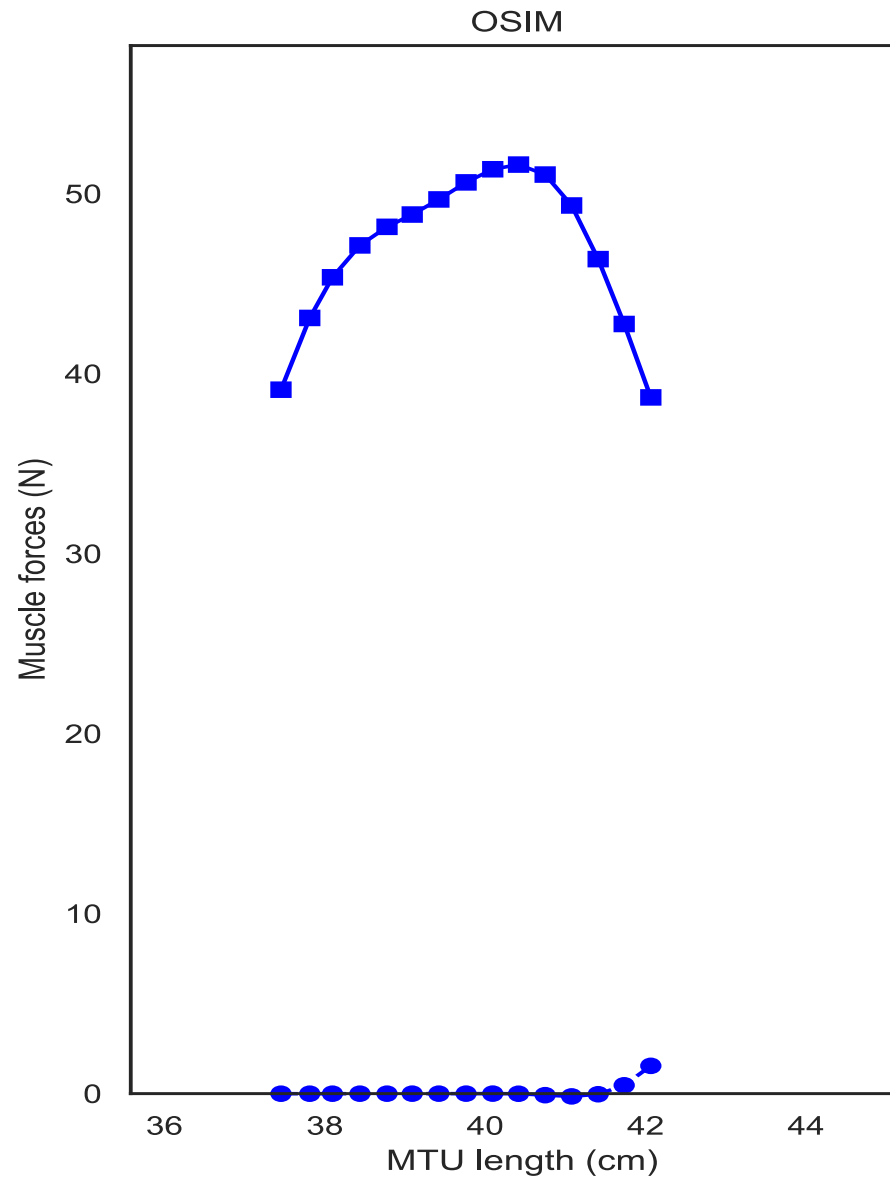
# Muscle specific force comparison before/after optimization

FDPM



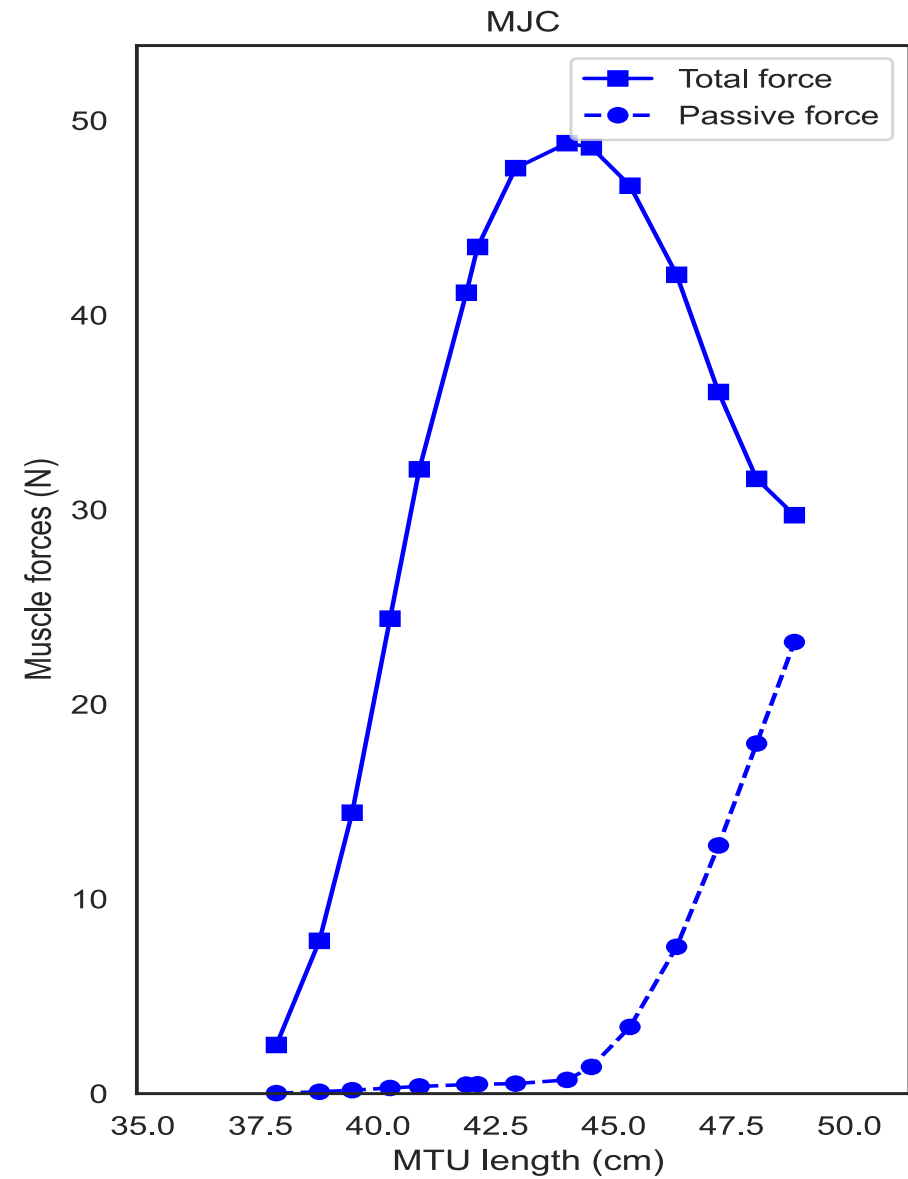
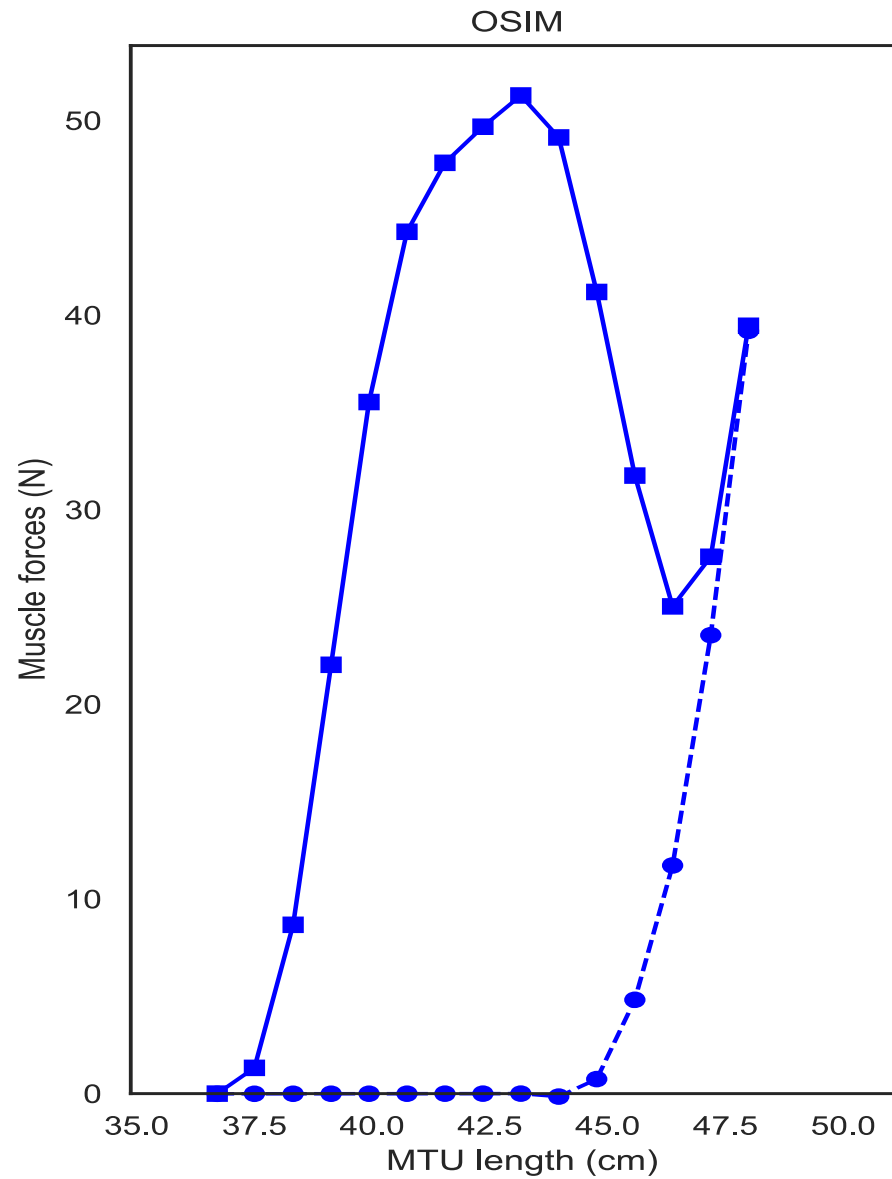
# Muscle specific force comparison before/after optimization

FDPR



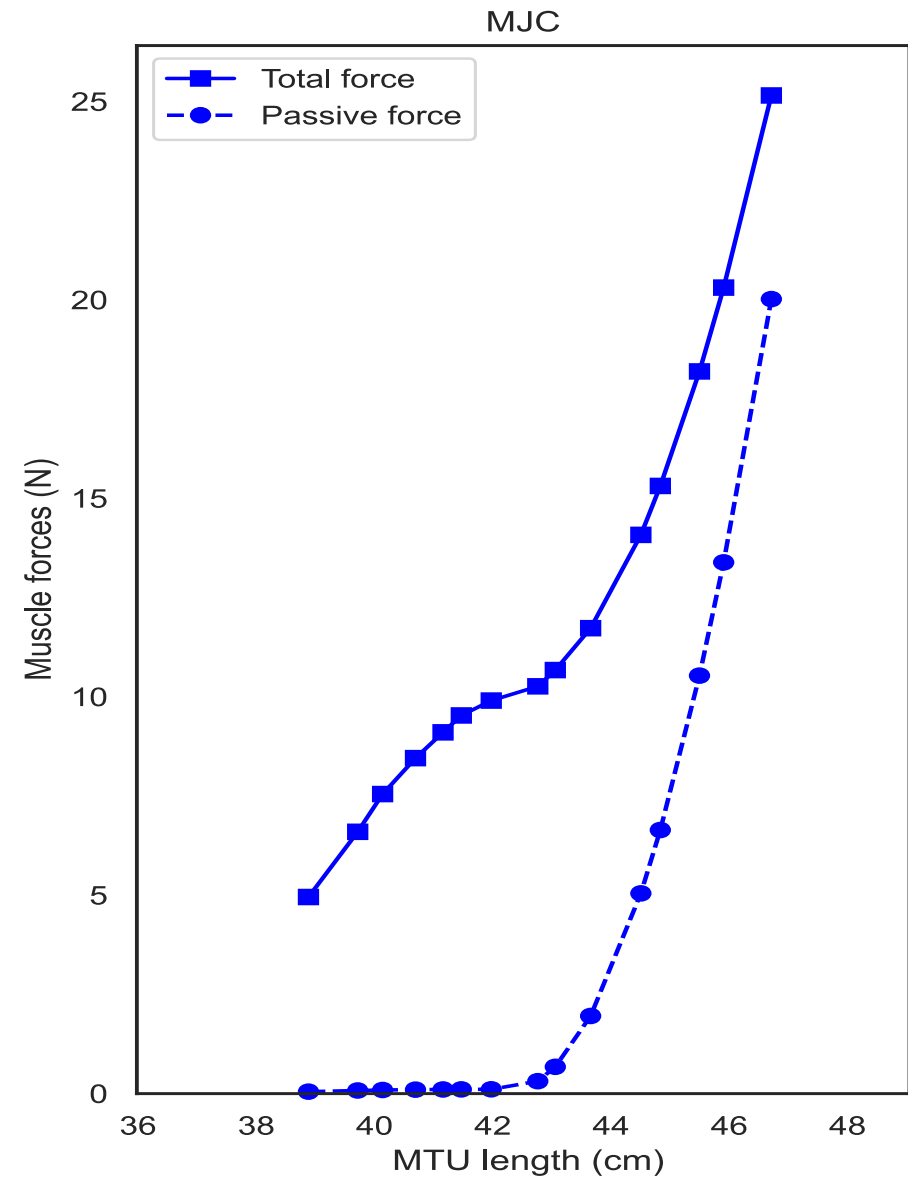
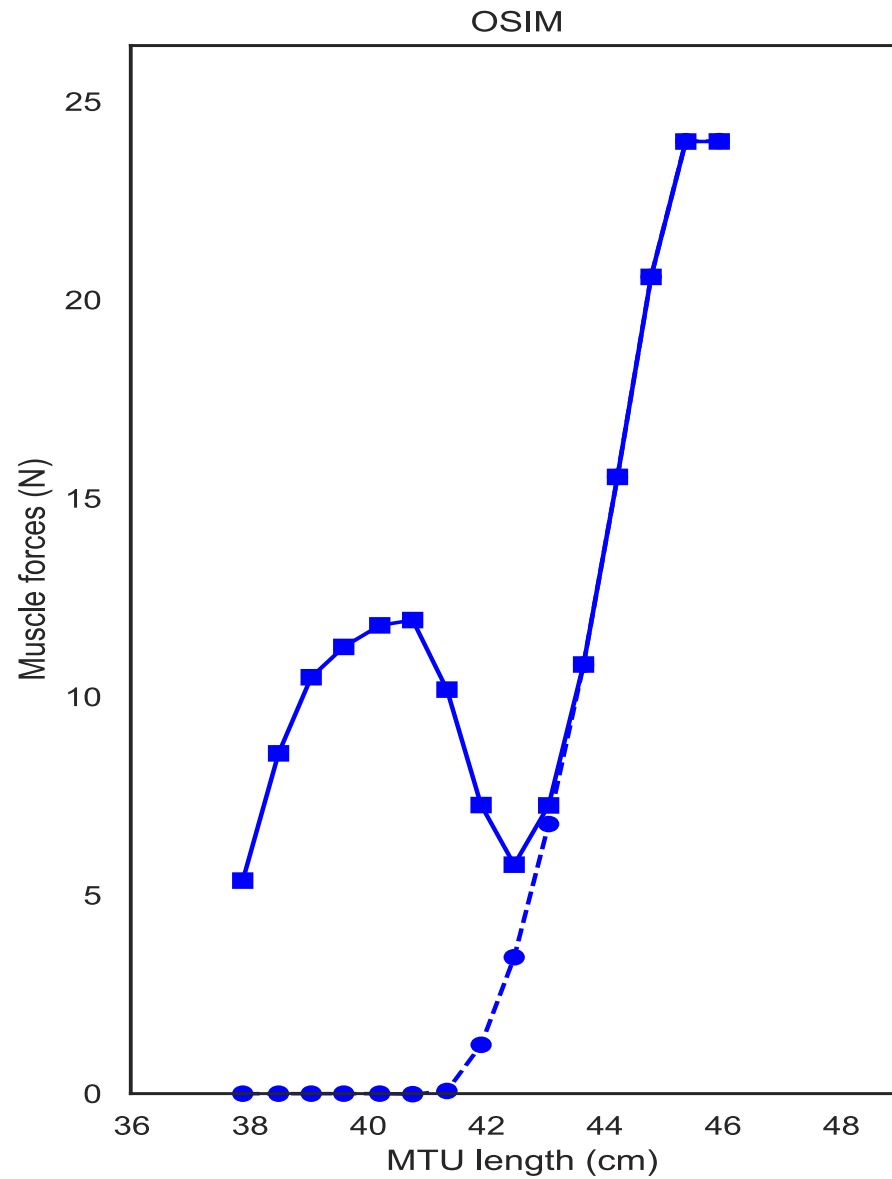
# Muscle specific force comparison before/after optimization

FDSI



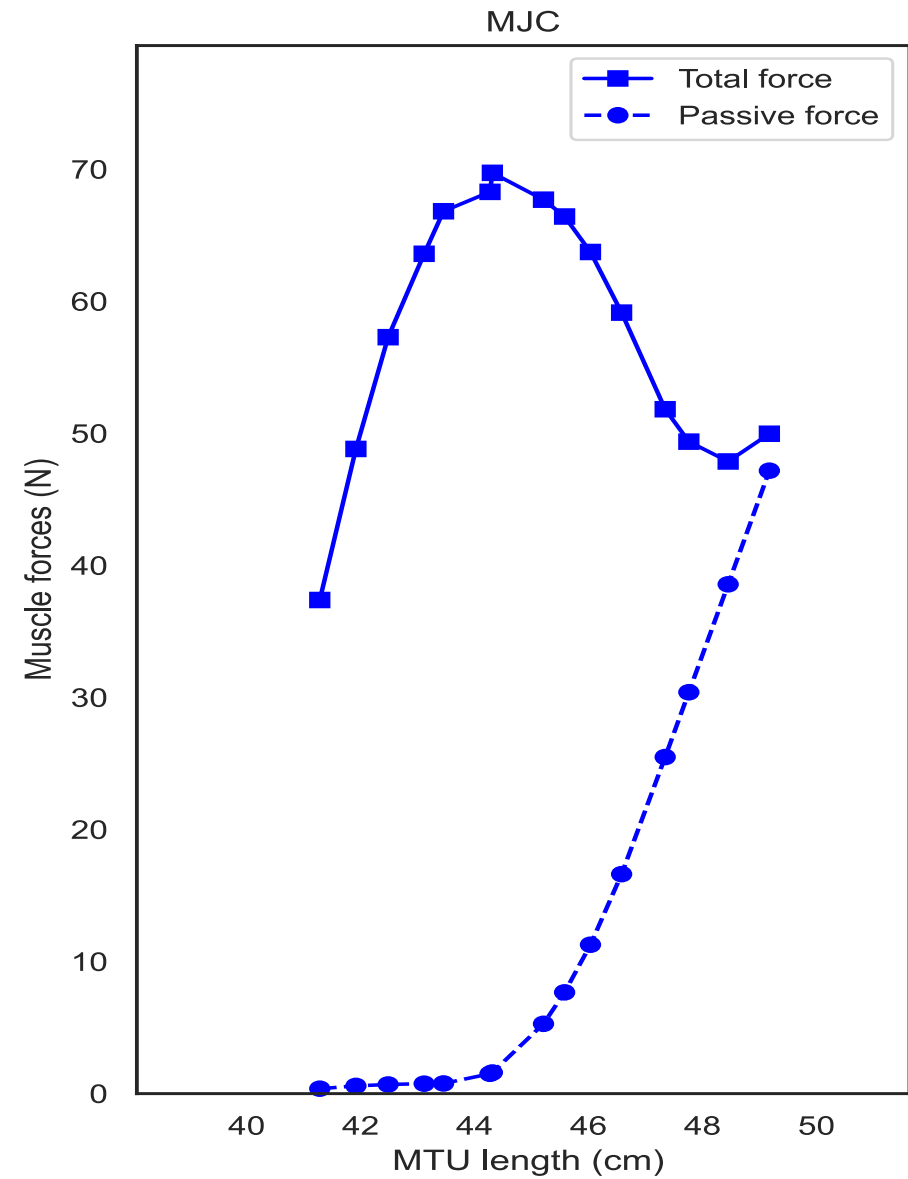
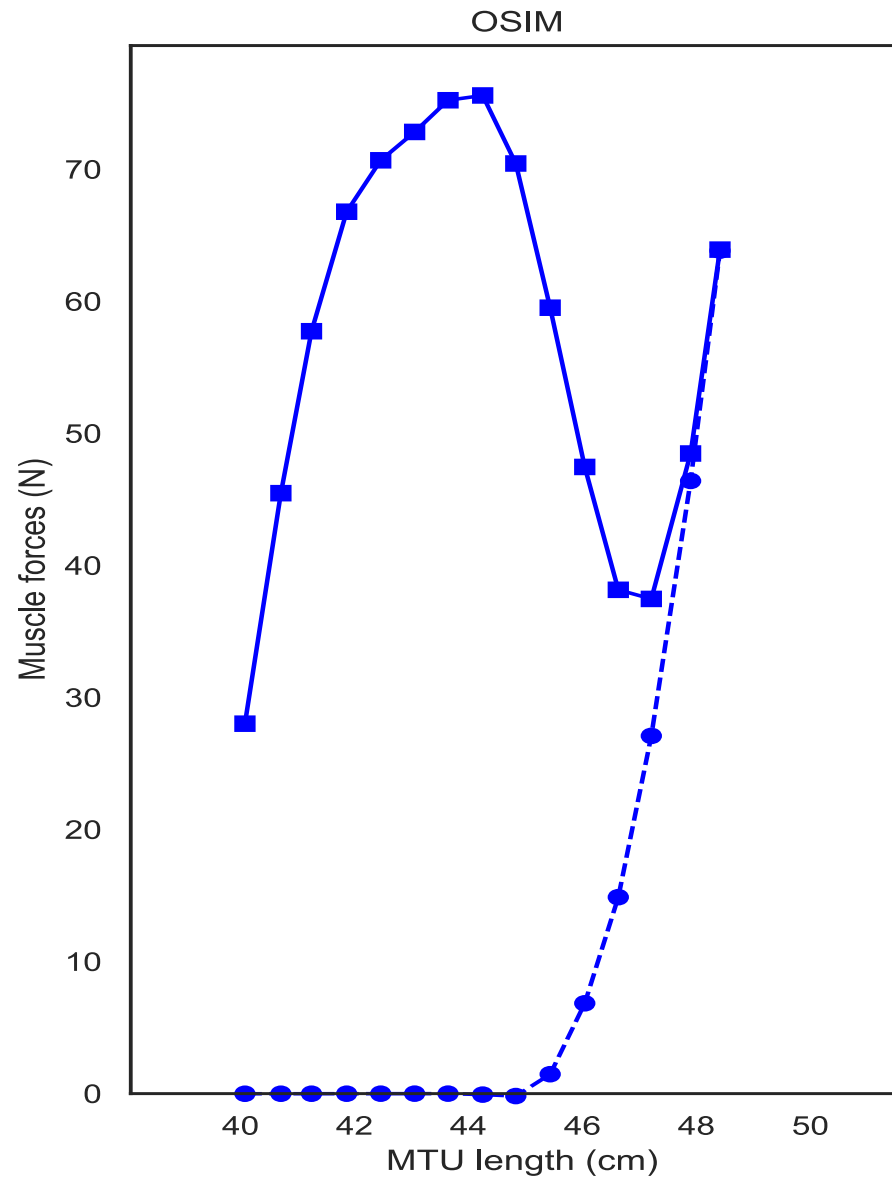
# Muscle specific force comparison before/after optimization

FDSL



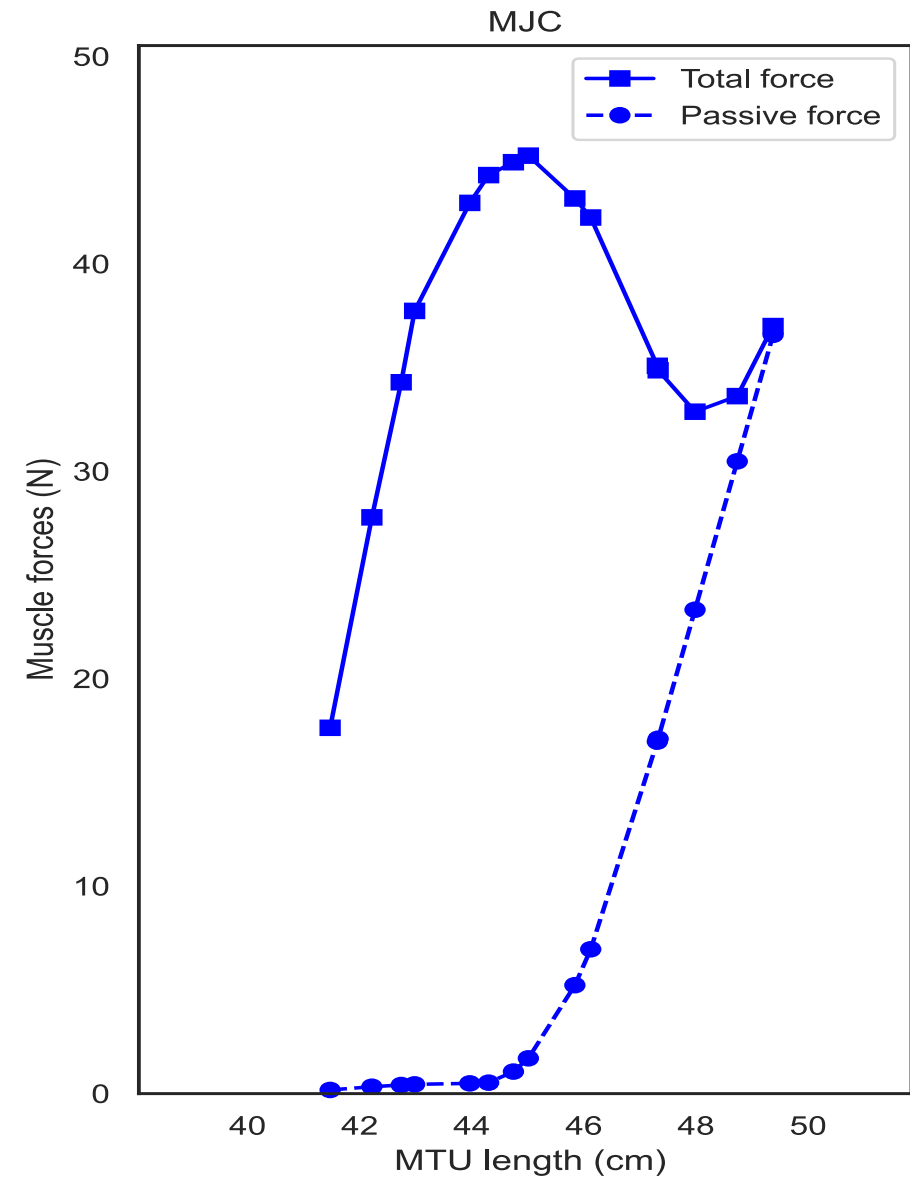
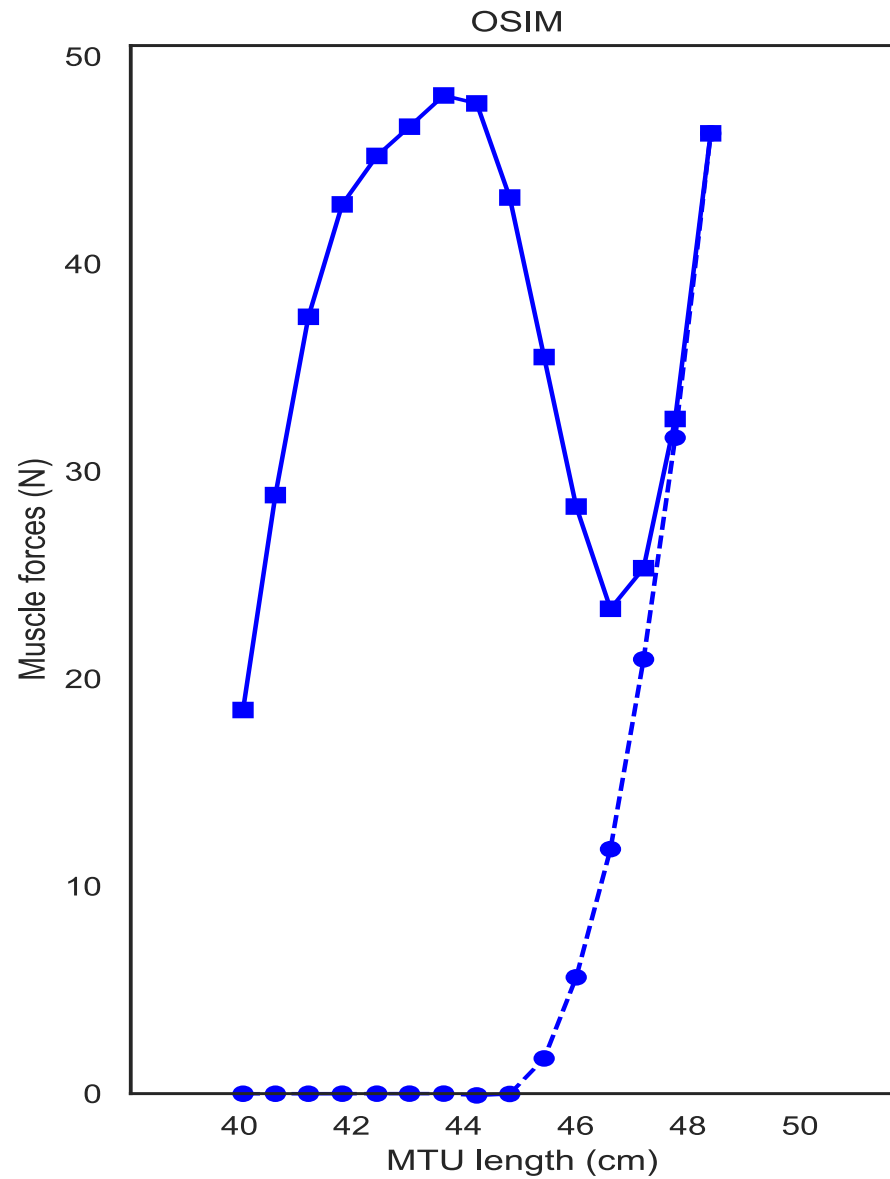
# Muscle specific force comparison before/after optimization

FDSM



# Muscle specific force comparison before/after optimization

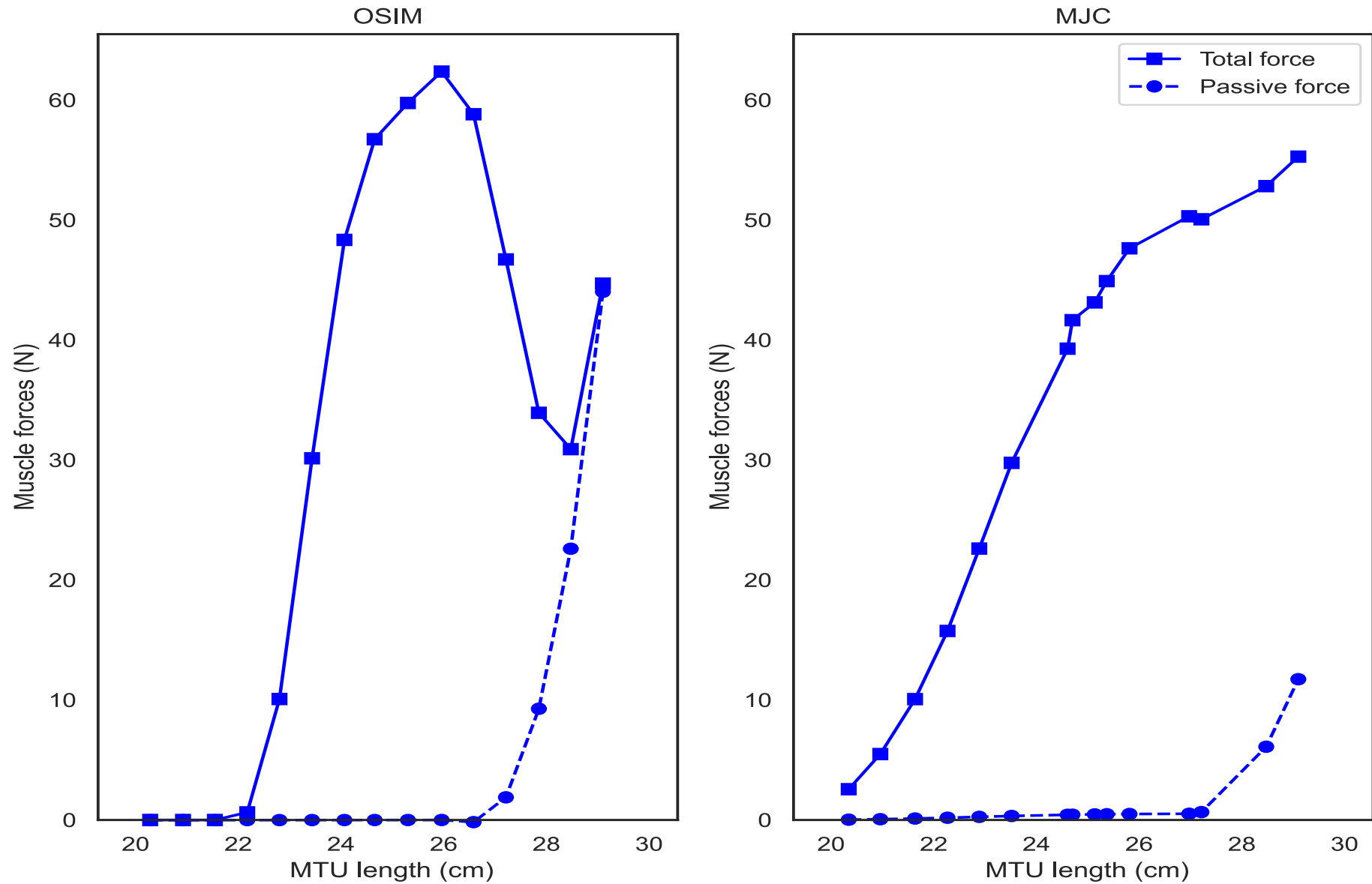
FDSR





# Muscle specific force comparison before/after optimization

FPL



# Muscle specific force comparison before/after optimization

PL

